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# TECNOLOGIE E PROGETTAZIONE DI SISTEMI INFORMATICI E DI TELECOMUNICAZIONI

1

Per l'articolazione Informatica degli Istituti Tecnici settore Tecnologico



## Soluzioni Eserciziario

## TPS

**Classe III - informatica**

**Prof. Gallina Roberto**

EDIZIONE OPENSCHOOL

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# Codifica dell'informazione


**1 Alfabeto sorgente e codice . . . . . 7**
1.1 Codifica lunghezza fissa e variabile

**2 Sistema binario . . . . . 13**
2.1 Base 2
2.2 Base 2 - Modulo e Segno
2.3 Base 2 - Complemento a 2

# 1

# Alfabeto sorgente e codice

## 1.1 Codifica lunghezza fissa e variabile

Dato il seguente alfabeto sorgente T e alfabeto codifica E,

- determinare qual è la lunghezza minima per una codifica a lunghezza fissa
- fornire una possibile codifica a lunghezza fissa
- fornire una possibile codifica a lunghezza variabile
- fornire una possibile codifica a lunghezza fissa raggruppando gli elementi in base a un criterio definito dallo studente

*Non è necessario specificare tutte le combinazioni possibili, ma è necessario far capire come è costruita la codifica*

1

**Esercizio guidato**
$T = \{LUN, MAR, MER, GIO, VEN\}$
$E = \{G, H\}$

Poichè $|T| = 5$ e $|E| = 2$, applicando la formula posso calcolare la lunghezza minima per una codifica a lunghezza fissa: $m = \lceil log_{|E|}(|T|) \rceil = \lceil log_2(5) \rceil \approx 2.3 \rightarrow 3$
Infatti $|E|^m = 2^3 = 8 > |T| = 5$

Posso proporre una possibile codifica a lunghezza fissa, in cui ogni codifica è lunghezza 3 caratteri dell'alfabeto codifica:

•LUN = GGG
•MAR = GGH
•MER = GHG
•GIO = GHH
•VEN = HGG

Posso proporre una possibile codifica a lunghezza varibile, in cui ogni codifica è lunghezza massima di 3 caratteri dell'alfabeto codifica:

•LUN = G
•MAR = GH
•MER = HG
•GIO = HH
•VEN = H

Posso proporre una possibile codifica a lunghezza fissa, dopo aver raggruppato gli elementi in base allo loro iniziale (M o altro). Ossia:

$$T_M = \{MAR, MER\} \qquad T_{ALTRO} = \{LUN, GIO, VEN\}$$

Avendo due gruppi avrà bisogno di $\lceil log_{|E|}(numero\ gruppi) \rceil = \lceil log_2(2) \rceil = 1$ cifre per rappresentare il gruppo; quindi:

•gli elementi del gruppo $T_M$ termineranno con $G$
•gli elementi del gruppo $T_{ALTRO}$ termineranno con $H$

Poichè il gruppo più grande ha 3 elementi avrò bisogno di ulteriori $\lceil log_2(3) \rceil \approx 1.58 \rightarrow 2$ cifre per rappresentare gli elementi del gruppo:

•LUN = GGH
•MAR = GGG
•MER = GHG
•GIO = GHH
•VEN = HGH

**2**

$T = \{1,\ 2,\ 3,\ 4,\ 5\}$
$E = \{A, B\}$

**SOLUZIONE** Soluzione non disponibile

**3**

$T = \{insieme\ dei\ primi\ cinque\ numeri\ naturali\}$
$E = \{A, B\}$

**SOLUZIONE** Soluzione non disponibile

**4**

$T = \{insieme\ dei\ colori\ primari\}$
$E = \{X, Y\}$

**SOLUZIONE** Soluzione non disponibile

**5**

$T = \{ALFA,\ BETA,\ GAMMA\}$
$E = \{P, Q\}$

**SOLUZIONE** Soluzione non disponibile

**6**

$T = \{insieme\ delle\ prime\ tre\ lettere\ dell'alfabeto\ greco\}$
$E = \{1, 2\}$

**SOLUZIONE** Soluzione non disponibile

**7**

$T = \{insieme\ dei\ numeri\ pari\ minori\ di\ 10\}$
$E = \{M, N\}$

**SOLUZIONE** Soluzione non disponibile

**8**

$T = \{A,\ B,\ C,\ D,\ E\}$
$E = \{1, 2\}$

**SOLUZIONE** Soluzione non disponibile

**9**

$T = \{insieme\ dei\ punti\ cardinali\ principali\}$
$E = \{N, S\}$

**SOLUZIONE** Soluzione non disponibile

**10**

$T = \{\rightarrow,\ \leftarrow,\ \uparrow,\ \downarrow\}$
$E = \{1, 2\}$

**SOLUZIONE** Soluzione non disponibile

**11**

$T = \{insieme\ delle\ frecce\ direzionali\ principali\}$
$E = \{A, B\}$

**SOLUZIONE** Soluzione non disponibile

**12** ●●

$T = \{1,\ 2,\ 3,\ 4,\ 5,\ 6,\ 7\}$
$E = \{A,B,C\}$

**SOLUZIONE** Soluzione non disponibile

**13** ●●

$T = \{insieme\ dei\ giorni\ della\ settimana\ esclusa\ domenica\}$
$E = \{G,H,I\}$

**SOLUZIONE** Soluzione non disponibile

**14** ●●

$T = \{insieme\ dei\ colori\ del\ semaforo\}$
$E = \{X,Y,Z\}$

**SOLUZIONE** Soluzione non disponibile

**15** ●●

$T = \{10,\ 20,\ 30,\ 40,\ 50,\ 60\}$
$E = \{M,N,O\}$

**SOLUZIONE** Soluzione non disponibile

**16** ●●

$T = \{insieme\ delle\ prime\ sette\ lettere\ dell'alfabeto\ latino\}$
$E = \{1,2,3\}$

**SOLUZIONE** Soluzione non disponibile

**17** ●●

$T = \{\alpha,\ \beta,\ \gamma,\ \delta,\ \varepsilon\}$
$E = \{1,2,3\}$

**SOLUZIONE** Soluzione non disponibile

**18** ●●

$T = \{insieme\ dei\ primi\ quattro\ numeri\ dispari\}$
$E = \{A,B,C\}$

**SOLUZIONE** Soluzione non disponibile

**19** ●●

$T = \{\infty,\ \emptyset,\ \forall\}$
$E = \{X,x\}$

**SOLUZIONE** Soluzione non disponibile

**20** ●●

$T = \{insieme\ dei\ multipli\ di\ 2\ minori\ di\ 12\}$
$E = \{P,Q\}$

**SOLUZIONE** Soluzione non disponibile

**21** ●●

$T = \{\rightarrow,\ \leftarrow,\ \uparrow,\ \downarrow,\ \nearrow\}$
$E = \{1,2,3\}$

**SOLUZIONE** Soluzione non disponibile

**22** ●●

$T = \{insieme\ dei\ simboli\ logici\ di\ base\}$
$E = \{A, B, C\}$

**SOLUZIONE** Soluzione non disponibile

**23** ●●

$T = \{\infty,\ \emptyset,\ \forall,\ \exists\}$
$E = \{X, Y, Z\}$

**SOLUZIONE** Soluzione non disponibile

**24** ●●

$T = \{insieme\ dei\ risultati\ possibili\ di\ un\ dado\}$
$E = \{1, 2, 3\}$

**SOLUZIONE** Soluzione non disponibile

**25** ●●

$T = \{insieme\ delle\ vocali\ e\ delle\ prime\ due\ consonanti\}$
$E = \{A, B, C\}$

**SOLUZIONE** Soluzione non disponibile

**26** ●●●

$T = \{insieme\ dei\ colori\ dell'arcobaleno\}$
$E = \{X, Y, Z, W\}$

**SOLUZIONE** Soluzione non disponibile

**27** ●●●

$T = \{insieme\ delle\ prime\ nove\ lettere\ dell'alfabeto\ latino\}$
$E = \{1, 2, 3, 4\}$

**SOLUZIONE** Soluzione non disponibile

**28** ●●●

$T = \{\alpha,\ \beta,\ \gamma,\ \delta,\ \varepsilon,\ \zeta,\ \eta\}$
$E = \{1, 2, 3, 4\}$

**SOLUZIONE** Soluzione non disponibile

**29** ●●●

$T = \{insieme\ dei\ primi\ cinque\ numeri\ dispari\}$
$E = \{A, B, C, D\}$

**SOLUZIONE** Soluzione non disponibile

**30** ●●●

$T = \{insieme\ dei\ mesi\ dell'anno\}$
$E = \{X, Y, Z, W\}$

**SOLUZIONE** Soluzione non disponibile

**31** ●●●

$T = \{insieme\ dei\ numeri\ naturali\ minori\ di\ 10\}$
$E = \{G, H, I, J\}$

**SOLUZIONE** Soluzione non disponibile

**32**

$T = \{insieme\ dei\ multipli\ di\ 3\ minori\ di\ 25\}$
$E = \{P, Q, R, S\}$

**SOLUZIONE** Soluzione non disponibile

**33**

$T = \{\rightarrow, \leftarrow, \uparrow, \downarrow, \nearrow, \searrow, \swarrow\}$
$E = \{1, 2, 3, 4\}$

**SOLUZIONE** Soluzione non disponibile

**34**

$T = \{\infty, \emptyset, \forall, \exists, \neg, \wedge\}$
$E = \{X, Y, Z, W\}$

**SOLUZIONE** Soluzione non disponibile

**35**

$T = \{insieme\ di\ tutte\ le\ lettere\ dell'alfabeto\ italiano\}$
$E = \{A, B, C, D\}$

**SOLUZIONE** Soluzione non disponibile

# 2

# Sistema binario

## 2.1 Base 2

Convertire i seguenti numeri dalla base 10 alla base 2:

1 **Esercizio guidato**

80

$80_{10} = 1010000_2$

$80 \div 2 = 40$ con resto 0
$40 \div 2 = 20$ con resto 0
$20 \div 2 = 10$ con resto 0
$10 \div 2 = 5$ con resto 0
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

2 2

**SOLUZIONE**

$2_{10} = 10_2$

$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

3 3

**SOLUZIONE**

$3_{10} = 11_2$

$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**4** 4

**SOLUZIONE**

$4_{10} = 100_2$
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**5** 5

**SOLUZIONE**

$5_{10} = 101_2$
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**6** 6

**SOLUZIONE**

$6_{10} = 110_2$
$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**7** 7

**SOLUZIONE**

$7_{10} = 111_2$
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**8** 8

**SOLUZIONE**

$8_{10} = 1000_2$
$8 \div 2 = 4$ con resto 0
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**9** 9

**SOLUZIONE**

$9_{10} = 1001_2$
$9 \div 2 = 4$ con resto 1
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**10** 10

**SOLUZIONE**

$10_{10} = 1010_2$
$10 \div 2 = 5$ con resto 0
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**11** 11

**SOLUZIONE**

$11_{10} = 1011_2$
$11 \div 2 = 5$ con resto 1
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**12** 12

**SOLUZIONE**

$12_{10} = 1100_2$
$12 \div 2 = 6$ con resto 0
$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**13** 13

**SOLUZIONE**

$13_{10} = 1101_2$
$13 \div 2 = 6$ con resto 1
$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**14** 14

**SOLUZIONE**

$14_{10} = 1110_2$
$14 \div 2 = 7$ con resto 0
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**15** 15

**SOLUZIONE**

$15_{10} = 1111_2$

$15 \div 2 = 7$ con resto 1
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**16** ●

20

**SOLUZIONE**

$20_{10} = 10100_2$
$20 \div 2 = 10$ con resto 0
$10 \div 2 = 5$ con resto 0
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**17** ●●

18

**SOLUZIONE**

$18_{10} = 10010_2$
$18 \div 2 = 9$ con resto 0
$9 \div 2 = 4$ con resto 1
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**18** ●●

23

**SOLUZIONE**

$23_{10} = 10111_2$
$23 \div 2 = 11$ con resto 1
$11 \div 2 = 5$ con resto 1
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**19** ●●

35

**SOLUZIONE**

$35_{10} = 100011_2$
$35 \div 2 = 17$ con resto 1
$17 \div 2 = 8$ con resto 1
$8 \div 2 = 4$ con resto 0
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**20** ●●

41

**SOLUZIONE**

$41_{10} = 101001_2$
$41 \div 2 = 20$ con resto 1
$20 \div 2 = 10$ con resto 0
$10 \div 2 = 5$ con resto 0
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**21** ●●

52

**SOLUZIONE**

$52_{10} = 110100_2$
$52 \div 2 = 26$ con resto 0
$26 \div 2 = 13$ con resto 0
$13 \div 2 = 6$ con resto 1
$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**22** ●●

58

**SOLUZIONE**

$58_{10} = 111010_2$
$58 \div 2 = 29$ con resto 0
$29 \div 2 = 14$ con resto 1
$14 \div 2 = 7$ con resto 0
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**23** ●●

67

**SOLUZIONE**

$67_{10} = 1000011_2$
$67 \div 2 = 33$ con resto 1
$33 \div 2 = 16$ con resto 1
$16 \div 2 = 8$ con resto 0
$8 \div 2 = 4$ con resto 0
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**24** ●●

74

**SOLUZIONE**

$74_{10} = 1001010_2$
$74 \div 2 = 37$ con resto 0
$37 \div 2 = 18$ con resto 1
$18 \div 2 = 9$ con resto 0

$9 \div 2 = 4$ con resto 1
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**25** ●●

77

**SOLUZIONE**

$77_{10} = 1001101_2$
$77 \div 2 = 38$ con resto 1
$38 \div 2 = 19$ con resto 0
$19 \div 2 = 9$ con resto 1
$9 \div 2 = 4$ con resto 1
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**26** ●●

89

**SOLUZIONE**

$89_{10} = 1011001_2$
$89 \div 2 = 44$ con resto 1
$44 \div 2 = 22$ con resto 0
$22 \div 2 = 11$ con resto 0
$11 \div 2 = 5$ con resto 1
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**27** ●●

96

**SOLUZIONE**

$96_{10} = 1100000_2$
$96 \div 2 = 48$ con resto 0
$48 \div 2 = 24$ con resto 0
$24 \div 2 = 12$ con resto 0
$12 \div 2 = 6$ con resto 0
$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**28** ●●

100

**SOLUZIONE**

$100_{10} = 1100100_2$
$100 \div 2 = 50$ con resto 0
$50 \div 2 = 25$ con resto 0
$25 \div 2 = 12$ con resto 1
$12 \div 2 = 6$ con resto 0

$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**29** ●●

111

**SOLUZIONE**

$111_{10} = 1101111_2$
$111 \div 2 = 55$ con resto 1
$55 \div 2 = 27$ con resto 1
$27 \div 2 = 13$ con resto 1
$13 \div 2 = 6$ con resto 1
$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**30** ●●

120

**SOLUZIONE**

$120_{10} = 1111000_2$
$120 \div 2 = 60$ con resto 0
$60 \div 2 = 30$ con resto 0
$30 \div 2 = 15$ con resto 0
$15 \div 2 = 7$ con resto 1
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**31** ●●

127

**SOLUZIONE**

$127_{10} = 1111111_2$
$127 \div 2 = 63$ con resto 1
$63 \div 2 = 31$ con resto 1
$31 \div 2 = 15$ con resto 1
$15 \div 2 = 7$ con resto 1
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**32** ●●●

128

**SOLUZIONE**

$128_{10} = 10000000_2$
$128 \div 2 = 64$ con resto 0
$64 \div 2 = 32$ con resto 0
$32 \div 2 = 16$ con resto 0
$16 \div 2 = 8$ con resto 0
$8 \div 2 = 4$ con resto 0

$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**33**

141

**SOLUZIONE**

$141_{10} = 10001101_2$
$141 \div 2 = 70$ con resto 1
$70 \div 2 = 35$ con resto 0
$35 \div 2 = 17$ con resto 1
$17 \div 2 = 8$ con resto 1
$8 \div 2 = 4$ con resto 0
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**34**

156

**SOLUZIONE**

$156_{10} = 10011100_2$
$156 \div 2 = 78$ con resto 0
$78 \div 2 = 39$ con resto 0
$39 \div 2 = 19$ con resto 1
$19 \div 2 = 9$ con resto 1
$9 \div 2 = 4$ con resto 1
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**35**

164

**SOLUZIONE**

$164_{10} = 10100100_2$
$164 \div 2 = 82$ con resto 0
$82 \div 2 = 41$ con resto 0
$41 \div 2 = 20$ con resto 1
$20 \div 2 = 10$ con resto 0
$10 \div 2 = 5$ con resto 0
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**36**

173

**SOLUZIONE**

$173_{10} = 10101101_2$
$173 \div 2 = 86$ con resto 1
$86 \div 2 = 43$ con resto 0

$43 \div 2 = 21$ con resto 1
$21 \div 2 = 10$ con resto 1
$10 \div 2 = 5$ con resto 0
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**37** ●●●

176

**SOLUZIONE**

$176_{10} = 10110000_2$
$176 \div 2 = 88$ con resto 0
$88 \div 2 = 44$ con resto 0
$44 \div 2 = 22$ con resto 0
$22 \div 2 = 11$ con resto 0
$11 \div 2 = 5$ con resto 1
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**38** ●●●

187

**SOLUZIONE**

$187_{10} = 10111011_2$
$187 \div 2 = 93$ con resto 1
$93 \div 2 = 46$ con resto 1
$46 \div 2 = 23$ con resto 0
$23 \div 2 = 11$ con resto 1
$11 \div 2 = 5$ con resto 1
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**39** ●●●

199

**SOLUZIONE**

$199_{10} = 11000111_2$
$199 \div 2 = 99$ con resto 1
$99 \div 2 = 49$ con resto 1
$49 \div 2 = 24$ con resto 1
$24 \div 2 = 12$ con resto 0
$12 \div 2 = 6$ con resto 0
$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**40** ●●●

212

**SOLUZIONE**

$212_{10} = 11010100_2$
$212 \div 2 = 106$ con resto 0
$106 \div 2 = 53$ con resto 0
$53 \div 2 = 26$ con resto 1
$26 \div 2 = 13$ con resto 0
$13 \div 2 = 6$ con resto 1
$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**41** ●●●

219

**SOLUZIONE**

$219_{10} = 11011011_2$
$219 \div 2 = 109$ con resto 1
$109 \div 2 = 54$ con resto 1
$54 \div 2 = 27$ con resto 0
$27 \div 2 = 13$ con resto 1
$13 \div 2 = 6$ con resto 1
$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**42** ●●●

233

**SOLUZIONE**

$233_{10} = 11101001_2$
$233 \div 2 = 116$ con resto 1
$116 \div 2 = 58$ con resto 0
$58 \div 2 = 29$ con resto 0
$29 \div 2 = 14$ con resto 1
$14 \div 2 = 7$ con resto 0
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**43** ●●●

240

**SOLUZIONE**

$240_{10} = 11110000_2$
$240 \div 2 = 120$ con resto 0
$120 \div 2 = 60$ con resto 0
$60 \div 2 = 30$ con resto 0
$30 \div 2 = 15$ con resto 0
$15 \div 2 = 7$ con resto 1
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**44** ●●●

246

**SOLUZIONE**

$246_{10} = 11110110_2$
$246 \div 2 = 123$ con resto 0
$123 \div 2 = 61$ con resto 1
$61 \div 2 = 30$ con resto 1
$30 \div 2 = 15$ con resto 0
$15 \div 2 = 7$ con resto 1
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**45** ●●●

250

**SOLUZIONE**

$250_{10} = 11111010_2$
$250 \div 2 = 125$ con resto 0
$125 \div 2 = 62$ con resto 1
$62 \div 2 = 31$ con resto 0
$31 \div 2 = 15$ con resto 1
$15 \div 2 = 7$ con resto 1
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**46** ●●●

255

**SOLUZIONE**

$255_{10} = 11111111_2$
$255 \div 2 = 127$ con resto 1
$127 \div 2 = 63$ con resto 1
$63 \div 2 = 31$ con resto 1
$31 \div 2 = 15$ con resto 1
$15 \div 2 = 7$ con resto 1
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

Convertire i seguenti numeri dalla base 2 alla base 10:

**47** **Esercizio guidato**

10101

$10101_2 = 21_{10}$

$1 \cdot 2^4 = 16$

$0 \cdot 2^3 = 0$

$1 \cdot 2^2 = 4$

$0 \cdot 2^1 = 0$

$1 \cdot 2^0 = 1$

$16 + 0 + 4 + 0 + 1 = 21$

**48** 10

**SOLUZIONE**

$10_2 = 2_{10}$

$1 \cdot 2^1 = 2$

$0 \cdot 2^0 = 0$

$2 + 0 = 2$

**49** 11

**SOLUZIONE**

$11_2 = 3_{10}$

$1 \cdot 2^1 = 2$

$1 \cdot 2^0 = 1$

$2 + 1 = 3$

**50** 100

**SOLUZIONE**

$100_2 = 4_{10}$

$1 \cdot 2^2 = 4$

$0 \cdot 2^1 = 0$

$0 \cdot 2^0 = 0$

$4 + 0 + 0 = 4$

**51** 101

**SOLUZIONE**

$101_2 = 5_{10}$

$1 \cdot 2^2 = 4$

$0 \cdot 2^1 = 0$

$1 \cdot 2^0 = 1$

$4 + 0 + 1 = 5$

**52** 110

**SOLUZIONE**

$110_2 = 6_{10}$
$1 \cdot 2^2 = 4$
$1 \cdot 2^1 = 2$
$0 \cdot 2^0 = 0$
$4+2+0 = 6$

**53** 111

**SOLUZIONE**

$111_2 = 7_{10}$
$1 \cdot 2^2 = 4$
$1 \cdot 2^1 = 2$
$1 \cdot 2^0 = 1$
$4+2+1 = 7$

**54** 1000

**SOLUZIONE**

$1000_2 = 8_{10}$
$1 \cdot 2^3 = 8$
$0 \cdot 2^2 = 0$
$0 \cdot 2^1 = 0$
$0 \cdot 2^0 = 0$
$8+0+0+0 = 8$

**55** 1001

**SOLUZIONE**

$1001_2 = 9_{10}$
$1 \cdot 2^3 = 8$
$0 \cdot 2^2 = 0$
$0 \cdot 2^1 = 0$
$1 \cdot 2^0 = 1$
$8+0+0+1 = 9$

**56** 1010

**SOLUZIONE**

$1010_2 = 10_{10}$
$1 \cdot 2^3 = 8$
$0 \cdot 2^2 = 0$
$1 \cdot 2^1 = 2$
$0 \cdot 2^0 = 0$
$8+0+2+0 = 10$

**57** 1011

**SOLUZIONE**

$1011_2 = 11_{10}$

$1 \cdot 2^3 = 8$
$0 \cdot 2^2 = 0$
$1 \cdot 2^1 = 2$
$1 \cdot 2^0 = 1$
$8+0+2+1 = 11$

**58** 1100

**SOLUZIONE**

$1100_2 = 12_{10}$
$1 \cdot 2^3 = 8$
$1 \cdot 2^2 = 4$
$0 \cdot 2^1 = 0$
$0 \cdot 2^0 = 0$
$8+4+0+0 = 12$

**59** 1101

**SOLUZIONE**

$1101_2 = 13_{10}$
$1 \cdot 2^3 = 8$
$1 \cdot 2^2 = 4$
$0 \cdot 2^1 = 0$
$1 \cdot 2^0 = 1$
$8+4+0+1 = 13$

**60** 1110

**SOLUZIONE**

$1110_2 = 14_{10}$
$1 \cdot 2^3 = 8$
$1 \cdot 2^2 = 4$
$1 \cdot 2^1 = 2$
$0 \cdot 2^0 = 0$
$8+4+2+0 = 14$

**61** 1111

**SOLUZIONE**

$1111_2 = 15_{10}$
$1 \cdot 2^3 = 8$
$1 \cdot 2^2 = 4$
$1 \cdot 2^1 = 2$
$1 \cdot 2^0 = 1$
$8+4+2+1 = 15$

**62** 10000

**SOLUZIONE**

$10000_2 = 16_{10}$
$1 \cdot 2^4 = 16$
$0 \cdot 2^3 = 0$
$0 \cdot 2^2 = 0$
$0 \cdot 2^1 = 0$
$0 \cdot 2^0 = 0$
$16+0+0+0+0 = 16$

**63** ●●

10001

**SOLUZIONE**

$10001_2 = 17_{10}$
$1 \cdot 2^4 = 16$
$0 \cdot 2^3 = 0$
$0 \cdot 2^2 = 0$
$0 \cdot 2^1 = 0$
$1 \cdot 2^0 = 1$
$16+0+0+0+1 = 17$

**64** ●●

10011

**SOLUZIONE**

$10011_2 = 19_{10}$
$1 \cdot 2^4 = 16$
$0 \cdot 2^3 = 0$
$0 \cdot 2^2 = 0$
$1 \cdot 2^1 = 2$
$1 \cdot 2^0 = 1$
$16+0+0+2+1 = 19$

**65** ●●

10101

**SOLUZIONE**

$10101_2 = 21_{10}$
$1 \cdot 2^4 = 16$
$0 \cdot 2^3 = 0$
$1 \cdot 2^2 = 4$
$0 \cdot 2^1 = 0$
$1 \cdot 2^0 = 1$
$16+0+4+0+1 = 21$

**66** ●●

10110

**SOLUZIONE**

$10110_2 = 22_{10}$
$1 \cdot 2^4 = 16$
$0 \cdot 2^3 = 0$
$1 \cdot 2^2 = 4$
$1 \cdot 2^1 = 2$

$0 \cdot 2^0 = 0$
$16+0+4+2+0 = 22$

**67** ●●

11000

**SOLUZIONE**

$11000_2 = 24_{10}$
$1 \cdot 2^4 = 16$
$1 \cdot 2^3 = 8$
$0 \cdot 2^2 = 0$
$0 \cdot 2^1 = 0$
$0 \cdot 2^0 = 0$
$16+8+0+0+0 = 24$

**68** ●●

11010

**SOLUZIONE**

$11010_2 = 26_{10}$
$1 \cdot 2^4 = 16$
$1 \cdot 2^3 = 8$
$0 \cdot 2^2 = 0$
$1 \cdot 2^1 = 2$
$0 \cdot 2^0 = 0$
$16+8+0+2+0 = 26$

**69** ●●

11100

**SOLUZIONE**

$11100_2 = 28_{10}$
$1 \cdot 2^4 = 16$
$1 \cdot 2^3 = 8$
$1 \cdot 2^2 = 4$
$0 \cdot 2^1 = 0$
$0 \cdot 2^0 = 0$
$16+8+4+0+0 = 28$

**70** ●●

11111

**SOLUZIONE**

$11111_2 = 31_{10}$
$1 \cdot 2^4 = 16$
$1 \cdot 2^3 = 8$
$1 \cdot 2^2 = 4$
$1 \cdot 2^1 = 2$
$1 \cdot 2^0 = 1$
$16+8+4+2+1 = 31$

**71** ●●

100000

**SOLUZIONE**

$100000_2 = 32_{10}$
$1 \cdot 2^5 = 32$
$0 \cdot 2^4 = 0$
$0 \cdot 2^3 = 0$
$0 \cdot 2^2 = 0$
$0 \cdot 2^1 = 0$
$0 \cdot 2^0 = 0$
$32+0+0+0+0+0 = 32$

**72** ●●

100101

**SOLUZIONE**

$100101_2 = 37_{10}$
$1 \cdot 2^5 = 32$
$0 \cdot 2^4 = 0$
$0 \cdot 2^3 = 0$
$1 \cdot 2^2 = 4$
$0 \cdot 2^1 = 0$
$1 \cdot 2^0 = 1$
$32+0+0+4+0+1 = 37$

**73** ●●

101010

**SOLUZIONE**

$101010_2 = 42_{10}$
$1 \cdot 2^5 = 32$
$0 \cdot 2^4 = 0$
$1 \cdot 2^3 = 8$
$0 \cdot 2^2 = 0$
$1 \cdot 2^1 = 2$
$0 \cdot 2^0 = 0$
$32+0+8+0+2+0 = 42$

**74** ●●

101101

**SOLUZIONE**

$101101_2 = 45_{10}$
$1 \cdot 2^5 = 32$
$0 \cdot 2^4 = 0$
$1 \cdot 2^3 = 8$
$1 \cdot 2^2 = 4$
$0 \cdot 2^1 = 0$
$1 \cdot 2^0 = 1$
$32+0+8+4+0+1 = 45$

**75** ●●

110011

**SOLUZIONE**

$110011_2 = 51_{10}$
$1 \cdot 2^5 = 32$
$1 \cdot 2^4 = 16$
$0 \cdot 2^3 = 0$
$0 \cdot 2^2 = 0$
$1 \cdot 2^1 = 2$
$1 \cdot 2^0 = 1$
$32 + 16 + 0 + 0 + 2 + 1 = 51$

**76** ●●

111000

**SOLUZIONE**

$111000_2 = 56_{10}$
$1 \cdot 2^5 = 32$
$1 \cdot 2^4 = 16$
$1 \cdot 2^3 = 8$
$0 \cdot 2^2 = 0$
$0 \cdot 2^1 = 0$
$0 \cdot 2^0 = 0$
$32 + 16 + 8 + 0 + 0 + 0 = 56$

**77** ●●●

1000000

**SOLUZIONE**

$1000000_2 = 64_{10}$
$1 \cdot 2^6 = 64$
$0 \cdot 2^5 = 0$
$0 \cdot 2^4 = 0$
$0 \cdot 2^3 = 0$
$0 \cdot 2^2 = 0$
$0 \cdot 2^1 = 0$
$0 \cdot 2^0 = 0$
$64 + 0 + 0 + 0 + 0 + 0 + 0 = 64$

**78** ●●●

1010101

**SOLUZIONE**

$1010101_2 = 85_{10}$
$1 \cdot 2^6 = 64$
$0 \cdot 2^5 = 0$
$1 \cdot 2^4 = 16$
$0 \cdot 2^3 = 0$
$1 \cdot 2^2 = 4$
$0 \cdot 2^1 = 0$
$1 \cdot 2^0 = 1$
$64 + 0 + 16 + 0 + 4 + 0 + 1 = 85$

**79** ●●●

1011010

**SOLUZIONE**

$1011010_2 = 90_{10}$
$1 \cdot 2^6 = 64$
$0 \cdot 2^5 = 0$
$1 \cdot 2^4 = 16$
$1 \cdot 2^3 = 8$
$0 \cdot 2^2 = 0$
$1 \cdot 2^1 = 2$
$0 \cdot 2^0 = 0$
$64 + 0 + 16 + 8 + 0 + 2 + 0 = 90$

**80**

1100110

**SOLUZIONE**

$1100110_2 = 102_{10}$
$1 \cdot 2^6 = 64$
$1 \cdot 2^5 = 32$
$0 \cdot 2^4 = 0$
$0 \cdot 2^3 = 0$
$1 \cdot 2^2 = 4$
$1 \cdot 2^1 = 2$
$0 \cdot 2^0 = 0$
$64 + 32 + 0 + 0 + 4 + 2 + 0 = 102$

**81**

1101101

**SOLUZIONE**

$1101101_2 = 109_{10}$
$1 \cdot 2^6 = 64$
$1 \cdot 2^5 = 32$
$0 \cdot 2^4 = 0$
$1 \cdot 2^3 = 8$
$1 \cdot 2^2 = 4$
$0 \cdot 2^1 = 0$
$1 \cdot 2^0 = 1$
$64 + 32 + 0 + 8 + 4 + 0 + 1 = 109$

**82**

1110111

**SOLUZIONE**

$1110111_2 = 119_{10}$
$1 \cdot 2^6 = 64$
$1 \cdot 2^5 = 32$
$1 \cdot 2^4 = 16$
$0 \cdot 2^3 = 0$
$1 \cdot 2^2 = 4$
$1 \cdot 2^1 = 2$
$1 \cdot 2^0 = 1$
$64 + 32 + 16 + 0 + 4 + 2 + 1 = 119$

**83** 1111000

**SOLUZIONE**

$1111000_2 = 120_{10}$
$1 \cdot 2^6 = 64$
$1 \cdot 2^5 = 32$
$1 \cdot 2^4 = 16$
$1 \cdot 2^3 = 8$
$0 \cdot 2^2 = 0$
$0 \cdot 2^1 = 0$
$0 \cdot 2^0 = 0$
$64 + 32 + 16 + 8 + 0 + 0 + 0 = 120$

**84** 10000001

**SOLUZIONE**

$10000001_2 = 129_{10}$
$1 \cdot 2^7 = 128$
$0 \cdot 2^6 = 0$
$0 \cdot 2^5 = 0$
$0 \cdot 2^4 = 0$
$0 \cdot 2^3 = 0$
$0 \cdot 2^2 = 0$
$0 \cdot 2^1 = 0$
$1 \cdot 2^0 = 1$
$128 + 0 + 0 + 0 + 0 + 0 + 0 + 1 = 129$

**85** 10011001

**SOLUZIONE**

$10011001_2 = 153_{10}$
$1 \cdot 2^7 = 128$
$0 \cdot 2^6 = 0$
$0 \cdot 2^5 = 0$
$1 \cdot 2^4 = 16$
$1 \cdot 2^3 = 8$
$0 \cdot 2^2 = 0$
$0 \cdot 2^1 = 0$
$1 \cdot 2^0 = 1$
$128 + 0 + 0 + 16 + 8 + 0 + 0 + 1 = 153$

**86** 10101010

**SOLUZIONE**

$10101010_2 = 170_{10}$
$1 \cdot 2^7 = 128$
$0 \cdot 2^6 = 0$
$1 \cdot 2^5 = 32$
$0 \cdot 2^4 = 0$

$1 \cdot 2^3 = 8$
$0 \cdot 2^2 = 0$
$1 \cdot 2^1 = 2$
$0 \cdot 2^0 = 0$
$128 + 0 + 32 + 0 + 8 + 0 + 2 + 0 = 170$

**87** ●●●

10110011

**SOLUZIONE**

$10110011_2 = 179_{10}$
$1 \cdot 2^7 = 128$
$0 \cdot 2^6 = 0$
$1 \cdot 2^5 = 32$
$1 \cdot 2^4 = 16$
$0 \cdot 2^3 = 0$
$0 \cdot 2^2 = 0$
$1 \cdot 2^1 = 2$
$1 \cdot 2^0 = 1$
$128 + 0 + 32 + 16 + 0 + 0 + 2 + 1 = 179$

**88** ●●●

11001100

**SOLUZIONE**

$11001100_2 = 204_{10}$
$1 \cdot 2^7 = 128$
$1 \cdot 2^6 = 64$
$0 \cdot 2^5 = 0$
$0 \cdot 2^4 = 0$
$1 \cdot 2^3 = 8$
$1 \cdot 2^2 = 4$
$0 \cdot 2^1 = 0$
$0 \cdot 2^0 = 0$
$128 + 64 + 0 + 0 + 8 + 4 + 0 + 0 = 204$

**89** ●●●

11010101

**SOLUZIONE**

$11010101_2 = 213_{10}$
$1 \cdot 2^7 = 128$
$1 \cdot 2^6 = 64$
$0 \cdot 2^5 = 0$
$1 \cdot 2^4 = 16$
$0 \cdot 2^3 = 0$
$1 \cdot 2^2 = 4$
$0 \cdot 2^1 = 0$
$1 \cdot 2^0 = 1$
$128 + 64 + 0 + 16 + 0 + 4 + 0 + 1 = 213$

**90** ●●●

11100011

**SOLUZIONE**

$11100011_2 = 227_{10}$
$1 \cdot 2^7 = 128$
$1 \cdot 2^6 = 64$
$1 \cdot 2^5 = 32$
$0 \cdot 2^4 = 0$
$0 \cdot 2^3 = 0$
$0 \cdot 2^2 = 0$
$1 \cdot 2^1 = 2$
$1 \cdot 2^0 = 1$
$128 + 64 + 32 + 0 + 0 + 0 + 2 + 1 = 227$

**91** ●●●

11110000

**SOLUZIONE**

$11110000_2 = 240_{10}$
$1 \cdot 2^7 = 128$
$1 \cdot 2^6 = 64$
$1 \cdot 2^5 = 32$
$1 \cdot 2^4 = 16$
$0 \cdot 2^3 = 0$
$0 \cdot 2^2 = 0$
$0 \cdot 2^1 = 0$
$0 \cdot 2^0 = 0$
$128 + 64 + 32 + 16 + 0 + 0 + 0 + 0 = 240$

**92** ●●●

11111111

**SOLUZIONE**

$11111111_2 = 255_{10}$
$1 \cdot 2^7 = 128$
$1 \cdot 2^6 = 64$
$1 \cdot 2^5 = 32$
$1 \cdot 2^4 = 16$
$1 \cdot 2^3 = 8$
$1 \cdot 2^2 = 4$
$1 \cdot 2^1 = 2$
$1 \cdot 2^0 = 1$
$128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255$

## 2.2 Base 2 - Modulo e Segno

Convertire i seguenti numeri dalla base 10 alla base 2 usando la codifica modulo e segno:

1 **Esercizio guidato**

Convertire il numero -71 dalla base 10 alla base 2

$-71_{10} = 1\,1000111_2$

$71 \div 2 = 35$ con resto 1

$35 \div 2 = 17$ con resto 1

$17 \div 2 = 8$ con resto 1

$8 \div 2 = 4$ con resto 0

$4 \div 2 = 2$ con resto 0

$2 \div 2 = 1$ con resto 0

$1 \div 2 = 0$ con resto 1

2 2

**SOLUZIONE**

$2_{10} = 0\,10_2$

$2 \div 2 = 1$ con resto 0

$1 \div 2 = 0$ con resto 1

3 -3

**SOLUZIONE**

$-3_{10} = 1\,11_2$

$3 \div 2 = 1$ con resto 1

$1 \div 2 = 0$ con resto 1

4 -5

**SOLUZIONE**

$-5_{10} = 1\,101_2$

$5 \div 2 = 2$ con resto 1

$2 \div 2 = 1$ con resto 0

$1 \div 2 = 0$ con resto 1

5 7

**SOLUZIONE**

$7_{10} = 0\,111_2$

$7 \div 2 = 3$ con resto 1

$3 \div 2 = 1$ con resto 1

$1 \div 2 = 0$ con resto 1

6 -6

**SOLUZIONE**

$-6_{10} = 1\,110_2$

$6 \div 2 = 3$ con resto 0

$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**7**

8

**SOLUZIONE**

$8_{10} = 0\,1000_2$
$8 \div 2 = 4$ con resto 0
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**8**

-11

**SOLUZIONE**

$-11_{10} = 1\,1011_2$
$11 \div 2 = 5$ con resto 1
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**9**

-15

**SOLUZIONE**

$-15_{10} = 1\,1111_2$
$15 \div 2 = 7$ con resto 1
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**10**

14

**SOLUZIONE**

$14_{10} = 0\,1110_2$
$14 \div 2 = 7$ con resto 0
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**11**

16

**SOLUZIONE**

$16_{10} = 0\,10000_2$
$16 \div 2 = 8$ con resto 0
$8 \div 2 = 4$ con resto 0
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**12** ●●

-27

**SOLUZIONE**

$-27_{10} = 1\,11011_2$
$27 \div 2 = 13$ con resto 1
$13 \div 2 = 6$ con resto 1
$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**13** ●●

45

**SOLUZIONE**

$45_{10} = 0\,101101_2$
$45 \div 2 = 22$ con resto 1
$22 \div 2 = 11$ con resto 0
$11 \div 2 = 5$ con resto 1
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**14** ●●

-58

**SOLUZIONE**

$-58_{10} = 1\,111010_2$
$58 \div 2 = 29$ con resto 0
$29 \div 2 = 14$ con resto 1
$14 \div 2 = 7$ con resto 0
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**15** ●●

64

**SOLUZIONE**

$64_{10} = 0\,1000000_2$
$64 \div 2 = 32$ con resto 0
$32 \div 2 = 16$ con resto 0
$16 \div 2 = 8$ con resto 0
$8 \div 2 = 4$ con resto 0
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**16** ●●

-77

**SOLUZIONE**

$-77_{10} = 1\,1001101_2$
$77 \div 2 = 38$ con resto 1

$38 \div 2 = 19$ con resto 0
$19 \div 2 = 9$ con resto 1
$9 \div 2 = 4$ con resto 1
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**17** ●●

100

**SOLUZIONE**

$100_{10} = 0\,1100100_2$
$100 \div 2 = 50$ con resto 0
$50 \div 2 = 25$ con resto 0
$25 \div 2 = 12$ con resto 1
$12 \div 2 = 6$ con resto 0
$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**18** ●●

-111

**SOLUZIONE**

$-111_{10} = 1\,1101111_2$
$111 \div 2 = 55$ con resto 1
$55 \div 2 = 27$ con resto 1
$27 \div 2 = 13$ con resto 1
$13 \div 2 = 6$ con resto 1
$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**19** ●●

-96

**SOLUZIONE**

$-96_{10} = 1\,1100000_2$
$96 \div 2 = 48$ con resto 0
$48 \div 2 = 24$ con resto 0
$24 \div 2 = 12$ con resto 0
$12 \div 2 = 6$ con resto 0
$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**20** ●●●

128

**SOLUZIONE**

$128_{10} = 0\,10000000_2$
$128 \div 2 = 64$ con resto 0
$64 \div 2 = 32$ con resto 0

$32 \div 2 = 16$ con resto 0
$16 \div 2 = 8$ con resto 0
$8 \div 2 = 4$ con resto 0
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**21**

-164

**SOLUZIONE**

$-164_{10} = 1\,10100100_2$
$164 \div 2 = 82$ con resto 0
$82 \div 2 = 41$ con resto 0
$41 \div 2 = 20$ con resto 1
$20 \div 2 = 10$ con resto 0
$10 \div 2 = 5$ con resto 0
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**22**

187

**SOLUZIONE**

$187_{10} = 0\,10111011_2$
$187 \div 2 = 93$ con resto 1
$93 \div 2 = 46$ con resto 1
$46 \div 2 = 23$ con resto 0
$23 \div 2 = 11$ con resto 1
$11 \div 2 = 5$ con resto 1
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**23**

-212

**SOLUZIONE**

$-212_{10} = 1\,11010100_2$
$212 \div 2 = 106$ con resto 0
$106 \div 2 = 53$ con resto 0
$53 \div 2 = 26$ con resto 1
$26 \div 2 = 13$ con resto 0
$13 \div 2 = 6$ con resto 1
$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**24**

233

**SOLUZIONE**

$233_{10} = 0\,11101001_2$
$233 \div 2 = 116$ con resto 1
$116 \div 2 = 58$ con resto 0
$58 \div 2 = 29$ con resto 0
$29 \div 2 = 14$ con resto 1
$14 \div 2 = 7$ con resto 0
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**25** ●●●

-246

**SOLUZIONE**

$-246_{10} = 1\,11110110_2$
$246 \div 2 = 123$ con resto 0
$123 \div 2 = 61$ con resto 1
$61 \div 2 = 30$ con resto 1
$30 \div 2 = 15$ con resto 0
$15 \div 2 = 7$ con resto 1
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**26** ●●●

240

**SOLUZIONE**

$240_{10} = 0\,11110000_2$
$240 \div 2 = 120$ con resto 0
$120 \div 2 = 60$ con resto 0
$60 \div 2 = 30$ con resto 0
$30 \div 2 = 15$ con resto 0
$15 \div 2 = 7$ con resto 1
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**27** ●●●

-255

**SOLUZIONE**

$-255_{10} = 1\,11111111_2$
$255 \div 2 = 127$ con resto 1
$127 \div 2 = 63$ con resto 1
$63 \div 2 = 31$ con resto 1
$31 \div 2 = 15$ con resto 1
$15 \div 2 = 7$ con resto 1
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**28**

199

**SOLUZIONE**

$199_{10} = 0\,11000111_2$
$199 \div 2 = 99$ con resto 1
$99 \div 2 = 49$ con resto 1
$49 \div 2 = 24$ con resto 1
$24 \div 2 = 12$ con resto 0
$12 \div 2 = 6$ con resto 0
$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**29**

-141

**SOLUZIONE**

$-141_{10} = 1\,10001101_2$
$141 \div 2 = 70$ con resto 1
$70 \div 2 = 35$ con resto 0
$35 \div 2 = 17$ con resto 1
$17 \div 2 = 8$ con resto 1
$8 \div 2 = 4$ con resto 0
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

## 2.3 Base 2 - Complemento a 2

Convertire i seguenti numeri dalla base 10 alla base 2 usando la codifica complemento a 2:

1 **Esercizio guidato**

Convertire il numero -71 dalla base 10 alla base 2

$-71_{10} = 1\,0111001_2$

$71 \div 2 = 35$ con resto 1

$35 \div 2 = 17$ con resto 1

$17 \div 2 = 8$ con resto 1

$8 \div 2 = 4$ con resto 0

$4 \div 2 = 2$ con resto 0

$2 \div 2 = 1$ con resto 0

$1 \div 2 = 0$ con resto 1

1000111

0111000

0111001

2 -2

**SOLUZIONE**

$-2_{10} = 1\,10_2$

$2 \div 2 = 1$ con resto 0

$1 \div 2 = 0$ con resto 1

10

01

10

3 -5

**SOLUZIONE**

$-5_{10} = 1\,011_2$

$5 \div 2 = 2$ con resto 1

$2 \div 2 = 1$ con resto 0

$1 \div 2 = 0$ con resto 1

101

010

011

4 -7

**SOLUZIONE**

$-7_{10} = 1\,001_2$

$7 \div 2 = 3$ con resto 1

$3 \div 2 = 1$ con resto 1

$1 \div 2 = 0$ con resto 1

111

000

001

**5** 6

**SOLUZIONE**

$6_{10} = 0\,110_2$
$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**6** -8

**SOLUZIONE**

$-8_{10} = 1\,1000_2$
$8 \div 2 = 4$ con resto 0
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1
1000
0111
1000

**7** 11

**SOLUZIONE**

$11_{10} = 0\,1011_2$
$11 \div 2 = 5$ con resto 1
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**8** 15

**SOLUZIONE**

$15_{10} = 0\,1111_2$
$15 \div 2 = 7$ con resto 1
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**9** -15

**SOLUZIONE**

$-15_{10} = 1\,0001_2$
$15 \div 2 = 7$ con resto 1
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1
1111
0000
0001

**10** ●●

-16

**SOLUZIONE**

$-16_{10} = 1\,10000_2$
$16 \div 2 = 8$ con resto 0
$8 \div 2 = 4$ con resto 0
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1
10000
01111
10000

**11** ●●

-27

**SOLUZIONE**

$-27_{10} = 1\,00101_2$
$27 \div 2 = 13$ con resto 1
$13 \div 2 = 6$ con resto 1
$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1
11011
00100
00101

**12** ●●

45

**SOLUZIONE**

$45_{10} = 0\,101101_2$
$45 \div 2 = 22$ con resto 1
$22 \div 2 = 11$ con resto 0
$11 \div 2 = 5$ con resto 1
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**13** ●●

-58

**SOLUZIONE**

$-58_{10} = 1\,000110_2$
$58 \div 2 = 29$ con resto 0
$29 \div 2 = 14$ con resto 1
$14 \div 2 = 7$ con resto 0
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1
111010
000101

000110

**14** ●●

64

**SOLUZIONE**

$64_{10} = 0\,1000000_2$
$64 \div 2 = 32$ con resto 0
$32 \div 2 = 16$ con resto 0
$16 \div 2 = 8$ con resto 0
$8 \div 2 = 4$ con resto 0
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1

**15** ●●

-77

**SOLUZIONE**

$-77_{10} = 1\,0110011_2$
$77 \div 2 = 38$ con resto 1
$38 \div 2 = 19$ con resto 0
$19 \div 2 = 9$ con resto 1
$9 \div 2 = 4$ con resto 1
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1
1001101
0110010
0110011

**16** ●●

100

**SOLUZIONE**

$100_{10} = 0\,1100100_2$
$100 \div 2 = 50$ con resto 0
$50 \div 2 = 25$ con resto 0
$25 \div 2 = 12$ con resto 1
$12 \div 2 = 6$ con resto 0
$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**17** ●●

-111

**SOLUZIONE**

$-111_{10} = 1\,0010001_2$
$111 \div 2 = 55$ con resto 1
$55 \div 2 = 27$ con resto 1
$27 \div 2 = 13$ con resto 1
$13 \div 2 = 6$ con resto 1

$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1
1101111
0010000
0010001

**18** ●●

-127

**SOLUZIONE**

$-127_{10} = 1\,0000001_2$
$127 \div 2 = 63$ con resto 1
$63 \div 2 = 31$ con resto 1
$31 \div 2 = 15$ con resto 1
$15 \div 2 = 7$ con resto 1
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1
1111111
0000000
0000001

**19** ●●●

-128

**SOLUZIONE**

$-128_{10} = 1\,10000000_2$
$128 \div 2 = 64$ con resto 0
$64 \div 2 = 32$ con resto 0
$32 \div 2 = 16$ con resto 0
$16 \div 2 = 8$ con resto 0
$8 \div 2 = 4$ con resto 0
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1
10000000
01111111
10000000

**20** ●●●

164

**SOLUZIONE**

$164_{10} = 0\,10100100_2$
$164 \div 2 = 82$ con resto 0
$82 \div 2 = 41$ con resto 0
$41 \div 2 = 20$ con resto 1
$20 \div 2 = 10$ con resto 0
$10 \div 2 = 5$ con resto 0
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0

$1 \div 2 = 0$ con resto 1

**21** ●●●

-187

**SOLUZIONE**

$-187_{10} = 1\,01000101_2$
$187 \div 2 = 93$ con resto 1
$93 \div 2 = 46$ con resto 1
$46 \div 2 = 23$ con resto 0
$23 \div 2 = 11$ con resto 1
$11 \div 2 = 5$ con resto 1
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1
10111011
01000100
01000101

**22** ●●●

212

**SOLUZIONE**

$212_{10} = 0\,11010100_2$
$212 \div 2 = 106$ con resto 0
$106 \div 2 = 53$ con resto 0
$53 \div 2 = 26$ con resto 1
$26 \div 2 = 13$ con resto 0
$13 \div 2 = 6$ con resto 1
$6 \div 2 = 3$ con resto 0
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**23** ●●●

-233

**SOLUZIONE**

$-233_{10} = 1\,00010111_2$
$233 \div 2 = 116$ con resto 1
$116 \div 2 = 58$ con resto 0
$58 \div 2 = 29$ con resto 0
$29 \div 2 = 14$ con resto 1
$14 \div 2 = 7$ con resto 0
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1
11101001
00010110
00010111

**24** ●●●

246

**SOLUZIONE**

$246_{10} = 0\,11110110_2$
$246 \div 2 = 123$ con resto 0
$123 \div 2 = 61$ con resto 1
$61 \div 2 = 30$ con resto 1
$30 \div 2 = 15$ con resto 0
$15 \div 2 = 7$ con resto 1
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**25** ●●●

-240

**SOLUZIONE**

$-240_{10} = 1\,00010000_2$
$240 \div 2 = 120$ con resto 0
$120 \div 2 = 60$ con resto 0
$60 \div 2 = 30$ con resto 0
$30 \div 2 = 15$ con resto 0
$15 \div 2 = 7$ con resto 1
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1
11110000
00001111
00010000

**26** ●●●

255

**SOLUZIONE**

$255_{10} = 0\,11111111_2$
$255 \div 2 = 127$ con resto 1
$127 \div 2 = 63$ con resto 1
$63 \div 2 = 31$ con resto 1
$31 \div 2 = 15$ con resto 1
$15 \div 2 = 7$ con resto 1
$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1

**27** ●●●

-255

**SOLUZIONE**

$-255_{10} = 1\,00000001_2$
$255 \div 2 = 127$ con resto 1
$127 \div 2 = 63$ con resto 1
$63 \div 2 = 31$ con resto 1
$31 \div 2 = 15$ con resto 1
$15 \div 2 = 7$ con resto 1

$7 \div 2 = 3$ con resto 1
$3 \div 2 = 1$ con resto 1
$1 \div 2 = 0$ con resto 1
11111111
00000000
00000001

# II

# Operazioni sui dati


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# 3

# Operazioni binarie

## 3.1 Somma

Effettua le seguenti somme in binario:

1

**Esercizio guidato**

41 + 18

Come prima cosa convertiamo i due numeri da decimale a binario

$41_{10} = 101001_2$

$41 \div 2 = 20$ con resto 1

$20 \div 2 = 10$ con resto 0

$10 \div 2 = 5$ con resto 0

$5 \div 2 = 2$ con resto 1

$2 \div 2 = 1$ con resto 0

$1 \div 2 = 0$ con resto 1

$18_{10} = 10010_2$

$18 \div 2 = 9$ con resto 0

$9 \div 2 = 4$ con resto 1

$4 \div 2 = 2$ con resto 0

$2 \div 2 = 1$ con resto 0

$1 \div 2 = 0$ con resto 1

A questo punto eseguire la somma bit a bit, ricordando che 1+1 fa 0 con riporto di 1 e 1+1+1 fa 1 con riporto di 1

$41_{10} = 101001_2$

$18_{10} = 10010_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | | carry |
| 1 | 0 | 1 | 0 | 0 | 1 | + |
| | 1 | 0 | 0 | 1 | 0 | = |
| 1 | 1 | 1 | 0 | 1 | 1 | |

**2** 12 + 8

**SOLUZIONE**

$12_{10} = 1100_2$
$8_{10} = 1000_2$

| | | | | | |
|---|---|---|---|---|---|
| 1 | | | | | carry |
| | 1 | 1 | 0 | 0 | + |
| | 1 | 0 | 0 | 0 | = |
| 1 | 0 | 1 | 0 | 0 | |

**3** 23 + 15

**SOLUZIONE**

$23_{10} = 10111_2$
$15_{10} = 1111_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | | carry |
| | 1 | 0 | 1 | 1 | 1 | + |
| | | 1 | 1 | 1 | 1 | = |
| 1 | 0 | 0 | 1 | 1 | 0 | |

**4** 41 + 7

**SOLUZIONE**

$41_{10} = 101001_2$
$7_{10} = 111_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| | 1 | 1 | 1 | 1 | | carry |
| 1 | 0 | 1 | 0 | 0 | 1 | + |
| | | | 1 | 1 | 1 | = |
| 1 | 1 | 0 | 0 | 0 | 0 | |

**5** 30 + 20

**SOLUZIONE**

$30_{10} = 11110_2$
$20_{10} = 10100_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 | 1 | 1 | | | | carry |
| | 1 | 1 | 1 | 1 | 0 | + |
| | 1 | 0 | 1 | 0 | 0 | = |
| 1 | 1 | 0 | 0 | 1 | 0 | |

**6** 14 + 26

**SOLUZIONE**

$14_{10} = 1110_2$
$26_{10} = 11010_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | | | carry |
| | | 1 | 1 | 1 | 0 | + |
| | 1 | 1 | 0 | 1 | 0 | = |
| 1 | 0 | 1 | 0 | 0 | 0 | |

**7**

50 + 9

**SOLUZIONE**

$50_{10} = 110010_2$
$9_{10} = 1001_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | | carry |
| 1 | 1 | 0 | 0 | 1 | 0 | + |
| | | 1 | 0 | 0 | 1 | = |
| 1 | 1 | 1 | 0 | 1 | 1 | |

**8**

18 + 12

**SOLUZIONE**

$18_{10} = 10010_2$
$12_{10} = 1100_2$

| | | | | | |
|---|---|---|---|---|---|
| | | | | | carry |
| 1 | 0 | 0 | 1 | 0 | + |
| | 1 | 1 | 0 | 0 | = |
| 1 | 1 | 1 | 1 | 0 | |

**9**

33 + 17

**SOLUZIONE**

$33_{10} = 100001_2$
$17_{10} = 10001_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | 1 | | carry |
| 1 | 0 | 0 | 0 | 0 | 1 | + |
| | 1 | 0 | 0 | 0 | 1 | = |
| 1 | 1 | 0 | 0 | 1 | 0 | |

**10**

25 + 25

**SOLUZIONE**

$25_{10} = 11001_2$
$25_{10} = 11001_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 | 1 | | | 1 | | carry |
| | 1 | 1 | 0 | 0 | 1 | + |
| | 1 | 1 | 0 | 0 | 1 | = |
| 1 | 1 | 0 | 0 | 1 | 0 | |

**11**

44 + 6

**SOLUZIONE**

$44_{10} = 101100_2$
$6_{10} = 110_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| | 1 | 1 | | | | carry |
| 1 | 0 | 1 | 1 | 0 | 0 | + |
| | | | 1 | 1 | 0 | = |
| 1 | 1 | 0 | 0 | 1 | 0 | |

**12**

34 + 29

**SOLUZIONE**

$34_{10} = 100010_2$
$29_{10} = 11101_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | | | | carry |
| 1 | 0 | 0 | 0 | 1 | 0 | + |
| | 1 | 1 | 1 | 0 | 1 | = |
| 1 | 1 | 1 | 1 | 1 | 1 | |

**13**

47 + 38

**SOLUZIONE**

$47_{10} = 101111_2$
$38_{10} = 100110_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | | 1 | 1 | 1 | | | carry |
| | 1 | 0 | 1 | 1 | 1 | 1 | + |
| | 1 | 0 | 0 | 1 | 1 | 0 | = |
| 1 | 0 | 1 | 0 | 1 | 0 | 1 | |

**14**

56 + 27

**SOLUZIONE**

$56_{10} = 111000_2$
$27_{10} = 11011_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | | | | | carry |
| | 1 | 1 | 1 | 0 | 0 | 0 | + |
| | | 1 | 1 | 0 | 1 | 1 | = |
| 1 | 0 | 1 | 0 | 0 | 1 | 1 | |

**15**

31 + 33

**SOLUZIONE**

$31_{10} = 11111_2$
$33_{10} = 100001_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | | carry |
| | | 1 | 1 | 1 | 1 | 1 | + |
| | 1 | 0 | 0 | 0 | 0 | 1 | = |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | |

**16** ●●

68 + 15

**SOLUZIONE**

$68_{10} = 1000100_2$
$15_{10} = 1111_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1 | 1 | | | | carry |
| 1 | 0 | 0 | 0 | 1 | 0 | 0 | + |
| | | | 1 | 1 | 1 | 1 | = |
| 1 | 0 | 1 | 0 | 0 | 1 | 1 | |

**17** ●●

42 + 39

**SOLUZIONE**

$42_{10} = 101010_2$
$39_{10} = 100111_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | | 1 | 1 | 1 | | | carry |
| | 1 | 0 | 1 | 0 | 1 | 0 | + |
| | 1 | 0 | 0 | 1 | 1 | 1 | = |
| 1 | 0 | 1 | 0 | 0 | 0 | 1 | |

**18** ●●

55 + 28

**SOLUZIONE**

$55_{10} = 110111_2$
$28_{10} = 11100_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | | | | carry |
| | 1 | 1 | 0 | 1 | 1 | 1 | + |
| | | 1 | 1 | 1 | 0 | 0 | = |
| 1 | 0 | 1 | 0 | 0 | 1 | 1 | |

**19** ●●

73 + 19

**SOLUZIONE**

$73_{10} = 1001001_2$
$19_{10} = 10011_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | 1 | 1 | | carry |
| 1 | 0 | 0 | 1 | 0 | 0 | 1 | + |
| | | 1 | 0 | 0 | 1 | 1 | = |
| 1 | 0 | 1 | 1 | 1 | 0 | 0 | |

**20** ●●

26 + 47

**SOLUZIONE**

$26_{10} = 11010_2$
$47_{10} = 101111_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | | | carry |
| | | 1 | 1 | 0 | 1 | 0 | + |
| | 1 | 0 | 1 | 1 | 1 | 1 | = |
| 1 | 0 | 0 | 1 | 0 | 0 | 1 | |

**21**

38 + 44

**SOLUZIONE**

$38_{10} = 100110_2$
$44_{10} = 101100_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | | 1 | 1 | | | | carry |
| | 1 | 0 | 0 | 1 | 1 | 0 | + |
| | 1 | 0 | 1 | 1 | 0 | 0 | = |
| 1 | 0 | 1 | 0 | 0 | 1 | 0 | |

**22**

59 + 23

**SOLUZIONE**

$59_{10} = 111011_2$
$23_{10} = 10111_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | | carry |
| | 1 | 1 | 1 | 0 | 1 | 1 | + |
| | | 1 | 0 | 1 | 1 | 1 | = |
| 1 | 0 | 1 | 0 | 0 | 1 | 0 | |

**23**

64 + 29

**SOLUZIONE**

$64_{10} = 1000000_2$
$29_{10} = 11101_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | | | | carry |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | + |
| | | 1 | 1 | 1 | 0 | 1 | = |
| 1 | 0 | 1 | 1 | 1 | 0 | 1 | |

**24**

45 + 37

**SOLUZIONE**

$45_{10} = 101101_2$
$37_{10} = 100101_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | | 1 | 1 | | 1 | | carry |
| | 1 | 0 | 1 | 1 | 0 | 1 | + |
| | 1 | 0 | 0 | 1 | 0 | 1 | = |
| 1 | 0 | 1 | 0 | 0 | 1 | 0 | |

**25**

52 + 39

**SOLUZIONE**

$52_{10} = 110100_2$
$39_{10} = 100111_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | | | 1 | | | | carry |
| | 1 | 1 | 0 | 1 | 0 | 0 | + |
| | 1 | 0 | 0 | 1 | 1 | 1 | = |
| 1 | 0 | 1 | 1 | 0 | 1 | 1 | |

**26**

67 + 26

**SOLUZIONE**

$67_{10} = 1000011_2$
$26_{10} = 11010_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | | 1 | | | carry |
| 1 | 0 | 0 | 0 | 0 | 1 | 1 | + |
| | | 1 | 1 | 0 | 1 | 0 | = |
| 1 | 0 | 1 | 1 | 1 | 0 | 1 | |

**27**

87 + 63

**SOLUZIONE**

$87_{10} = 1010111_2$
$63_{10} = 111111_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | | carry |
| | 1 | 0 | 1 | 0 | 1 | 1 | 1 | + |
| | | 1 | 1 | 1 | 1 | 1 | 1 | = |
| 1 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | |

**28**

94 + 78

**SOLUZIONE**

$94_{10} = 1011110_2$
$78_{10} = 1001110_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | | 1 | 1 | 1 | 1 | | | carry |
| | 1 | 0 | 1 | 1 | 1 | 1 | 0 | + |
| | 1 | 0 | 0 | 1 | 1 | 1 | 0 | = |
| 1 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | |

**29**

69 + 85

**SOLUZIONE**

$69_{10} = 1000101_2$
$85_{10} = 1010101_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | | | | 1 | | 1 | | carry |
| | 1 | 0 | 0 | 0 | 1 | 0 | 1 | + |
| | 1 | 0 | 1 | 0 | 1 | 0 | 1 | = |
| 1 | 0 | 0 | 1 | 1 | 0 | 1 | 0 | |

**30**

76 + 67

**SOLUZIONE**

$76_{10} = 1001100_2$
$67_{10} = 1000011_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | carry |
| | 1 | 0 | 0 | 1 | 1 | 0 | 0 | + |
| | 1 | 0 | 0 | 0 | 0 | 1 | 1 | = |
| 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | |

**31**

88 + 59

**SOLUZIONE**

$88_{10} = 1011000_2$
$59_{10} = 111011_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | | | | | carry |
| | 1 | 0 | 1 | 1 | 0 | 0 | 0 | + |
| | | 1 | 1 | 1 | 0 | 1 | 1 | = |
| 1 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | |

**32**

95 + 48

**SOLUZIONE**

$95_{10} = 1011111_2$
$48_{10} = 110000_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | | | | | | carry |
| | 1 | 0 | 1 | 1 | 1 | 1 | 1 | + |
| | | 1 | 1 | 0 | 0 | 0 | 0 | = |
| 1 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | |

**33**

72 + 89

**SOLUZIONE**

$72_{10} = 1001000_2$
$89_{10} = 1011001_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | | 1 | 1 | | | | | carry |
| | 1 | 0 | 0 | 1 | 0 | 0 | 0 | + |
| | 1 | 0 | 1 | 1 | 0 | 0 | 1 | = |
| 1 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | |

**34** ●●●

83 + 79

**SOLUZIONE**

$83_{10} = 1010011_2$
$79_{10} = 1001111_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | | 1 | 1 | 1 | 1 | 1 | | carry |
| | 1 | 0 | 1 | 0 | 0 | 1 | 1 | + |
| | 1 | 0 | 0 | 1 | 1 | 1 | 1 | = |
| 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | |

**35** ●●●

66 + 97

**SOLUZIONE**

$66_{10} = 1000010_2$
$97_{10} = 1100001_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | carry |
| | 1 | 0 | 0 | 0 | 0 | 1 | 0 | + |
| | 1 | 1 | 0 | 0 | 0 | 0 | 1 | = |
| 1 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | |

**36** ●●●

78 + 84

**SOLUZIONE**

$78_{10} = 1001110_2$
$84_{10} = 1010100_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | | 1 | 1 | 1 | | | | carry |
| | 1 | 0 | 0 | 1 | 1 | 1 | 0 | + |
| | 1 | 0 | 1 | 0 | 1 | 0 | 0 | = |
| 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | |

**37** ●●●

91 + 69

**SOLUZIONE**

$91_{10} = 1011011_2$
$69_{10} = 1000101_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | | 1 | 1 | 1 | 1 | 1 | | carry |
| | 1 | 0 | 1 | 1 | 0 | 1 | 1 | + |
| | 1 | 0 | 0 | 0 | 1 | 0 | 1 | = |
| 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | |

**38** ●●●

85 + 76

**SOLUZIONE**

$85_{10} = 1010101_2$
$76_{10} = 1001100_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | | 1 | 1 | 1 | | | | carry |
| | 1 | 0 | 1 | 0 | 1 | 0 | 1 | + |
| | 1 | 0 | 0 | 1 | 1 | 0 | 0 | = |
| 1 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | |

**39** ●●●

93 + 58

**SOLUZIONE**

$93_{10} = 1011101_2$
$58_{10} = 111010_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | | | | | carry |
| | 1 | 0 | 1 | 1 | 1 | 0 | 1 | + |
| | | 1 | 1 | 1 | 0 | 1 | 0 | = |
| 1 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | |

**40** ●●●

74 + 88

**SOLUZIONE**

$74_{10} = 1001010_2$
$88_{10} = 1011000_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | | 1 | 1 | | | | | carry |
| | 1 | 0 | 0 | 1 | 0 | 1 | 0 | + |
| | 1 | 0 | 1 | 1 | 0 | 0 | 0 | = |
| 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | |

**41** ●●●

89 + 73

**SOLUZIONE**

$89_{10} = 1011001_2$
$73_{10} = 1001001_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | | 1 | 1 | | | 1 | | carry |
| | 1 | 0 | 1 | 1 | 0 | 0 | 1 | + |
| | 1 | 0 | 0 | 1 | 0 | 0 | 1 | = |
| 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | |

## 3.2 Sottrazione

Effettua le seguenti sottrazioni in binario:

**1** ●

**Esercizio guidato**

41 - 18

Come prima cosa convertiamo i due numeri da decimale a binario

$41_{10} = 101001_2$
$41 \div 2 = 20$ con resto 1
$20 \div 2 = 10$ con resto 0
$10 \div 2 = 5$ con resto 0
$5 \div 2 = 2$ con resto 1
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1
$18_{10} = 10010_2$
$18 \div 2 = 9$ con resto 0
$9 \div 2 = 4$ con resto 1
$4 \div 2 = 2$ con resto 0
$2 \div 2 = 1$ con resto 0
$1 \div 2 = 0$ con resto 1
A questo punto eseguire la differenza bit a bit, ricordando che 0-1 fa 1 con il prestito di 1 dalla cifra precedente
$41_{10} = 101001_2$
$18_{10} = 010010_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 | $_1$0 | 1 | $_1$0 | $_1$0 | 1 | - |
| 1 | | 1 | 1 | | | prestito |
| 0 | 1 | 0 | 0 | 1 | 0 | = |
| | 1 | 0 | 1 | 1 | 1 | |

**2** 15 - 7

**SOLUZIONE**

$15_{10} = 1111_2$
$7_{10} = 0111_2$

| | | | | |
|---|---|---|---|---|
| 1 | 1 | 1 | 1 | - |
| | | | | prestito |
| 0 | 1 | 1 | 1 | = |
| 1 | 0 | 0 | 0 | |

**3** 23 - 10

**SOLUZIONE**

$23_{10} = 10111_2$
$10_{10} = 01010_2$

| | | | | | |
|---|---|---|---|---|---|
| 1 | $_1$0 | 1 | 1 | 1 | - |
| 1 | | | | | prestito |
| 0 | 1 | 0 | 1 | 0 | = |
| | 1 | 1 | 0 | 1 | |

**4** 30 - 12

**SOLUZIONE**

$30_{10} = 11110_2$
$12_{10} = 01100_2$

| | | | | | |
|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 0 | - |
| | | | | | prestito |
| 0 | 1 | 1 | 0 | 0 | = |
| 1 | 0 | 0 | 1 | 0 | |

**5** 18 - 9

**SOLUZIONE**

$18_{10} = 10010_2$
$9_{10} = 01001_2$

| | | | | | |
|---|---|---|---|---|---|
| 1 | $_1$0 | $_1$0 | 1 | $_1$0 | - |
| 1 | 1 | | 1 | | prestito |
| 0 | 1 | 0 | 0 | 1 | = |
| | 1 | 0 | 0 | 1 | |

**6** 25 - 13

**SOLUZIONE**

$25_{10} = 11001_2$
$13_{10} = 01101_2$

| | | | | | |
|---|---|---|---|---|---|
| 1 | $_1$1 | $_1$0 | 0 | 1 | - |
| 1 | 1 | | | | prestito |
| 0 | 1 | 1 | 0 | 1 | = |
| | 1 | 1 | 0 | 0 | |

**7**

40 - 20

**SOLUZIONE**

$40_{10} = 101000_2$
$20_{10} = 010100_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 | $_1$0 | 1 | $_1$0 | 0 | 0 | - |
| 1 | | 1 | | | | prestito |
| 0 | 1 | 0 | 1 | 0 | 0 | = |
| | 1 | 0 | 1 | 0 | 0 | |

**8**

36 - 14

**SOLUZIONE**

$36_{10} = 100100_2$
$14_{10} = 001110_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 | $_1$0 | $_1$0 | $_1$1 | $_1$0 | 0 | - |
| 1 | 1 | 1 | 1 | | | prestito |
| 0 | 0 | 1 | 1 | 1 | 0 | = |
| | 1 | 0 | 1 | 1 | 0 | |

**9**

50 - 25

**SOLUZIONE**

$50_{10} = 110010_2$
$25_{10} = 011001_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 | $_1$1 | $_1$0 | $_1$0 | 1 | $_1$0 | - |
| 1 | 1 | 1 | | 1 | | prestito |
| 0 | 1 | 1 | 0 | 0 | 1 | = |
| | 1 | 1 | 0 | 0 | 1 | |

**10**

28 - 16

**SOLUZIONE**

$28_{10} = 11100_2$
$16_{10} = 10000_2$

| | | | | | |
|---|---|---|---|---|---|
| 1 | 1 | 1 | 0 | 0 | - |
| | | | | | prestito |
| 1 | 0 | 0 | 0 | 0 | = |
| | 1 | 1 | 0 | 0 | |

**11**

45 - 23

**SOLUZIONE**

$45_{10} = 101101_2$
$23_{10} = 010111_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 | $_1$0 | 1 | $_1$1 | $_1$0 | 1 | - |
| 1 | | 1 | 1 | | | prestito |
| 0 | 1 | 0 | 1 | 1 | 1 | = |
| | 1 | 0 | 1 | 1 | 0 | |

**12**

42 - 17

**SOLUZIONE**

$42_{10} = 101010_2$
$17_{10} = 010001_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 | $_1$0 | 1 | $_1$0 | 1 | $_1$0 | - |
| 1 | | 1 | | 1 | | prestito |
| 0 | 1 | 0 | 0 | 0 | 1 | = |
| | 1 | 1 | 0 | 0 | 1 | |

**13**

53 - 28

**SOLUZIONE**

$53_{10} = 110101_2$
$28_{10} = 011100_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 | $_1$1 | $_1$0 | 1 | 0 | 1 | - |
| 1 | 1 | | | | | prestito |
| 0 | 1 | 1 | 1 | 0 | 0 | = |
| | 1 | 1 | 0 | 0 | 1 | |

**14**

61 - 35

**SOLUZIONE**

$61_{10} = 111101_2$
$35_{10} = 100011_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| $_1$1 | 1 | 1 | 1 | $_1$0 | 1 | - |
| | | | 1 | | 1 | prestito |
| 1 | 0 | 0 | 0 | 1 | 1 | = |
| | 1 | 1 | 0 | 1 | 0 | |

**15**

70 - 44

**SOLUZIONE**

$70_{10} = 1000110_2$
$44_{10} = 0101100_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | $_1$0 | $_1$0 | $_1$0 | 1 | 1 | 0 | - |
| 1 | 1 | 1 | | | | | prestito |
| 0 | 1 | 0 | 1 | 1 | 0 | 0 | = |
| | | 1 | 1 | 0 | 1 | 0 | |

**16** ●●

85 - 39

**SOLUZIONE**

$85_{10} = 1010101_2$
$39_{10} = 0100111_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | $_1$0 | 1 | $_1$0 | $_1$1 | $_1$0 | 1 | - |
| 1 | | 1 | 1 | 1 | | | prestito |
| 0 | 1 | 0 | 0 | 1 | 1 | 1 | = |
| | 1 | 0 | 1 | 1 | 1 | 0 | |

**17** ●●

56 - 29

**SOLUZIONE**

$56_{10} = 111000_2$
$29_{10} = 011101_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 | $_1$1 | $_1$1 | $_1$0 | $_1$0 | $_1$0 | - |
| 1 | 1 | 1 | 1 | 1 | | prestito |
| 0 | 1 | 1 | 1 | 0 | 1 | = |
| | 1 | 1 | 0 | 1 | 1 | |

**18** ●●

64 - 37

**SOLUZIONE**

$64_{10} = 1000000_2$
$37_{10} = 0100101_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | $_1$0 | $_1$0 | $_1$0 | $_1$0 | $_1$0 | $_1$0 | - |
| 1 | 1 | 1 | 1 | 1 | 1 | | prestito |
| 0 | 1 | 0 | 0 | 1 | 0 | 1 | = |
| | | 1 | 1 | 0 | 1 | 1 | |

**19** ●●

72 - 46

**SOLUZIONE**

$72_{10} = 1001000_2$
$46_{10} = 0101110_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | $_1$0 | $_1$0 | $_1$1 | $_1$0 | $_1$0 | 0 | - |
| 1 | 1 | 1 | 1 | 1 | | | prestito |
| 0 | 1 | 0 | 1 | 1 | 1 | 0 | = |
| | | 1 | 1 | 0 | 1 | 0 | |

**20** ●●

83 - 55

**SOLUZIONE**

$83_{10} = 1010011_2$
$55_{10} = 0110111_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | $_1$0 | $_1$1 | $_1$0 | $_1$0 | 1 | 1 | - |
| 1 | 1 | 1 | 1 | | | | prestito |
| 0 | 1 | 1 | 0 | 1 | 1 | 1 | = |
| | | 1 | 1 | 1 | 0 | 0 | |

**21**

91 - 64

**SOLUZIONE**

$91_{10} = 1011011_2$
$64_{10} = 1000000_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 0 | 1 | 1 | - |
| | | | | | | | prestito |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | = |
| | | 1 | 1 | 0 | 1 | 1 | |

**22**

47 - 29

**SOLUZIONE**

$47_{10} = 101111_2$
$29_{10} = 011101_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| 1 | $_1$0 | 1 | 1 | 1 | 1 | - |
| 1 | | | | | | prestito |
| 0 | 1 | 1 | 1 | 0 | 1 | = |
| | 1 | 0 | 0 | 1 | 0 | |

**23**

58 - 33

**SOLUZIONE**

$58_{10} = 111010_2$
$33_{10} = 100001_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| $_1$1 | 1 | 1 | $_1$0 | 1 | $_1$0 | - |
| | | 1 | | 1 | 1 | prestito |
| 1 | 0 | 0 | 0 | 0 | 1 | = |
| | 1 | 1 | 0 | 0 | 1 | |

**24**

66 - 48

**SOLUZIONE**

$66_{10} = 1000010_2$
$48_{10} = 0110000_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | $_1$0 | $_1$0 | 0 | 0 | 1 | 0 | - |
| 1 | 1 | | | | | | prestito |
| 0 | 1 | 1 | 0 | 0 | 0 | 0 | = |
| | | 1 | 0 | 0 | 1 | 0 | |

**25**

75 - 57

**SOLUZIONE**

$75_{10} = 1001011_2$
$57_{10} = 0111001_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | $_1$0 | $_1$0 | 1 | 0 | 1 | 1 | - |
| 1 | 1 | | | | | | prestito |
| 0 | 1 | 1 | 1 | 0 | 0 | 1 | = |
| | | 1 | 0 | 0 | 1 | 0 | |

**26**

88 - 69

**SOLUZIONE**

$88_{10} = 1011000_2$
$69_{10} = 1000101_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| $_1$1 | $_1$0 | 1 | 1 | $_1$0 | $_1$0 | $_1$0 | - |
| 1 | | | 1 | 1 | 1 | 1 | prestito |
| 1 | 0 | 0 | 0 | 1 | 0 | 1 | = |
| | | 1 | 0 | 0 | 1 | 1 | |

**27**

100 - 67

**SOLUZIONE**

$100_{10} = 1100100_2$
$67_{10} = 1000011_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| $_1$1 | 1 | $_1$0 | $_1$0 | 1 | $_1$0 | $_1$0 | - |
| | 1 | 1 | | 1 | 1 | 1 | prestito |
| 1 | 0 | 0 | 0 | 0 | 1 | 1 | = |
| | 1 | 0 | 0 | 0 | 0 | 1 | |

**28**

123 - 89

**SOLUZIONE**

$123_{10} = 1111011_2$
$89_{10} = 1011001_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 0 | 1 | 1 | - |
| | | | | | | | prestito |
| 1 | 0 | 1 | 1 | 0 | 0 | 1 | = |
| | 1 | 0 | 0 | 0 | 1 | 0 | |

**29**

145 - 98

**SOLUZIONE**

$145_{10} = 10010001_2$
$98_{10} = 01100010_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | $_1$0 | $_1$0 | 1 | $_1$0 | $_1$0 | $_1$0 | 1 | - |
| 1 | 1 | | 1 | 1 | 1 | | | prestito |
| 0 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | = |
| | | 1 | 0 | 1 | 1 | 1 | 1 | |

**30**

132 - 76

**SOLUZIONE**

$132_{10} = 10000100_2$
$76_{10} = 01001100_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | $_1$0 | $_1$0 | $_1$0 | $_1$0 | 1 | 0 | 0 | - |
| 1 | 1 | 1 | 1 | | | | | prestito |
| 0 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | = |
| | | 1 | 1 | 1 | 0 | 0 | 0 | |

**31**

156 - 99

**SOLUZIONE**

$156_{10} = 10011100_2$
$99_{10} = 01100011_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | $_1$0 | $_1$0 | 1 | 1 | 1 | $_1$0 | $_1$0 | - |
| 1 | 1 | | | | 1 | 1 | | prestito |
| 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | = |
| | | 1 | 1 | 1 | 0 | 0 | 1 | |

**32**

178 - 129

**SOLUZIONE**

$178_{10} = 10110010_2$
$129_{10} = 10000001_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $_1$1 | $_1$0 | 1 | 1 | $_1$0 | $_1$0 | 1 | $_1$0 | - |
| 1 | | | 1 | 1 | | 1 | 1 | prestito |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | = |
| | | 1 | 1 | 0 | 0 | 0 | 1 | |

**33**

200 - 137

**SOLUZIONE**

$200_{10} = 11001000_2$
$137_{10} = 10001001_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $_1$1 | 1 | $_1$0 | $_1$0 | $_1$1 | $_1$0 | $_1$0 | $_1$0 | - |
| | 1 | 1 | 1 | 1 | 1 | 1 | 1 | prestito |
| 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | = |
| | | 1 | 1 | 1 | 1 | 1 | 1 | |

**34**

184 - 156

**SOLUZIONE**

$184_{10} = 10111000_2$
$156_{10} = 10011100_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $_1$1 | $_1$0 | 1 | $_1$1 | $_1$1 | $_1$0 | 0 | 0 | - |
| 1 | | 1 | 1 | 1 | | | 1 | prestito |
| 1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | = |
| | | | 1 | 1 | 1 | 0 | 0 | |

**35**

215 - 178

**SOLUZIONE**

$215_{10} = 11010111_2$
$178_{10} = 10110010_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $_1$1 | 1 | $_1$0 | 1 | 0 | 1 | 1 | 1 | - |
| | 1 | | | | | | 1 | prestito |
| 1 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | = |
| | | 1 | 0 | 0 | 1 | 0 | 1 | |

**36**

230 - 189

**SOLUZIONE**

$230_{10} = 11100110_2$
$189_{10} = 10111101_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $_1$1 | 1 | $_1$1 | $_1$0 | $_1$0 | $_1$1 | 1 | $_1$0 | - |
| | 1 | 1 | 1 | 1 | | 1 | 1 | prestito |
| 1 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | = |
| | | 1 | 0 | 1 | 0 | 0 | 1 | |

**37**

167 - 129

**SOLUZIONE**

$167_{10} = 10100111_2$
$129_{10} = 10000001_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | - |
| | | | | | | | | prestito |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | = |
| | | 1 | 0 | 0 | 1 | 1 | 0 | |

**38**

192 - 145

**SOLUZIONE**

$192_{10} = 11000000_2$
$145_{10} = 10010001_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $_1$1 | 1 | $_1$0 | $_1$0 | $_1$0 | $_1$0 | $_1$0 | $_1$0 | - |
| | 1 | 1 | 1 | 1 | 1 | 1 | 1 | prestito |
| 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | = |
| | | 1 | 0 | 1 | 1 | 1 | 1 | |

**39**

224 - 187

**SOLUZIONE**

$224_{10} = 11100000_2$
$187_{10} = 10111011_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $_1$1 | 1 | $_1$1 | $_1$0 | $_1$0 | $_1$0 | $_1$0 | $_1$0 | - |
| | 1 | 1 | 1 | 1 | 1 | 1 | 1 | prestito |
| 1 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | = |
| | | 1 | 0 | 0 | 1 | 0 | 1 | |

**40**

250 - 199

**SOLUZIONE**

$250_{10} = 11111010_2$
$199_{10} = 11000111_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| $_1$1 | $_1$1 | 1 | 1 | 1 | $_1$0 | $_1$1 | $_1$0 | - |
| 1 | | | | 1 | 1 | 1 | 1 | prestito |
| 1 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | = |
| | | 1 | 1 | 0 | 0 | 1 | 1 | |

**41**

243 - 195

**SOLUZIONE**

$243_{10} = 11110011_2$
$195_{10} = 11000011_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | - |
| | | | | | | | | prestito |
| 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | = |
| | | 1 | 1 | 0 | 0 | 0 | 0 | |

## 3.3 Moltiplicazione

Effettua le seguenti moltiplicazioni in binario:

1 **Esercizio guidato**

41 x 9

Come prima cosa convertiamo i due numeri da decimale a binario

$41_{10} = 101001_2$

$41 \div 2 = 20$ con resto 1

$20 \div 2 = 10$ con resto 0

$10 \div 2 = 5$ con resto 0

$5 \div 2 = 2$ con resto 1

$2 \div 2 = 1$ con resto 0

$1 \div 2 = 0$ con resto 1

$9_{10} = 1001_2$

$9 \div 2 = 4$ con resto 1

$4 \div 2 = 2$ con resto 0

$2 \div 2 = 1$ con resto 0

$1 \div 2 = 0$ con resto 1

Andiamo ad eseguire la moltiplicazione del primo numero per ogni bit del secondo, ricordandosi di spostarsi di una posizione ogni volta.

Al termine di deve eseguire la somma in colonna, qual'ora le righe risultimo molte conviene fare due righe per volta

$41_{10} = 101001_2$

$9_{10} = 1001_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | 1 | 0 | 1 | 0 | 0 | 1 | * |
| | | | | | 1 | 0 | 0 | 1 | = |
| | | | 1 | 0 | 1 | 0 | 0 | 1 | |
| | | 0 | 0 | 0 | 0 | 0 | 0 | - | |
| | 0 | 0 | 0 | 0 | 0 | 0 | - | - | |
| 1 | 0 | 1 | 0 | 0 | 1 | - | - | - | |
| 1 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | |

**2** 3 x 4

**SOLUZIONE**

$3_{10} = 11_2$
$4_{10} = 100_2$

| | | | | |
|---|---|---|---|---|
| | | 1 | 1 | * |
| | 1 | 0 | 0 | = |
| | | 0 | 0 | |
| | 0 | 0 | - | |
| 1 | 1 | - | - | |
| 1 | 1 | 0 | 0 | |

**3** 5 x 6

**SOLUZIONE**

$5_{10} = 101_2$
$6_{10} = 110_2$

| | | | | | |
|---|---|---|---|---|---|
| | | 1 | 0 | 1 | * |
| | | 1 | 1 | 0 | = |
| | | 0 | 0 | 0 | |
| | 1 | 0 | 1 | - | |
| 1 | 0 | 1 | - | - | |
| 1 | 1 | 1 | 1 | 0 | |

**4** 7 x 8

**SOLUZIONE**

$7_{10} = 111_2$
$8_{10} = 1000_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | 1 | 1 | 1 | * |
| | | 1 | 0 | 0 | 0 | = |
| | | | 0 | 0 | 0 | |
| | | 0 | 0 | 0 | - | |
| | 0 | 0 | 0 | - | - | |
| 1 | 1 | 1 | - | - | - | |
| 1 | 1 | 1 | 0 | 0 | 0 | |

**5** 9 x 2

**SOLUZIONE**

$9_{10} = 1001_2$
$2_{10} = 10_2$

| | | | | | |
|---|---|---|---|---|---|
| | 1 | 0 | 0 | 1 | * |
| | | | 1 | 0 | = |
| | 0 | 0 | 0 | 0 | |
| 1 | 0 | 0 | 1 | - | |
| 1 | 0 | 0 | 1 | 0 | |

**6**

6 x 7

**SOLUZIONE**

$6_{10} = 110_2$
$7_{10} = 111_2$

| | | | | | |
|---|---|---|---|---|---|
| | | 1 | 1 | 0 | * |
| | | 1 | 1 | 1 | = |
| | | 1 | 1 | 0 | |
| | 1 | 1 | 0 | - | |
| 1 | 1 | 0 | - | - | |
| 1 | 0 | 1 | 0 | 1 | 0 |

**7**

8 x 5

**SOLUZIONE**

$8_{10} = 1000_2$
$5_{10} = 101_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| | | 1 | 0 | 0 | 0 | * |
| | | | 1 | 0 | 1 | = |
| | | 1 | 0 | 0 | 0 | |
| | 0 | 0 | 0 | 0 | - | |
| 1 | 0 | 0 | 0 | - | - | |
| 1 | 0 | 1 | 0 | 0 | 0 | |

**8**

4 x 9

**SOLUZIONE**

$4_{10} = 100_2$
$9_{10} = 1001_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| | | | 1 | 0 | 0 | * |
| | | 1 | 0 | 0 | 1 | = |
| | | | 1 | 0 | 0 | |
| | | 0 | 0 | 0 | - | |
| | 0 | 0 | 0 | - | - | |
| 1 | 0 | 0 | - | - | - | |
| 1 | 0 | 0 | 1 | 0 | 0 | |

**9**

10 x 3

**SOLUZIONE**

$10_{10} = 1010_2$

$3_{10} = 11_2$

| | | | | | |
|---|---|---|---|---|---|
| | 1 | 0 | 1 | 0 | * |
| | | | 1 | 1 | = |
| | 1 | 0 | 1 | 0 | |
| 1 | 0 | 1 | 0 | - | |
| 1 | 1 | 1 | 1 | 0 | |

**10**

11 x 2

**SOLUZIONE**

$11_{10} = 1011_2$
$2_{10} = 10_2$

| | | | | | |
|---|---|---|---|---|---|
| | 1 | 0 | 1 | 1 | * |
| | | | 1 | 0 | = |
| | 0 | 0 | 0 | 0 | |
| 1 | 0 | 1 | 1 | - | |
| 1 | 0 | 1 | 1 | 0 | |

**11**

12 x 4

**SOLUZIONE**

$12_{10} = 1100_2$
$4_{10} = 100_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| | | 1 | 1 | 0 | 0 | * |
| | | | 1 | 0 | 0 | = |
| | | 0 | 0 | 0 | 0 | |
| | 0 | 0 | 0 | 0 | - | |
| 1 | 1 | 0 | 0 | - | - | |
| 1 | 1 | 0 | 0 | 0 | 0 | |

**12**

13 x 4

**SOLUZIONE**

$13_{10} = 1101_2$
$4_{10} = 100_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| | | 1 | 1 | 0 | 1 | * |
| | | | 1 | 0 | 0 | = |
| | | 0 | 0 | 0 | 0 | |
| | 0 | 0 | 0 | 0 | - | |
| 1 | 1 | 0 | 1 | - | - | |
| 1 | 1 | 0 | 1 | 0 | 0 | |

**13**

15 x 6

**SOLUZIONE**

$15_{10} = 1111_2$

$6_{10} = 110_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | 1 | 1 | 1 | 1 | * |
| | | | | 1 | 1 | 0 | = |
| | | | 0 | 0 | 0 | 0 | |
| | | 1 | 1 | 1 | 1 | - | |
| | 1 | 1 | 1 | 1 | - | - | |
| 1 | 0 | 1 | 1 | 0 | 1 | 0 | |

**14**

18 x 5

**SOLUZIONE**

$18_{10} = 10010_2$
$5_{10} = 101_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1 | 0 | 0 | 1 | 0 | * |
| | | | | 1 | 0 | 1 | = |
| | | 1 | 0 | 0 | 1 | 0 | |
| | 0 | 0 | 0 | 0 | 0 | - | |
| 1 | 0 | 0 | 1 | 0 | - | - | |
| 1 | 0 | 1 | 1 | 0 | 1 | 0 | |

**15**

22 x 3

**SOLUZIONE**

$22_{10} = 10110_2$
$3_{10} = 11_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1 | 0 | 1 | 1 | 0 | * |
| | | | | | 1 | 1 | = |
| | | 1 | 0 | 1 | 1 | 0 | |
| | 1 | 0 | 1 | 1 | 0 | - | |
| 1 | 0 | 0 | 0 | 0 | 1 | 0 | |

**16**

24 x 4

**SOLUZIONE**

$24_{10} = 11000_2$
$4_{10} = 100_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1 | 1 | 0 | 0 | 0 | * |
| | | | | 1 | 0 | 0 | = |
| | | 0 | 0 | 0 | 0 | 0 | |
| | 0 | 0 | 0 | 0 | 0 | - | |
| 1 | 1 | 0 | 0 | 0 | - | - | |
| 1 | 1 | 0 | 0 | 0 | 0 | 0 | |

**17**

16 x 7

**SOLUZIONE**

$16_{10} = 10000_2$
$7_{10} = 111_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| | | 1 | 0 | 0 | 0 | 0 | * |
| | | | | 1 | 1 | 1 | = |
| | | 1 | 0 | 0 | 0 | 0 | |
| | 1 | 0 | 0 | 0 | 0 | - | |
| 1 | 0 | 0 | 0 | 0 | - | - | |
| 1 | 1 | 1 | 0 | 0 | 0 | 0 | |

**18**

19 x 5

**SOLUZIONE**

$19_{10} = 10011_2$
$5_{10} = 101_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| | | 1 | 0 | 0 | 1 | 1 | * |
| | | | | 1 | 0 | 1 | = |
| | | 1 | 0 | 0 | 1 | 1 | |
| | 0 | 0 | 0 | 0 | 0 | - | |
| 1 | 0 | 0 | 1 | 1 | - | - | |
| 1 | 0 | 1 | 1 | 1 | 1 | 1 | |

**19**

21 x 6

**SOLUZIONE**

$21_{10} = 10101_2$
$6_{10} = 110_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| | | 1 | 0 | 1 | 0 | 1 | * |
| | | | | 1 | 1 | 0 | = |
| | | 0 | 0 | 0 | 0 | 0 | |
| | 1 | 0 | 1 | 0 | 1 | - | |
| 1 | 0 | 1 | 0 | 1 | - | - | |
| 1 | 1 | 1 | 1 | 1 | 1 | 0 | |

**20**

25 x 4

**SOLUZIONE**

$25_{10} = 11001_2$
$4_{10} = 100_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| | | 1 | 1 | 0 | 0 | 1 | * |
| | | | | 1 | 0 | 0 | = |
| | | 0 | 0 | 0 | 0 | 0 | |
| | 0 | 0 | 0 | 0 | 0 | - | |
| 1 | 1 | 0 | 0 | 1 | - | - | |
| 1 | 1 | 0 | 0 | 1 | 0 | 0 | |

**21**

28 x 3

**SOLUZIONE**

$28_{10} = 11100_2$
$3_{10} = 11_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1 | 1 | 1 | 0 | 0 | * |
| | | | | | 1 | 1 | = |
| | | 1 | 1 | 1 | 0 | 0 | |
| | 1 | 1 | 1 | 0 | 0 | - | |
| 1 | 0 | 1 | 0 | 1 | 0 | 0 | |

**22**

14 x 8

**SOLUZIONE**

$14_{10} = 1110_2$
$8_{10} = 1000_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | | 1 | 1 | 1 | 0 | * |
| | | | 1 | 0 | 0 | 0 | = |
| | | | 0 | 0 | 0 | 0 | |
| | | 0 | 0 | 0 | 0 | - | |
| | 0 | 0 | 0 | 0 | - | - | |
| 1 | 1 | 1 | 0 | - | - | - | |
| 1 | 1 | 1 | 0 | 0 | 0 | 0 | |

**23**

17 x 6

**SOLUZIONE**

$17_{10} = 10001_2$
$6_{10} = 110_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1 | 0 | 0 | 0 | 1 | * |
| | | | | 1 | 1 | 0 | = |
| | | 0 | 0 | 0 | 0 | 0 | |
| | 1 | 0 | 0 | 0 | 1 | - | |
| 1 | 0 | 0 | 0 | 1 | - | - | |
| 1 | 1 | 0 | 0 | 1 | 1 | 0 | |

**24**

23 x 5

**SOLUZIONE**

$23_{10} = 10111_2$
$5_{10} = 101_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | | 1 | 0 | 1 | 1 | 1 | * |
| | | | | 1 | 0 | 1 | = |
| | | 1 | 0 | 1 | 1 | 1 | |
| | 0 | 0 | 0 | 0 | 0 | - | |
| 1 | 0 | 1 | 1 | 1 | - | - | |
| 1 | 1 | 1 | 0 | 0 | 1 | 1 | |

**25** 26 x 4

**SOLUZIONE**

$26_{10} = 11010_2$
$4_{10} = 100_2$

| | | | | | | |
|---|---|---|---|---|---|---|
| | | 1 | 1 | 0 | 1 | 0 | * |
| | | | | 1 | 0 | 0 | = |
| | | 0 | 0 | 0 | 0 | 0 | |
| | 0 | 0 | 0 | 0 | 0 | - | |
| 1 | 1 | 0 | 1 | 0 | - | - | |
| 1 | 1 | 0 | 1 | 0 | 0 | 0 | |

**26** 34 x 12

**SOLUZIONE**

$34_{10} = 100010_2$
$12_{10} = 1100_2$

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | | | 1 | 0 | 0 | 0 | 1 | 0 | * |
| | | | | | 1 | 1 | 0 | 0 | = |
| | | | 0 | 0 | 0 | 0 | 0 | 0 | |
| | | 0 | 0 | 0 | 0 | 0 | 0 | - | |
| | 1 | 0 | 0 | 0 | 1 | 0 | - | - | |
| 1 | 0 | 0 | 0 | 1 | 0 | - | - | - | |
| 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | |

**27** 45 x 13

**SOLUZIONE**

$45_{10} = 101101_2$
$13_{10} = 1101_2$

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | 1 | 0 | 1 | 1 | 0 | 1 | * |
| | | | | | | 1 | 1 | 0 | 1 | = |
| | | | | 1 | 0 | 1 | 1 | 0 | 1 | |
| | | | 0 | 0 | 0 | 0 | 0 | 0 | - | |
| | | 1 | 0 | 1 | 1 | 0 | 1 | - | - | |
| | 1 | 0 | 1 | 1 | 0 | 1 | - | - | - | |
| 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | |

**28** 56 x 14

**SOLUZIONE**

$56_{10} = 111000_2$
$14_{10} = 1110_2$

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | | 1 | 1 | 1 | 0 | 0 | 0 | * |
| | | | | | | 1 | 1 | 1 | 0 | = |
| | | | | 0 | 0 | 0 | 0 | 0 | 0 | |
| | | | 1 | 1 | 1 | 0 | 0 | 0 | - | |
| | | 1 | 1 | 1 | 0 | 0 | 0 | - | - | |
| | 1 | 1 | 1 | 0 | 0 | 0 | - | - | - | |
| 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | |

**29**

67 x 15

**SOLUZIONE**

$67_{10} = 1000011_2$
$15_{10} = 1111_2$

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | 1 | 0 | 0 | 0 | 0 | 1 | 1 | * |
| | | | | | | 1 | 1 | 1 | 1 | = |
| | | | 1 | 0 | 0 | 0 | 0 | 1 | 1 | |
| | | 1 | 0 | 0 | 0 | 0 | 1 | 1 | - | |
| | 1 | 0 | 0 | 0 | 0 | 1 | 1 | - | - | |
| 1 | 0 | 0 | 0 | 0 | 1 | 1 | - | - | - | |
| 1 | 1 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | |

**30**

78 x 12

**SOLUZIONE**

$78_{10} = 1001110_2$
$12_{10} = 1100_2$

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | 1 | 0 | 0 | 1 | 1 | 1 | 0 | * |
| | | | | | | 1 | 1 | 0 | 0 | = |
| | | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | |
| | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | - | |
| | 1 | 0 | 0 | 1 | 1 | 1 | 0 | - | - | |
| 1 | 0 | 0 | 1 | 1 | 1 | 0 | - | - | - | |
| 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | |

**31**

89 x 11

**SOLUZIONE**

$89_{10} = 1011001_2$
$11_{10} = 1011_2$

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | | 1 | 0 | 1 | 1 | 0 | 0 | 1 | * |
| | | | | | | 1 | 0 | 1 | 1 | = |
| | | | 1 | 0 | 1 | 1 | 0 | 0 | 1 | |
| | | 1 | 0 | 1 | 1 | 0 | 0 | 1 | - | |
| | 0 | 0 | 0 | 0 | 0 | 0 | 0 | - | - | |
| 1 | 0 | 1 | 1 | 0 | 0 | 1 | - | - | - | |
| 1 | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | |

**32** ●●●

92 x 13

**SOLUZIONE**

$92_{10} = 1011100_2$
$13_{10} = 1101_2$

| | | | | 1 | 0 | 1 | 1 | 1 | 0 | 0 | * |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | 1 | 1 | 0 | 1 | = |
| | | | | 1 | 0 | 1 | 1 | 1 | 0 | 0 | |
| | | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | - | |
| | | 1 | 0 | 1 | 1 | 1 | 0 | 0 | - | - | |
| | 1 | 0 | 1 | 1 | 1 | 0 | 0 | - | - | - | |
| 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | |

**33** ●●●

84 x 16

**SOLUZIONE**

$84_{10} = 1010100_2$
$16_{10} = 10000_2$

| | | | | 1 | 0 | 1 | 0 | 1 | 0 | 0 | * |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 1 | 0 | 0 | 0 | 0 | = |
| | | | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | |
| | | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | - | |
| | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | - | - | |
| | 0 | 0 | 0 | 0 | 0 | 0 | 0 | - | - | - | |
| 1 | 0 | 1 | 0 | 1 | 0 | 0 | - | - | - | - | |
| 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | |

**34** ●●●

73 x 18

**SOLUZIONE**

$73_{10} = 1001001_2$
$18_{10} = 10010_2$

| | | | | 1 | 0 | 0 | 1 | 0 | 0 | 1 | * |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | 1 | 0 | 0 | 1 | 0 | = |
| | | | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | |
| | | | 1 | 0 | 0 | 1 | 0 | 0 | 1 | - | |
| | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | - | - | |
| | 0 | 0 | 0 | 0 | 0 | 0 | 0 | - | - | - | |
| 1 | 0 | 0 | 1 | 0 | 0 | 1 | - | - | - | - | |
| 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | |

**35** ●●●

65 x 19

**SOLUZIONE**

$65_{10} = 1000001_2$
$19_{10} = 10011_2$

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | * |
| | | | | | | 1 | 0 | 0 | 1 | 1 | = |
| | | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | |
| | | | 1 | 0 | 0 | 0 | 0 | 0 | 1 | - | |
| | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | - | - | |
| | 0 | 0 | 0 | 0 | 0 | 0 | 0 | - | - | - | |
| 1 | 0 | 0 | 0 | 0 | 0 | 1 | - | - | - | - | |
| 1 | 0 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | |

**36**

58 x 17

**SOLUZIONE**

$58_{10} = 111010_2$
$17_{10} = 10001_2$

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | 1 | 1 | 1 | 0 | 1 | 0 | * |
| | | | | | 1 | 0 | 0 | 0 | 1 | = |
| | | | | 1 | 1 | 1 | 0 | 1 | 0 | |
| | | | 0 | 0 | 0 | 0 | 0 | 0 | - | |
| | | 0 | 0 | 0 | 0 | 0 | 0 | - | - | |
| | 0 | 0 | 0 | 0 | 0 | 0 | - | - | - | |
| 1 | 1 | 1 | 0 | 1 | 0 | - | - | - | - | |
| 1 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 0 | |

**37**

49 x 21

**SOLUZIONE**

$49_{10} = 110001_2$
$21_{10} = 10101_2$

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | 1 | 1 | 0 | 0 | 0 | 1 | * |
| | | | | | | 1 | 0 | 1 | 0 | 1 | = |
| | | | | | 1 | 1 | 0 | 0 | 0 | 1 | |
| | | | | 0 | 0 | 0 | 0 | 0 | 0 | - | |
| | | | 1 | 1 | 0 | 0 | 0 | 1 | - | - | |
| | | 0 | 0 | 0 | 0 | 0 | 0 | - | - | - | |
| | 1 | 1 | 0 | 0 | 0 | 1 | - | - | - | - | |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | |

**38**

37 x 24

**SOLUZIONE**

$37_{10} = 100101_2$
$24_{10} = 11000_2$

| | | | | 1 | 0 | 0 | 1 | 0 | 1 | * |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | 1 | 1 | 0 | 0 | 0 | = |
| | | | | 0 | 0 | 0 | 0 | 0 | 0 | |
| | | | 0 | 0 | 0 | 0 | 0 | 0 | - | |
| | | 0 | 0 | 0 | 0 | 0 | 0 | - | - | |
| | 1 | 0 | 0 | 1 | 0 | 1 | - | - | - | |
| 1 | 0 | 0 | 1 | 0 | 1 | - | - | - | - | |
| 1 | 1 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | |

**39**

46 x 22

**SOLUZIONE**

$46_{10} = 101110_2$
$22_{10} = 10110_2$

| | | | | 1 | 0 | 1 | 1 | 1 | 0 | * |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | 1 | 0 | 1 | 1 | 0 | = |
| | | | | 0 | 0 | 0 | 0 | 0 | 0 | |
| | | | 1 | 0 | 1 | 1 | 1 | 0 | - | |
| | | 1 | 0 | 1 | 1 | 1 | 0 | - | - | |
| | 0 | 0 | 0 | 0 | 0 | 0 | - | - | - | |
| 1 | 0 | 1 | 1 | 1 | 0 | - | - | - | - | |
| 1 | 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 0 | |

## 3.4 Divisione

Effettua le seguenti divisioni in binario:

1

**Esercizio guidato**

41 : 9

Come prima cosa convertiamo i due numeri da decimale a binario

$41_{10} = 101001_2$

$41 \div 2 = 20$ con resto 1

$20 \div 2 = 10$ con resto 0

$10 \div 2 = 5$ con resto 0

$5 \div 2 = 2$ con resto 1

$2 \div 2 = 1$ con resto 0

$1 \div 2 = 0$ con resto 1

$9_{10} = 1001_2$

$9 \div 2 = 4$ con resto 1

$4 \div 2 = 2$ con resto 0

$2 \div 2 = 1$ con resto 0

$1 \div 2 = 0$ con resto 1

Abbasso un bit alla volta fino a quando ottengo un numero maggiore o uguale del divisore, per ogni bit che abbasso mi segno uno 0 aggiunto un numero adatto mi segno un 1, scrivo il divisore sotto al numero stando attento di allineare le colonne e effettuo la sottrazione; ripeto le operazioni fino all'ultimo bit;

Il risultato sarà composto da quoziente e resto.

$41_{10} = 101001_2$

$9_{10} = 1001_2$

| 1 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | |
| 1 | 0 | 1 | | | | | | | | | |
| 1 | 0 | 1 | 0 | | | | | | | | |
| 1 | 0 | 0 | 1 | | | | | | | | |
| | | | 1 | 0 | | | | | | | |
| | | | 1 | 0 | 1 | | | | | | |
| | | | 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 |

**2**

12 : 3

**SOLUZIONE**

$12_{10} = 1100_2$
$3_{10} = 11_2$

| 1 | 1 | 0 | 0 | 1 | 1 | | |
|---|---|---|---|---|---|---|---|
| 1 | | | | | | | |
| 1 | 1 | | | | | | |
| 1 | 1 | | | | | | |
| | 0 | 0 | | | | | |
| | 0 | 0 | 0 | | | | |
| | 0 | 0 | 0 | 0 | 1 | 0 | 0 |

**3**

20 : 4

**SOLUZIONE**

$20_{10} = 10100_2$
$4_{10} = 100_2$

| 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | |
| 1 | 0 | | | | | | | | |
| 1 | 0 | 1 | | | | | | | |
| 1 | 0 | 0 | | | | | | | |
| | | 1 | 0 | | | | | | |
| | | 1 | 0 | 0 | | | | | |
| | | 1 | 0 | 0 | | | | | |
| | | | | 0 | 0 | 0 | 1 | 0 | 1 |

**4**

18 : 2

**SOLUZIONE**

$18_{10} = 10010_2$
$2_{10} = 10_2$

| 1 | 0 | 0 | 1 | 0 | 1 | 0 | | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | |
| 1 | 0 | | | | | | | | |
| 1 | 0 | | | | | | | | |
| | 0 | 0 | | | | | | | |
| | 0 | 0 | 1 | | | | | | |
| | 0 | 0 | 1 | 0 | | | | | |
| | | | 1 | 0 | | | | | |
| | | | | 0 | 0 | 1 | 0 | 0 | 1 |

**5**

24 : 6

**SOLUZIONE**

$24_{10} = 11000_2$
$6_{10} = 110_2$

| 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | |
| 1 | 1 | | | | | | | | |
| 1 | 1 | 0 | | | | | | | |
| 1 | 1 | 0 | | | | | | | |
| | | 0 | 0 | | | | | | |
| | | 0 | 0 | 0 | | | | | |
| | | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |

**6**

30 : 5

**SOLUZIONE**

$30_{10} = 11110_2$
$5_{10} = 101_2$

| 1 | 1 | 1 | 1 | 0 | 1 | 0 | 1 | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | |
| 1 | 1 | | | | | | | | |
| 1 | 1 | 1 | | | | | | | |
| 1 | 0 | 1 | | | | | | | |
| | 1 | 0 | 1 | | | | | | |
| | 1 | 0 | 1 | | | | | | |
| | | | 0 | 0 | | | | | |
| | | | 0 | 0 | 0 | 0 | 1 | 1 | 0 |

**7**

16 : 4

**SOLUZIONE**

$16_{10} = 10000_2$
$4_{10} = 100_2$

| 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | |
| 1 | 0 | | | | | | | | |
| 1 | 0 | 0 | | | | | | | |
| 1 | 0 | 0 | | | | | | | |
| | | 0 | 0 | | | | | | |
| | | 0 | 0 | 0 | | | | | |
| | | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |

**8**

21 : 7

**SOLUZIONE**

$21_{10} = 10101_2$
$7_{10} = 111_2$

| 1 | 0 | 1 | 0 | 1 | 1 | 1 | 1 | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | |
| 1 | 0 | | | | | | | | |
| 1 | 0 | 1 | | | | | | | |
| 1 | 0 | 1 | 0 | | | | | | |
| | 1 | 1 | 1 | | | | | | |
| | | 1 | 1 | 1 | | | | | |
| | | 1 | 1 | 1 | | | | | |
| | | | | 0 | 0 | 0 | 0 | 1 | 1 |

**9**

28 : 7

**SOLUZIONE**

$28_{10} = 11100_2$
$7_{10} = 111_2$

| 1 | 1 | 1 | 0 | 0 | 1 | 1 | 1 | | |
|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | |
| 1 | 1 | | | | | | | | |
| 1 | 1 | 1 | | | | | | | |
| 1 | 1 | 1 | | | | | | | |
| | | 0 | 0 | | | | | | |
| | | 0 | 0 | 0 | | | | | |
| | | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |

**10**

36 : 6

**SOLUZIONE**

$36_{10} = 100100_2$
$6_{10} = 110_2$

| 1 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | |
| 1 | 0 | 0 | | | | | | | | | |
| 1 | 0 | 0 | 1 | | | | | | | | |
| | 1 | 1 | 0 | | | | | | | | |
| | | 1 | 1 | 0 | | | | | | | |
| | | 1 | 1 | 0 | | | | | | | |
| | | | | 0 | 0 | | | | | | |
| | | | | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 |

**11**

40 : 8

**SOLUZIONE**

$40_{10} = 101000_2$
$8_{10} = 1000_2$

| 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | |
| 1 | 0 | 1 | | | | | | | | | | |
| 1 | 0 | 1 | 0 | | | | | | | | | |
| 1 | 0 | 0 | 0 | | | | | | | | | |
| | | 1 | 0 | 0 | | | | | | | | |
| | | 1 | 0 | 0 | 0 | | | | | | | |
| | | 1 | 0 | 0 | 0 | | | | | | | |
| | | | | | 0 | 0 | 0 | 0 | 1 | 0 | 1 | |

**12** ●●

48 : 6

**SOLUZIONE**

$48_{10} = 110000_2$
$6_{10} = 110_2$

| 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | |
| 1 | 1 | 0 | | | | | | | | | |
| 1 | 1 | 0 | | | | | | | | | |
| | | 0 | 0 | | | | | | | | |
| | | 0 | 0 | 0 | | | | | | | |
| | | 0 | 0 | 0 | 0 | | | | | | |
| | | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |

**13** ●●

56 : 7

**SOLUZIONE**

$56_{10} = 111000_2$
$7_{10} = 111_2$

| 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | |
| 1 | 1 | 1 | | | | | | | | | |
| 1 | 1 | 1 | | | | | | | | | |
| | | 0 | 0 | | | | | | | | |
| | | 0 | 0 | 0 | | | | | | | |
| | | 0 | 0 | 0 | 0 | | | | | | |
| | | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 |

**14** ●●

63 : 9

**SOLUZIONE**

$63_{10} = 111111_2$
$9_{10} = 1001_2$

| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | |
| 1 | 1 | 1 | | | | | | | | | |
| 1 | 1 | 1 | 1 | | | | | | | | |
| 1 | 0 | 0 | 1 | | | | | | | | |
| | 1 | 1 | 0 | 1 | | | | | | | |
| | 1 | 0 | 0 | 1 | | | | | | | |
| | | 1 | 0 | 0 | 1 | | | | | | |
| | | 1 | 0 | 0 | 1 | | | | | | |
| | | | | | 0 | 0 | 0 | 0 | 1 | 1 | 1 |

**15** ●●

72 : 8

**SOLUZIONE**

$72_{10} = 1001000_2$
$8_{10} = 1000_2$

| 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | | |
| 1 | 0 | 0 | | | | | | | | | | | |
| 1 | 0 | 0 | 1 | | | | | | | | | | |
| 1 | 0 | 0 | 0 | | | | | | | | | | |
| | | | 1 | 0 | | | | | | | | | |
| | | | 1 | 0 | 0 | | | | | | | | |
| | | | 1 | 0 | 0 | 0 | | | | | | | |
| | | | 1 | 0 | 0 | 0 | | | | | | | |
| | | | | | | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |

**16** ●●

84 : 7

**SOLUZIONE**

$84_{10} = 1010100_2$
$7_{10} = 111_2$

| 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | |
| 1 | 0 | 1 | | | | | | | | | | |
| 1 | 0 | 1 | 0 | | | | | | | | | |
| | 1 | 1 | 1 | | | | | | | | | |
| | | 1 | 1 | 1 | | | | | | | | |
| | | 1 | 1 | 1 | | | | | | | | |
| | | | | 0 | 0 | | | | | | | |
| | | | | 0 | 0 | 0 | | | | | | |
| | | | | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |

**17** ●●

90 : 9

**SOLUZIONE**

$90_{10} = 1011010_2$
$9_{10} = 1001_2$

| 1 | 0 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | | |
| 1 | 0 | 1 | | | | | | | | | | | |
| 1 | 0 | 1 | 1 | | | | | | | | | | |
| 1 | 0 | 0 | 1 | | | | | | | | | | |
| | | 1 | 0 | 0 | | | | | | | | | |
| | | 1 | 0 | 0 | 1 | | | | | | | | |
| | | 1 | 0 | 0 | 1 | | | | | | | | |
| | | | | | 0 | 0 | | | | | | | |
| | | | | | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |

**18** ●●

96 : 8

**SOLUZIONE**

$96_{10} = 1100000_2$
$8_{10} = 1000_2$

| 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | | | |
| 1 | 1 | 0 | | | | | | | | | | | |
| 1 | 1 | 0 | 0 | | | | | | | | | | |
| 1 | 0 | 0 | 0 | | | | | | | | | | |
| | 1 | 0 | 0 | 0 | | | | | | | | | |
| | 1 | 0 | 0 | 0 | | | | | | | | | |
| | | | | 0 | 0 | | | | | | | | |
| | | | | 0 | 0 | 0 | | | | | | | |
| | | | | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |

**19** ●●

108 : 9

**SOLUZIONE**

$108_{10} = 1101100_2$
$9_{10} = 1001_2$

| 1 | 1 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | | | |
| 1 | 1 | 0 | | | | | | | | | | | |
| 1 | 1 | 0 | 1 | | | | | | | | | | |
| 1 | 0 | 0 | 1 | | | | | | | | | | |
| | 1 | 0 | 0 | 1 | | | | | | | | | |
| | 1 | 0 | 0 | 1 | | | | | | | | | |
| | | | | 0 | 0 | | | | | | | | |
| | | | | 0 | 0 | 0 | | | | | | | |
| | | | | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |

**20** ●●

120 : 10

**SOLUZIONE**

$120_{10} = 1111000_2$
$10_{10} = 1010_2$

| 1 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | |
| 1 | 1 | | | | | | | | | |
| 1 | 1 | 1 | | | | | | | | |
| 1 | 1 | 1 | 1 | | | | | | | |
| 1 | 0 | 1 | 0 | | | | | | | |
| | 1 | 0 | 1 | 0 | | | | | | |
| | 1 | 0 | 1 | 0 | | | | | | |
| | | | | 0 | 0 | | | | | |
| | | | | 0 | 0 | 0 | | | | |
| | | | | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |

**21** ●●

132 : 11

**SOLUZIONE**

$132_{10} = 10000100_2$
$11_{10} = 1011_2$

| 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | |
| 1 | 0 | 0 | | | | | | | | | |
| 1 | 0 | 0 | 0 | | | | | | | | |
| 1 | 0 | 0 | 0 | 0 | | | | | | | |
| | 1 | 0 | 1 | 1 | | | | | | | |
| | | 1 | 0 | 1 | 1 | | | | | | |
| | | 1 | 0 | 1 | 1 | | | | | | |
| | | | | | 0 | 0 | | | | | | | |
| | | | | | 0 | 0 | 0 | | | | | | |
| | | | | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |

**22** ●●

144 : 12

**SOLUZIONE**

$144_{10} = 10010000_2$
$12_{10} = 1100_2$

| 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | | | | |
| 1 | 0 | 0 | | | | | | | | | | | | | |
| 1 | 0 | 0 | 1 | | | | | | | | | | | | |
| 1 | 0 | 0 | 1 | 0 | | | | | | | | | | | |
| | 1 | 1 | 0 | 0 | | | | | | | | | | | |
| | | 1 | 1 | 0 | 0 | | | | | | | | | | |
| | | 1 | 1 | 0 | 0 | | | | | | | | | | |
| | | | | | 0 | 0 | | | | | | | | | |
| | | | | | 0 | 0 | 0 | | | | | | | | |
| | | | | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |

**23**

150 : 15

**SOLUZIONE**

$150_{10} = 10010110_2$
$15_{10} = 1111_2$

| 1 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | | | | |
| 1 | 0 | 0 | | | | | | | | | | | | | |
| 1 | 0 | 0 | 1 | | | | | | | | | | | | |
| 1 | 0 | 0 | 1 | 0 | | | | | | | | | | | |
| | 1 | 1 | 1 | 1 | | | | | | | | | | | |
| | | | 1 | 1 | 1 | | | | | | | | | | |
| | | | 1 | 1 | 1 | 1 | | | | | | | | | |
| | | | 1 | 1 | 1 | 1 | | | | | | | | | |
| | | | | | | 0 | 0 | | | | | | | | |
| | | | | | | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |

**24**

168 : 12

**SOLUZIONE**

$168_{10} = 10101000_2$
$12_{10} = 1100_2$

| 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | | | | |
| 1 | 0 | 1 | | | | | | | | | | | | | |
| 1 | 0 | 1 | 0 | | | | | | | | | | | | |
| 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | |
| | 1 | 1 | 0 | 0 | | | | | | | | | | | |
| | 1 | 0 | 0 | 1 | 0 | | | | | | | | | | |
| | | 1 | 1 | 0 | 0 | | | | | | | | | | |
| | | | 1 | 1 | 0 | 0 | | | | | | | | | |
| | | | 1 | 1 | 0 | 0 | | | | | | | | | |
| | | | | | | 0 | 0 | | | | | | | | |
| | | | | | | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 |

**25** 192 : 16

**SOLUZIONE**

$192_{10} = 11000000_2$
$16_{10} = 10000_2$

| 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | | |
| 1 | 1 | 0 | | | | | | | | | | |
| 1 | 1 | 0 | 0 | | | | | | | | | |
| 1 | 1 | 0 | 0 | 0 | | | | | | | | |
| 1 | 0 | 0 | 0 | 0 | | | | | | | | |
| | 1 | 0 | 0 | 0 | 0 | | | | | | | |
| | 1 | 0 | 0 | 0 | 0 | | | | | | | |
| | | | | | 0 | 0 | | | | | | |
| | | | | | 0 | 0 | 0 | | | | | |
| | | | | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 |

**26** 234 : 13

**SOLUZIONE**

$234_{10} = 11101010_2$
$13_{10} = 1101_2$

| 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 0 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | |
| 1 | 1 | 1 | | | | | | | | | |
| 1 | 1 | 1 | 0 | | | | | | | | |
| 1 | 1 | 0 | 1 | | | | | | | | |
| | | | 1 | 1 | | | | | | | |
| | | | 1 | 1 | 0 | | | | | | |
| | | | 1 | 1 | 0 | 1 | | | | | |
| | | | 1 | 1 | 0 | 1 | | | | | |
| | | | | | | 0 | 0 | | | | |
| | | | | | | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 |

**27** 276 : 12

**SOLUZIONE**

$276_{10} = 100010100_2$
$12_{10} = 1100_2$

| 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | |
| 1 | 0 | 0 | | | | | | | | | | |
| 1 | 0 | 0 | 0 | | | | | | | | | |
| 1 | 0 | 0 | 0 | 1 | | | | | | | | |
| | 1 | 1 | 0 | 0 | | | | | | | | |
| | | 1 | 0 | 1 | 0 | | | | | | | |
| | | 1 | 0 | 1 | 0 | 1 | | | | | | |
| | | | 1 | 1 | 0 | 0 | | | | | | |
| | | | 1 | 0 | 0 | 1 | 0 | | | | | |
| | | | | 1 | 1 | 0 | 0 | | | | | |
| | | | | | 1 | 1 | 0 | 0 | | | | |
| | | | | | 1 | 1 | 0 | 0 | | | | |
| | | | | | | | | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 |

**28** ●●●

315 : 15

**SOLUZIONE**

$315_{10} = 100111011_2$
$15_{10} = 1111_2$

| 1 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 1 | 1 | 1 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | |
| 1 | 0 | 0 | | | | | | | | | | |
| 1 | 0 | 0 | 1 | | | | | | | | | |
| 1 | 0 | 0 | 1 | 1 | | | | | | | | |
| | 1 | 1 | 1 | 1 | | | | | | | | |
| | | 1 | 0 | 0 | 1 | | | | | | | |
| | | 1 | 0 | 0 | 1 | 0 | | | | | | |
| | | | 1 | 1 | 1 | 1 | | | | | | |
| | | | | | 1 | 1 | 1 | | | | | |
| | | | | | 1 | 1 | 1 | 1 | | | | |
| | | | | | 1 | 1 | 1 | 1 | | | | |
| | | | | | | | | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 |

**29** ●●●

348 : 12

**SOLUZIONE**

$348_{10} = 101011100_2$
$12_{10} = 1100_2$

| 1 | 0 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | 0 | | | | | | | | | | | | | | |
| 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | |
| | 1 | 1 | 0 | 0 | | | | | | | | | | | | | |
| | 1 | 0 | 0 | 1 | 1 | | | | | | | | | | | | |
| | | 1 | 1 | 0 | 0 | | | | | | | | | | | | |
| | | | 1 | 1 | 1 | 1 | | | | | | | | | | | |
| | | | 1 | 1 | 0 | 0 | | | | | | | | | | | |
| | | | | | 1 | 1 | 0 | | | | | | | | | | |
| | | | | | 1 | 1 | 0 | 0 | | | | | | | | | |
| | | | | | 1 | 1 | 0 | 0 | | | | | | | | | |
| | | | | | | | | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 1 |

**30** ●●●

390 : 13

**SOLUZIONE**

$390_{10} = 110000110_2$

$13_{10} = 1101_2$

| 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | | | | | | | |
| 1 | 1 | 0 | | | | | | | | | | | | | | | |
| 1 | 1 | 0 | 0 | | | | | | | | | | | | | | |
| 1 | 1 | 0 | 0 | 0 | | | | | | | | | | | | | |
| | 1 | 1 | 0 | 1 | | | | | | | | | | | | | |
| | 1 | 0 | 1 | 1 | 0 | | | | | | | | | | | | |
| | | 1 | 1 | 0 | 1 | | | | | | | | | | | | |
| | | 1 | 0 | 0 | 1 | 1 | | | | | | | | | | | |
| | | | 1 | 1 | 0 | 1 | | | | | | | | | | | |
| | | | | 1 | 1 | 0 | 1 | | | | | | | | | | |
| | | | | 1 | 1 | 0 | 1 | | | | | | | | | | |
| | | | | | | | 0 | 0 | | | | | | | | | |
| | | | | | | | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 |

**31** ●●●

425 : 17

**SOLUZIONE**

$425_{10} = 110101001_2$

$17_{10} = 10001_2$

| 1 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | | | | | | | |
| 1 | 1 | 0 | | | | | | | | | | | | | | | |
| 1 | 1 | 0 | 1 | | | | | | | | | | | | | | |
| 1 | 1 | 0 | 1 | 0 | | | | | | | | | | | | | |
| 1 | 0 | 0 | 0 | 1 | | | | | | | | | | | | | |
| | 1 | 0 | 0 | 1 | 1 | | | | | | | | | | | | |
| | 1 | 0 | 0 | 0 | 1 | | | | | | | | | | | | |
| | | | | 1 | 0 | 0 | | | | | | | | | | | |
| | | | | 1 | 0 | 0 | 0 | | | | | | | | | | |
| | | | | 1 | 0 | 0 | 0 | 1 | | | | | | | | | |
| | | | | 1 | 0 | 0 | 0 | 1 | | | | | | | | | |
| | | | | | | | | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 |

**32** 468 : 18

**SOLUZIONE**

$468_{10} = 111010100_2$

$18_{10} = 10010_2$

| 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | | | | | | | |
| 1 | 1 | 1 | | | | | | | | | | | | | | | |
| 1 | 1 | 1 | 0 | | | | | | | | | | | | | | |
| 1 | 1 | 1 | 0 | 1 | | | | | | | | | | | | | |
| 1 | 0 | 0 | 1 | 0 | | | | | | | | | | | | | |
| | 1 | 0 | 1 | 1 | 0 | | | | | | | | | | | | |
| | 1 | 0 | 0 | 1 | 0 | | | | | | | | | | | | |
| | | | 1 | 0 | 0 | 1 | | | | | | | | | | | |
| | | | 1 | 0 | 0 | 1 | 0 | | | | | | | | | | |
| | | | 1 | 0 | 0 | 1 | 0 | | | | | | | | | | |
| | | | | | | | 0 | 0 | | | | | | | | | |
| | | | | | | | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 0 | |

**33** 504 : 21

**SOLUZIONE**

$504_{10} = 111111000_2$

$21_{10} = 10101_2$

| 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | | | | | | | | |
| 1 | 1 | 1 | | | | | | | | | | | | | | | | |
| 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | |
| 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | |
| 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | |
| | 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | |
| | 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | |
| | | | | | 0 | 0 | | | | | | | | | | | | |
| | | | | | 0 | 0 | 0 | | | | | | | | | | | |
| | | | | | 0 | 0 | 0 | 0 | | | | | | | | | | |
| | | | | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | |

**34** ●●●

540 : 18

**SOLUZIONE**

$540_{10} = 1000011100_2$
$18_{10} = 10010_2$

| 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | | | | | | | | | |
| 1 | 0 | 0 | | | | | | | | | | | | | | | | | | |
| 1 | 0 | 0 | 0 | | | | | | | | | | | | | | | | | |
| 1 | 0 | 0 | 0 | 0 | | | | | | | | | | | | | | | | |
| 1 | 0 | 0 | 0 | 0 | 1 | | | | | | | | | | | | | | | |
| | 1 | 0 | 0 | 1 | 0 | | | | | | | | | | | | | | | |
| | | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | |
| | | 1 | 0 | 0 | 1 | 0 | | | | | | | | | | | | | | |
| | | | 1 | 1 | 0 | 1 | 1 | | | | | | | | | | | | | |
| | | | 1 | 0 | 0 | 1 | 0 | | | | | | | | | | | | | |
| | | | | 1 | 0 | 0 | 1 | 0 | | | | | | | | | | | | |
| | | | | 1 | 0 | 0 | 1 | 0 | | | | | | | | | | | | |
| | | | | | | | | 0 | 0 | | | | | | | | | | | |
| | | | | | | | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | |

**35** ●●●

576 : 24

**SOLUZIONE**

$576_{10} = 1001000000_2$
$24_{10} = 11000_2$

| 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | | | |
| 1 | 0 | 0 | | | | | | | | | | | | |
| 1 | 0 | 0 | 1 | | | | | | | | | | | |
| 1 | 0 | 0 | 1 | 0 | | | | | | | | | | |
| 1 | 0 | 0 | 1 | 0 | 0 | | | | | | | | | |
| | 1 | 1 | 0 | 0 | 0 | | | | | | | | | |
| | | 1 | 1 | 0 | 0 | 0 | | | | | | | | |
| | | 1 | 1 | 0 | 0 | 0 | | | | | | | | |
| | | | | | | 0 | 0 | | | | | | | |
| | | | | | | 0 | 0 | 0 | | | | | | |
| | | | | | | 0 | 0 | 0 | 0 | | | | | |
| | | | | | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

**36** 612 : 17

**SOLUZIONE**

$612_{10} = 1001100100_2$
$17_{10} = 10001_2$

| 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | | | |
| 1 | 0 | 0 | | | | | | | | | | | | |
| 1 | 0 | 0 | 1 | | | | | | | | | | | |
| 1 | 0 | 0 | 1 | 1 | | | | | | | | | | |
| 1 | 0 | 0 | 0 | 1 | | | | | | | | | | |
| | | | 1 | 0 | 0 | | | | | | | | | |
| | | | 1 | 0 | 0 | 0 | | | | | | | | |
| | | | 1 | 0 | 0 | 0 | 1 | | | | | | | |
| | | | 1 | 0 | 0 | 0 | 1 | | | | | | | |
| | | | | | | | 0 | 0 | | | | | | |
| | | | | | | | 0 | 0 | 0 | | | | | |
| | | | | | | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

| 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

**37** 648 : 24

**SOLUZIONE**

$648_{10} = 1010001000_2$
$24_{10} = 11000_2$

| 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | | | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | 0 | | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | 0 | 0 | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | 0 | 0 | 0 | | | | | | | | | | | | | | |
| | 1 | 1 | 0 | 0 | 0 | | | | | | | | | | | | | | |
| | 1 | 0 | 0 | 0 | 0 | 1 | | | | | | | | | | | | | |
| | | 1 | 1 | 0 | 0 | 0 | | | | | | | | | | | | | |
| | | | 1 | 0 | 0 | 1 | 0 | | | | | | | | | | | | |
| | | | 1 | 0 | 0 | 1 | 0 | 0 | | | | | | | | | | | |
| | | | | 1 | 1 | 0 | 0 | 0 | | | | | | | | | | | |
| | | | | | 1 | 1 | 0 | 0 | 0 | | | | | | | | | | |
| | | | | | 1 | 1 | 0 | 0 | 0 | | | | | | | | | | |
| | | | | | | | | | | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 |

**38** ●●●

684 : 19

**SOLUZIONE**

$684_{10} = 1010101100_2$
$19_{10} = 10011_2$

| 1 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | | | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | 0 | | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | | |
| 1 | 0 | 0 | 1 | 1 | | | | | | | | | | | | | | | |
| | | | 1 | 0 | 0 | | | | | | | | | | | | | | |
| | | | 1 | 0 | 0 | 1 | | | | | | | | | | | | | |
| | | | 1 | 0 | 0 | 1 | 1 | | | | | | | | | | | | |
| | | | 1 | 0 | 0 | 1 | 1 | | | | | | | | | | | | |
| | | | | | | | 0 | 0 | | | | | | | | | | | |
| | | | | | | | 0 | 0 | 0 | | | | | | | | | | |
| | | | | | | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 |

**39** ●●●

756 : 21

**SOLUZIONE**

$756_{10} = 1011110100_2$
$21_{10} = 10101_2$

| 1 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | | | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | 1 | | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | 1 | 1 | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | | |
| | | | 1 | 0 | 1 | | | | | | | | | | | | | | |
| | | | 1 | 0 | 1 | 0 | | | | | | | | | | | | | |
| | | | 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | | |
| | | | 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | | |
| | | | | | | | 0 | 0 | | | | | | | | | | | |
| | | | | | | | 0 | 0 | 0 | | | | | | | | | | |
| | | | | | | | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 0 |

# 4

# Rilevazione e correzione errori

## 4.1 Bit di parità

Il bit di parità è un bit aggiunto alla sequenza di dati affinché il numero complessivo di bit pari a 1 sia pari (parità pari) o dispari (parità dispari).
Consente di rilevare errori di singolo bit durante la trasmissione: se un bit viene alterato, la parità della parola ricevuta non corrisponde più a quella attesa e l'errore viene segnalato.
Non consente però di correggere l'errore né di rilevare errori su un numero pari di bit.

Effettua il calcolo del bit di parità per le seguenti parole binarie:

1

**Esercizio guidato**
Aggiungi il bit di parità *pari* alla parola 1011001.
Parola dati: 1011001
Numero di bit a 1 nella parola dati: 4
Tipo di parità richiesta: **pari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 10110010

| 1 | 0 | 1 | 1 | 0 | 0 | 1 | |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |

**2** 1011 pari

**SOLUZIONE**

Parola dati: 1011
Numero di bit a 1 nella parola dati: 3
Tipo di parità richiesta: **pari**
Il bit di parità è: 1
Parola codificata (dati + bit di parità): 10111

| | | | | |
|---|---|---|---|---|
| 1 | 0 | 1 | 1 | |
| 1 | 0 | 1 | 1 | 1 |

**3** 1100 pari

**SOLUZIONE**

Parola dati: 1100
Numero di bit a 1 nella parola dati: 2
Tipo di parità richiesta: **pari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 11000

| | | | | |
|---|---|---|---|---|
| 1 | 1 | 0 | 0 | |
| 1 | 1 | 0 | 0 | 0 |

**4** 0011 pari

**SOLUZIONE**

Parola dati: 0011
Numero di bit a 1 nella parola dati: 2
Tipo di parità richiesta: **pari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 00110

| | | | | |
|---|---|---|---|---|
| 0 | 0 | 1 | 1 | |
| 0 | 0 | 1 | 1 | 0 |

**5** 1111 pari

**SOLUZIONE**

Parola dati: 1111
Numero di bit a 1 nella parola dati: 4
Tipo di parità richiesta: **pari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 11110

| | | | | |
|---|---|---|---|---|
| 1 | 1 | 1 | 1 | |
| 1 | 1 | 1 | 1 | 0 |

**6** 1000 pari

**SOLUZIONE**

Parola dati: 1000
Numero di bit a 1 nella parola dati: 1
Tipo di parità richiesta: **pari**

Il bit di parità è: 1
Parola codificata (dati + bit di parità): 10001

| 1 | 0 | 0 | 0 | |
|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 1 |

**7** 1110 pari

**SOLUZIONE**

Parola dati: 1110
Numero di bit a 1 nella parola dati: 3
Tipo di parità richiesta: **pari**
Il bit di parità è: 1
Parola codificata (dati + bit di parità): 11101

| 1 | 1 | 1 | 0 | |
|---|---|---|---|---|
| 1 | 1 | 1 | 0 | 1 |

**8** 0110 pari

**SOLUZIONE**

Parola dati: 0110
Numero di bit a 1 nella parola dati: 2
Tipo di parità richiesta: **pari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 01100

| 0 | 1 | 1 | 0 | |
|---|---|---|---|---|
| 0 | 1 | 1 | 0 | 0 |

**9** 1001 pari

**SOLUZIONE**

Parola dati: 1001
Numero di bit a 1 nella parola dati: 2
Tipo di parità richiesta: **pari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 10010

| 1 | 0 | 0 | 1 | |
|---|---|---|---|---|
| 1 | 0 | 0 | 1 | 0 |

**10** 1101 pari

**SOLUZIONE**

Parola dati: 1101
Numero di bit a 1 nella parola dati: 3
Tipo di parità richiesta: **pari**
Il bit di parità è: 1
Parola codificata (dati + bit di parità): 11011

| 1 | 1 | 0 | 1 | |
|---|---|---|---|---|
| 1 | 1 | 0 | 1 | 1 |

**11** ●

10111 pari

**SOLUZIONE**

Parola dati: 10111
Numero di bit a 1 nella parola dati: 4
Tipo di parità richiesta: **pari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 101110

| 1 | 0 | 1 | 1 | 1 | |
|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 1 | 0 |

**12** ●●

1011001 pari

**SOLUZIONE**

Parola dati: 1011001
Numero di bit a 1 nella parola dati: 4
Tipo di parità richiesta: **pari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 10110010

| 1 | 0 | 1 | 1 | 0 | 0 | 1 | |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |

**13** ●●

11001100 pari

**SOLUZIONE**

Parola dati: 11001100
Numero di bit a 1 nella parola dati: 4
Tipo di parità richiesta: **pari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 110011000

| 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 0 |

**14** ●●

10101010 dispari

**SOLUZIONE**

Parola dati: 10101010
Numero di bit a 1 nella parola dati: 4
Tipo di parità richiesta: **dispari**
Il bit di parità è: 1
Parola codificata (dati + bit di parità): 101010101

| 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |

**15** ●●

11110000 dispari

**SOLUZIONE**

Parola dati: 11110000
Numero di bit a 1 nella parola dati: 4
Tipo di parità richiesta: **dispari**

Il bit di parità è: 1
Parola codificata (dati + bit di parità): 111100001

| 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 |

**16** ●●

10011001 pari

**SOLUZIONE**

Parola dati: 10011001
Numero di bit a 1 nella parola dati: 4
Tipo di parità richiesta: **pari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 100110010

| 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |

**17** ●●

11010010 pari

**SOLUZIONE**

Parola dati: 11010010
Numero di bit a 1 nella parola dati: 4
Tipo di parità richiesta: **pari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 110100100

| 1 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 |

**18** ●●

00111001 dispari

**SOLUZIONE**

Parola dati: 00111001
Numero di bit a 1 nella parola dati: 4
Tipo di parità richiesta: **dispari**
Il bit di parità è: 1
Parola codificata (dati + bit di parità): 001110011

| 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 | 1 |

**19** ●●

10100111 pari

**SOLUZIONE**

Parola dati: 10100111
Numero di bit a 1 nella parola dati: 5
Tipo di parità richiesta: **pari**
Il bit di parità è: 1
Parola codificata (dati + bit di parità): 101001111

| 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 1 |

**20** 11101010 dispari

**SOLUZIONE**

Parola dati: 11101010
Numero di bit a 1 nella parola dati: 5
Tipo di parità richiesta: **dispari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 111010100

| 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 0 | 1 | 0 | 1 | 0 | 0 |

**21** 01010101 pari

**SOLUZIONE**

Parola dati: 01010101
Numero di bit a 1 nella parola dati: 4
Tipo di parità richiesta: **pari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 010101010

| 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | |
|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |

**22** 10010011 dispari

**SOLUZIONE**

Parola dati: 10010011
Numero di bit a 1 nella parola dati: 4
Tipo di parità richiesta: **dispari**
Il bit di parità è: 1
Parola codificata (dati + bit di parità): 100100111

| 1 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 1 |

**23** 01100110 pari

**SOLUZIONE**

Parola dati: 01100110
Numero di bit a 1 nella parola dati: 4
Tipo di parità richiesta: **pari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 011001100

| 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | |
|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 |

**24** 11001101 dispari

**SOLUZIONE**

Parola dati: 11001101
Numero di bit a 1 nella parola dati: 5
Tipo di parità richiesta: **dispari**

Il bit di parità è: 0
Parola codificata (dati + bit di parità): 110011010

| 1 | 1 | 0 | 0 | 1 | 1 | 0 | 1 | |
|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 1 | 1 | 0 | 1 | 0 |

**25** ●●

10111000 pari

**SOLUZIONE**

Parola dati: 10111000
Numero di bit a 1 nella parola dati: 4
Tipo di parità richiesta: **pari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 101110000

| 1 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | |
|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |

**26** ●●

00101101 dispari

**SOLUZIONE**

Parola dati: 00101101
Numero di bit a 1 nella parola dati: 4
Tipo di parità richiesta: **dispari**
Il bit di parità è: 1
Parola codificata (dati + bit di parità): 001011011

| 0 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | 1 |

**27** ●●●

11010011101100 dispari

**SOLUZIONE**

Parola dati: 11010011101100
Numero di bit a 1 nella parola dati: 8
Tipo di parità richiesta: **dispari**
Il bit di parità è: 1
Parola codificata (dati + bit di parità): 110100111011001

| 1 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 1 |

**28** ●●●

10101010101010 pari

**SOLUZIONE**

Parola dati: 10101010101010
Numero di bit a 1 nella parola dati: 7
Tipo di parità richiesta: **pari**
Il bit di parità è: 1
Parola codificata (dati + bit di parità): 101010101010101

| 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |

**29** 1111000011110000 dispari

**SOLUZIONE**

Parola dati: 1111000011110000
Numero di bit a 1 nella parola dati: 8
Tipo di parità richiesta: **dispari**
Il bit di parità è: 1
Parola codificata (dati + bit di parità): 11110000111100001

| 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 1 |

**30** 1001100110011001 pari

**SOLUZIONE**

Parola dati: 1001100110011001
Numero di bit a 1 nella parola dati: 8
Tipo di parità richiesta: **pari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 10011001100110010

| 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |

**31** 11011011011011 dispari

**SOLUZIONE**

Parola dati: 11011011011011
Numero di bit a 1 nella parola dati: 10
Tipo di parità richiesta: **dispari**
Il bit di parità è: 1
Parola codificata (dati + bit di parità): 110110110110111

| 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 1 | |

**32** 1010110100110101 pari

**SOLUZIONE**

Parola dati: 1010110100110101
Numero di bit a 1 nella parola dati: 9
Tipo di parità richiesta: **pari**
Il bit di parità è: 1
Parola codificata (dati + bit di parità): 10101101001101011

| 1 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 1 | 0 | 1 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 1 | 0 | 1 | 1 |

**33** 1110001110001110 dispari

**SOLUZIONE**

Parola dati: 1110001110001110
Numero di bit a 1 nella parola dati: 9
Tipo di parità richiesta: **dispari**

Il bit di parità è: 0
Parola codificata (dati + bit di parità): 11100011100011100

| 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 0 |

**34** ●●●

1001001100110011 pari

**SOLUZIONE**

Parola dati: 1001001100110011
Numero di bit a 1 nella parola dati: 8
Tipo di parità richiesta: **pari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 10010011001100110

| 1 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 |

**35** ●●●

0101010101010101 dispari

**SOLUZIONE**

Parola dati: 0101010101010101
Numero di bit a 1 nella parola dati: 8
Tipo di parità richiesta: **dispari**
Il bit di parità è: 1
Parola codificata (dati + bit di parità): 01010101010101011

| 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 1 |

**36** ●●●

1100110011001100 pari

**SOLUZIONE**

Parola dati: 1100110011001100
Numero di bit a 1 nella parola dati: 8
Tipo di parità richiesta: **pari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 11001100110011000

| 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 0 |

**37** ●●●

1011011011011011 dispari

**SOLUZIONE**

Parola dati: 1011011011011011
Numero di bit a 1 nella parola dati: 11
Tipo di parità richiesta: **dispari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 10110110110110110

| 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 |

**38** 0011001100110011 pari

**SOLUZIONE**

Parola dati: 0011001100110011
Numero di bit a 1 nella parola dati: 8
Tipo di parità richiesta: **pari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 00110011001100110

| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 |

**39** 1111111100000000 dispari

**SOLUZIONE**

Parola dati: 1111111100000000
Numero di bit a 1 nella parola dati: 8
Tipo di parità richiesta: **dispari**
Il bit di parità è: 1
Parola codificata (dati + bit di parità): 11111111000000001

| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |

**40** 1010101001010101 pari

**SOLUZIONE**

Parola dati: 1010101001010101
Numero di bit a 1 nella parola dati: 8
Tipo di parità richiesta: **pari**
Il bit di parità è: 0
Parola codificata (dati + bit di parità): 10101010010101010

| 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 |

**41** 1100110000111100 dispari

**SOLUZIONE**

Parola dati: 1100110000111100
Numero di bit a 1 nella parola dati: 8
Tipo di parità richiesta: **dispari**
Il bit di parità è: 1
Parola codificata (dati + bit di parità): 11001100001111001

| 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 | 1 |

## 4.2 CRC

Effettua il calcolo del CRC per le seguenti parole binarie dato un generatore:

1

**Esercizio guidato**

Calcola il CRC del messaggio 1101001110 usando il generatore 101.

Lunghezza generatore 3 bit

Si aggiungono 2 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | | | | |
| 1 | 1 | 0 | | | | | | | | | | | | |
| 1 | 0 | 1 | | | | | | | | | | | | |
| | 1 | 1 | 1 | | | | | | | | | | | |
| | 1 | 0 | 1 | | | | | | | | | | | |
| | | 1 | 0 | 0 | | | | | | | | | | |
| | | 1 | 0 | 1 | | | | | | | | | | |
| | | | 0 | 1 | 0 | | | | | | | | | |
| | | | | 1 | 0 | 1 | | | | | | | | |
| | | | | 1 | 0 | 1 | | | | | | | | |
| | | | | | 0 | 0 | 1 | | | | | | | |
| | | | | | | 0 | 1 | 1 | | | | | | |
| | | | | | | | 1 | 1 | 0 | | | | | |
| | | | | | | | 1 | 0 | 1 | | | | | |
| | | | | | | | | 1 | 1 | 0 | | | | |
| | | | | | | | | 1 | 0 | 1 | | | | |
| | | | | | | | | | 1 | 1 | 0 | | | |
| | | | | | | | | | 1 | 0 | 1 | | | |
| | | | | | | | | | | 0 | 1 | 1 | | |

Messaggio trasmesso (dati + FCS): 1101001110 **11**

**2**

Parola : 1101
Generatore : 101

**SOLUZIONE**

Lunghezza generatore 3 bit
Si aggiungono 2 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 |
|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | |
| 1 | 1 | | | | | | | |
| 1 | 1 | 0 | | | | | | |
| 1 | 0 | 1 | | | | | | |
| | 1 | 1 | 1 | | | | | |
| | 1 | 0 | 1 | | | | | |
| | | 1 | 0 | 0 | | | | |
| | | 1 | 0 | 1 | | | | |
| | | | 0 | 1 | 0 | | | |
| | | | 0 | 1 | 0 | | | |

Messaggio trasmesso (dati + FCS): 1101 **10**

**3**

Parola : 1011
Generatore : 110

**SOLUZIONE**

Lunghezza generatore 3 bit
Si aggiungono 2 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 |
|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | |
| 1 | 0 | | | | | | | |
| 1 | 0 | 1 | | | | | | |
| 1 | 1 | 0 | | | | | | |
| | 1 | 1 | 1 | | | | | |
| | 1 | 1 | 0 | | | | | |
| | | 0 | 1 | 0 | | | | |
| | | | 1 | 0 | 0 | | | |
| | | | 1 | 1 | 0 | | | |
| | | | 0 | 1 | 0 | | | |

Messaggio trasmesso (dati + FCS): 1011 **10**

**4**

Parola : 1110
Generatore : 101

**SOLUZIONE**

Lunghezza generatore 3 bit
Si aggiungono 2 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 1 |
|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | |
| 1 | 1 | | | | | | | |
| 1 | 1 | 1 | | | | | | |
| 1 | 0 | 1 | | | | | | |
| | 1 | 0 | 0 | | | | | |
| | 1 | 0 | 1 | | | | | |
| | | 0 | 1 | 0 | | | | |
| | | | 1 | 0 | 0 | | | |
| | | | 1 | 0 | 1 | | | |
| | | | 0 | 0 | 1 | | | |

Messaggio trasmesso (dati + FCS): 1110 **01**

**5**

Parola : 1001
Generatore : 111

**SOLUZIONE**

Lunghezza generatore 3 bit
Si aggiungono 2 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 0 | 0 | 1 | 0 | 0 | 1 | 1 | 1 |
|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | |
| 1 | 0 | | | | | | | |
| 1 | 0 | 0 | | | | | | |
| 1 | 1 | 1 | | | | | | |
| | 1 | 1 | 1 | | | | | |
| | 1 | 1 | 1 | | | | | |
| | | 0 | 0 | 0 | | | | |
| | | | 0 | 0 | 0 | | | |
| | | | 0 | 0 | 0 | | | |

Messaggio trasmesso (dati + FCS): 1001 **00**

**6**

Parola : 1100
Generatore : 101

**SOLUZIONE**

Lunghezza generatore 3 bit
Si aggiungono 2 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 1 |
|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | |
| 1 | 1 | | | | | | | |
| 1 | 1 | 0 | | | | | | |
| 1 | 0 | 1 | | | | | | |
| | 1 | 1 | 0 | | | | | |
| | 1 | 0 | 1 | | | | | |
| | | 1 | 1 | 0 | | | | |
| | | 1 | 0 | 1 | | | | |
| | | | 1 | 1 | 0 | | | |
| | | | 1 | 0 | 1 | | | |
| | | | 0 | 1 | 1 | | | |

Messaggio trasmesso (dati + FCS): 1100 **11**

**7**

Parola : 1010
Generatore : 101

**SOLUZIONE**

Lunghezza generatore 3 bit
Si aggiungono 2 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 1 |
|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | |
| 1 | 0 | | | | | | | |
| 1 | 0 | 1 | | | | | | |
| 1 | 0 | 1 | | | | | | |
| | 0 | 0 | 0 | | | | | |
| | | 0 | 0 | 0 | | | | |
| | | | 0 | 0 | 0 | | | |
| | | | 0 | 0 | 0 | | | |

Messaggio trasmesso (dati + FCS): 1010 **00**

**8**

Parola : 1111
Generatore : 1101

**SOLUZIONE**

Lunghezza generatore 4 bit
Si aggiungono 3 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | |
| 1 | 1 | | | | | | | | | |
| 1 | 1 | 1 | | | | | | | | |
| 1 | 1 | 1 | 1 | | | | | | | |
| 1 | 1 | 0 | 1 | | | | | | | |
| | 0 | 1 | 0 | 0 | | | | | | |
| | | 1 | 0 | 0 | 0 | | | | | |
| | | 1 | 1 | 0 | 1 | | | | | |
| | | | 1 | 0 | 1 | 0 | | | | |
| | | | 1 | 1 | 0 | 1 | | | | |
| | | | 0 | 1 | 1 | 1 | | | | |

Messaggio trasmesso (dati + FCS): 1111 **111**

**9**

Parola : 1000
Generatore : 1011

**SOLUZIONE**

Lunghezza generatore 4 bit
Si aggiungono 3 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | |
| 1 | 0 | | | | | | | | | |
| 1 | 0 | 0 | | | | | | | | |
| 1 | 0 | 0 | 0 | | | | | | | |
| 1 | 0 | 1 | 1 | | | | | | | |
| | 0 | 1 | 1 | 0 | | | | | | |
| | | 1 | 1 | 0 | 0 | | | | | |
| | | 1 | 0 | 1 | 1 | | | | | |
| | | | 1 | 1 | 1 | 0 | | | | |
| | | | 1 | 0 | 1 | 1 | | | | |
| | | | 0 | 1 | 0 | 1 | | | | |

Messaggio trasmesso (dati + FCS): 1000 **101**

**10**

Parola : 0110
Generatore : 1001

**SOLUZIONE**

Lunghezza generatore 4 bit
Si aggiungono 3 zeri in coda al messaggio prima di procedere con il calcolo

| 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | | | | | | | | | | |
| 0 | 1 | | | | | | | | | |
| 0 | 1 | 1 | | | | | | | | |
| 0 | 1 | 1 | 0 | | | | | | | |
| | 1 | 1 | 0 | 0 | | | | | | |
| | 1 | 0 | 0 | 1 | | | | | | |
| | | 1 | 0 | 1 | 0 | | | | | |
| | | 1 | 0 | 0 | 1 | | | | | |
| | | | 0 | 1 | 1 | 0 | | | | |
| | | | 0 | 1 | 1 | 0 | | | | |

Messaggio trasmesso (dati + FCS): 0110 **110**

**11**

Parola : 0101
Generatore : 1101

**SOLUZIONE**

Lunghezza generatore 4 bit
Si aggiungono 3 zeri in coda al messaggio prima di procedere con il calcolo

| 0 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|
| 0 | | | | | | | | | | |
| 0 | 1 | | | | | | | | | |
| 0 | 1 | 0 | | | | | | | | |
| 0 | 1 | 0 | 1 | | | | | | | |
| | 1 | 0 | 1 | 0 | | | | | | |
| | 1 | 1 | 0 | 1 | | | | | | |
| | | 1 | 1 | 1 | 0 | | | | | |
| | | 1 | 1 | 0 | 1 | | | | | |
| | | | 0 | 1 | 1 | 0 | | | | |
| | | | 0 | 1 | 1 | 0 | | | | |

Messaggio trasmesso (dati + FCS): 0101 **110**

**12**

Parola : 10101
Generatore : 1011

**SOLUZIONE**

Lunghezza generatore 4 bit
Si aggiungono 3 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | |
| 1 | 0 | 1 | | | | | | | | | |
| 1 | 0 | 1 | 0 | | | | | | | | |
| 1 | 0 | 1 | 1 | | | | | | | | |
| | 0 | 0 | 1 | 1 | | | | | | | |
| | | 0 | 1 | 1 | 0 | | | | | | |
| | | | 1 | 1 | 0 | 0 | | | | | |
| | | | 1 | 0 | 1 | 1 | | | | | |
| | | | | 1 | 1 | 1 | 0 | | | | |
| | | | | 1 | 0 | 1 | 1 | | | | |
| | | | | 0 | 1 | 0 | 1 | | | | |

Messaggio trasmesso (dati + FCS): 10101 **101**

**13**

Parola : 1101001
Generatore : 1011

**SOLUZIONE**

Lunghezza generatore 4 bit
Si aggiungono 3 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 1 | 0 | 1 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | | | |
| 1 | 1 | 0 | | | | | | | | | | | |
| 1 | 1 | 0 | 1 | | | | | | | | | | |
| 1 | 0 | 1 | 1 | | | | | | | | | | |
| | 1 | 1 | 0 | 0 | | | | | | | | | |
| | 1 | 0 | 1 | 1 | | | | | | | | | |
| | | 1 | 1 | 1 | 0 | | | | | | | | |
| | | 1 | 0 | 1 | 1 | | | | | | | | |
| | | | 1 | 0 | 1 | 1 | | | | | | | |
| | | | 1 | 0 | 1 | 1 | | | | | | | |
| | | | | 0 | 0 | 0 | 0 | | | | | | |
| | | | | | 0 | 0 | 0 | 0 | | | | | |
| | | | | | | 0 | 0 | 0 | 0 | | | | |
| | | | | | | 0 | 0 | 0 | 0 | | | | |

Messaggio trasmesso (dati + FCS): 1101001 **000**

**14**

Parola : 1010100
Generatore : 1101

**SOLUZIONE**

Lunghezza generatore 4 bit
Si aggiungono 3 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | | |
| 1 | 0 | 1 | | | | | | | | | | | |
| 1 | 0 | 1 | 0 | | | | | | | | | | |
| 1 | 1 | 0 | 1 | | | | | | | | | | |
| | 1 | 1 | 1 | 1 | | | | | | | | | |
| | 1 | 1 | 0 | 1 | | | | | | | | | |
| | | 0 | 1 | 0 | 0 | | | | | | | | |
| | | | 1 | 0 | 0 | 0 | | | | | | | |
| | | | 1 | 1 | 0 | 1 | | | | | | | |
| | | | | 1 | 0 | 1 | 0 | | | | | | |
| | | | | 1 | 1 | 0 | 1 | | | | | | |
| | | | | | 1 | 1 | 1 | 0 | | | | | |
| | | | | | 1 | 1 | 0 | 1 | | | | | |
| | | | | | | 0 | 1 | 1 | 0 | | | | |
| | | | | | | 0 | 1 | 1 | 0 | | | | |

Messaggio trasmesso (dati + FCS): 1010100 **110**

**15** ●●

Parola : 1100110
Generatore : 1001

**SOLUZIONE**

Lunghezza generatore 4 bit
Si aggiungono 3 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | | | |
| 1 | 1 | 0 | | | | | | | | | | | |
| 1 | 1 | 0 | 0 | | | | | | | | | | |
| 1 | 0 | 0 | 1 | | | | | | | | | | |
| | 1 | 0 | 1 | 1 | | | | | | | | | |
| | 1 | 0 | 0 | 1 | | | | | | | | | |
| | | 0 | 1 | 0 | 1 | | | | | | | | |
| | | | 1 | 0 | 1 | 0 | | | | | | | |
| | | | 1 | 0 | 0 | 1 | | | | | | | |
| | | | | 0 | 1 | 1 | 0 | | | | | | |
| | | | | | 1 | 1 | 0 | 0 | | | | | |
| | | | | | 1 | 0 | 0 | 1 | | | | | |
| | | | | | | 1 | 0 | 1 | 0 | | | | |
| | | | | | | 1 | 0 | 0 | 1 | | | | |
| | | | | | | | 0 | 0 | 1 | 1 | | | |

Messaggio trasmesso (dati + FCS): 1100110 **011**

**16** ●●

Parola : 1010111
Generatore : 1101

**SOLUZIONE**

Lunghezza generatore 4 bit
Si aggiungono 3 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 0 | 1 | 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | | |
| 1 | 0 | 1 | | | | | | | | | | | |
| 1 | 0 | 1 | 0 | | | | | | | | | | |
| 1 | 1 | 0 | 1 | | | | | | | | | | |
| | 1 | 1 | 1 | 1 | | | | | | | | | |
| | 1 | 1 | 0 | 1 | | | | | | | | | |
| | | 0 | 1 | 0 | 1 | | | | | | | | |
| | | | 1 | 0 | 1 | 1 | | | | | | | |
| | | | 1 | 1 | 0 | 1 | | | | | | | |
| | | | | 1 | 1 | 0 | 0 | | | | | | |
| | | | | 1 | 1 | 0 | 1 | | | | | | |
| | | | | | 0 | 0 | 1 | 0 | | | | | |
| | | | | | | 0 | 1 | 0 | 0 | | | | |
| | | | | | | | 0 | 1 | 0 | 0 | | | |
| | | | | | | | 0 | 1 | 0 | 0 | | | |

Messaggio trasmesso (dati + FCS): 1010111 **100**

**17**

Parola : 1100100
Generatore : 1001

**SOLUZIONE**

Lunghezza generatore 4 bit
Si aggiungono 3 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | | | |
| 1 | 1 | 0 | | | | | | | | | | | |
| 1 | 1 | 0 | 0 | | | | | | | | | | |
| 1 | 0 | 0 | 1 | | | | | | | | | | |
| | 1 | 0 | 1 | 1 | | | | | | | | | |
| | 1 | 0 | 0 | 1 | | | | | | | | | |
| | | 0 | 1 | 0 | 0 | | | | | | | | |
| | | | 1 | 0 | 0 | 0 | | | | | | | |
| | | | 1 | 0 | 0 | 1 | | | | | | | |
| | | | | 0 | 0 | 1 | 0 | | | | | | |
| | | | | | 0 | 1 | 0 | 0 | | | | | |
| | | | | | | 1 | 0 | 0 | 0 | | | | |
| | | | | | | 1 | 0 | 0 | 1 | | | | |
| | | | | | | | 0 | 0 | 0 | 1 | | | |

Messaggio trasmesso (dati + FCS): 1100100 **001**

**18**

Parola : 10110011
Generatore : 1101

**SOLUZIONE**

Lunghezza generatore 4 bit
Si aggiungono 3 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | | | |
| 1 | 0 | 1 | | | | | | | | | | | | |
| 1 | 0 | 1 | 1 | | | | | | | | | | | |
| 1 | 1 | 0 | 1 | | | | | | | | | | | |
| | 1 | 1 | 0 | 0 | | | | | | | | | | |
| | 1 | 1 | 0 | 1 | | | | | | | | | | |
| | | 0 | 0 | 1 | 0 | | | | | | | | | |
| | | | 0 | 1 | 0 | 1 | | | | | | | | |
| | | | | 1 | 0 | 1 | 1 | | | | | | | |
| | | | | 1 | 1 | 0 | 1 | | | | | | | |
| | | | | | 1 | 1 | 0 | 0 | | | | | | |
| | | | | | 1 | 1 | 0 | 1 | | | | | | |
| | | | | | | 0 | 0 | 1 | 0 | | | | | |
| | | | | | | | 0 | 1 | 0 | 0 | | | | |
| | | | | | | | | 0 | 1 | 0 | 0 | | | |

Messaggio trasmesso (dati + FCS): 10110011 **100**

**19** ●●

Parola : 11100110
Generatore : 1111

**SOLUZIONE**

Lunghezza generatore 4 bit
Si aggiungono 3 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | | | | |
| 1 | 1 | 1 | | | | | | | | | | | | |
| 1 | 1 | 1 | 0 | | | | | | | | | | | |
| 1 | 1 | 1 | 1 | | | | | | | | | | | |
| | 0 | 0 | 1 | 0 | | | | | | | | | | |
| | | 0 | 1 | 0 | 1 | | | | | | | | | |
| | | | 1 | 0 | 1 | 1 | | | | | | | | |
| | | | 1 | 1 | 1 | 1 | | | | | | | | |
| | | | | 1 | 0 | 0 | 0 | | | | | | | |
| | | | | 1 | 1 | 1 | 1 | | | | | | | |
| | | | | | 1 | 1 | 1 | 0 | | | | | | |
| | | | | | 1 | 1 | 1 | 1 | | | | | | |
| | | | | | | 0 | 0 | 1 | 0 | | | | | |
| | | | | | | | 0 | 1 | 0 | 0 | | | | |
| | | | | | | | | 0 | 1 | 0 | 0 | | | |

Messaggio trasmesso (dati + FCS): 11100110 **100**

**20** ●●

Parola : 10010110
Generatore : 10111

**SOLUZIONE**

Lunghezza generatore 5 bit
Si aggiungono 4 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | | | | | |
| 1 | 0 | 0 | | | | | | | | | | | | | | |
| 1 | 0 | 0 | 1 | | | | | | | | | | | | | |
| 1 | 0 | 0 | 1 | 0 | | | | | | | | | | | | |
| 1 | 0 | 1 | 1 | 1 | | | | | | | | | | | | |
| | 0 | 1 | 0 | 1 | 1 | | | | | | | | | | | |
| | | 1 | 0 | 1 | 1 | 1 | | | | | | | | | | |
| | | 1 | 0 | 1 | 1 | 1 | | | | | | | | | | |
| | | | 0 | 0 | 0 | 0 | 0 | | | | | | | | | |
| | | | | 0 | 0 | 0 | 0 | 0 | | | | | | | | |
| | | | | | 0 | 0 | 0 | 0 | 0 | | | | | | | |
| | | | | | | 0 | 0 | 0 | 0 | 0 | | | | | | |
| | | | | | | | 0 | 0 | 0 | 0 | 0 | | | | | |
| | | | | | | | | 0 | 0 | 0 | 0 | 0 | | | | |
| | | | | | | | | 0 | 0 | 0 | 0 | 0 | | | | |

Messaggio trasmesso (dati + FCS): 10010110 **0000**

**21**

Parola : 11011001
Generatore : 10011

**SOLUZIONE**

Lunghezza generatore 5 bit
Si aggiungono 4 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 1 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | | | | | | |
| 1 | 1 | 0 | | | | | | | | | | | | | | |
| 1 | 1 | 0 | 1 | | | | | | | | | | | | | |
| 1 | 1 | 0 | 1 | 1 | | | | | | | | | | | | |
| 1 | 0 | 0 | 1 | 1 | | | | | | | | | | | | |
| | 1 | 0 | 0 | 0 | 0 | | | | | | | | | | | |
| | 1 | 0 | 0 | 1 | 1 | | | | | | | | | | | |
| | | 0 | 0 | 1 | 1 | 0 | | | | | | | | | | |
| | | | 0 | 1 | 1 | 0 | 1 | | | | | | | | | |
| | | | | 1 | 1 | 0 | 1 | 0 | | | | | | | | |
| | | | | 1 | 0 | 0 | 1 | 1 | | | | | | | | |
| | | | | | 1 | 0 | 0 | 1 | 0 | | | | | | | |
| | | | | | 1 | 0 | 0 | 1 | 1 | | | | | | | |
| | | | | | | 0 | 0 | 0 | 1 | 0 | | | | | | |
| | | | | | | | 0 | 0 | 1 | 0 | 0 | | | | | |
| | | | | | | | | 0 | 0 | 1 | 0 | 0 | | | | |

Messaggio trasmesso (dati + FCS): 11011001 **0100**

**22**

Parola : 1101001100

Generatore : 10111

**SOLUZIONE**

Lunghezza generatore 5 bit
Si aggiungono 4 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | | | | | | | | |
| 1 | 1 | 0 | | | | | | | | | | | | | | | | |
| 1 | 1 | 0 | 1 | | | | | | | | | | | | | | | |
| 1 | 1 | 0 | 1 | 0 | | | | | | | | | | | | | | |
| 1 | 0 | 1 | 1 | 1 | | | | | | | | | | | | | | |
| | 1 | 1 | 0 | 1 | 0 | | | | | | | | | | | | | |
| | 1 | 0 | 1 | 1 | 1 | | | | | | | | | | | | | |
| | | 1 | 1 | 0 | 1 | 1 | | | | | | | | | | | | |
| | | 1 | 0 | 1 | 1 | 1 | | | | | | | | | | | | |
| | | | 1 | 1 | 0 | 0 | 1 | | | | | | | | | | | |
| | | | 1 | 0 | 1 | 1 | 1 | | | | | | | | | | | |
| | | | | 1 | 1 | 1 | 0 | 0 | | | | | | | | | | |
| | | | | 1 | 0 | 1 | 1 | 1 | | | | | | | | | | |
| | | | | | 1 | 0 | 1 | 1 | 0 | | | | | | | | | |
| | | | | | 1 | 0 | 1 | 1 | 1 | | | | | | | | | |
| | | | | | | 0 | 0 | 0 | 1 | 0 | | | | | | | | |
| | | | | | | | 0 | 0 | 1 | 0 | 0 | | | | | | | |
| | | | | | | | | 0 | 1 | 0 | 0 | 0 | | | | | | |
| | | | | | | | | | 1 | 0 | 0 | 0 | 0 | | | | | |
| | | | | | | | | | 1 | 0 | 1 | 1 | 1 | | | | | |
| | | | | | | | | | 0 | 0 | 1 | 1 | 1 | | | | | |

Messaggio trasmesso (dati + FCS): 1101001100 **0111**

**23**

Parola : 1010101010
Generatore : 11011

**SOLUZIONE**

Lunghezza generatore 5 bit
Si aggiungono 4 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | 0 | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | |
| 1 | 1 | 0 | 1 | 1 | | | | | | | | | | | | | | |
| | 1 | 1 | 1 | 0 | 0 | | | | | | | | | | | | | |
| | 1 | 1 | 0 | 1 | 1 | | | | | | | | | | | | | |
| | | 0 | 1 | 1 | 1 | 1 | | | | | | | | | | | | |
| | | | 1 | 1 | 1 | 1 | 0 | | | | | | | | | | | |
| | | | 1 | 1 | 0 | 1 | 1 | | | | | | | | | | | |
| | | | | 0 | 1 | 0 | 1 | 1 | | | | | | | | | | |
| | | | | | 1 | 0 | 1 | 1 | 0 | | | | | | | | | |
| | | | | | 1 | 1 | 0 | 1 | 1 | | | | | | | | | |
| | | | | | | 1 | 1 | 0 | 1 | 0 | | | | | | | | |
| | | | | | | 1 | 1 | 0 | 1 | 1 | | | | | | | | |
| | | | | | | | 0 | 0 | 0 | 1 | 0 | | | | | | | |
| | | | | | | | | 0 | 0 | 1 | 0 | 0 | | | | | | |
| | | | | | | | | | 0 | 1 | 0 | 0 | 0 | | | | | |
| | | | | | | | | | 0 | 1 | 0 | 0 | 0 | | | | | |

Messaggio trasmesso (dati + FCS): 1010101010 **1000**

**24**

Parola : 11110010000
Generatore : 10111

**SOLUZIONE**

Lunghezza generatore 5 bit
Si aggiungono 4 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | | | | | | | | | |
| 1 | 1 | 1 | | | | | | | | | | | | | | | | | |
| 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | |
| 1 | 1 | 1 | 1 | 0 | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | 1 | 1 | | | | | | | | | | | | | | | |
| | 1 | 0 | 0 | 1 | 0 | | | | | | | | | | | | | | |
| | 1 | 0 | 1 | 1 | 1 | | | | | | | | | | | | | | |
| | | 0 | 1 | 0 | 1 | 1 | | | | | | | | | | | | | |
| | | | 1 | 0 | 1 | 1 | 0 | | | | | | | | | | | | |
| | | | 1 | 0 | 1 | 1 | 1 | | | | | | | | | | | | |
| | | | | 0 | 0 | 0 | 1 | 0 | | | | | | | | | | | |
| | | | | | 0 | 0 | 1 | 0 | 0 | | | | | | | | | | |
| | | | | | | 0 | 1 | 0 | 0 | 0 | | | | | | | | | |
| | | | | | | | 1 | 0 | 0 | 0 | 0 | | | | | | | | |
| | | | | | | | 1 | 0 | 1 | 1 | 1 | | | | | | | | |
| | | | | | | | | 0 | 1 | 1 | 1 | 0 | | | | | | | |
| | | | | | | | | | 1 | 1 | 1 | 0 | 0 | | | | | | |
| | | | | | | | | | 1 | 0 | 1 | 1 | 1 | | | | | | |
| | | | | | | | | | | 1 | 0 | 1 | 1 | 0 | | | | | |
| | | | | | | | | | | 1 | 0 | 1 | 1 | 1 | | | | | |
| | | | | | | | | | | | 0 | 0 | 0 | 0 | 1 | | | | |

Messaggio trasmesso (dati + FCS): 11110010000 **0001**

**25**

Parola : 10011011001
Generatore : 11001

**SOLUZIONE**

Lunghezza generatore 5 bit
Si aggiungono 4 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | | | | | | | | |
| 1 | 0 | 0 | | | | | | | | | | | | | | | | | |
| 1 | 0 | 0 | 1 | | | | | | | | | | | | | | | | |
| 1 | 0 | 0 | 1 | 1 | | | | | | | | | | | | | | | |
| 1 | 1 | 0 | 0 | 1 | | | | | | | | | | | | | | | |
| | 1 | 0 | 1 | 0 | 0 | | | | | | | | | | | | | | |
| | 1 | 1 | 0 | 0 | 1 | | | | | | | | | | | | | | |
| | | 1 | 1 | 0 | 1 | 1 | | | | | | | | | | | | | |
| | | 1 | 1 | 0 | 0 | 1 | | | | | | | | | | | | | |
| | | | 0 | 0 | 1 | 0 | 1 | | | | | | | | | | | | |
| | | | | 0 | 1 | 0 | 1 | 0 | | | | | | | | | | | |
| | | | | | 1 | 0 | 1 | 0 | 0 | | | | | | | | | | |
| | | | | | 1 | 1 | 0 | 0 | 1 | | | | | | | | | | |
| | | | | | | 1 | 1 | 0 | 1 | 1 | | | | | | | | | |
| | | | | | | 1 | 1 | 0 | 0 | 1 | | | | | | | | | |
| | | | | | | | 0 | 0 | 1 | 0 | 0 | | | | | | | | |
| | | | | | | | | 0 | 1 | 0 | 0 | 0 | | | | | | | |
| | | | | | | | | | 1 | 0 | 0 | 0 | 0 | | | | | | |
| | | | | | | | | | 1 | 1 | 0 | 0 | 1 | | | | | | |
| | | | | | | | | | | 1 | 0 | 0 | 1 | 0 | | | | | |
| | | | | | | | | | | 1 | 1 | 0 | 0 | 1 | | | | | |
| | | | | | | | | | | 0 | 1 | 0 | 1 | 1 | | | | | |

Messaggio trasmesso (dati + FCS): 10011011001 **1011**

**26**

Parola : 11011011011
Generatore : 10101

**SOLUZIONE**

Lunghezza generatore 5 bit
Si aggiungono 4 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | | | | | | | | | |
| 1 | 1 | 0 | | | | | | | | | | | | | | | | | |
| 1 | 1 | 0 | 1 | | | | | | | | | | | | | | | | |
| 1 | 1 | 0 | 1 | 1 | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | | |
| | 1 | 1 | 1 | 0 | 0 | | | | | | | | | | | | | | |
| | 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | |
| | | 1 | 0 | 0 | 1 | 1 | | | | | | | | | | | | | |
| | | 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | |
| | | | 0 | 1 | 1 | 0 | 1 | | | | | | | | | | | | |
| | | | | 1 | 1 | 0 | 1 | 0 | | | | | | | | | | | |
| | | | | 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | |
| | | | | | 1 | 1 | 1 | 1 | 1 | | | | | | | | | | |
| | | | | | 1 | 0 | 1 | 0 | 1 | | | | | | | | | | |
| | | | | | | 1 | 0 | 1 | 0 | 1 | | | | | | | | | |
| | | | | | | 1 | 0 | 1 | 0 | 1 | | | | | | | | | |
| | | | | | | | 0 | 0 | 0 | 0 | 0 | | | | | | | | |
| | | | | | | | | 0 | 0 | 0 | 0 | 0 | | | | | | | |
| | | | | | | | | | 0 | 0 | 0 | 0 | 0 | | | | | | |
| | | | | | | | | | | 0 | 0 | 0 | 0 | 0 | | | | | |
| | | | | | | | | | | | 0 | 0 | 0 | 0 | 0 | | | | |

Messaggio trasmesso (dati + FCS): 11011011011 **0000**

**27**

Parola : 101010101010
Generatore : 11001

**SOLUZIONE**

Lunghezza generatore 5 bit
Si aggiungono 4 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 0 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | | | | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | 0 | | | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | | | |
| 1 | 1 | 0 | 0 | 1 | | | | | | | | | | | | | | | | |
| | 1 | 1 | 0 | 0 | 0 | | | | | | | | | | | | | | | |
| | 1 | 1 | 0 | 0 | 1 | | | | | | | | | | | | | | | |
| | | 0 | 0 | 0 | 1 | 1 | | | | | | | | | | | | | | |
| | | | 0 | 0 | 1 | 1 | 0 | | | | | | | | | | | | | |
| | | | | 0 | 1 | 1 | 0 | 1 | | | | | | | | | | | | |
| | | | | | 1 | 1 | 0 | 1 | 0 | | | | | | | | | | | |
| | | | | | 1 | 1 | 0 | 0 | 1 | | | | | | | | | | | |
| | | | | | | 0 | 0 | 1 | 1 | 1 | | | | | | | | | | |
| | | | | | | | 0 | 1 | 1 | 1 | 0 | | | | | | | | | |
| | | | | | | | | 1 | 1 | 1 | 0 | 0 | | | | | | | | |
| | | | | | | | | 1 | 1 | 0 | 0 | 1 | | | | | | | | |
| | | | | | | | | | 0 | 1 | 0 | 1 | 0 | | | | | | | |
| | | | | | | | | | | 1 | 0 | 1 | 0 | 0 | | | | | | |
| | | | | | | | | | | 1 | 1 | 0 | 0 | 1 | | | | | | |
| | | | | | | | | | | | 1 | 1 | 0 | 1 | 0 | | | | | |
| | | | | | | | | | | | 1 | 1 | 0 | 0 | 1 | | | | | |
| | | | | | | | | | | | 0 | 0 | 0 | 1 | 1 | | | | | |

Messaggio trasmesso (dati + FCS): 101010101010 **0011**

**28**

Parola : 111100010000
Generatore : 10101

**SOLUZIONE**

Lunghezza generatore 5 bit
Si aggiungono 4 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 1 | 1 | 1 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | | | | |
| 1 | 1 | | | | | | | | | | | | | | | | | | |
| 1 | 1 | 1 | | | | | | | | | | | | | | | | | |
| 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | |
| 1 | 1 | 1 | 1 | 0 | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | | |
| | 1 | 0 | 1 | 1 | 0 | | | | | | | | | | | | | | |
| | 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | |
| | | 0 | 0 | 1 | 1 | 0 | | | | | | | | | | | | | |
| | | | 0 | 1 | 1 | 0 | 1 | | | | | | | | | | | | |
| | | | | 1 | 1 | 0 | 1 | 0 | | | | | | | | | | | |
| | | | | 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | |
| | | | | | 1 | 1 | 1 | 1 | 0 | | | | | | | | | | |
| | | | | | 1 | 0 | 1 | 0 | 1 | | | | | | | | | | |
| | | | | | | 1 | 0 | 1 | 1 | 0 | | | | | | | | | |
| | | | | | | 1 | 0 | 1 | 0 | 1 | | | | | | | | | |
| | | | | | | | 0 | 0 | 1 | 1 | 0 | | | | | | | | |
| | | | | | | | | 0 | 1 | 1 | 0 | 0 | | | | | | | |
| | | | | | | | | | 1 | 1 | 0 | 0 | 0 | | | | | | |
| | | | | | | | | | 1 | 0 | 1 | 0 | 1 | | | | | | |
| | | | | | | | | | | 1 | 1 | 0 | 1 | 0 | | | | | |
| | | | | | | | | | | 1 | 0 | 1 | 0 | 1 | | | | | |
| | | | | | | | | | | | 1 | 1 | 1 | 1 | 0 | | | | |
| | | | | | | | | | | | 1 | 0 | 1 | 0 | 1 | | | | |
| | | | | | | | | | | | 0 | 1 | 0 | 1 | 1 | | | | |

Messaggio trasmesso (dati + FCS): 111100010000 **1011**

**29**

Parola : 100110011001
Generatore : 101111

**SOLUZIONE**

Lunghezza generatore 6 bit
Si aggiungono 5 zeri in coda al messaggio prima di procedere con il calcolo

| 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 | 1 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | | | | | | | | | | | | | | | | | | | | |
| 1 | 0 | | | | | | | | | | | | | | | | | | | | | |
| 1 | 0 | 0 | | | | | | | | | | | | | | | | | | | | |
| 1 | 0 | 0 | 1 | | | | | | | | | | | | | | | | | | | |
| 1 | 0 | 0 | 1 | 1 | | | | | | | | | | | | | | | | | | |
| 1 | 0 | 0 | 1 | 1 | 0 | | | | | | | | | | | | | | | | | |
| 1 | 0 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | | | |
| | 0 | 1 | 0 | 0 | 1 | 0 | | | | | | | | | | | | | | | | |
| | | 1 | 0 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | | |
| | | 1 | 0 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | | | |
| | | | 0 | 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | | | | |
| | | | | 1 | 0 | 1 | 0 | 1 | 0 | | | | | | | | | | | | | |
| | | | | 1 | 0 | 1 | 1 | 1 | 1 | | | | | | | | | | | | | |
| | | | | | 0 | 0 | 1 | 0 | 1 | 0 | | | | | | | | | | | | |
| | | | | | | 0 | 1 | 0 | 1 | 0 | 1 | | | | | | | | | | | |
| | | | | | | | 1 | 0 | 1 | 0 | 1 | 0 | | | | | | | | | | |
| | | | | | | | 1 | 0 | 1 | 1 | 1 | 1 | | | | | | | | | | |
| | | | | | | | | 0 | 0 | 1 | 0 | 1 | 0 | | | | | | | | | |
| | | | | | | | | | 0 | 1 | 0 | 1 | 0 | 0 | | | | | | | | |
| | | | | | | | | | | 1 | 0 | 1 | 0 | 0 | 0 | | | | | | | |
| | | | | | | | | | | 1 | 0 | 1 | 1 | 1 | 1 | | | | | | | |
| | | | | | | | | | | | 0 | 0 | 1 | 1 | 1 | 0 | | | | | | |
| | | | | | | | | | | | 0 | 0 | 1 | 1 | 1 | 0 | | | | | | |

Messaggio trasmesso (dati + FCS): 100110011001 **01110**

## 4.3 LRC

Calcola il LRC delle seguenti parole:

1 **Esercizio guidato**

Calcola il LRC delle seguenti parole (7 bit ciascuna): $45, 67, 89$.

$45 = 0101101_2$
$67 = 1000011_2$
$89 = 1011001_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 1 | 1 | 0 | 1 | 0 |
| 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 |
| 1 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 | 1 | 1 | 1 |

La colonna 8 contiene i bit di parità di riga, la riga 4 contiene i bit di parità di colonna.

2 45, 67, 89

**SOLUZIONE**

$45 = 0101101_2$
$67 = 1000011_2$
$89 = 1011001_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 1 | 1 | 0 | 1 | 0 |
| 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 |
| 1 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 | 1 | 1 | 1 |

3 12, 34, 56

**SOLUZIONE**

$12 = 0001100_2$
$34 = 0100010_2$
$56 = 0111000_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 | 1 | 1 | 0 | 1 |

4 100, 20, 50

**SOLUZIONE**

$100 = 1100100_2$
$20 = 0010100_2$
$50 = 0110010_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |
| 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |

5 10, 20, 30

**SOLUZIONE**

$10 = 0001010_2$
$20 = 0010100_2$
$30 = 0011110_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 |
| 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

**6**

120, 100, 50

**SOLUZIONE**

$120 = 1111000_2$
$100 = 1100100_2$
$50 = 0110010_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 1 | 1 | 1 | 0 | 0 |

**7**

5, 15, 25, 35

**SOLUZIONE**

$5 = 0000101_2$
$15 = 0001111_2$
$25 = 0011001_2$
$35 = 0100011_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| 0 | 1 | 0 | 0 | 0 | 1 | 1 | 1 |
| 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 |

**8**

60, 70, 80

**SOLUZIONE**

$60 = 0111100_2$
$70 = 1000110_2$
$80 = 1010000_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 1 | 0 | 1 |
| 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |

**9**

1, 2, 3, 4, 5

**SOLUZIONE**

$1 = 0000001_2$

$2 = 0000010_2$
$3 = 0000011_2$
$4 = 0000100_2$
$5 = 0000101_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 |
| 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |

**10** ●

127, 0, 64

**SOLUZIONE**

$127 = 1111111_2$
$0 = 0000000_2$
$64 = 1000000_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 1 | 1 | 1 | 1 | 1 | 1 | 0 |

**11** ●

33, 66, 99

**SOLUZIONE**

$33 = 0100001_2$
$66 = 1000010_2$
$99 = 1100011_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 |
| 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 1 | 1 | 0 | 0 | 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

**12** ● ●

45, 67, 89, 12, 34

**SOLUZIONE**

$45 = 0101101_2$
$67 = 1000011_2$
$89 = 1011001_2$
$12 = 0001100_2$
$34 = 0100010_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 1 | 1 | 0 | 1 | 0 |
| 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 |
| 1 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |

**13** 100, 50, 25, 75, 125

**SOLUZIONE**

$100 = 1100100_2$
$50 = 0110010_2$
$25 = 0011001_2$
$75 = 1001011_2$
$125 = 1111101_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| 1 | 0 | 0 | 1 | 0 | 1 | 1 | 0 |
| 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 |
| 1 | 1 | 1 | 1 | 0 | 0 | 1 | 1 |

**14** 10, 20, 30, 40, 50, 60

**SOLUZIONE**

$10 = 0001010_2$
$20 = 0010100_2$
$30 = 0011110_2$
$40 = 0101000_2$
$50 = 0110010_2$
$60 = 0111100_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 |
| 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 |
| 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 1 | 1 | 0 | 1 |

**15** 20, 150, 100, 50, 25

**SOLUZIONE**

Errore: ogni parola deve essere compresa tra 0 e 127 (7 bit).

**16** 7, 14, 21, 28, 35, 42

**SOLUZIONE**

$7 = 0000111_2$
$14 = 0001110_2$
$21 = 0010101_2$
$28 = 0011100_2$
$35 = 0100011_2$
$42 = 0101010_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 | 1 | 0 | 1 |
| 0 | 0 | 1 | 0 | 1 | 0 | 1 | 1 |
| 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 |
| 0 | 1 | 0 | 0 | 0 | 1 | 1 | 1 |
| 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 |

**17** ●●

64, 32, 16, 8, 4, 2

**SOLUZIONE**

$64 = 1000000_2$
$32 = 0100000_2$
$16 = 0010000_2$
$8 = 0001000_2$
$4 = 0000100_2$
$2 = 0000010_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 0 | 0 |

**18** ●●

127, 64, 32, 16, 8

**SOLUZIONE**

$127 = 1111111_2$
$64 = 1000000_2$
$32 = 0100000_2$
$16 = 0010000_2$
$8 = 0001000_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |

**19** ●●

11, 22, 33, 44, 55, 66

**SOLUZIONE**

$11 = 0001011_2$
$22 = 0010110_2$
$33 = 0100001_2$
$44 = 0101100_2$
$55 = 0110111_2$
$66 = 1000010_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
| 0 | 0 | 1 | 0 | 1 | 1 | 0 | 1 |
| 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 | 1 | 0 | 0 | 1 |
| 0 | 1 | 1 | 0 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 1 | 1 | 0 | 0 | 1 | 0 | 1 | 0 |

**20** ●●

90, 80, 70, 60, 50

**SOLUZIONE**

$90 = 1011010_2$
$80 = 1010000_2$
$70 = 1000110_2$
$60 = 0111100_2$
$50 = 0110010_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 0 | 1 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |
| 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |

**21** ●●

1, 3, 5, 7, 9, 11

**SOLUZIONE**

$1 = 0000001_2$
$3 = 0000011_2$
$5 = 0000101_2$
$7 = 0000111_2$
$9 = 0001001_2$
$11 = 0001011_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |
| 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |
| 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 |
| 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 |

**22** ●●●

45, 67, 89, 12, 34, 56, 78

**SOLUZIONE**

$45 = 0101101_2$
$67 = 1000011_2$
$89 = 1011001_2$
$12 = 0001100_2$
$34 = 0100010_2$
$56 = 0111000_2$

$78 = 1001110_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 1 | 0 | 1 | 1 | 0 | 1 | 0 |
| 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 |
| 1 | 0 | 1 | 1 | 0 | 0 | 1 | 0 |
| 0 | 0 | 0 | 1 | 1 | 0 | 0 | 0 |
| 0 | 1 | 0 | 0 | 0 | 1 | 0 | 0 |
| 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 |
| 1 | 0 | 0 | 1 | 1 | 1 | 0 | 0 |
| 1 | 1 | 0 | 1 | 1 | 1 | 1 | 0 |

**23** ●●●

100, 50, 25, 75, 125, 10, 20

**SOLUZIONE**

$100 = 1100100_2$
$50 = 0110010_2$
$25 = 0011001_2$
$75 = 1001011_2$
$125 = 1111101_2$
$10 = 0001010_2$
$20 = 0010100_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 0 | 0 | 1 | 0 | 0 | 1 |
| 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |
| 0 | 0 | 1 | 1 | 0 | 0 | 1 | 1 |
| 1 | 0 | 0 | 1 | 0 | 1 | 1 | 0 |
| 1 | 1 | 1 | 1 | 1 | 0 | 1 | 0 |
| 0 | 0 | 0 | 1 | 0 | 1 | 0 | 0 |
| 0 | 0 | 1 | 0 | 1 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 | 1 | 1 | 1 | 1 |

**24** ●●●

7, 14, 21, 28, 35, 42, 49, 56

**SOLUZIONE**

$7 = 0000111_2$
$14 = 0001110_2$
$21 = 0010101_2$
$28 = 0011100_2$
$35 = 0100011_2$
$42 = 0101010_2$
$49 = 0110001_2$
$56 = 0111000_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 | 1 | 0 | 1 |
| 0 | 0 | 1 | 0 | 1 | 0 | 1 | 1 |
| 0 | 0 | 1 | 1 | 1 | 0 | 0 | 1 |
| 0 | 1 | 0 | 0 | 0 | 1 | 1 | 1 |
| 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 |
| 0 | 1 | 1 | 0 | 0 | 0 | 1 | 1 |
| 0 | 1 | 1 | 1 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

**25** 64, 32, 16, 8, 4, 2, 1

**SOLUZIONE**

$64 = 1000000_2$
$32 = 0100000_2$
$16 = 0010000_2$
$8 = 0001000_2$
$4 = 0000100_2$
$2 = 0000010_2$
$1 = 0000001_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 |
| 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |

**26** 127, 64, 32, 16, 8, 4, 2, 1

**SOLUZIONE**

$127 = 1111111_2$
$64 = 1000000_2$
$32 = 0100000_2$
$16 = 0010000_2$
$8 = 0001000_2$
$4 = 0000100_2$
$2 = 0000010_2$
$1 = 0000001_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 1 | 0 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0 | 0 | 1 | 0 | 1 |
| 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

**27** 11, 22, 33, 44, 55, 66, 77

**SOLUZIONE**

$11 = 0001011_2$
$22 = 0010110_2$
$33 = 0100001_2$
$44 = 0101100_2$
$55 = 0110111_2$
$66 = 1000010_2$

$77 = 1001101_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
| 0 | 0 | 1 | 0 | 1 | 1 | 0 | 1 |
| 0 | 1 | 0 | 0 | 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 | 1 | 0 | 0 | 1 |
| 0 | 1 | 1 | 0 | 1 | 1 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 | 1 | 0 | 0 |
| 1 | 0 | 0 | 1 | 1 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 |

**28** ●●●

90, 80, 70, 60, 50, 40, 30

**SOLUZIONE**

$90 = 1011010_2$
$80 = 1010000_2$
$70 = 1000110_2$
$60 = 0111100_2$
$50 = 0110010_2$
$40 = 0101000_2$
$30 = 0011110_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | 0 | 1 | 1 | 0 | 1 | 0 | 0 |
| 1 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 1 | 1 | 0 | 1 |
| 0 | 1 | 1 | 1 | 1 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 | 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 1 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 1 | 1 | 0 | 0 |
| 1 | 1 | 1 | 0 | 1 | 0 | 0 | 0 |

**29** ●●●

1, 3, 5, 7, 9, 11, 13, 15

**SOLUZIONE**

$1 = 0000001_2$
$3 = 0000011_2$
$5 = 0000101_2$
$7 = 0000111_2$
$9 = 0001001_2$
$11 = 0001011_2$
$13 = 0001101_2$
$15 = 0001111_2$

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 1 | 1 |
| 0 | 0 | 0 | 0 | 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 | 1 | 0 | 1 | 0 |
| 0 | 0 | 0 | 0 | 1 | 1 | 1 | 1 |
| 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 |
| 0 | 0 | 0 | 1 | 0 | 1 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 | 0 | 1 | 1 |
| 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

6.2 Memoria a partizione variabili
6.3 Allocazione delle pagine

## 5.1 FCFS, SJF, SRTF

Effettua lo schedeuling dei seguenti processi, usando l'algoritmo FCFS, SJF e SRTF:

**1**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 3 |
| P3 | 0 | 2 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | 0 | 6 |
| P2 | 0 | 3 | - - - - - | - | 6 | 9 |
| P3 | 0 | 2 | - - - - - | - - - - | 9 | 11 |
| | | | | | 5.00 | 8.67 |

**SJF**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 6 | - - - - - | | 5 | 11 |
| P2 | 0 | 3 | - - | | 2 | 5 |
| P3 | 0 | 2 | | | 0 | 2 |
| | | | | | 2.33 | 6.00 |

**SRTF**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 6 | - - - - - | | 5 | 11 |
| P2 | 0 | 3 | - - | | 2 | 5 |
| P3 | 0 | 2 | | | 0 | 2 |
| | | | | | 2.33 | 6.00 |

2

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 5 |
| P2 | 0 | 2 |
| P3 | 1 | 3 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 5 | | | 0 | 5 |
| P2 | 0 | 2 | - - - - - | | 5 | 7 |
| P3 | 1 | 3 | - - - - | - - | 6 | 9 |
| | | | | | 3.67 | 7.00 |

**SJF**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 5 | - - - - - | | 5 | 10 |
| P2 | 0 | 2 | | | 0 | 2 |
| P3 | 1 | 3 | - | | 1 | 4 |
| | | | | | 2.00 | 5.33 |

**SRTF**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 5 | - - - - - | | 5 | 10 |
| P2 | 0 | 2 | | | 0 | 2 |
| P3 | 1 | 3 | - | | 1 | 4 |
| | | | | | 2.00 | 5.33 |

3

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 1 | 2 |
| P3 | 2 | 3 |

**SOLUZIONE**

**FCFS**

| P | A | D | | T. Att. | T. Compl. |
|---|---|---|---|---|---|
| P1 | 0 | 6 | | 0 | 6 |
| P2 | 1 | 2 | - - - - - | 5 | 7 |
| P3 | 2 | 3 | - - - - - - - | 6 | 9 |
| | | | | 3.67 | 7.33 |

**SJF**

| P | A | D | | T. Att. | T. Compl. |
|---|---|---|---|---|---|
| P1 | 0 | 6 | | 0 | 6 |
| P2 | 1 | 2 | - - - - - | 5 | 7 |
| P3 | 2 | 3 | - - - - - - - | 6 | 9 |
| | | | | 3.67 | 7.33 |

**SRTF**

| P | A | D | | T. Att. | T. Compl. |
|---|---|---|---|---|---|
| P1 | 0 | 6 | - - - - - | 5 | 11 |
| P2 | 1 | 2 | | 0 | 2 |
| P3 | 2 | 3 | - | 1 | 4 |
| | | | | 2.00 | 5.67 |

**4**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 2 |
| P2 | 0 | 4 |
| P3 | 0 | 6 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 2 | | | 0 | 2 |
| P2 | 0 | 4 | - - | | 2 | 6 |
| P3 | 0 | 6 | - - - - - | - | 6 | 12 |
| | | | | | 2.67 | 6.67 |

**SJF**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 2 | | | 0 | 2 |
| P2 | 0 | 4 | - - | | 2 | 6 |
| P3 | 0 | 6 | - - - - - | - | 6 | 12 |
| | | | | | 2.67 | 6.67 |

**SRTF**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 2 | | | 0 | 2 |
| P2 | 0 | 4 | - - | | 2 | 6 |
| P3 | 0 | 6 | - - - - - | - | 6 | 12 |
| | | | | | 2.67 | 6.67 |

5

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 4 |
| P2 | 0 | 4 |
| P3 | 1 | 2 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 4 | | | 0 | 4 |
| P2 | 0 | 4 | - - - - | | 4 | 8 |
| P3 | 1 | 2 | - - - - | - - - | 7 | 9 |
| | | | | | 3.67 | 7.00 |

**SJF**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 4 | | | 0 | 4 |
| P2 | 0 | 4 | - - - - - | - | 6 | 10 |
| P3 | 1 | 2 | - - - | | 3 | 5 |
| | | | | | 3.00 | 6.33 |

**SRTF**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 4 | - - | | 2 | 6 |
| P2 | 0 | 4 | - - - - - | - | 6 | 10 |
| P3 | 1 | 2 | | | 0 | 2 |
| | | | | | 2.67 | 6.00 |

**6**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 1 | 2 |
| P3 | 2 | 3 |
| P4 | 2 | 4 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | 0 | 6 |
| P2 | 1 | 2 | - - - - | - | | 5 | 7 |
| P3 | 2 | 3 | - - - | - - - | | 6 | 9 |
| P4 | 2 | 4 | - - - | - - - - - | - | 9 | 13 |
| | | | | | | 5.00 | 8.75 |

**SJF**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | 0 | 6 |
| P2 | 1 | 2 | - - - - | - | | 5 | 7 |
| P3 | 2 | 3 | - - - | - - - | | 6 | 9 |
| P4 | 2 | 4 | - - - | - - - - - | - | 9 | 13 |
| | | | | | | 5.00 | 8.75 |

**SRTF**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | - - - - | - - - - - | | 9 | 15 |
| P2 | 1 | 2 | | | | 0 | 2 |
| P3 | 2 | 3 | - | | | 1 | 4 |
| P4 | 2 | 4 | - - - | - | | 4 | 8 |
| | | | | | | 3.50 | 7.25 |

7

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 5 |
| P2 | 0 | 3 |
| P3 | 1 | 2 |
| P4 | 1 | 3 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 5 | | | 0 | 5 |
| P2 | 0 | 3 | - - - - - | | 5 | 8 |
| P3 | 1 | 2 | - - - - | - - - | 7 | 9 |
| P4 | 1 | 3 | - - - - | - - - - - | 9 | 12 |
| | | | | | 5.25 | 8.50 |

**SJF**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 5 | - - - - - | - - - | 8 | 13 |
| P2 | 0 | 3 | | | 0 | 3 |
| P3 | 1 | 2 | - - | | 2 | 4 |
| P4 | 1 | 3 | - - - - | | 4 | 7 |
| | | | | | 3.50 | 6.75 |

**SRTF**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 5 | - - - - - | - - - | 8 | 13 |
| P2 | 0 | 3 | | | 0 | 3 |
| P3 | 1 | 2 | - - | | 2 | 4 |
| P4 | 1 | 3 | - - - - | | 4 | 7 |
| | | | | | 3.50 | 6.75 |

**8**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 1 | 4 |
| P3 | 2 | 2 |
| P4 | 0 | 3 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | 0 | 6 |
| P2 | 1 | 4 | - - - - | - - - - | | 8 | 12 |
| P3 | 2 | 2 | - - | - - - - - | - - - | 11 | 13 |
| P4 | 0 | 3 | - - - - - | - | | 6 | 9 |
| | | | | | | 6.25 | 10.00 |

**SJF**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | - - - - - | - - - - | | 9 | 15 |
| P2 | 1 | 4 | - - - - | | | 4 | 8 |
| P3 | 2 | 2 | - | | | 1 | 3 |
| P4 | 0 | 3 | | | | 0 | 3 |
| | | | | | | 3.50 | 7.25 |

**SRTF**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | - - - - - | - - - - | | 9 | 15 |
| P2 | 1 | 4 | - - - - | | | 4 | 8 |
| P3 | 2 | 2 | - | | | 1 | 3 |
| P4 | 0 | 3 | | | | 0 | 3 |
| | | | | | | 3.50 | 7.25 |

**9**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 5 |
| P2 | 0 | 6 |
| P3 | 1 | 2 |
| P4 | 1 | 3 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 5 | | | | 0 | 5 |
| P2 | 0 | 6 | - - - - - | | | 5 | 11 |
| P3 | 1 | 2 | - - - - | - - - - - | - | 10 | 12 |
| P4 | 1 | 3 | - - - - | - - - - - | - - - | 12 | 15 |
| | | | | | | 6.75 | 10.75 |

**SJF**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 5 | | | | 0 | 5 |
| P2 | 0 | 6 | - - - - - | - - - - - | | 10 | 16 |
| P3 | 1 | 2 | - - - - | | | 4 | 6 |
| P4 | 1 | 3 | - - - - | - - | | 6 | 9 |
| | | | | | | 5.00 | 9.00 |

**SRTF**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 5 | - - - - - | - | | 5 | 10 |
| P2 | 0 | 6 | - - - - - | - - - - - | | 10 | 16 |
| P3 | 1 | 2 | | | | 0 | 2 |
| P4 | 1 | 3 | - - | | | 2 | 5 |
| | | | | | | 4.25 | 8.25 |

**10**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 4 |
| P2 | 0 | 3 |
| P3 | 1 | 2 |
| P4 | 2 | 3 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 4 | | | 0 | 4 |
| P2 | 0 | 3 | - - - - | | 4 | 7 |
| P3 | 1 | 2 | - - - - | - - | 6 | 8 |
| P4 | 2 | 3 | - - - | - - - - | 7 | 10 |
| | | | | | 4.25 | 7.25 |

**SJF**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 4 | - - - - | - - - | 8 | 12 |
| P2 | 0 | 3 | | | 0 | 3 |
| P3 | 1 | 2 | - - | | 2 | 4 |
| P4 | 2 | 3 | - - - | | 3 | 6 |
| | | | | | 3.25 | 6.25 |

**SRTF**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 4 | - - - - | - - - | 8 | 12 |
| P2 | 0 | 3 | | | 0 | 3 |
| P3 | 1 | 2 | - - | | 2 | 4 |
| P4 | 2 | 3 | - - - | | 3 | 6 |
| | | | | | 3.25 | 6.25 |

**11**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 3 |
| P3 | 0 | 2 |
| P4 | 1 | 4 |
| P5 | 2 | 3 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | 0 | 6 |
| P2 | 0 | 3 | - - - - - | - | | 6 | 9 |
| P3 | 0 | 2 | - - - - - | - - - - | | 9 | 11 |
| P4 | 1 | 4 | - - - - | - - - - - | - | 10 | 14 |
| P5 | 2 | 3 | - - - | - - - - - | - - - - - | 13 | 16 |
| | | | | | | 7.60 | 11.20 |

**SJF**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | - - - - - | - - - - - | - - | 12 | 18 |
| P2 | 0 | 3 | - - | | | 2 | 5 |
| P3 | 0 | 2 | | | | 0 | 2 |
| P4 | 1 | 4 | - - - - | - - - | | 7 | 11 |
| P5 | 2 | 3 | - - - | | | 3 | 6 |
| | | | | | | 4.80 | 8.40 |

**SRTF**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | - - - - - | - - - - - | - - | 12 | 18 |
| P2 | 0 | 3 | - - | | | 2 | 5 |
| P3 | 0 | 2 | | | | 0 | 2 |
| P4 | 1 | 4 | - - - - | - - - | | 7 | 11 |
| P5 | 2 | 3 | - - - | | | 3 | 6 |
| | | | | | | 4.80 | 8.40 |

12

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 7 |
| P2 | 0 | 6 |
| P3 | 1 | 2 |
| P4 | 2 | 3 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 7 | | | | 0 | 7 |
| P2 | 0 | 6 | - - - - - | - - | | 7 | 13 |
| P3 | 1 | 2 | - - - - | - - - - - | - - - | 12 | 14 |
| P4 | 2 | 3 | - - - | - - - - - | - - - - - | 13 | 16 |
| | | | | | | 8.00 | 12.50 |

**SJF**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 7 | - - - - - | - - - - - | - | 11 | 18 |
| P2 | 0 | 6 | | | | 0 | 6 |
| P3 | 1 | 2 | - - - - | - | | 5 | 7 |
| P4 | 2 | 3 | - - - | - - - | | 6 | 9 |
| | | | | | | 5.50 | 10.00 |

**SRTF**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 7 | - - - - - | - - - - - | - | 11 | 18 |
| P2 | 0 | 6 | - - - - - | - | | 5 | 11 |
| P3 | 1 | 2 | | | | 0 | 2 |
| P4 | 2 | 3 | - | | | 1 | 4 |
| | | | | | | 4.25 | 8.75 |

**13**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 5 |
| P2 | 0 | 4 |
| P3 | 1 | 2 |
| P4 | 1 | 3 |
| P5 | 2 | 4 |

**SOLUZIONE**

**FCFS**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 5 | 0 | 5 |
| P2 | 0 | 4 | 5 | 9 |
| P3 | 1 | 2 | 8 | 10 |
| P4 | 1 | 3 | 10 | 13 |
| P5 | 2 | 4 | 12 | 16 |
| | | | 7.00 | 10.60 |

**SJF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 5 | 13 | 18 |
| P2 | 0 | 4 | 0 | 4 |
| P3 | 1 | 2 | 3 | 5 |
| P4 | 1 | 3 | 5 | 8 |
| P5 | 2 | 4 | 7 | 11 |
| | | | 5.60 | 9.20 |

**SRTF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 5 | 13 | 18 |
| P2 | 0 | 4 | 2 | 6 |
| P3 | 1 | 2 | 0 | 2 |
| P4 | 1 | 3 | 5 | 8 |
| P5 | 2 | 4 | 7 | 11 |
| | | | 5.40 | 9.00 |

**14**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 8 |
| P2 | 1 | 2 |
| P3 | 2 | 3 |
| P4 | 3 | 2 |
| P5 | 4 | 3 |
| P6 | 1 | 2 |

**SOLUZIONE**

**FCFS**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 8 | 0 | 8 |
| P2 | 1 | 2 | 7 | 9 |
| P3 | 2 | 3 | 10 | 13 |
| P4 | 3 | 2 | 12 | 14 |
| P5 | 4 | 3 | 13 | 16 |
| P6 | 1 | 2 | 9 | 11 |
| | | | 8.50 | 11.83 |

**SJF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 8 | 0 | 8 |
| P2 | 1 | 2 | 7 | 9 |
| P3 | 2 | 3 | 12 | 15 |
| P4 | 3 | 2 | 7 | 9 |
| P5 | 4 | 3 | 13 | 16 |
| P6 | 1 | 2 | 11 | 13 |
| | | | 8.33 | 11.67 |

**SRTF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 8 | 12 | 20 |
| P2 | 1 | 2 | 0 | 2 |
| P3 | 2 | 3 | 5 | 8 |
| P4 | 3 | 2 | 2 | 4 |
| P5 | 4 | 3 | 6 | 9 |
| P6 | 1 | 2 | 2 | 4 |
| | | | 4.50 | 7.83 |

**15**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 7 |
| P2 | 0 | 6 |
| P3 | 1 | 4 |
| P4 | 2 | 3 |
| P5 | 3 | 4 |
| P6 | 3 | 1 |

**SOLUZIONE**

**FCFS**

| P | A | D | T. Att. | T. Com |
|---|---|---|---|---|
| P1 | 0 | 7 | 0 | 7 |
| P2 | 0 | 6 | 7 | 13 |
| P3 | 1 | 4 | 12 | 16 |
| P4 | 2 | 3 | 15 | 18 |
| P5 | 3 | 4 | 17 | 21 |
| P6 | 3 | 1 | 21 | 22 |
| | | | 12.00 | 16.17 |

**SJF**

| P | A | D | T. Att. | T. Com |
|---|---|---|---|---|
| P1 | 0 | 7 | 18 | 25 |
| P2 | 0 | 6 | 0 | 6 |
| P3 | 1 | 4 | 9 | 13 |
| P4 | 2 | 3 | 5 | 8 |
| P5 | 3 | 4 | 11 | 15 |
| P6 | 3 | 1 | 3 | 4 |
| | | | 7.67 | 11.83 |

**SRTF**

| P | A | D | T. Att. | T. Com |
|---|---|---|---|---|
| P1 | 0 | 7 | 18 | 25 |
| P2 | 0 | 6 | 12 | 18 |
| P3 | 1 | 4 | 1 | 5 |
| P4 | 2 | 3 | 4 | 7 |
| P5 | 3 | 4 | 6 | 10 |
| P6 | 3 | 1 | 0 | 1 |
| | | | 6.83 | 11.00 |

**16**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 5 |
| P2 | 0 | 4 |
| P3 | 0 | 3 |
| P4 | 1 | 2 |
| P5 | 2 | 4 |
| P6 | 2 | 1 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 5 | | | | 0 | 5 |
| P2 | 0 | 4 | - - - - - | | | 5 | 9 |
| P3 | 0 | 3 | - - - - - | - - - - - | | 9 | 12 |
| P4 | 1 | 2 | - - - - | - - - - - | - - | 11 | 13 |
| P5 | 2 | 4 | - - - | - - - - - | - - - - | 12 | 16 |
| P6 | 2 | 1 | - - - | - - - - - | - - - - - - - - | 16 | 17 |
| | | | | | | 8.83 | 12.00 |

**SJF**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 5 | - - - - - | - - - - - | - - - - | 14 | 19 |
| P2 | 0 | 4 | - - - - - | - | | 6 | 10 |
| P3 | 0 | 3 | | | | 0 | 3 |
| P4 | 1 | 2 | - - - | | | 3 | 5 |
| P5 | 2 | 4 | - - - | - - - - - | | 8 | 12 |
| P6 | 2 | 1 | - | | | 1 | 2 |
| | | | | | | 5.33 | 8.50 |

**SRTF**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 5 | - - - - - | - - - - - | - - - - | 14 | 19 |
| P2 | 0 | 4 | - - - - - | - | | 6 | 10 |
| P3 | 0 | 3 | | | | 0 | 3 |
| P4 | 1 | 2 | - - - | | | 3 | 5 |
| P5 | 2 | 4 | - - - | - - - - - | | 8 | 12 |
| P6 | 2 | 1 | - | | | 1 | 2 |
| | | | | | | 5.33 | 8.50 |

**17**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 9 |
| P2 | 1 | 2 |
| P3 | 1 | 3 |
| P4 | 3 | 4 |
| P5 | 2 | 3 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 9 | | | | | 0 | 9 |
| P2 | 1 | 2 | - - - - | - - - - | | | 8 | 10 |
| P3 | 1 | 3 | - - - - | - - - - - | - | | 10 | 13 |
| P4 | 3 | 4 | - - | - - - - - | - - - - - | - - | 14 | 18 |
| P5 | 2 | 3 | - - - | - - - - - | - - - - | | 12 | 15 |
| | | | | | | | 8.80 | 13.00 |

**SJF**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 9 | | | | | 0 | 9 |
| P2 | 1 | 2 | - - - - | - - - - | | | 8 | 10 |
| P3 | 1 | 3 | - - - - | - - - - - | - | | 10 | 13 |
| P4 | 3 | 4 | - - | - - - - - | - - - - - | - - | 14 | 18 |
| P5 | 2 | 3 | - - - | - - - - - | - - - - | | 12 | 15 |
| | | | | | | | 8.80 | 13.00 |

**SRTF**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 9 | - - - - | - - - - - | - - - | | 12 | 21 |
| P2 | 1 | 2 | | | | | 0 | 2 |
| P3 | 1 | 3 | - - | | | | 2 | 5 |
| P4 | 3 | 4 | - - | - - - - | | | 6 | 10 |
| P5 | 2 | 3 | - - - | - | | | 4 | 7 |
| | | | | | | | 4.80 | 9.00 |

**18**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 4 |
| P3 | 2 | 2 |
| P4 | 2 | 3 |
| P5 | 3 | 5 |
| P6 | 1 | 4 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 0 | 6 |
| P2 | 0 | 4 | | | | | 6 | 10 |
| P3 | 2 | 2 | | | | | 12 | 14 |
| P4 | 2 | 3 | | | | | 14 | 17 |
| P5 | 3 | 5 | | | | | 16 | 21 |
| P6 | 1 | 4 | | | | | 9 | 13 |
| | | | | | | | 9.50 | 13.50 |

**SJF**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 18 | 24 |
| P2 | 0 | 4 | | | | | 0 | 4 |
| P3 | 2 | 2 | | | | | 2 | 4 |
| P4 | 2 | 3 | | | | | 4 | 7 |
| P5 | 3 | 5 | | | | | 10 | 15 |
| P6 | 1 | 4 | | | | | 8 | 12 |
| | | | | | | | 7.00 | 11.00 |

**SRTF**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 18 | 24 |
| P2 | 0 | 4 | | | | | 0 | 4 |
| P3 | 2 | 2 | | | | | 2 | 4 |
| P4 | 2 | 3 | | | | | 4 | 7 |
| P5 | 3 | 5 | | | | | 10 | 15 |
| P6 | 1 | 4 | | | | | 8 | 12 |
| | | | | | | | 7.00 | 11.00 |

**19**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 5 |
| P2 | 0 | 4 |
| P3 | 0 | 3 |
| P4 | 2 | 2 |
| P5 | 2 | 4 |
| P6 | 4 | 3 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 5 | ■ | ■ | ■ | ■ | 0 | 5 |
| P2 | 0 | 4 | - | - | - | - | 5 | 9 |
| P3 | 0 | 3 | - | - | - | - | 9 | 12 |
| P4 | 2 | 2 | | - | - | - | 10 | 12 |
| P5 | 2 | 4 | | - | - | - | 12 | 16 |
| P6 | 4 | 3 | | | - | - | 14 | 17 |
| | | | | | | | 8.33 | 11.83 |

**SJF**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 5 | - | - | - | ■ | 16 | 21 |
| P2 | 0 | 4 | - | - | - | - | 8 | 12 |
| P3 | 0 | 3 | ■ | ■ | ■ | | 0 | 3 |
| P4 | 2 | 2 | | - | ■ | | 1 | 3 |
| P5 | 2 | 4 | | - | - | ■ | 10 | 14 |
| P6 | 4 | 3 | | | - | - | 1 | 4 |
| | | | | | | | 6.00 | 9.50 |

**SRTF**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 5 | - | - | - | ■ | 16 | 21 |
| P2 | 0 | 4 | - | - | - | - | 8 | 12 |
| P3 | 0 | 3 | ■ | ■ | ■ | | 0 | 3 |
| P4 | 2 | 2 | | - | ■ | | 1 | 3 |
| P5 | 2 | 4 | | - | - | ■ | 10 | 14 |
| P6 | 4 | 3 | | | - | - | 1 | 4 |
| | | | | | | | 6.00 | 9.50 |

20

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 10 |
| P2 | 1 | 2 |
| P3 | 2 | 3 |
| P4 | 3 | 2 |
| P5 | 4 | 4 |
| P6 | 1 | 1 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 10 | | | | | 0 | 10 |
| P2 | 1 | 2 | - - - - | - - - - - | | | 9 | 11 |
| P3 | 2 | 3 | - - - | - - - - - - - - | | | 11 | 14 |
| P4 | 3 | 2 | - - | - - - - - - - - - - | - | | 13 | 15 |
| P5 | 4 | 4 | - | - - - - - - - - - - - - | - - - | | 14 | 18 |
| P6 | 1 | 1 | - - - - | - - - - - - - - | - - | | 11 | 12 |
| | | | | | | | 9.67 | 13.33 |

**SJF**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 10 | | | | | 0 | 10 |
| P2 | 1 | 2 | - - - - | - - - - - - - - | - | | 10 | 12 |
| P3 | 2 | 3 | - - - | - - - - - - - - - - - - - | | | 13 | 16 |
| P4 | 3 | 2 | - - | - - - - - - - - - - - | | | 10 | 12 |
| P5 | 4 | 4 | - | - - - - - - - - - - - - | - - - | | 14 | 18 |
| P6 | 1 | 1 | - - - - | - - - - - - - - | | | 9 | 10 |
| | | | | | | | 9.33 | 13.00 |

**SRTF**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 10 | - - - - | - - - - - - - - | - - - | | 12 | 22 |
| P2 | 1 | 2 | - | | | | 1 | 3 |
| P3 | 2 | 3 | - - - | - | | | 4 | 7 |
| P4 | 3 | 2 | - | | | | 1 | 3 |
| P5 | 4 | 4 | - | - - - - - | | | 5 | 9 |
| P6 | 1 | 1 | | | | | 0 | 1 |
| | | | | | | | 3.83 | 7.50 |

21

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 8 |
| P2 | 0 | 7 |
| P3 | 1 | 2 |
| P4 | 2 | 3 |
| P5 | 3 | 2 |
| P6 | 2 | 2 |
| P7 | 1 | 3 |

**SOLUZIONE**

**FCFS**

| P | A | D | T. Att. | T. |
|---|---|---|---|---|
| P1 | 0 | 8 | 0 | |
| P2 | 0 | 7 | 8 | |
| P3 | 1 | 2 | 14 | |
| P4 | 2 | 3 | 18 | |
| P5 | 3 | 2 | 22 | |
| P6 | 2 | 2 | 21 | |
| P7 | 1 | 3 | 16 | |
| | | | 14.14 | |

**SJF**

| P | A | D | T. Att. | T. |
|---|---|---|---|---|
| P1 | 0 | 8 | 19 | |
| P2 | 0 | 7 | 0 | |
| P3 | 1 | 2 | 6 | |
| P4 | 2 | 3 | 11 | |
| P5 | 3 | 2 | 6 | |
| P6 | 2 | 2 | 9 | |
| P7 | 1 | 3 | 15 | |
| | | | 9.43 | |

**SRTF**

| P | A | D | T. Att. | T. |
|---|---|---|---|---|
| P1 | 0 | 8 | 19 | |
| P2 | 0 | 7 | 12 | |
| P3 | 1 | 2 | 0 | |
| P4 | 2 | 3 | 8 | |
| P5 | 3 | 2 | 2 | |
| P6 | 2 | 2 | 1 | |
| P7 | 1 | 3 | 6 | |
| | | | 6.86 | |

22

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 4 |
| P3 | 0 | 5 |
| P4 | 2 | 2 |
| P5 | 2 | 3 |
| P6 | 2 | 4 |
| P7 | 3 | 1 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | | | | T. Att. | T. Comp |
|---|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | | 0 | 6 |
| P2 | 0 | 4 | - - - - - | | | | | 6 | 10 |
| P3 | 0 | 5 | - - - - - | - - - - - | | | | 10 | 15 |
| P4 | 2 | 2 | - - - | - - - - - | - - - - - | | | 13 | 15 |
| P5 | 2 | 3 | - - - | - - - - - | - - - - - | - - | | 15 | 18 |
| P6 | 2 | 4 | - - - | - - - - - | - - - - - | - - - - - | | 18 | 22 |
| P7 | 3 | 1 | - - | - - - - - | - - - - - | - - - - - | - - - - | 21 | 22 |
| | | | | | | | | 11.86 | 15.43 |

**SJF**

| P | A | D | | | | | | T. Att. | T. Comp |
|---|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | - - - - - | - - - - - | - - - - - | - - - - | | 19 | 25 |
| P2 | 0 | 4 | | | | | | 0 | 4 |
| P3 | 0 | 5 | - - - - - | - - - - - | - - - - | | | 14 | 19 |
| P4 | 2 | 2 | - - - | | | | | 3 | 5 |
| P5 | 2 | 3 | - - - | - - | | | | 5 | 8 |
| P6 | 2 | 4 | - - - | - - - - - | | | | 8 | 12 |
| P7 | 3 | 1 | - | | | | | 1 | 2 |
| | | | | | | | | 7.14 | 10.71 |

**SRTF**

| P | A | D | | | | | | T. Att. | T. Comp |
|---|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | - - - - - | - - - - - | - - - - - | - - - - | | 19 | 25 |
| P2 | 0 | 4 | | | | | | 0 | 4 |
| P3 | 0 | 5 | - - - - - | - - - - - | - - - - | | | 14 | 19 |
| P4 | 2 | 2 | - - - | | | | | 3 | 5 |
| P5 | 2 | 3 | - - - | - - | | | | 5 | 8 |
| P6 | 2 | 4 | - - - | - - - - - | | | | 8 | 12 |
| P7 | 3 | 1 | - | | | | | 1 | 2 |
| | | | | | | | | 7.14 | 10.71 |

**23**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 7 |
| P2 | 1 | 2 |
| P3 | 2 | 2 |
| P4 | 3 | 3 |
| P5 | 4 | 2 |
| P6 | 5 | 3 |
| P7 | 1 | 1 |
| P8 | 2 | 1 |

**SOLUZIONE**

**FCFS**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 7 | 0 | 7 |
| P2 | 1 | 2 | 6 | 8 |
| P3 | 2 | 2 | 8 | 10 |
| P4 | 3 | 3 | 10 | 13 |
| P5 | 4 | 2 | 12 | 14 |
| P6 | 5 | 3 | 13 | 16 |
| P7 | 1 | 1 | 8 | 9 |
| P8 | 2 | 1 | 10 | 11 |
| | | | 8.38 | 11.00 |

**SJF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 7 | 0 | 7 |
| P2 | 1 | 2 | 8 | 10 |
| P3 | 2 | 2 | 9 | 11 |
| P4 | 3 | 3 | 12 | 15 |
| P5 | 4 | 2 | 9 | 11 |
| P6 | 5 | 3 | 13 | 16 |
| P7 | 1 | 1 | 6 | 7 |
| P8 | 2 | 1 | 6 | 7 |
| | | | 7.88 | 10.50 |

**SRTF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 7 | 14 | 21 |
| P2 | 1 | 2 | 2 | 4 |
| P3 | 2 | 2 | 3 | 5 |
| P4 | 3 | 3 | 6 | 9 |
| P5 | 4 | 2 | 3 | 5 |
| P6 | 5 | 3 | 7 | 10 |
| P7 | 1 | 1 | 0 | 1 |
| P8 | 2 | 1 | 0 | 1 |
| | | | 4.38 | 7.00 |

24

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 5 |
| P3 | 2 | 2 |
| P4 | 3 | 3 |
| P5 | 4 | 2 |
| P6 | 4 | 4 |
| P7 | 1 | 1 |
| P8 | 2 | 1 |

**SOLUZIONE**

**FCFS**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 0 | 6 |
| P2 | 0 | 5 | 6 | 11 |
| P3 | 2 | 2 | 10 | 12 |
| P4 | 3 | 3 | 12 | 15 |
| P5 | 4 | 2 | 14 | 16 |
| P6 | 4 | 4 | 16 | 20 |
| P7 | 1 | 1 | 10 | 11 |
| P8 | 2 | 1 | 12 | 13 |
| | | | 10.00 | 13.00 |

**SJF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 18 | 24 |
| P2 | 0 | 5 | 0 | 5 |
| P3 | 2 | 2 | 5 | 7 |
| P4 | 3 | 3 | 8 | 11 |
| P5 | 4 | 2 | 5 | 7 |
| P6 | 4 | 4 | 10 | 14 |
| P7 | 1 | 1 | 4 | 5 |
| P8 | 2 | 1 | 4 | 5 |
| | | | 6.75 | 9.75 |

**SRTF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 18 | 24 |
| P2 | 0 | 5 | 9 | 14 |
| P3 | 2 | 2 | 1 | 3 |
| P4 | 3 | 3 | 4 | 7 |
| P5 | 4 | 2 | 1 | 3 |
| P6 | 4 | 4 | 10 | 14 |
| P7 | 1 | 1 | 0 | 1 |
| P8 | 2 | 1 | 0 | 1 |
| | | | 5.38 | 8.38 |

25

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 5 |
| P3 | 0 | 4 |
| P4 | 1 | 3 |
| P5 | 2 | 2 |
| P6 | 3 | 2 |
| P7 | 4 | 1 |
| P8 | 0 | 1 |

**SOLUZIONE**

**FCFS**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 0 | 6 |
| P2 | 0 | 5 | 6 | 11 |
| P3 | 0 | 4 | 11 | 15 |
| P4 | 1 | 3 | 15 | 18 |
| P5 | 2 | 2 | 17 | 19 |
| P6 | 3 | 2 | 18 | 20 |
| P7 | 4 | 1 | 19 | 20 |
| P8 | 0 | 1 | 15 | 16 |
| | | | 12.62 | 15.62 |

**SJF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 18 | 24 |
| P2 | 0 | 5 | 13 | 18 |
| P3 | 0 | 4 | 9 | 13 |
| P4 | 1 | 3 | 0 | 3 |
| P5 | 2 | 2 | 3 | 5 |
| P6 | 3 | 2 | 4 | 6 |
| P7 | 4 | 1 | 0 | 1 |
| P8 | 0 | 1 | 0 | 1 |
| | | | 5.88 | 8.88 |

**SRTF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 18 | 24 |
| P2 | 0 | 5 | 13 | 18 |
| P3 | 0 | 4 | 9 | 13 |
| P4 | 1 | 3 | 0 | 3 |
| P5 | 2 | 2 | 3 | 5 |
| P6 | 3 | 2 | 4 | 6 |
| P7 | 4 | 1 | 0 | 1 |
| P8 | 0 | 1 | 0 | 1 |
| | | | 5.88 | 8.88 |

**26**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 5 |
| P3 | 0 | 4 |
| P4 | 1 | 2 |
| P5 | 2 | 3 |
| P6 | 3 | 2 |
| P7 | 2 | 3 |
| P8 | 1 | 3 |

**SOLUZIONE**

**FCFS**

| P | A | D | T. Att. |
|---|---|---|---|
| P1 | 0 | 6 | 0 |
| P2 | 0 | 5 | 6 |
| P3 | 0 | 4 | 11 |
| P4 | 1 | 2 | 14 |
| P5 | 2 | 3 | 18 |
| P6 | 3 | 2 | 23 |
| P7 | 2 | 3 | 21 |
| P8 | 1 | 3 | 16 |
| | | | 13.62 |

**SJF**

| P | A | D | T. Att. |
|---|---|---|---|
| P1 | 0 | 6 | 22 |
| P2 | 0 | 5 | 17 |
| P3 | 0 | 4 | 0 |
| P4 | 1 | 2 | 3 |
| P5 | 2 | 3 | 6 |
| P6 | 3 | 2 | 3 |
| P7 | 2 | 3 | 9 |
| P8 | 1 | 3 | 13 |
| | | | 9.12 |

**SRTF**

| P | A | D | T. Att. |
|---|---|---|---|
| P1 | 0 | 6 | 22 |
| P2 | 0 | 5 | 17 |
| P3 | 0 | 4 | 4 |
| P4 | 1 | 2 | 0 |
| P5 | 2 | 3 | 9 |
| P6 | 3 | 2 | 0 |
| P7 | 2 | 3 | 12 |
| P8 | 1 | 3 | 7 |
| | | | 8.88 |

**27**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 5 |
| P3 | 0 | 3 |
| P4 | 0 | 2 |
| P5 | 2 | 1 |
| P6 | 2 | 2 |
| P7 | 4 | 3 |
| P8 | 4 | 1 |

**SOLUZIONE**

**FCFS**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 0 | 6 |
| P2 | 0 | 5 | 6 | 11 |
| P3 | 0 | 3 | 11 | 14 |
| P4 | 0 | 2 | 14 | 16 |
| P5 | 2 | 1 | 14 | 15 |
| P6 | 2 | 2 | 15 | 17 |
| P7 | 4 | 3 | 15 | 18 |
| P8 | 4 | 1 | 18 | 19 |
| | | | 11.62 | 14.50 |

**SJF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 17 | 23 |
| P2 | 0 | 5 | 12 | 17 |
| P3 | 0 | 3 | 6 | 9 |
| P4 | 0 | 2 | 0 | 2 |
| P5 | 2 | 1 | 0 | 1 |
| P6 | 2 | 2 | 1 | 3 |
| P7 | 4 | 3 | 5 | 8 |
| P8 | 4 | 1 | 1 | 2 |
| | | | 5.25 | 8.12 |

**SRTF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 17 | 23 |
| P2 | 0 | 5 | 12 | 17 |
| P3 | 0 | 3 | 6 | 9 |
| P4 | 0 | 2 | 0 | 2 |
| P5 | 2 | 1 | 0 | 1 |
| P6 | 2 | 2 | 1 | 3 |
| P7 | 4 | 3 | 5 | 8 |
| P8 | 4 | 1 | 1 | 2 |
| | | | 5.25 | 8.12 |

28

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 8 |
| P2 | 1 | 2 |
| P3 | 2 | 2 |
| P4 | 3 | 3 |
| P5 | 4 | 2 |
| P6 | 5 | 3 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 8 | | | | | 0 | 8 |
| P2 | 1 | 2 | - - - - | - - - | | | 7 | 9 |
| P3 | 2 | 2 | - - - | - - - - - - | | | 8 | 10 |
| P4 | 3 | 3 | - - | - - - - - - | - - | | 9 | 12 |
| P5 | 4 | 2 | - | - - - - - | - - - - - | | 11 | 13 |
| P6 | 5 | 3 | | - - - - - | - - - - - | - - | 12 | 15 |
| | | | | | | | 7.83 | 11.17 |

**SJF**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 8 | | | | | 0 | 8 |
| P2 | 1 | 2 | - - - - | - - - | | | 7 | 9 |
| P3 | 2 | 2 | - - - | - - - - - - | | | 8 | 10 |
| P4 | 3 | 3 | - - | - - - - - - | - - - - | | 11 | 14 |
| P5 | 4 | 2 | - | - - - - - - | - - | | 8 | 10 |
| P6 | 5 | 3 | | - - - - - | - - - - - | - - | 12 | 15 |
| | | | | | | | 7.67 | 11.00 |

**SRTF**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 8 | - - - - | - - - - - | - - - | | 12 | 20 |
| P2 | 1 | 2 | | | | | 0 | 2 |
| P3 | 2 | 2 | - | | | | 1 | 3 |
| P4 | 3 | 3 | - - | - - | | | 4 | 7 |
| P5 | 4 | 2 | - | | | | 1 | 3 |
| P6 | 5 | 3 | | - - - - - | | | 5 | 8 |
| | | | | | | | 3.83 | 7.17 |

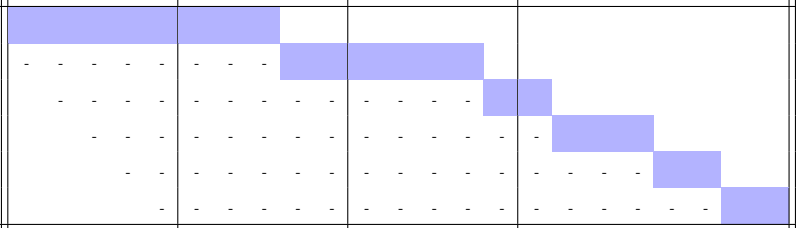
29

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 8 |
| P2 | 0 | 6 |
| P3 | 1 | 2 |
| P4 | 2 | 3 |
| P5 | 3 | 2 |
| P6 | 4 | 2 |

**SOLUZIONE**

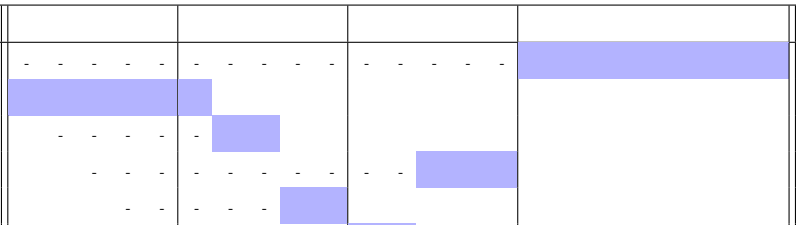
**FCFS**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 8 | 0 | 8 |
| P2 | 0 | 6 | 8 | 14 |
| P3 | 1 | 2 | 13 | 15 |
| P4 | 2 | 3 | 14 | 17 |
| P5 | 3 | 2 | 16 | 18 |
| P6 | 4 | 2 | 17 | 19 |
| | | | 11.33 | 15.17 |



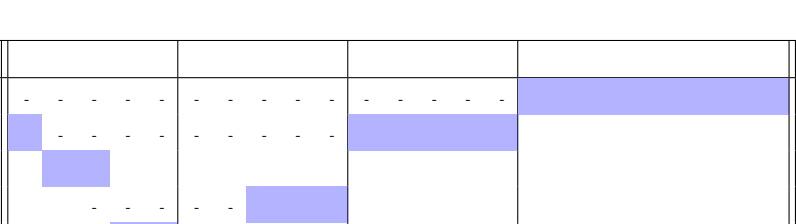
**SJF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 8 | 15 | 23 |
| P2 | 0 | 6 | 0 | 6 |
| P3 | 1 | 2 | 5 | 7 |
| P4 | 2 | 3 | 10 | 13 |
| P5 | 3 | 2 | 5 | 7 |
| P6 | 4 | 2 | 6 | 8 |
| | | | 6.83 | 10.67 |



**SRTF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 8 | 15 | 23 |
| P2 | 0 | 6 | 9 | 15 |
| P3 | 1 | 2 | 0 | 2 |
| P4 | 2 | 3 | 5 | 8 |
| P5 | 3 | 2 | 0 | 2 |
| P6 | 4 | 2 | 1 | 3 |
| | | | 5.00 | 8.83 |



30

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 4 |
| P3 | 0 | 3 |
| P4 | 1 | 2 |
| P5 | 1 | 1 |
| P6 | 1 | 3 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | 0 | 6 |
| P2 | 0 | 4 | - - - - - | - | | 6 | 10 |
| P3 | 0 | 3 | - - - - - | - - - - - | | 10 | 13 |
| P4 | 1 | 2 | - - - - | - - - - - | - - - | 12 | 14 |
| P5 | 1 | 1 | - - - - | - - - - - | - - - - - | 14 | 15 |
| P6 | 1 | 3 | - - - - | - - - - - | - - - - - - | 15 | 18 |
| | | | | | | 9.50 | 12.67 |

**SJF**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | - - - - - | - - - - - | - - - | 13 | 19 |
| P2 | 0 | 4 | - - - - - | - - - - | | 9 | 13 |
| P3 | 0 | 3 | | | | 0 | 3 |
| P4 | 1 | 2 | - - - | | | 3 | 5 |
| P5 | 1 | 1 | - - | | | 2 | 3 |
| P6 | 1 | 3 | - - - - | - | | 5 | 8 |
| | | | | | | 5.33 | 8.50 |

**SRTF**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | - - - - - | - - - - - | - - - | 13 | 19 |
| P2 | 0 | 4 | - - - - - | - - - - | | 9 | 13 |
| P3 | 0 | 3 | | - | | 1 | 4 |
| P4 | 1 | 2 | - - - | | | 3 | 5 |
| P5 | 1 | 1 | | | | 0 | 1 |
| P6 | 1 | 3 | - - - - | - | | 5 | 8 |
| | | | | | | 5.17 | 8.33 |

**31**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 5 |
| P3 | 0 | 4 |
| P4 | 0 | 3 |
| P5 | 2 | 1 |
| P6 | 2 | 2 |

**SOLUZIONE**

**FCFS**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 0 | 6 |
| P2 | 0 | 5 | 6 | 11 |
| P3 | 0 | 4 | 11 | 15 |
| P4 | 0 | 3 | 15 | 18 |
| P5 | 2 | 1 | 16 | 17 |
| P6 | 2 | 2 | 17 | 19 |
| | | | 10.83 | 14.33 |

**SJF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 15 | 21 |
| P2 | 0 | 5 | 10 | 15 |
| P3 | 0 | 4 | 6 | 10 |
| P4 | 0 | 3 | 0 | 3 |
| P5 | 2 | 1 | 1 | 2 |
| P6 | 2 | 2 | 2 | 4 |
| | | | 5.67 | 9.17 |

**SRTF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 15 | 21 |
| P2 | 0 | 5 | 10 | 15 |
| P3 | 0 | 4 | 6 | 10 |
| P4 | 0 | 3 | 0 | 3 |
| P5 | 2 | 1 | 1 | 2 |
| P6 | 2 | 2 | 2 | 4 |
| | | | 5.67 | 9.17 |

32

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 1 | 2 |
| P3 | 2 | 3 |
| P4 | 3 | 2 |
| P5 | 4 | 2 |
| P6 | 5 | 3 |
| P7 | 6 | 2 |

**SOLUZIONE**

**FCFS**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 0 | 6 |
| P2 | 1 | 2 | 5 | 7 |
| P3 | 2 | 3 | 6 | 9 |
| P4 | 3 | 2 | 8 | 10 |
| P5 | 4 | 2 | 9 | 11 |
| P6 | 5 | 3 | 10 | 13 |
| P7 | 6 | 2 | 12 | 14 |
| | | | 7.14 | 10.00 |

**SJF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 0 | 6 |
| P2 | 1 | 2 | 5 | 7 |
| P3 | 2 | 3 | 12 | 15 |
| P4 | 3 | 2 | 5 | 7 |
| P5 | 4 | 2 | 6 | 8 |
| P6 | 5 | 3 | 12 | 15 |
| P7 | 6 | 2 | 6 | 8 |
| | | | 6.57 | 9.43 |

**SRTF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 14 | 20 |
| P2 | 1 | 2 | 0 | 2 |
| P3 | 2 | 3 | 7 | 10 |
| P4 | 3 | 2 | 0 | 2 |
| P5 | 4 | 2 | 1 | 3 |
| P6 | 5 | 3 | 7 | 10 |
| P7 | 6 | 2 | 1 | 3 |
| | | | 4.29 | 7.14 |

33

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 5 |
| P3 | 0 | 4 |
| P4 | 1 | 1 |
| P5 | 3 | 3 |
| P6 | 5 | 2 |
| P7 | 4 | 2 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 0 | 6 |
| P2 | 0 | 5 | | | | | 6 | 11 |
| P3 | 0 | 4 | | | | | 11 | 15 |
| P4 | 1 | 1 | | | | | 14 | 15 |
| P5 | 3 | 3 | | | | | 13 | 16 |
| P6 | 5 | 2 | | | | | 16 | 18 |
| P7 | 4 | 2 | | | | | 15 | 17 |
| | | | | | | | 10.71 | 14.00 |

**SJF**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 17 | 23 |
| P2 | 0 | 5 | | | | | 12 | 17 |
| P3 | 0 | 4 | | | | | 0 | 4 |
| P4 | 1 | 1 | | | | | 3 | 4 |
| P5 | 3 | 3 | | | | | 6 | 9 |
| P6 | 5 | 2 | | | | | 2 | 4 |
| P7 | 4 | 2 | | | | | 1 | 3 |
| | | | | | | | 5.86 | 9.14 |

**SRTF**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 17 | 23 |
| P2 | 0 | 5 | | | | | 12 | 17 |
| P3 | 0 | 4 | | | | | 1 | 5 |
| P4 | 1 | 1 | | | | | 0 | 1 |
| P5 | 3 | 3 | | | | | 6 | 9 |
| P6 | 5 | 2 | | | | | 2 | 4 |
| P7 | 4 | 2 | | | | | 1 | 3 |
| | | | | | | | 5.57 | 8.86 |

34

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 5 |
| P2 | 0 | 4 |
| P3 | 1 | 2 |
| P4 | 2 | 2 |
| P5 | 3 | 3 |
| P6 | 4 | 2 |
| P7 | 5 | 2 |

**SOLUZIONE**

**FCFS**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 5 | 0 | 5 |
| P2 | 0 | 4 | 5 | 9 |
| P3 | 1 | 2 | 8 | 10 |
| P4 | 2 | 2 | 9 | 11 |
| P5 | 3 | 3 | 10 | 13 |
| P6 | 4 | 2 | 12 | 14 |
| P7 | 5 | 2 | 13 | 15 |
| | | | 8.14 | 11.00 |

**SJF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 5 | 15 | 20 |
| P2 | 0 | 4 | 0 | 4 |
| P3 | 1 | 2 | 3 | 5 |
| P4 | 2 | 2 | 4 | 6 |
| P5 | 3 | 3 | 9 | 12 |
| P6 | 4 | 2 | 4 | 6 |
| P7 | 5 | 2 | 5 | 7 |
| | | | 5.71 | 8.57 |

**SRTF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 5 | 15 | 20 |
| P2 | 0 | 4 | 8 | 12 |
| P3 | 1 | 2 | 0 | 2 |
| P4 | 2 | 2 | 1 | 3 |
| P5 | 3 | 3 | 9 | 12 |
| P6 | 4 | 2 | 1 | 3 |
| P7 | 5 | 2 | 2 | 4 |
| | | | 5.14 | 8.00 |

35

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 5 |
| P2 | 0 | 4 |
| P3 | 0 | 3 |
| P4 | 0 | 3 |
| P5 | 2 | 2 |
| P6 | 2 | 3 |
| P7 | 2 | 4 |
| P8 | 4 | 2 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | | | | T. Att. | T. Co |
|---|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 5 | | | | | | 0 | |
| P2 | 0 | 4 | - - - - - | | | | | 5 | |
| P3 | 0 | 3 | - - - - - | - - - - - | | | | 9 | 1 |
| P4 | 0 | 3 | - - - - - | - - - - - | - - | | | 12 | 1 |
| P5 | 2 | 2 | - - - | - - - - - | - - - - - | | | 13 | 1 |
| P6 | 2 | 3 | - - - | - - - - - | - - - - - | - - | | 15 | 1 |
| P7 | 2 | 4 | - - - | - - - - - | - - - - - | - - - - - | | 18 | 2 |
| P8 | 4 | 2 | - | - - - - - | - - - - - | - - - - - | - - - - | 20 | 2 |
| | | | | | | | | 11.50 | 14 |

**SJF**

| P | A | D | | | | | | T. Att. | T. Co |
|---|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 5 | - - - - - | - - - - - | - - - - - | - - - - - | - | 21 | 2 |
| P2 | 0 | 4 | - - - - - | - - - - - | - - - | | | 13 | 1 |
| P3 | 0 | 3 | | | | | | 0 | |
| P4 | 0 | 3 | - - - - - | - - | | | | 7 | 1 |
| P5 | 2 | 2 | - | | | | | 1 | |
| P6 | 2 | 3 | - - - | - - - - - | | | | 8 | 1 |
| P7 | 2 | 4 | - - - | - - - - - | - - - - - | - - | | 15 | 1 |
| P8 | 4 | 2 | - | | | | | 1 | |
| | | | | | | | | 8.25 | 11 |

**SRTF**

| P | A | D | | | | | | T. Att. | T. Co |
|---|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 5 | - - - - - | - - - - - | - - - - - | - - - - - | - | 21 | 2 |
| P2 | 0 | 4 | - - - - - | - - - - - | - - - | | | 13 | 1 |
| P3 | 0 | 3 | | | | | | 0 | |
| P4 | 0 | 3 | - - - - - | - - | | | | 7 | 1 |
| P5 | 2 | 2 | - | | | | | 1 | |
| P6 | 2 | 3 | - - - | - - - - - | | | | 8 | 1 |
| P7 | 2 | 4 | - - - | - - - - - | - - - - - | - - | | 15 | 1 |
| P8 | 4 | 2 | - | | | | | 1 | |
| | | | | | | | | 8.25 | 11 |

36

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 1 |
| P2 | 0 | 3 |
| P3 | 0 | 5 |
| P4 | 1 | 2 |
| P5 | 1 | 3 |
| P6 | 1 | 2 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 1 | | | | 0 | 1 |
| P2 | 0 | 3 | - | | | 1 | 4 |
| P3 | 0 | 5 | - - - - | | | 4 | 9 |
| P4 | 1 | 2 | - - - - | - - - - | | 8 | 10 |
| P5 | 1 | 3 | - - - - | - - - - - | - | 10 | 13 |
| P6 | 1 | 2 | - - - - | - - - - - | - - - - | 13 | 15 |
| | | | | | | 6.00 | 8.67 |

**SJF**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 1 | | | | 0 | 1 |
| P2 | 0 | 3 | - - - - - | | | 5 | 8 |
| P3 | 0 | 5 | - - - - - | - - - - - | - | 11 | 16 |
| P4 | 1 | 2 | | | | 0 | 2 |
| P5 | 1 | 3 | - - - - | - - - | | 7 | 10 |
| P6 | 1 | 2 | - - | | | 2 | 4 |
| | | | | | | 4.17 | 6.83 |

**SRTF**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 1 | | | | 0 | 1 |
| P2 | 0 | 3 | - - - - - | | | 5 | 8 |
| P3 | 0 | 5 | - - - - - | - - - - - | - | 11 | 16 |
| P4 | 1 | 2 | | | | 0 | 2 |
| P5 | 1 | 3 | - - - - | - - - | | 7 | 10 |
| P6 | 1 | 2 | - - | | | 2 | 4 |
| | | | | | | 4.17 | 6.83 |

37

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 5 |
| P3 | 1 | 2 |
| P4 | 2 | 2 |
| P5 | 3 | 3 |
| P6 | 4 | 2 |
| P7 | 5 | 2 |

**SOLUZIONE**

**FCFS**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 0 | 6 |
| P2 | 0 | 5 | 6 | 11 |
| P3 | 1 | 2 | 10 | 12 |
| P4 | 2 | 2 | 11 | 13 |
| P5 | 3 | 3 | 12 | 15 |
| P6 | 4 | 2 | 14 | 16 |
| P7 | 5 | 2 | 15 | 17 |
| | | | 9.71 | 12.86 |

**SJF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 16 | 22 |
| P2 | 0 | 5 | 0 | 5 |
| P3 | 1 | 2 | 4 | 6 |
| P4 | 2 | 2 | 5 | 7 |
| P5 | 3 | 3 | 10 | 13 |
| P6 | 4 | 2 | 5 | 7 |
| P7 | 5 | 2 | 6 | 8 |
| | | | 6.57 | 9.71 |

**SRTF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 16 | 22 |
| P2 | 0 | 5 | 11 | 16 |
| P3 | 1 | 2 | 0 | 2 |
| P4 | 2 | 2 | 1 | 3 |
| P5 | 3 | 3 | 6 | 9 |
| P6 | 4 | 2 | 1 | 3 |
| P7 | 5 | 2 | 2 | 4 |
| | | | 5.29 | 8.43 |

38

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 4 |
| P2 | 0 | 3 |
| P3 | 0 | 2 |
| P4 | 0 | 3 |
| P5 | 2 | 2 |
| P6 | 2 | 3 |
| P7 | 2 | 4 |

**SOLUZIONE**

**FCFS**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 4 | | | | | 0 | 4 |
| P2 | 0 | 3 | - - - - | | | | 4 | 7 |
| P3 | 0 | 2 | - - - - - - | - - | | | 7 | 9 |
| P4 | 0 | 3 | - - - - - - | - - - - | | | 9 | 12 |
| P5 | 2 | 2 | - - - | - - - - - - | - - | | 10 | 12 |
| P6 | 2 | 3 | - - - | - - - - - - | - - - - | | 12 | 15 |
| P7 | 2 | 4 | - - - | - - - - - | - - - - - - | - - | 15 | 19 |
| | | | | | | | 8.14 | 11.14 |

**SJF**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 4 | - - - - - - | - - - - - | - - - | | 13 | 17 |
| P2 | 0 | 3 | - - - - | | | | 4 | 7 |
| P3 | 0 | 2 | | | | | 0 | 2 |
| P4 | 0 | 3 | - - - - - - | - - | | | 7 | 10 |
| P5 | 2 | 2 | | | | | 0 | 2 |
| P6 | 2 | 3 | - - - | - - - - - - | | | 8 | 11 |
| P7 | 2 | 4 | - - - | - - - - - | - - - - - - | - - | 15 | 19 |
| | | | | | | | 6.71 | 9.71 |

**SRTF**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 4 | - - - - - - | - - - - - | - - - | | 13 | 17 |
| P2 | 0 | 3 | - - - - | | | | 4 | 7 |
| P3 | 0 | 2 | | | | | 0 | 2 |
| P4 | 0 | 3 | - - - - - - | - - | | | 7 | 10 |
| P5 | 2 | 2 | | | | | 0 | 2 |
| P6 | 2 | 3 | - - - | - - - - - - | | | 8 | 11 |
| P7 | 2 | 4 | - - - | - - - - - | - - - - - - | - - | 15 | 19 |
| | | | | | | | 6.71 | 9.71 |

39

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 3 |
| P2 | 0 | 1 |
| P3 | 0 | 5 |
| P4 | 1 | 3 |
| P5 | 1 | 2 |
| P6 | 1 | 4 |
| P7 | 2 | 2 |
| P8 | 2 | 3 |

**SOLUZIONE**

**FCFS**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 3 | 0 | 3 |
| P2 | 0 | 1 | 3 | 4 |
| P3 | 0 | 5 | 4 | 9 |
| P4 | 1 | 3 | 8 | 11 |
| P5 | 1 | 2 | 11 | 13 |
| P6 | 1 | 4 | 13 | 17 |
| P7 | 2 | 2 | 16 | 18 |
| P8 | 2 | 3 | 18 | 21 |
| | | | 9.12 | 12.00 |

**SJF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 3 | 5 | 8 |
| P2 | 0 | 1 | 0 | 1 |
| P3 | 0 | 5 | 18 | 23 |
| P4 | 1 | 3 | 7 | 10 |
| P5 | 1 | 2 | 0 | 2 |
| P6 | 1 | 4 | 13 | 17 |
| P7 | 2 | 2 | 1 | 3 |
| P8 | 2 | 3 | 9 | 12 |
| | | | 6.62 | 9.50 |

**SRTF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 3 | 5 | 8 |
| P2 | 0 | 1 | 0 | 1 |
| P3 | 0 | 5 | 18 | 23 |
| P4 | 1 | 3 | 7 | 10 |
| P5 | 1 | 2 | 0 | 2 |
| P6 | 1 | 4 | 13 | 17 |
| P7 | 2 | 2 | 1 | 3 |
| P8 | 2 | 3 | 9 | 12 |
| | | | 6.62 | 9.50 |

40

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 1 |
| P2 | 1 | 2 |
| P3 | 2 | 2 |
| P4 | 3 | 3 |
| P5 | 4 | 2 |
| P6 | 5 | 2 |
| P7 | 6 | 3 |
| P8 | 7 | 2 |

**SOLUZIONE**

**FCFS**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 1 | 0 | 1 |
| P2 | 1 | 2 | 0 | 2 |
| P3 | 2 | 2 | 1 | 3 |
| P4 | 3 | 3 | 2 | 5 |
| P5 | 4 | 2 | 4 | 6 |
| P6 | 5 | 2 | 5 | 7 |
| P7 | 6 | 3 | 6 | 9 |
| P8 | 7 | 2 | 8 | 10 |
| | | | 3.25 | 5.38 |

**SJF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 1 | 0 | 1 |
| P2 | 1 | 2 | 0 | 2 |
| P3 | 2 | 2 | 1 | 3 |
| P4 | 3 | 3 | 8 | 11 |
| P5 | 4 | 2 | 1 | 3 |
| P6 | 5 | 2 | 2 | 4 |
| P7 | 6 | 3 | 8 | 11 |
| P8 | 7 | 2 | 2 | 4 |
| | | | 2.75 | 4.88 |

**SRTF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 1 | 0 | 1 |
| P2 | 1 | 2 | 0 | 2 |
| P3 | 2 | 2 | 1 | 3 |
| P4 | 3 | 3 | 8 | 11 |
| P5 | 4 | 2 | 1 | 3 |
| P6 | 5 | 2 | 2 | 4 |
| P7 | 6 | 3 | 8 | 11 |
| P8 | 7 | 2 | 2 | 4 |
| | | | 2.75 | 4.88 |

**41**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 4 |
| P3 | 1 | 2 |
| P4 | 2 | 2 |
| P5 | 3 | 3 |
| P6 | 4 | 2 |
| P7 | 5 | 2 |
| P8 | 6 | 3 |

**SOLUZIONE**

**FCFS**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 0 | 6 |
| P2 | 0 | 4 | 6 | 10 |
| P3 | 1 | 2 | 9 | 11 |
| P4 | 2 | 2 | 10 | 12 |
| P5 | 3 | 3 | 11 | 14 |
| P6 | 4 | 2 | 13 | 15 |
| P7 | 5 | 2 | 14 | 16 |
| P8 | 6 | 3 | 15 | 18 |
| | | | 9.75 | 12.75 |

**SJF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 18 | 24 |
| P2 | 0 | 4 | 0 | 4 |
| P3 | 1 | 2 | 3 | 5 |
| P4 | 2 | 2 | 4 | 6 |
| P5 | 3 | 3 | 9 | 12 |
| P6 | 4 | 2 | 4 | 6 |
| P7 | 5 | 2 | 5 | 7 |
| P8 | 6 | 3 | 9 | 12 |
| | | | 6.50 | 9.50 |

**SRTF**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 18 | 24 |
| P2 | 0 | 4 | 8 | 12 |
| P3 | 1 | 2 | 0 | 2 |
| P4 | 2 | 2 | 1 | 3 |
| P5 | 3 | 3 | 9 | 12 |
| P6 | 4 | 2 | 1 | 3 |
| P7 | 5 | 2 | 2 | 4 |
| P8 | 6 | 3 | 9 | 12 |
| | | | 6.00 | 9.00 |

## 5.2 Round Robin

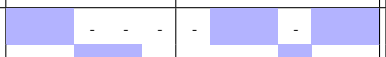
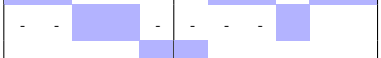
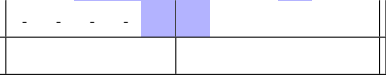
Effettua lo schedeuling dei seguenti processi, usando l'algoritmo Round Robin con t=2 e t=3

1

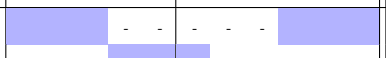
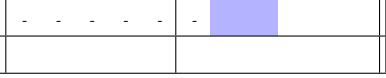
| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 3 |
| P3 | 0 | 2 |

**SOLUZIONE**

**RR2**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 6 |  | | 5 | 11 |
| P2 | 0 | 3 |  | | 6 | 9 |
| P3 | 0 | 2 |  | | 4 | 6 |
| | | | | | 5.00 | 8.67 |

**RR3**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 6 |  | | 5 | 11 |
| P2 | 0 | 3 |  | | 3 | 6 |
| P3 | 0 | 2 |  | | 6 | 8 |
| | | | | | 4.67 | 8.33 |

**2**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 5 |
| P2 | 0 | 2 |
| P3 | 1 | 3 |

**SOLUZIONE**

**RR2**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 5 | | | 5 | 10 |
| P2 | 0 | 2 | | | 2 | 4 |
| P3 | 1 | 3 | | | 5 | 8 |
| | | | | | 4.00 | 7.33 |

**RR3**

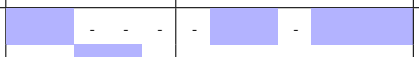
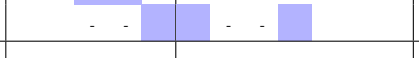
| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 5 | | | 5 | 10 |
| P2 | 0 | 2 | | | 3 | 5 |
| P3 | 1 | 3 | | | 4 | 7 |
| | | | | | 4.00 | 7.33 |

3

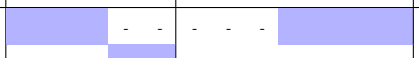
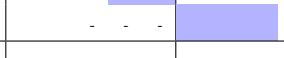
| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 7 |
| P2 | 1 | 2 |
| P3 | 2 | 3 |

**SOLUZIONE**

**RR2**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 7 |  | | 5 | 12 |
| P2 | 1 | 2 |  | | 1 | 3 |
| P3 | 2 | 3 |  | | 4 | 7 |
| | | | | | 3.33 | 7.33 |

**RR3**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 7 |  | | 5 | 12 |
| P2 | 1 | 2 |  | | 2 | 4 |
| P3 | 2 | 3 |  | | 3 | 6 |
| | | | | | 3.33 | 7.33 |

**4**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 2 |
| P2 | 0 | 4 |
| P3 | 0 | 6 |

**SOLUZIONE**

**RR2**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 2 | | | 0 | 2 |
| P2 | 0 | 4 | - - | - - | 4 | 8 |
| P3 | 0 | 6 | - - - - | - - | 6 | 12 |
| | | | | | 3.33 | 7.33 |

**RR3**

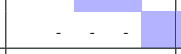
| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 2 | | | 0 | 2 |
| P2 | 0 | 4 | - - | - - - | 5 | 9 |
| P3 | 0 | 6 | - - - - - | - | 6 | 12 |
| | | | | | 3.67 | 7.67 |

5

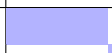
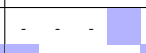
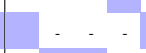
| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 4 |
| P2 | 0 | 4 |
| P3 | 1 | 2 |

**SOLUZIONE**

**RR2**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 4 |  |  | 4 | 8 |
| P2 | 0 | 4 |  |  | 6 | 10 |
| P3 | 1 | 2 |  |  | 3 | 5 |
| | | | | | 4.33 | 7.67 |

**RR3**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 4 |  |  | 5 | 9 |
| P2 | 0 | 4 |  |  | 6 | 10 |
| P3 | 1 | 2 |  |  | 5 | 7 |
| | | | | | 5.33 | 8.67 |

**6**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 1 | 2 |
| P3 | 2 | 2 |

**SOLUZIONE**

**RR2**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | 4 | 10 |
| P2 | 1 | 2 | | | 1 | 3 |
| P3 | 2 | 2 | | | 2 | 4 |
| | | | | | 2.33 | 5.67 |

**RR3**

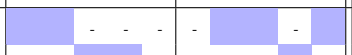
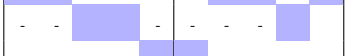
| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | 4 | 10 |
| P2 | 1 | 2 | | | 2 | 4 |
| P3 | 2 | 2 | | | 3 | 5 |
| | | | | | 3.00 | 6.33 |

7

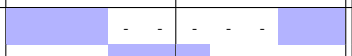
| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 5 |
| P2 | 0 | 3 |
| P3 | 1 | 2 |

**SOLUZIONE**

**RR2**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 5 |  | | 5 | 10 |
| P2 | 0 | 3 |  | | 6 | 9 |
| P3 | 1 | 2 |  | | 3 | 5 |
| | | | | | 4.67 | 8.00 |

**RR3**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 5 |  | | 5 | 10 |
| P2 | 0 | 3 |  | | 3 | 6 |
| P3 | 1 | 2 |  | | 5 | 7 |
| | | | | | 4.33 | 7.67 |

**8**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 1 | 4 |
| P3 | 2 | 2 |

**SOLUZIONE**

**RR2**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | 6 | 12 |
| P2 | 1 | 4 | | | 5 | 9 |
| P3 | 2 | 2 | | | 2 | 4 |
| | | | | | 4.33 | 8.33 |

**RR3**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | 5 | 11 |
| P2 | 1 | 4 | | | 7 | 11 |
| P3 | 2 | 2 | | | 4 | 6 |
| | | | | | 5.33 | 9.33 |

**9**

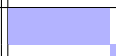
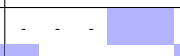
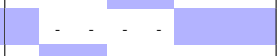
| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 5 |
| P2 | 0 | 6 |
| P3 | 1 | 2 |

**SOLUZIONE**

**RR2**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 5 |  |  | 6 | 11 |
| P2 | 0 | 6 |  |  | 7 | 13 |
| P3 | 1 | 2 |  | | 3 | 5 |
| | | | | | 5.33 | 9.67 |

**RR3**

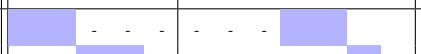
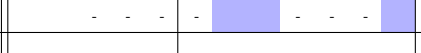
| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 5 |  |  | 5 | 10 |
| P2 | 0 | 6 |  |  | 7 | 13 |
| P3 | 1 | 2 |  |  | 5 | 7 |
| | | | | | 5.67 | 10.00 |

10

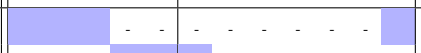
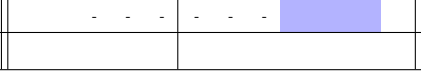
| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 4 |
| P2 | 0 | 3 |
| P3 | 1 | 2 |
| P4 | 2 | 3 |

**SOLUZIONE**

**RR2**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 4 |  | | 6 | 10 |
| P2 | 0 | 3 |  | | 8 | 11 |
| P3 | 1 | 2 |  | | 3 | 5 |
| P4 | 2 | 3 |  | | 7 | 10 |
| | | | | | 6.00 | 9.00 |

**RR3**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 4 |  | | 8 | 12 |
| P2 | 0 | 3 |  | | 3 | 6 |
| P3 | 1 | 2 |  | | 5 | 7 |
| P4 | 2 | 3 |  | | 6 | 9 |
| | | | | | 5.50 | 8.50 |

**11**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 3 |
| P3 | 0 | 2 |
| P4 | 1 | 4 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | 9 | 15 |
| P2 | 0 | 3 | | | | 8 | 11 |
| P3 | 0 | 2 | | | | 4 | 6 |
| P4 | 1 | 4 | | | | 8 | 12 |
| | | | | | | 7.25 | 11.00 |

**RR3**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | 8 | 14 |
| P2 | 0 | 3 | | | | 3 | 6 |
| P3 | 0 | 2 | | | | 6 | 8 |
| P4 | 1 | 4 | | | | 10 | 14 |
| | | | | | | 6.75 | 10.50 |

12

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 7 |
| P2 | 0 | 6 |
| P3 | 1 | 2 |
| P4 | 2 | 3 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 7 | | | | 11 | 18 |
| P2 | 0 | 6 | | | | 11 | 17 |
| P3 | 1 | 2 | | | | 3 | 5 |
| P4 | 2 | 3 | | | | 8 | 11 |
| | | | | | | 8.25 | 12.75 |

**RR3**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 7 | | | | 11 | 18 |
| P2 | 0 | 6 | | | | 11 | 17 |
| P3 | 1 | 2 | | | | 5 | 7 |
| P4 | 2 | 3 | | | | 6 | 9 |
| | | | | | | 8.25 | 12.75 |

13

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 5 |
| P2 | 0 | 4 |
| P3 | 1 | 2 |
| P4 | 1 | 3 |

**SOLUZIONE**

**RR2**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 5 | | | 9 | 14 |
| P2 | 0 | 4 | | | 8 | 12 |
| P3 | 1 | 2 | | | 3 | 5 |
| P4 | 1 | 3 | | | 9 | 12 |
| | | | | | 7.25 | 10.75 |

**RR3**

| P | A | D | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|
| P1 | 0 | 5 | | | 8 | 13 |
| P2 | 0 | 4 | | | 10 | 14 |
| P3 | 1 | 2 | | | 5 | 7 |
| P4 | 1 | 3 | | | 7 | 10 |
| | | | | | 7.50 | 11.00 |

14

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 8 |
| P2 | 1 | 2 |
| P3 | 2 | 3 |
| P4 | 3 | 2 |
| P5 | 4 | 3 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 8 | | | | 10 | 18 |
| P2 | 1 | 2 | | | | 1 | 3 |
| P3 | 2 | 3 | | | | 8 | 11 |
| P4 | 3 | 2 | | | | 5 | 7 |
| P5 | 4 | 3 | | | | 9 | 12 |
| | | | | | | 6.60 | 10.20 |

**RR3**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 8 | | | | 10 | 18 |
| P2 | 1 | 2 | | | | 2 | 4 |
| P3 | 2 | 3 | | | | 3 | 6 |
| P4 | 3 | 2 | | | | 5 | 7 |
| P5 | 4 | 3 | | | | 9 | 12 |
| | | | | | | 5.80 | 9.40 |

**15**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 7 |
| P2 | 0 | 6 |
| P3 | 1 | 4 |
| P4 | 2 | 3 |
| P5 | 3 | 4 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 7 | | | | | 17 | 24 |
| P2 | 0 | 6 | | | | | 17 | 23 |
| P3 | 1 | 4 | | | | | 11 | 15 |
| P4 | 2 | 3 | | | | | 12 | 15 |
| P5 | 3 | 4 | | | | | 14 | 18 |
| | | | | | | | 14.20 | 19.00 |

**RR3**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 7 | | | | | 17 | 24 |
| P2 | 0 | 6 | | | | | 15 | 21 |
| P3 | 1 | 4 | | | | | 17 | 21 |
| P4 | 2 | 3 | | | | | 7 | 10 |
| P5 | 3 | 4 | | | | | 16 | 20 |
| | | | | | | | 14.40 | 19.20 |

16

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 5 |
| P2 | 0 | 4 |
| P3 | 0 | 3 |
| P4 | 1 | 2 |
| P5 | 1 | 4 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 5 | | | | 13 | 18 |
| P2 | 0 | 4 | | | | 10 | 14 |
| P3 | 0 | 3 | | | | 12 | 15 |
| P4 | 1 | 2 | | | | 5 | 7 |
| P5 | 1 | 4 | | | | 12 | 16 |
| | | | | | | 10.40 | 14.00 |

**RR3**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 5 | | | | 11 | 16 |
| P2 | 0 | 4 | | | | 13 | 17 |
| P3 | 0 | 3 | | | | 6 | 9 |
| P4 | 1 | 2 | | | | 8 | 10 |
| P5 | 1 | 4 | | | | 13 | 17 |
| | | | | | | 10.20 | 13.80 |

**17**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 9 |
| P2 | 1 | 2 |
| P3 | 1 | 3 |
| P4 | 3 | 4 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 9 | | | | 9 | 18 |
| P2 | 1 | 2 | | | | 1 | 3 |
| P3 | 1 | 3 | | | | 7 | 10 |
| P4 | 3 | 4 | | | | 8 | 12 |
| | | | | | | 6.25 | 10.75 |

**RR3**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 9 | | | | 9 | 18 |
| P2 | 1 | 2 | | | | 2 | 4 |
| P3 | 1 | 3 | | | | 4 | 7 |
| P4 | 3 | 4 | | | | 8 | 12 |
| | | | | | | 5.75 | 10.25 |

**18**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 4 |
| P3 | 2 | 2 |
| P4 | 2 | 3 |
| P5 | 3 | 5 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | ■ - - - | - - - ■ | - - - - - | ■ | 11 | 17 |
| P2 | 0 | 4 | - - ■ - | - - - - - | - - ■ | | 10 | 14 |
| P3 | 2 | 2 | - - ■ | ■ | | | 2 | 4 |
| P4 | 2 | 3 | - - - | - ■ - - | - - - - ■ | | 10 | 13 |
| P5 | 3 | 5 | - - | - - - - - | ■ - - - | - - ■ | 12 | 17 |
| | | | | | | | 9.00 | 13.00 |

**RR3**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | ■ - - | - - - - - | - - - - ■ | ■ | 11 | 17 |
| P2 | 0 | 4 | - - - ■ | ■ - - - - | - - - - - | - - ■ | 14 | 18 |
| P3 | 2 | 2 | - - - | - ■ | | | 4 | 6 |
| P4 | 2 | 3 | - - - | - - - ■ | ■ | | 6 | 9 |
| P5 | 3 | 5 | - - | - - - - - | - ■ - | - - - ■ | 12 | 17 |
| | | | | | | | 9.40 | 13.40 |

19

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 5 |
| P2 | 0 | 4 |
| P3 | 0 | 3 |
| P4 | 2 | 2 |
| P5 | 2 | 4 |
| P6 | 4 | 3 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 5 | | | | | 15 | 20 |
| P2 | 0 | 4 | | | | | 12 | 16 |
| P3 | 0 | 3 | | | | | 14 | 17 |
| P4 | 2 | 2 | | | | | 4 | 6 |
| P5 | 2 | 4 | | | | | 13 | 17 |
| P6 | 4 | 3 | | | | | 14 | 17 |
| | | | | | | | 12.00 | 15.50 |

**RR3**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 5 | | | | | 11 | 16 |
| P2 | 0 | 4 | | | | | 16 | 20 |
| P3 | 0 | 3 | | | | | 6 | 9 |
| P4 | 2 | 2 | | | | | 7 | 9 |
| P5 | 2 | 4 | | | | | 15 | 19 |
| P6 | 4 | 3 | | | | | 12 | 15 |
| | | | | | | | 11.17 | 14.67 |

20

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 7 |
| P2 | 1 | 2 |
| P3 | 2 | 3 |
| P4 | 3 | 2 |
| P5 | 2 | 1 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 7 | | | | 8 | 15 |
| P2 | 1 | 2 | | | | 1 | 3 |
| P3 | 2 | 3 | | | | 7 | 10 |
| P4 | 3 | 2 | | | | 6 | 8 |
| P5 | 2 | 1 | | | | 4 | 5 |
| | | | | | | 5.20 | 8.20 |

**RR3**

| P | A | D | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|
| P1 | 0 | 7 | | | | 8 | 15 |
| P2 | 1 | 2 | | | | 2 | 4 |
| P3 | 2 | 3 | | | | 3 | 6 |
| P4 | 3 | 2 | | | | 6 | 8 |
| P5 | 2 | 1 | | | | 6 | 7 |
| | | | | | | 5.00 | 8.00 |

**21**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 7 |
| P3 | 1 | 2 |
| P4 | 2 | 3 |
| P5 | 3 | 2 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | | | | | | | | | | | | | | | | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | 11 | 17 |
| P2 | 0 | 7 | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | 13 | 20 |
| P3 | 1 | 2 | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | 3 | 5 |
| P4 | 2 | 3 | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | 10 | 13 |
| P5 | 3 | 2 | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | 7 | 9 |
| | | | | | | | | | | | | | | | | | | | | | | | | | 8.80 | 12.80 |

**RR3**

| P | A | D | | | | | | | | | | | | | | | | | | | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | 10 | 16 |
| P2 | 0 | 7 | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | 13 | 20 |
| P3 | 1 | 2 | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | 5 | 7 |
| P4 | 2 | 3 | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | 6 | 9 |
| P5 | 3 | 2 | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | ■ | 8 | 10 |
| | | | | | | | | | | | | | | | | | | | | | | | | | 8.40 | 12.40 |

22

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 4 |
| P3 | 0 | 5 |
| P4 | 2 | 2 |
| P5 | 2 | 3 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 13 | 19 |
| P2 | 0 | 4 | | | | | 10 | 14 |
| P3 | 0 | 5 | | | | | 15 | 20 |
| P4 | 2 | 2 | | | | | 4 | 6 |
| P5 | 2 | 3 | | | | | 12 | 15 |
| | | | | | | | 10.80 | 14.80 |

**RR3**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 11 | 17 |
| P2 | 0 | 4 | | | | | 14 | 18 |
| P3 | 0 | 5 | | | | | 15 | 20 |
| P4 | 2 | 2 | | | | | 7 | 9 |
| P5 | 2 | 3 | | | | | 9 | 12 |
| | | | | | | | 11.20 | 15.20 |

**23**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 7 |
| P2 | 1 | 2 |
| P3 | 2 | 2 |
| P4 | 3 | 3 |
| P5 | 4 | 2 |
| P6 | 5 | 3 |

**SOLUZIONE**

**RR2**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 7 | 12 | 19 |
| P2 | 1 | 2 | 1 | 3 |
| P3 | 2 | 2 | 2 | 4 |
| P4 | 3 | 3 | 11 | 14 |
| P5 | 4 | 2 | 6 | 8 |
| P6 | 5 | 3 | 10 | 13 |
| | | | 7.00 | 10.17 |

**RR3**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 7 | 12 | 19 |
| P2 | 1 | 2 | 2 | 4 |
| P3 | 2 | 2 | 3 | 5 |
| P4 | 3 | 3 | 4 | 7 |
| P5 | 4 | 2 | 9 | 11 |
| P6 | 5 | 3 | 10 | 13 |
| | | | 6.67 | 9.83 |

24

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 5 |
| P3 | 2 | 2 |
| P4 | 3 | 3 |
| P5 | 4 | 2 |
| P6 | 4 | 4 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 12 | 18 |
| P2 | 0 | 5 | | | | | 17 | 22 |
| P3 | 2 | 2 | | | | | 2 | 4 |
| P4 | 3 | 3 | | | | | 13 | 16 |
| P5 | 4 | 2 | | | | | 6 | 8 |
| P6 | 4 | 4 | | | | | 13 | 17 |
| | | | | | | | 10.50 | 14.17 |

**RR3**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 8 | 14 |
| P2 | 0 | 5 | | | | | 16 | 21 |
| P3 | 2 | 2 | | | | | 4 | 6 |
| P4 | 3 | 3 | | | | | 5 | 8 |
| P5 | 4 | 2 | | | | | 10 | 12 |
| P6 | 4 | 4 | | | | | 14 | 18 |
| | | | | | | | 9.50 | 13.17 |

25

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 5 |
| P3 | 0 | 4 |
| P4 | 1 | 3 |
| P5 | 2 | 2 |
| P6 | 3 | 2 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 15 | 21 |
| P2 | 0 | 5 | | | | | 17 | 22 |
| P3 | 0 | 4 | | | | | 14 | 18 |
| P4 | 1 | 3 | | | | | 15 | 18 |
| P5 | 2 | 2 | | | | | 6 | 8 |
| P6 | 3 | 2 | | | | | 9 | 11 |
| | | | | | | | 12.67 | 16.33 |

**RR3**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 13 | 19 |
| P2 | 0 | 5 | | | | | 16 | 21 |
| P3 | 0 | 4 | | | | | 18 | 22 |
| P4 | 1 | 3 | | | | | 8 | 11 |
| P5 | 2 | 2 | | | | | 10 | 12 |
| P6 | 3 | 2 | | | | | 11 | 13 |
| | | | | | | | 12.67 | 16.33 |

26

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 5 |
| P3 | 0 | 4 |
| P4 | 1 | 2 |
| P5 | 2 | 3 |
| P6 | 3 | 2 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 15 | 21 |
| P2 | 0 | 5 | | | | | 17 | 22 |
| P3 | 0 | 4 | | | | | 14 | 18 |
| P4 | 1 | 2 | | | | | 5 | 7 |
| P5 | 2 | 3 | | | | | 14 | 17 |
| P6 | 3 | 2 | | | | | 9 | 11 |
| | | | | | | | 12.33 | 16.00 |

**RR3**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 13 | 19 |
| P2 | 0 | 5 | | | | | 16 | 21 |
| P3 | 0 | 4 | | | | | 18 | 22 |
| P4 | 1 | 2 | | | | | 8 | 10 |
| P5 | 2 | 3 | | | | | 9 | 12 |
| P6 | 3 | 2 | | | | | 11 | 13 |
| | | | | | | | 12.50 | 16.17 |

27

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 2 |
| P3 | 0 | 4 |
| P4 | 0 | 3 |
| P5 | 2 | 2 |
| P6 | 2 | 3 |
| P7 | 4 | 2 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 16 | 22 |
| P2 | 0 | 2 | | | | | 2 | 4 |
| P3 | 0 | 4 | | | | | 14 | 18 |
| P4 | 0 | 3 | | | | | 16 | 19 |
| P5 | 2 | 2 | | | | | 6 | 8 |
| P6 | 2 | 3 | | | | | 15 | 18 |
| P7 | 4 | 2 | | | | | 10 | 12 |
| | | | | | | | 11.29 | 14.43 |

**RR3**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 13 | 19 |
| P2 | 0 | 2 | | | | | 3 | 5 |
| P3 | 0 | 4 | | | | | 18 | 22 |
| P4 | 0 | 3 | | | | | 8 | 11 |
| P5 | 2 | 2 | | | | | 9 | 11 |
| P6 | 2 | 3 | | | | | 11 | 14 |
| P7 | 4 | 2 | | | | | 15 | 17 |
| | | | | | | | 11.00 | 14.14 |

28

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 10 |
| P2 | 1 | 2 |
| P3 | 2 | 2 |
| P4 | 3 | 3 |
| P5 | 4 | 2 |
| P6 | 5 | 3 |
| P7 | 5 | 2 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 10 | | | | | 14 | 24 |
| P2 | 1 | 2 | | | | | 1 | 3 |
| P3 | 2 | 2 | | | | | 2 | 4 |
| P4 | 3 | 3 | | | | | 13 | 16 |
| P5 | 4 | 2 | | | | | 6 | 8 |
| P6 | 5 | 3 | | | | | 12 | 15 |
| P7 | 5 | 2 | | | | | 9 | 11 |
| | | | | | | | 8.14 | 11.57 |

**RR3**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 10 | | | | | 14 | 24 |
| P2 | 1 | 2 | | | | | 2 | 4 |
| P3 | 2 | 2 | | | | | 3 | 5 |
| P4 | 3 | 3 | | | | | 4 | 7 |
| P5 | 4 | 2 | | | | | 9 | 11 |
| P6 | 5 | 3 | | | | | 10 | 13 |
| P7 | 5 | 2 | | | | | 13 | 15 |
| | | | | | | | 7.86 | 11.29 |

**29**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 7 |
| P2 | 0 | 6 |
| P3 | 1 | 2 |
| P4 | 2 | 3 |
| P5 | 3 | 2 |
| P6 | 4 | 4 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 7 | | | | | 17 | 24 |
| P2 | 0 | 6 | | | | | 17 | 23 |
| P3 | 1 | 2 | | | | | 3 | 5 |
| P4 | 2 | 3 | | | | | 12 | 15 |
| P5 | 3 | 2 | | | | | 7 | 9 |
| P6 | 4 | 4 | | | | | 13 | 17 |
| | | | | | | | 11.50 | 15.50 |

**RR3**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 7 | | | | | 16 | 23 |
| P2 | 0 | 6 | | | | | 16 | 22 |
| P3 | 1 | 2 | | | | | 5 | 7 |
| P4 | 2 | 3 | | | | | 6 | 9 |
| P5 | 3 | 2 | | | | | 8 | 10 |
| P6 | 4 | 4 | | | | | 16 | 20 |
| | | | | | | | 11.17 | 15.17 |

30

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 5 |
| P2 | 0 | 3 |
| P3 | 0 | 2 |
| P4 | 1 | 3 |
| P5 | 1 | 2 |
| P6 | 1 | 4 |
| P7 | 3 | 2 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 5 | | | | | 16 | 21 |
| P2 | 0 | 3 | | | | | 14 | 17 |
| P3 | 0 | 2 | | | | | 4 | 6 |
| P4 | 1 | 3 | | | | | 14 | 17 |
| P5 | 1 | 2 | | | | | 7 | 9 |
| P6 | 1 | 4 | | | | | 15 | 19 |
| P7 | 3 | 2 | | | | | 11 | 13 |
| | | | | | | | 11.57 | 14.57 |

**RR3**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 5 | | | | | 15 | 20 |
| P2 | 0 | 3 | | | | | 3 | 6 |
| P3 | 0 | 2 | | | | | 6 | 8 |
| P4 | 1 | 3 | | | | | 7 | 10 |
| P5 | 1 | 2 | | | | | 10 | 12 |
| P6 | 1 | 4 | | | | | 16 | 20 |
| P7 | 3 | 2 | | | | | 13 | 15 |
| | | | | | | | 10.00 | 13.00 |

**31**

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 4 |
| P3 | 0 | 2 |
| P4 | 0 | 3 |
| P5 | 2 | 2 |
| P6 | 2 | 3 |
| P7 | 2 | 4 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 18 | 24 |
| P2 | 0 | 4 | | | | | 14 | 18 |
| P3 | 0 | 2 | | | | | 4 | 6 |
| P4 | 0 | 3 | | | | | 16 | 19 |
| P5 | 2 | 2 | | | | | 6 | 8 |
| P6 | 2 | 3 | | | | | 15 | 18 |
| P7 | 2 | 4 | | | | | 16 | 20 |
| | | | | | | | 12.71 | 16.14 |

**RR3**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 16 | 22 |
| P2 | 0 | 4 | | | | | 19 | 23 |
| P3 | 0 | 2 | | | | | 6 | 8 |
| P4 | 0 | 3 | | | | | 8 | 11 |
| P5 | 2 | 2 | | | | | 9 | 11 |
| P6 | 2 | 3 | | | | | 11 | 14 |
| P7 | 2 | 4 | | | | | 18 | 22 |
| | | | | | | | 12.43 | 15.86 |

32

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 1 | 2 |
| P3 | 2 | 3 |
| P4 | 3 | 2 |
| P5 | 4 | 2 |
| P6 | 5 | 3 |
| P7 | 6 | 2 |
| P8 | 6 | 3 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 15 | 21 |
| P2 | 1 | 2 | | | | | 1 | 3 |
| P3 | 2 | 3 | | | | | 14 | 17 |
| P4 | 3 | 2 | | | | | 5 | 7 |
| P5 | 4 | 2 | | | | | 6 | 8 |
| P6 | 5 | 3 | | | | | 14 | 17 |
| P7 | 6 | 2 | | | | | 8 | 10 |
| P8 | 6 | 3 | | | | | 14 | 17 |
| | | | | | | | 9.62 | 12.50 |

**RR3**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 7 | 13 |
| P2 | 1 | 2 | | | | | 2 | 4 |
| P3 | 2 | 3 | | | | | 3 | 6 |
| P4 | 3 | 2 | | | | | 5 | 7 |
| P5 | 4 | 2 | | | | | 9 | 11 |
| P6 | 5 | 3 | | | | | 10 | 13 |
| P7 | 6 | 2 | | | | | 12 | 14 |
| P8 | 6 | 3 | | | | | 14 | 17 |
| | | | | | | | 7.75 | 10.62 |

33

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 1 |
| P3 | 0 | 5 |
| P4 | 1 | 2 |
| P5 | 2 | 2 |
| P6 | 3 | 3 |
| P7 | 4 | 2 |
| P8 | 5 | 2 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 15 | 21 |
| P2 | 0 | 1 | | | | | 2 | 3 |
| P3 | 0 | 5 | | | | | 18 | 23 |
| P4 | 1 | 2 | | | | | 4 | 6 |
| P5 | 2 | 2 | | | | | 5 | 7 |
| P6 | 3 | 3 | | | | | 16 | 19 |
| P7 | 4 | 2 | | | | | 9 | 11 |
| P8 | 5 | 2 | | | | | 10 | 12 |
| | | | | | | | 9.88 | 12.75 |

**RR3**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 6 | | | | | 11 | 17 |
| P2 | 0 | 1 | | | | | 3 | 4 |
| P3 | 0 | 5 | | | | | 18 | 23 |
| P4 | 1 | 2 | | | | | 6 | 8 |
| P5 | 2 | 2 | | | | | 7 | 9 |
| P6 | 3 | 3 | | | | | 8 | 11 |
| P7 | 4 | 2 | | | | | 13 | 15 |
| P8 | 5 | 2 | | | | | 14 | 16 |
| | | | | | | | 10.00 | 12.88 |

34

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 5 |
| P2 | 0 | 6 |
| P3 | 1 | 2 |
| P4 | 2 | 2 |
| P5 | 3 | 3 |
| P6 | 4 | 2 |
| P7 | 5 | 2 |
| P8 | 6 | 3 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | | | T. Att. | T. Com |
|---|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 5 | ■ | - - - | - - - | ■ | - - - - - | - - - - - | ■ | 16 | 21 |
| P2 | 0 | 6 | - - | ■ | - | - - - - - | - - - - | ■ | - - - - - | - - | ■ | 18 | 24 |
| P3 | 1 | 2 | | - - - | ■ | ■ | | | | | | 3 | 5 |
| P4 | 2 | 2 | | | - - - | - | ■ | | | | | 4 | 6 |
| P5 | 3 | 3 | | | | - - | - - - - - | ■ | - - - | - - - - - | - | ■ | 16 | 19 |
| P6 | 4 | 2 | | | | - | - - - - - | - - | ■ | | | 8 | 10 |
| P7 | 5 | 2 | | | | | - - - - - | - - - - - | - | ■ | | 11 | 13 |
| P8 | 6 | 3 | | | | | - - - - | - - - - - | - - - | ■ | - - - - | ■ | 16 | 19 |
| | | | | | | | | | | | | | 11.50 | 14.62 |

**RR3**

| P | A | D | | | | | | T. Att. | T. Com |
|---|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 5 | ■ | - - | - - - - - | - - - | ■ | | | 10 | 15 |
| P2 | 0 | 6 | - - - | ■ | ■ | - - - - - | - - - - - | - - - - - | - - | ■ | 19 | 25 |
| P3 | 1 | 2 | | - - - - | - | ■ | | | | | | 5 | 7 |
| P4 | 2 | 2 | | | - - - | - - - | ■ | | | | | 6 | 8 |
| P5 | 3 | 3 | | | | - - | - - - - - | ■ | | | | 7 | 10 |
| P6 | 4 | 2 | | | | - | - - - - - | - - - - - | ■ | | | 11 | 13 |
| P7 | 5 | 2 | | | | | - - - - - | - - - - - | - - | ■ | | 12 | 14 |
| P8 | 6 | 3 | | | | | - - - - | - - - - - | - - - - | ■ | ■ | 13 | 16 |
| | | | | | | | | | | | | 10.38 | 13.50 |

35

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 6 |
| P2 | 0 | 1 |
| P3 | 1 | 1 |
| P4 | 1 | 2 |
| P5 | 2 | 2 |
| P6 | 2 | 3 |
| P7 | 3 | 1 |
| P8 | 4 | 2 |
| P9 | 4 | 3 |
| P10 | 4 | 2 |

**SOLUZIONE**

**RR2**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 16 | 22 |
| P2 | 0 | 1 | 2 | 3 |
| P3 | 1 | 1 | 2 | 3 |
| P4 | 1 | 2 | 3 | 5 |
| P5 | 2 | 2 | 4 | 6 |
| P6 | 2 | 3 | 15 | 18 |
| P7 | 3 | 1 | 9 | 10 |
| P8 | 4 | 2 | 9 | 11 |
| P9 | 4 | 3 | 16 | 19 |
| P10 | 4 | 2 | 13 | 15 |
| | | | 8.90 | 11.20 |

**RR3**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 6 | 10 | 16 |
| P2 | 0 | 1 | 3 | 4 |
| P3 | 1 | 1 | 3 | 4 |
| P4 | 1 | 2 | 4 | 6 |
| P5 | 2 | 2 | 5 | 7 |
| P6 | 2 | 3 | 7 | 10 |
| P7 | 3 | 1 | 9 | 10 |
| P8 | 4 | 2 | 12 | 14 |
| P9 | 4 | 3 | 14 | 17 |
| P10 | 4 | 2 | 17 | 19 |
| | | | 8.40 | 10.70 |

36

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 2 |
| P2 | 0 | 5 |
| P3 | 0 | 1 |
| P4 | 1 | 2 |
| P5 | 1 | 3 |
| P6 | 1 | 2 |
| P7 | 2 | 1 |
| P8 | 3 | 2 |
| P9 | 3 | 3 |

**SOLUZIONE**

**RR2**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 2 | 0 | 2 |
| P2 | 0 | 5 | 16 | 21 |
| P3 | 0 | 1 | 4 | 5 |
| P4 | 1 | 2 | 4 | 6 |
| P5 | 1 | 3 | 15 | 18 |
| P6 | 1 | 2 | 8 | 10 |
| P7 | 2 | 1 | 9 | 10 |
| P8 | 3 | 2 | 9 | 11 |
| P9 | 3 | 3 | 14 | 17 |
| | | | 8.78 | 11.11 |

**RR3**

| P | A | D | T. Att. | T. Compl. |
|---|---|---|---|---|
| P1 | 0 | 2 | 0 | 2 |
| P2 | 0 | 5 | 16 | 21 |
| P3 | 0 | 1 | 5 | 6 |
| P4 | 1 | 2 | 5 | 7 |
| P5 | 1 | 3 | 7 | 10 |
| P6 | 1 | 2 | 10 | 12 |
| P7 | 2 | 1 | 11 | 12 |
| P8 | 3 | 2 | 11 | 13 |
| P9 | 3 | 3 | 13 | 16 |
| | | | 8.67 | 11.00 |

37

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 2 |
| P2 | 0 | 6 |
| P3 | 1 | 2 |
| P4 | 2 | 2 |
| P5 | 3 | 3 |
| P6 | 4 | 2 |
| P7 | 5 | 2 |
| P8 | 6 | 3 |
| P9 | 7 | 2 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 2 | | | | | 0 | 2 |
| P2 | 0 | 6 | | | | | 17 | 23 |
| P3 | 1 | 2 | | | | | 3 | 5 |
| P4 | 2 | 2 | | | | | 4 | 6 |
| P5 | 3 | 3 | | | | | 15 | 18 |
| P6 | 4 | 2 | | | | | 6 | 8 |
| P7 | 5 | 2 | | | | | 9 | 11 |
| P8 | 6 | 3 | | | | | 15 | 18 |
| P9 | 7 | 2 | | | | | 11 | 13 |
| | | | | | | | 8.89 | 11.56 |

**RR3**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 2 | | | | | 0 | 2 |
| P2 | 0 | 6 | | | | | 13 | 19 |
| P3 | 1 | 2 | | | | | 4 | 6 |
| P4 | 2 | 2 | | | | | 5 | 7 |
| P5 | 3 | 3 | | | | | 6 | 9 |
| P6 | 4 | 2 | | | | | 8 | 10 |
| P7 | 5 | 2 | | | | | 9 | 11 |
| P8 | 6 | 3 | | | | | 13 | 16 |
| P9 | 7 | 2 | | | | | 15 | 17 |
| | | | | | | | 8.11 | 10.78 |

38

| Processo | Arrivo | Burst |
|---|---|---|
| P1 | 0 | 3 |
| P2 | 0 | 1 |
| P3 | 0 | 2 |
| P4 | 1 | 2 |
| P5 | 2 | 2 |
| P6 | 2 | 3 |
| P7 | 2 | 4 |
| P8 | 3 | 1 |
| P9 | 4 | 2 |
| P10 | 4 | 3 |

**SOLUZIONE**

**RR2**

| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 3 | | | | | 11 | 14 |
| P2 | 0 | 1 | | | | | 2 | 3 |
| P3 | 0 | 2 | | | | | 3 | 5 |
| P4 | 1 | 2 | | | | | 4 | 6 |
| P5 | 2 | 2 | | | | | 5 | 7 |
| P6 | 2 | 3 | | | | | 15 | 18 |
| P7 | 2 | 4 | | | | | 16 | 20 |
| P8 | 3 | 1 | | | | | 11 | 12 |
| P9 | 4 | 2 | | | | | 11 | 13 |
| P10 | 4 | 3 | | | | | 16 | 19 |
| | | | | | | | 9.40 | 11.70 |

**RR3**

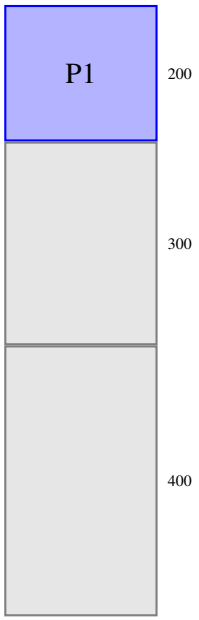
| P | A | D | | | | | T. Att. | T. Compl. |
|---|---|---|---|---|---|---|---|---|
| P1 | 0 | 3 | | | | | 0 | 3 |
| P2 | 0 | 1 | | | | | 3 | 4 |
| P3 | 0 | 2 | | | | | 4 | 6 |
| P4 | 1 | 2 | | | | | 5 | 7 |
| P5 | 2 | 2 | | | | | 6 | 8 |
| P6 | 2 | 3 | | | | | 8 | 11 |
| P7 | 2 | 4 | | | | | 17 | 21 |
| P8 | 3 | 1 | | | | | 13 | 14 |
| P9 | 4 | 2 | | | | | 13 | 15 |
| P10 | 4 | 3 | | | | | 15 | 18 |
| | | | | | | | 8.40 | 10.70 |

## 6.1 Memoria a partizione fisse

Mostra l'evoluzione della memoria dati i seguenti processi e la seguente memoria a partizioni fisse usando l'algoritmo FirstFit, BestFit e WorstFit

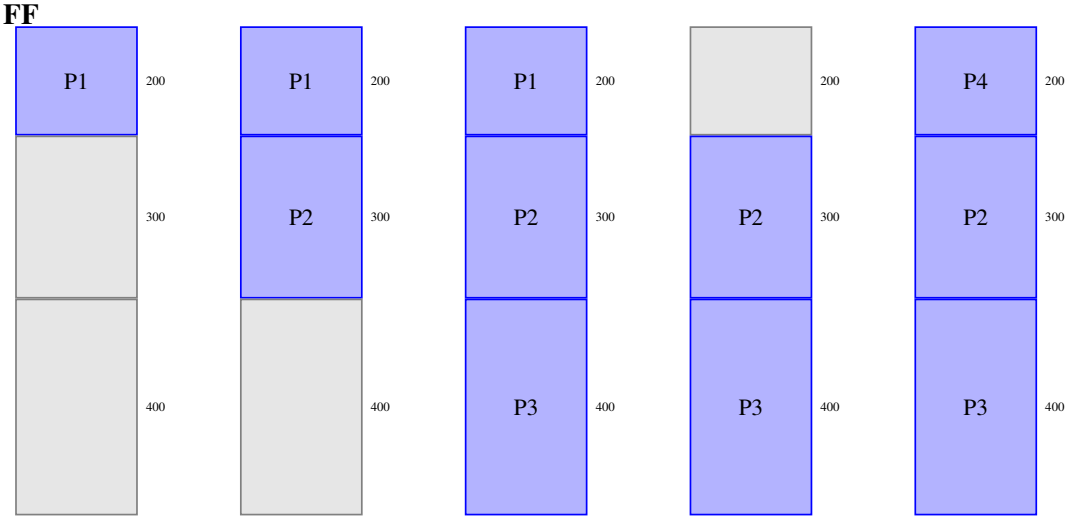
1

- P2 di dimensione 150Kb
- P3 di dimensione 250Kb
- P1 esce
- P4 di dimensione 180Kb

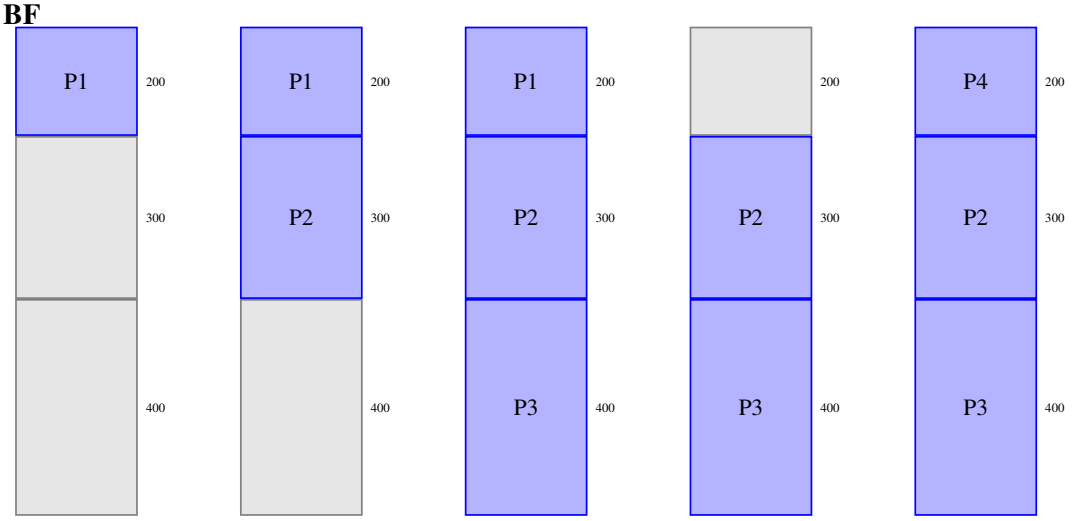



**SOLUZIONE**

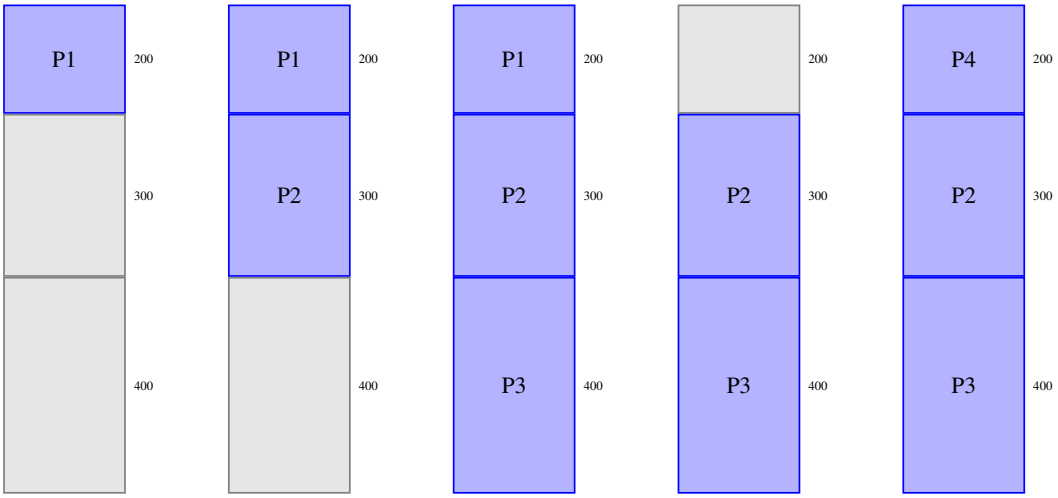
**FF**




**BF**

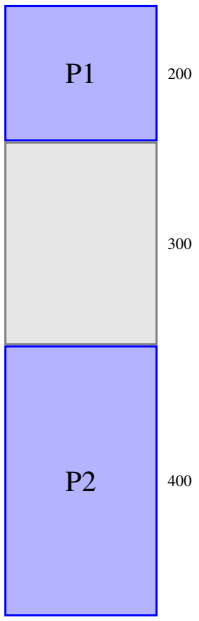



**WF**


P1
200
300
400
P1
200
P2
300
400
P1
200
P2
300
P3
400
200
P2
300
P3
400
P4
200
P2
300
P3
400

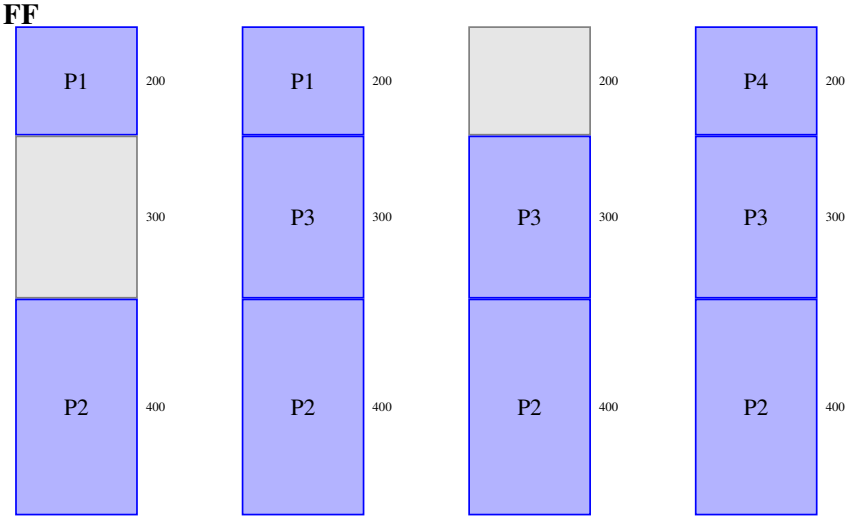
2

- P3 di dimensione 180Kb
- P1 esce
- P4 di dimensione 190Kb

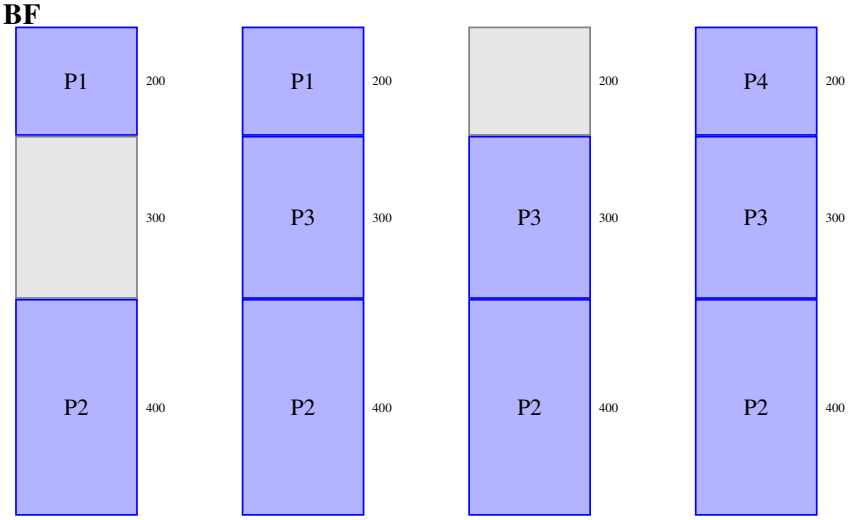



**SOLUZIONE**

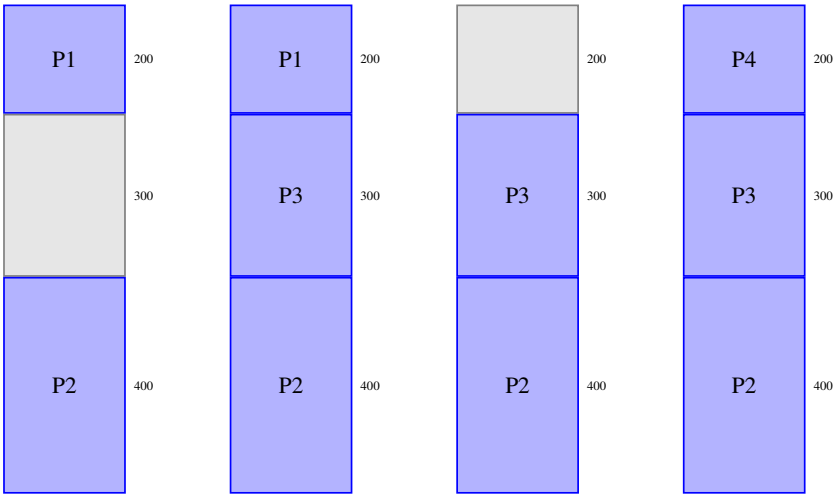
**FF**




**BF**

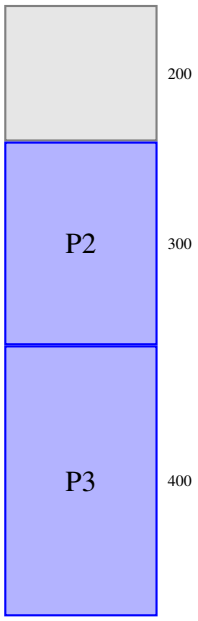



**WF**


P1
200
300
P2
400
P1
200
P3
300
P2
400
200
P3
300
P2
400
P4
200
P3
300
P2
400

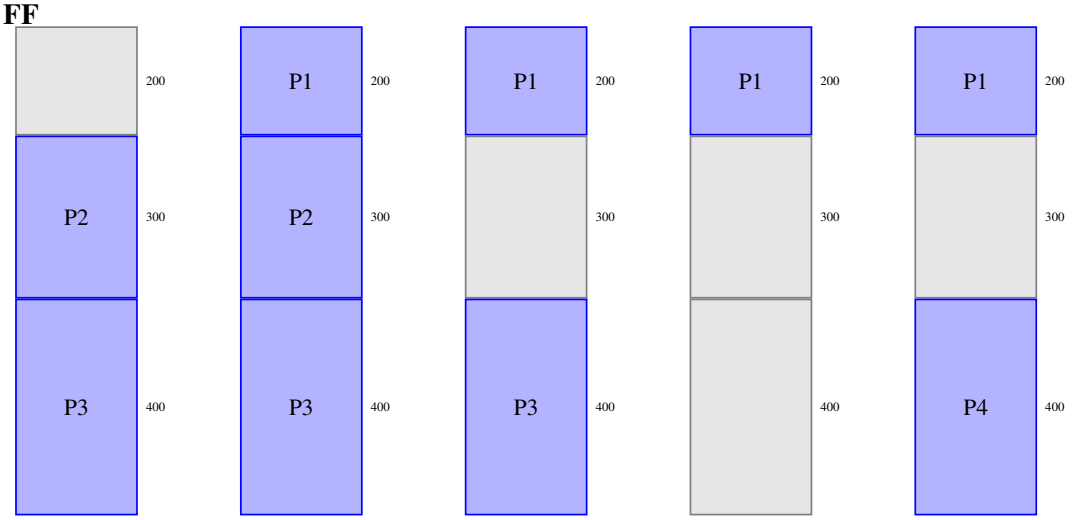
3

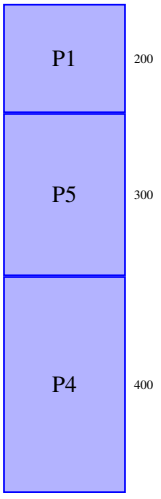
- P1 di dimensione 150Kb
- P2 esce
- P3 esce
- P4 di dimensione 250Kb
- P5 di dimensione 180Kb



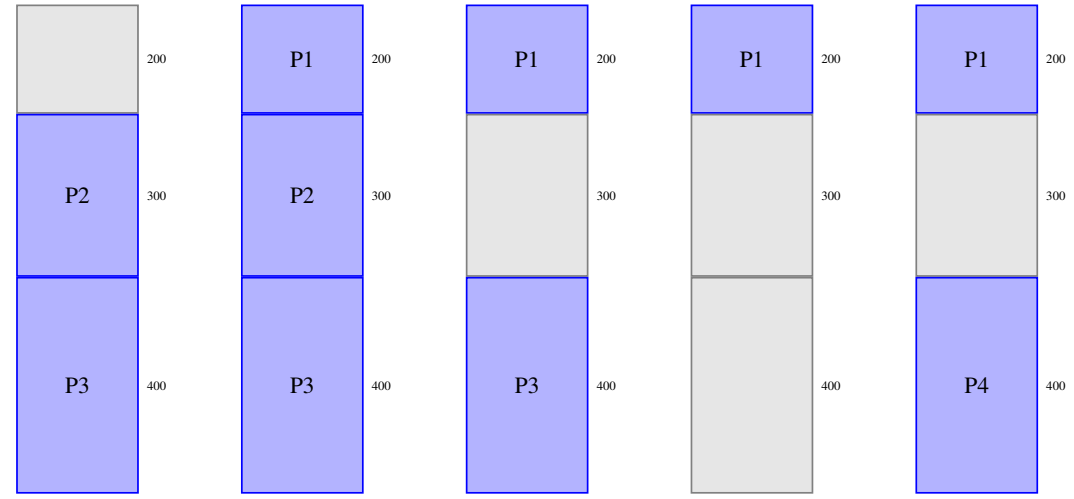

**SOLUZIONE**

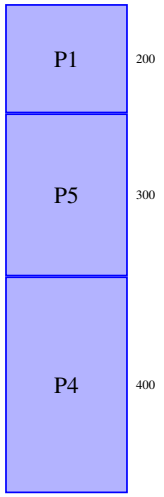
**FF**



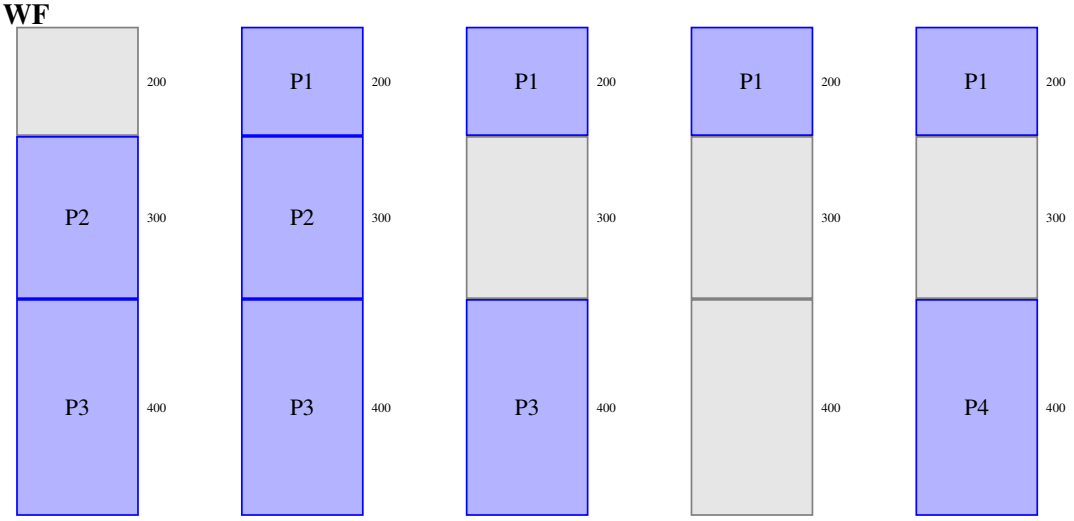



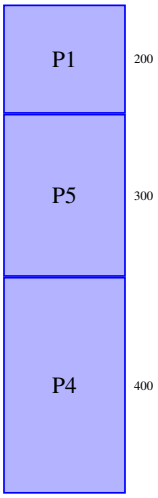

**BF**


200
P2
300
P3
400
P1
200
P2
300
P3
400
P1
200
300
P3
400
P1
200
300
400
P1
200
300
P4
400


P1
200
P5
300
P4
400

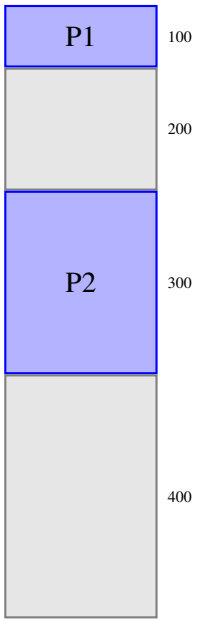
**WF**


200
P2
300
P3
400
P1
200
P2
300
P3
400
P1
200
300
P3
400
P1
200
300
400
P1
200
300
P4
400


P1
200
P5
300
P4
400

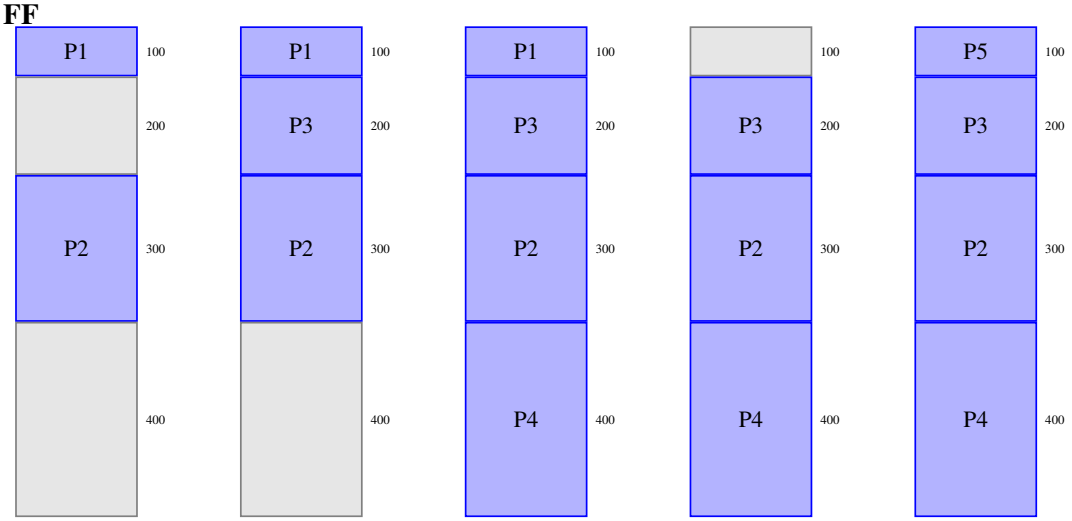
4

- P3 di dimensione 150Kb
- P4 di dimensione 180Kb
- P1 esce
- P5 di dimensione 90Kb

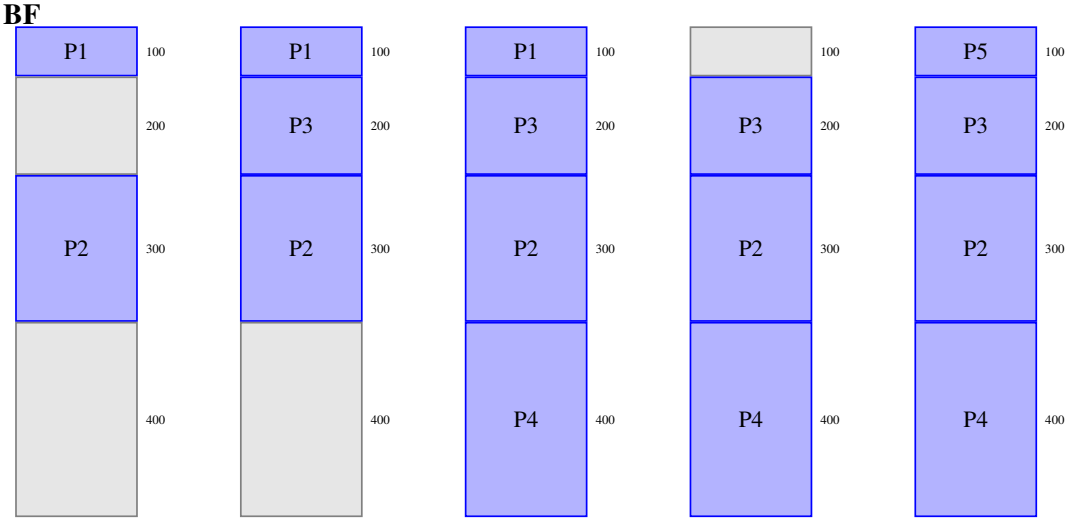



**SOLUZIONE**

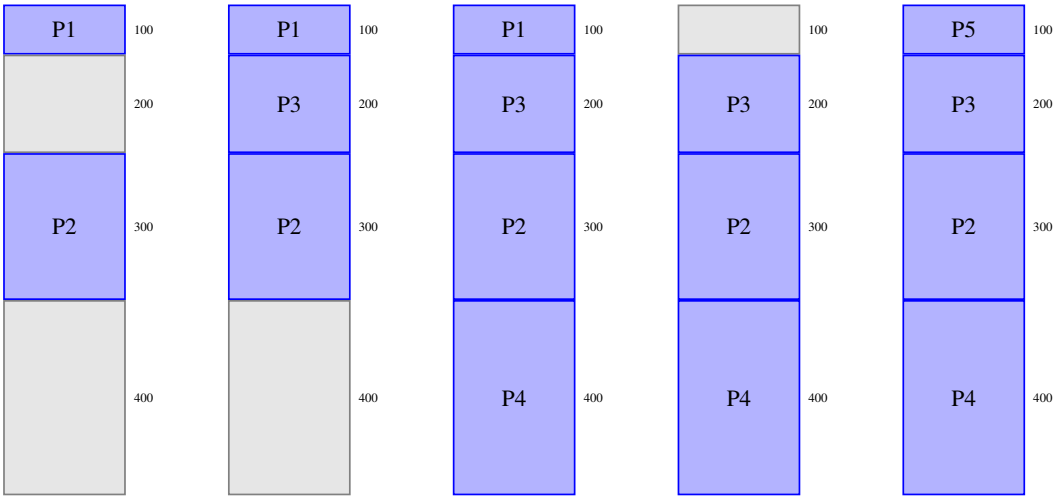
**FF**




**BF**

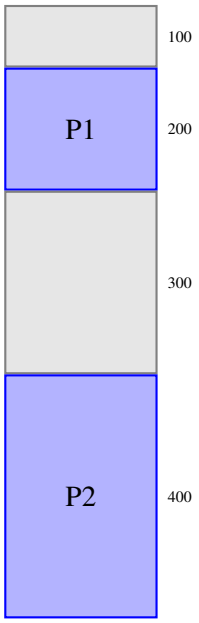



**WF**


P1
100
200
P2
300
400
P1
100
P3
200
P2
300
400
P1
100
P3
200
P2
300
P4
400
100
P3
200
P2
300
P4
400
P5
100
P3
200
P2
300
P4
400

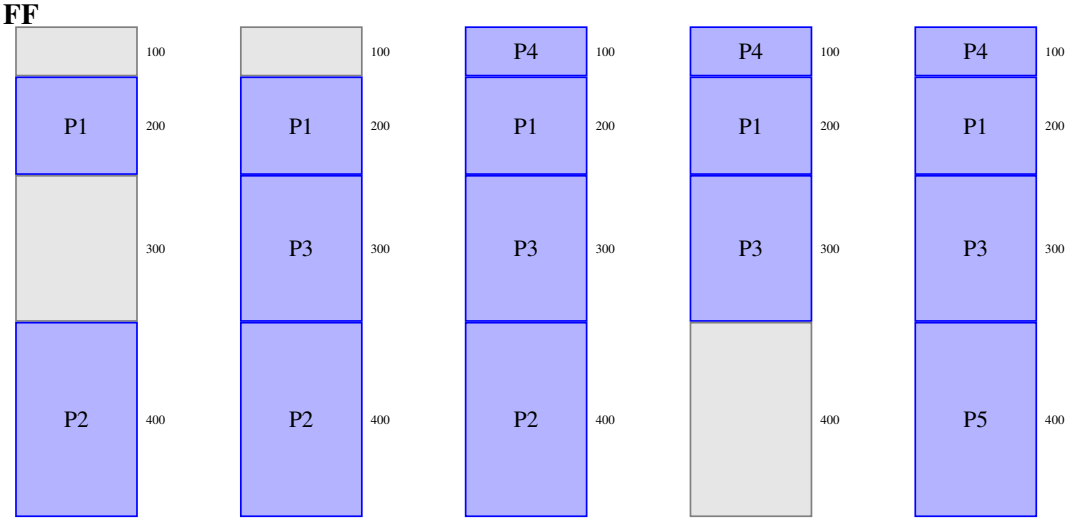
5

- P3 di dimensione 180Kb
- P4 di dimensione 90Kb
- P2 esce
- P5 di dimensione 190Kb

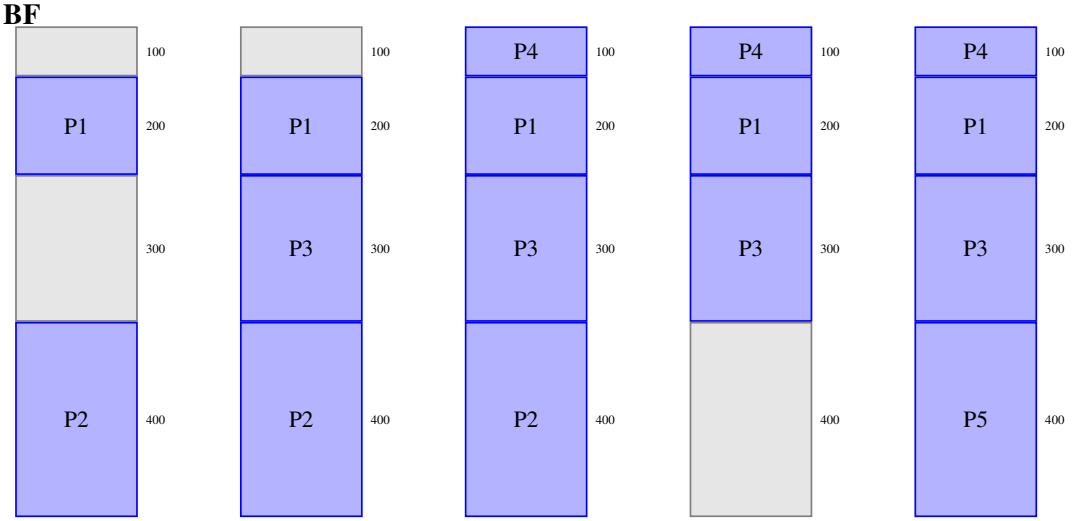



**SOLUZIONE**

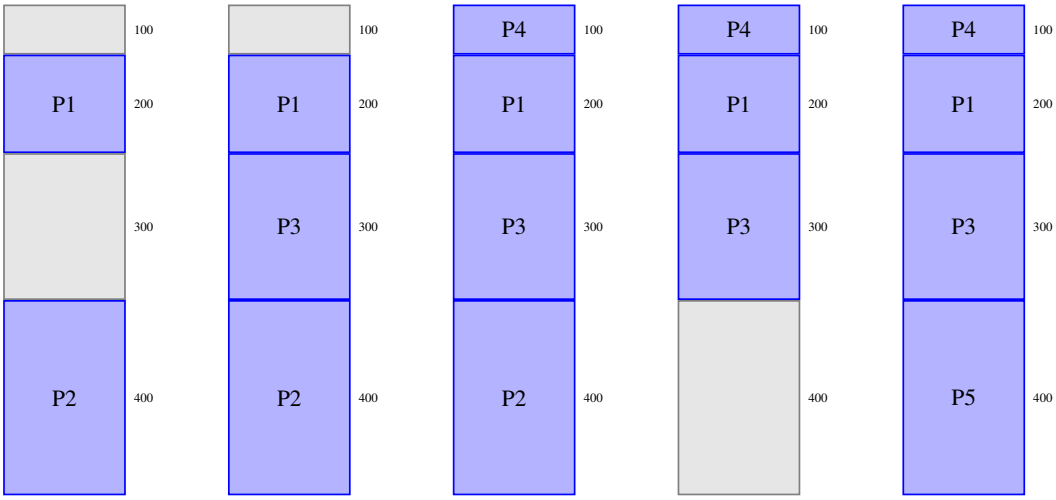
**FF**




**BF**

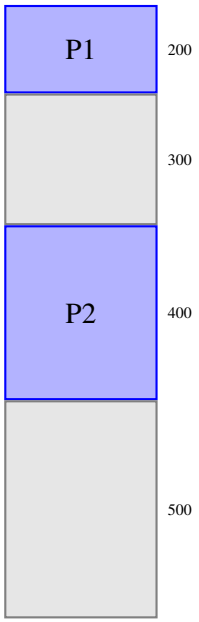



**WF**


100
P1
200
300
P2
400
100
P1
200
P3
300
P2
400
P4
100
P1
200
P3
300
P2
400
P4
100
P1
200
P3
300
400
P4
100
P1
200
P3
300
P5
400

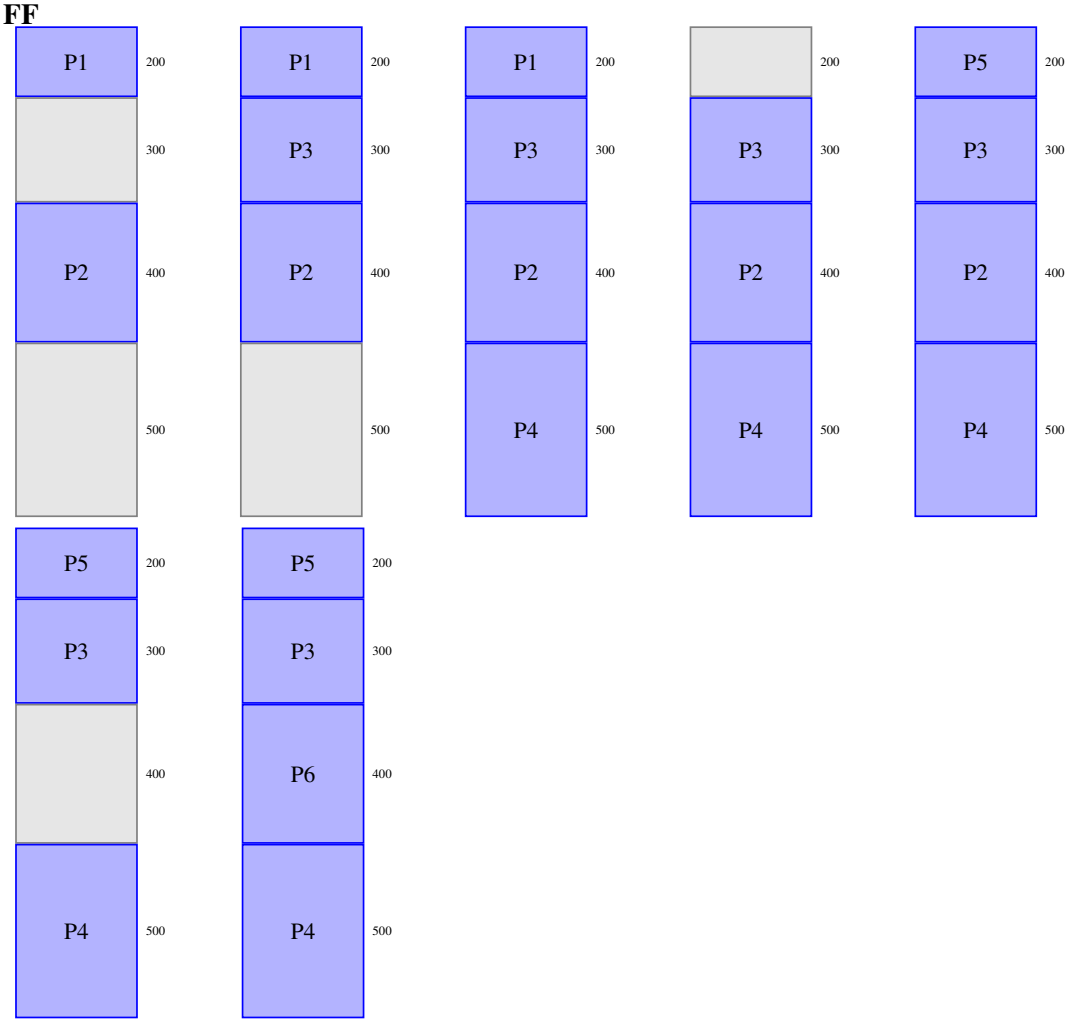
6

- •P3 di dimensione 250Kb
- •P4 di dimensione 280Kb
- •P1 esce
- •P5 di dimensione 190Kb
- •P2 esce
- •P6 di dimensione 380Kb

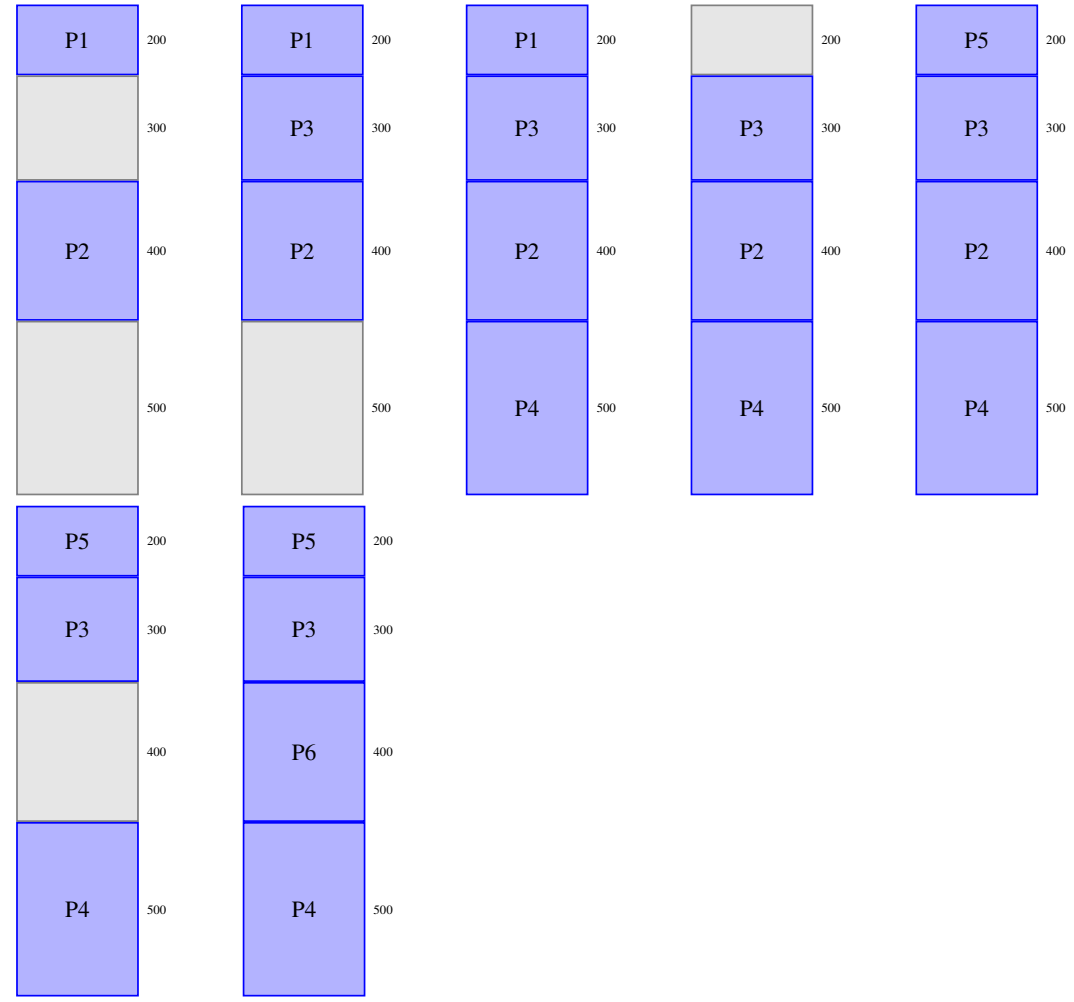



**SOLUZIONE**

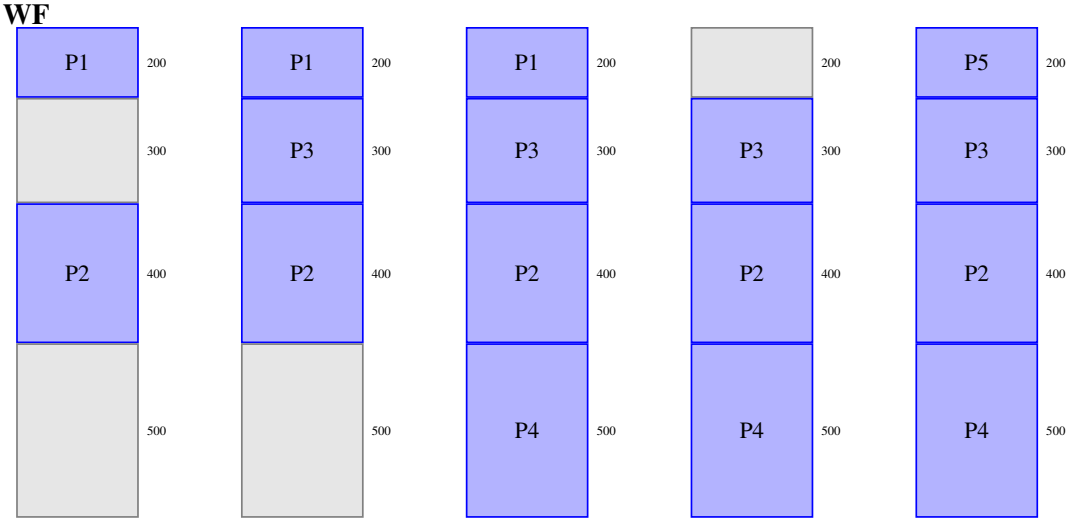
**FF**

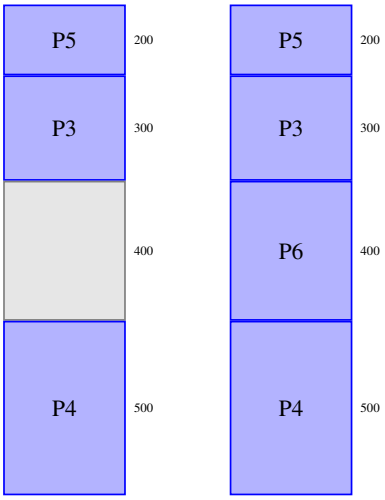



**BF**


P1
200
300
P2
400
500
P1
200
P3
300
P2
400
500
P1
200
P3
300
P2
400
P4
500
200
P3
300
P2
400
P4
500
P5
200
P3
300
P2
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P3
300
P6
400
P4
500

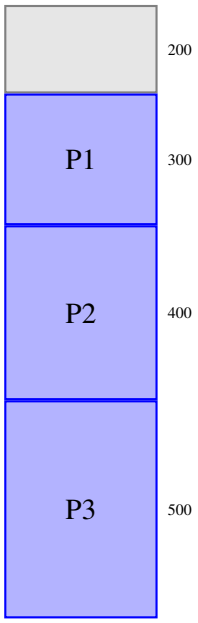
**WF**


P1
200
300
P2
400
500
P1
200
P3
300
P2
400
500
P1
200
P3
300
P2
400
P4
500
200
P3
300
P2
400
P4
500
P5
200
P3
300
P2
400
P4
500


P5
200
P3
300
400
P4
500
P5
200
P3
300
P6
400
P4
500

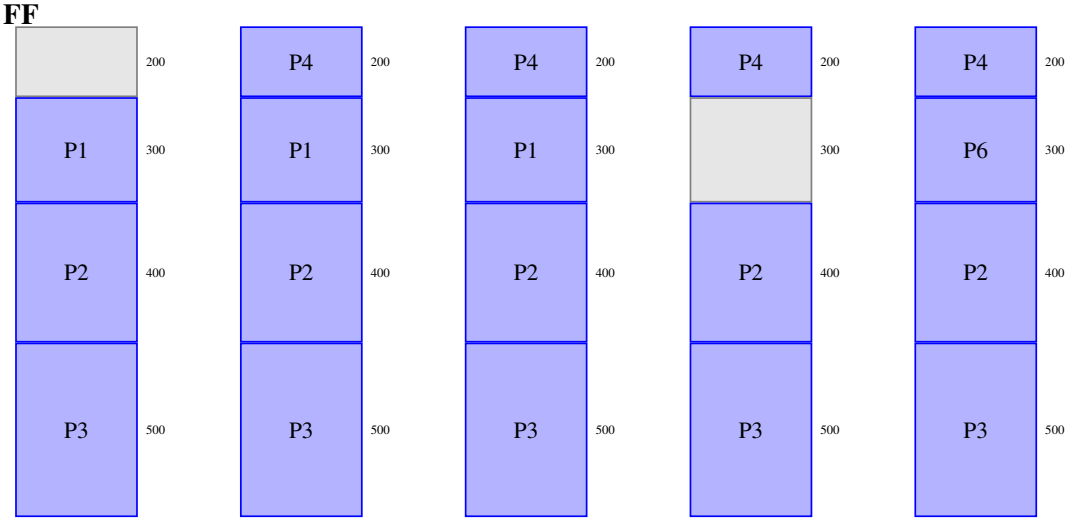
7

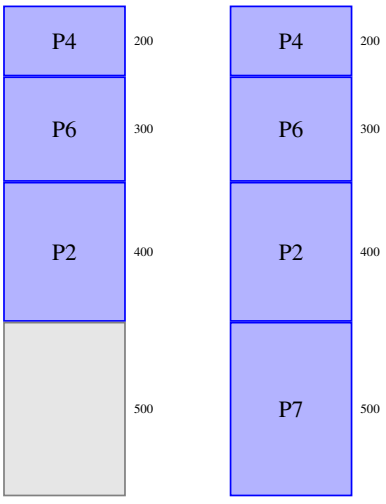
- •P4 di dimensione 150Kb
- •P5 di dimensione 250Kb
- •P1 esce
- •P6 di dimensione 190Kb
- •P3 esce
- •P7 di dimensione 350Kb



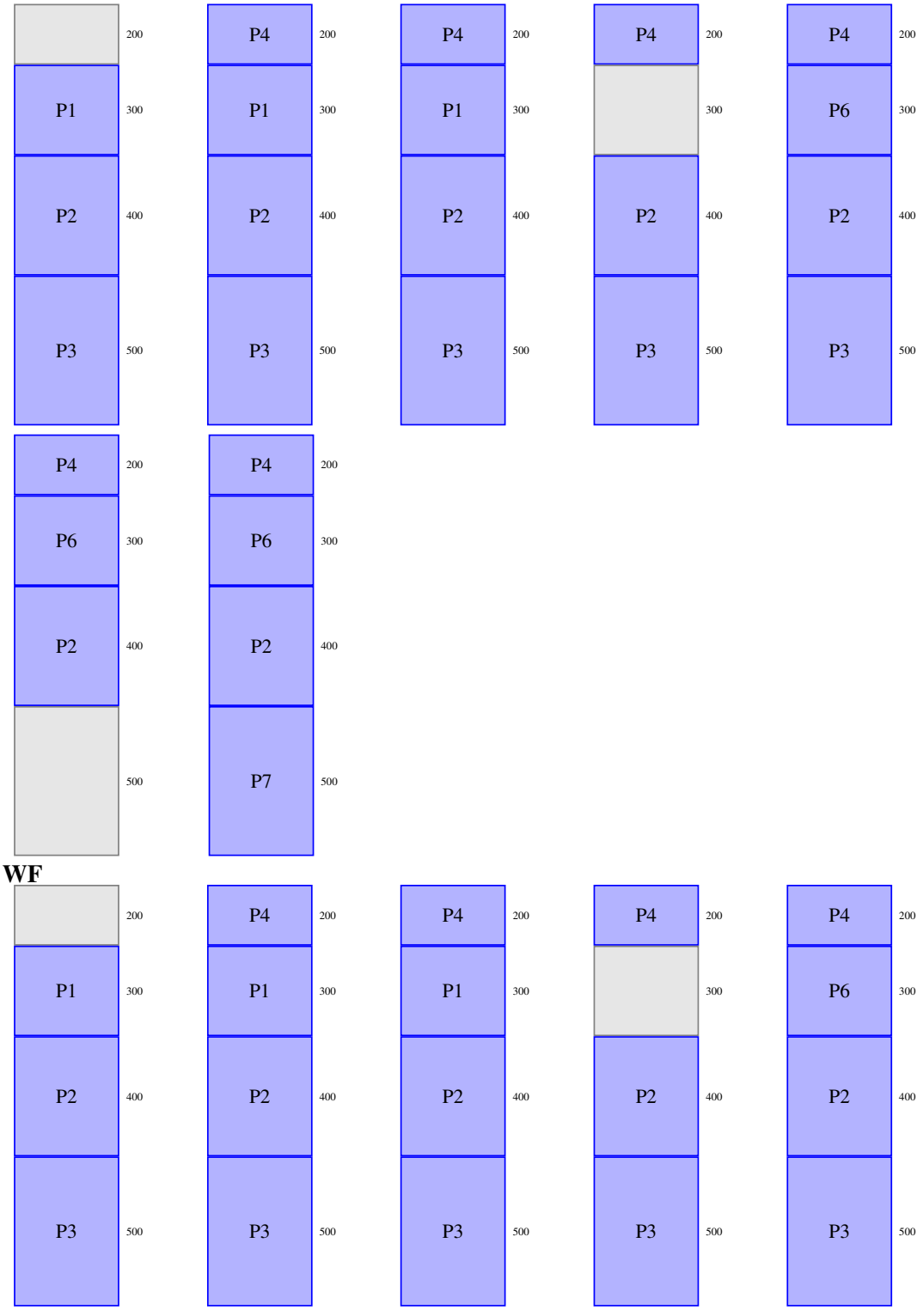

**SOLUZIONE**

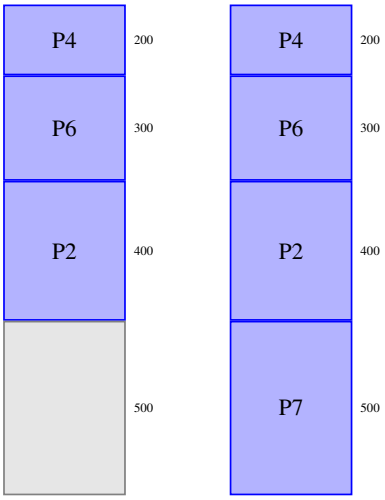
**FF**



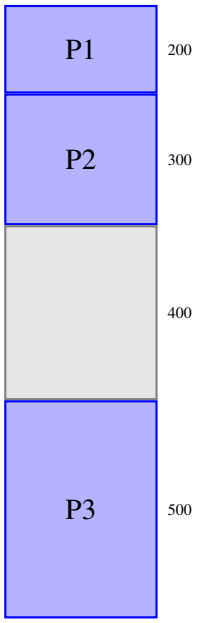

**BF**


200
P1
300
P2
400
P3
500
P4
P6
P7
WF


P4
200
P6
300
P2
400
500
P4
200
P6
300
P2
400
P7
500

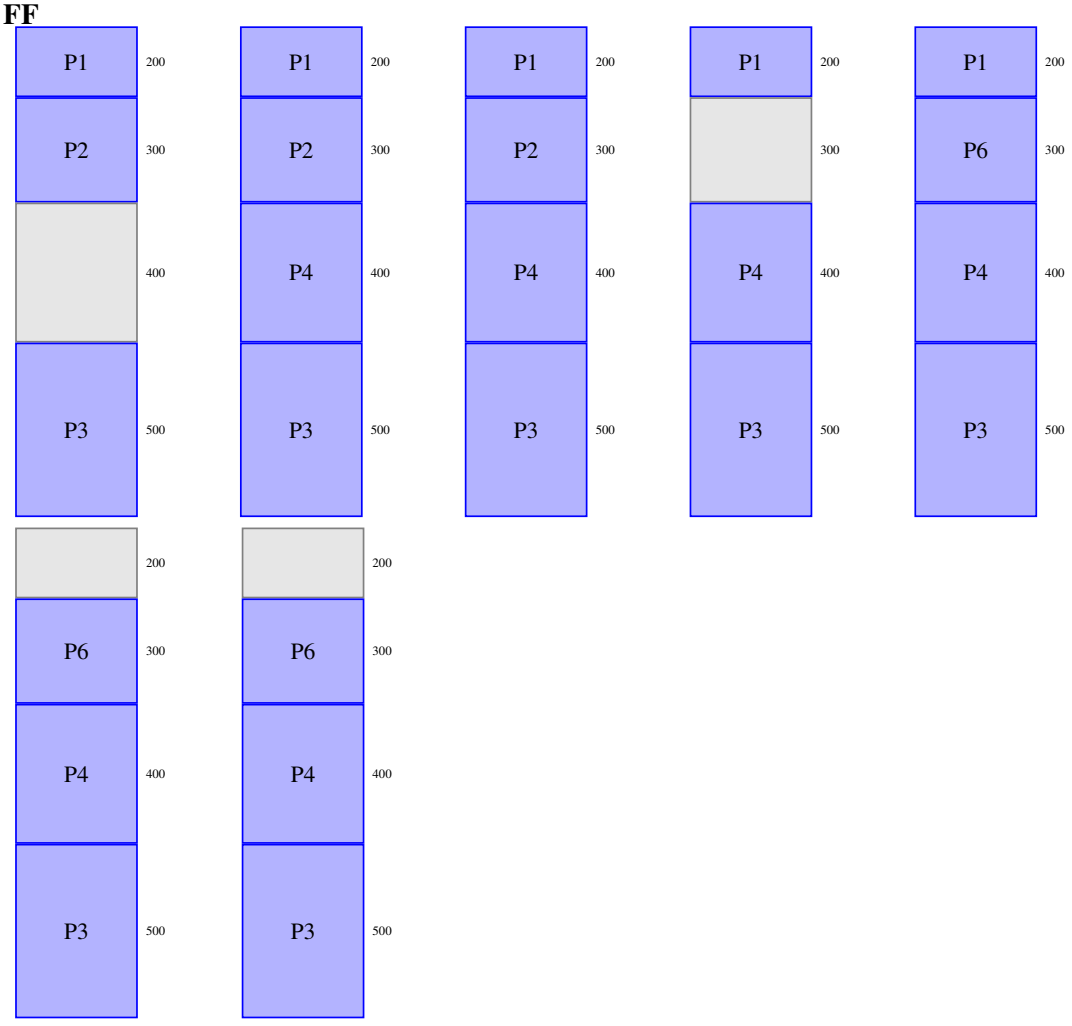
8

- P4 di dimensione 200Kb
- P5 di dimensione 300Kb
- P2 esce
- P6 di dimensione 250Kb
- P1 esce
- P7 di dimensione 450Kb

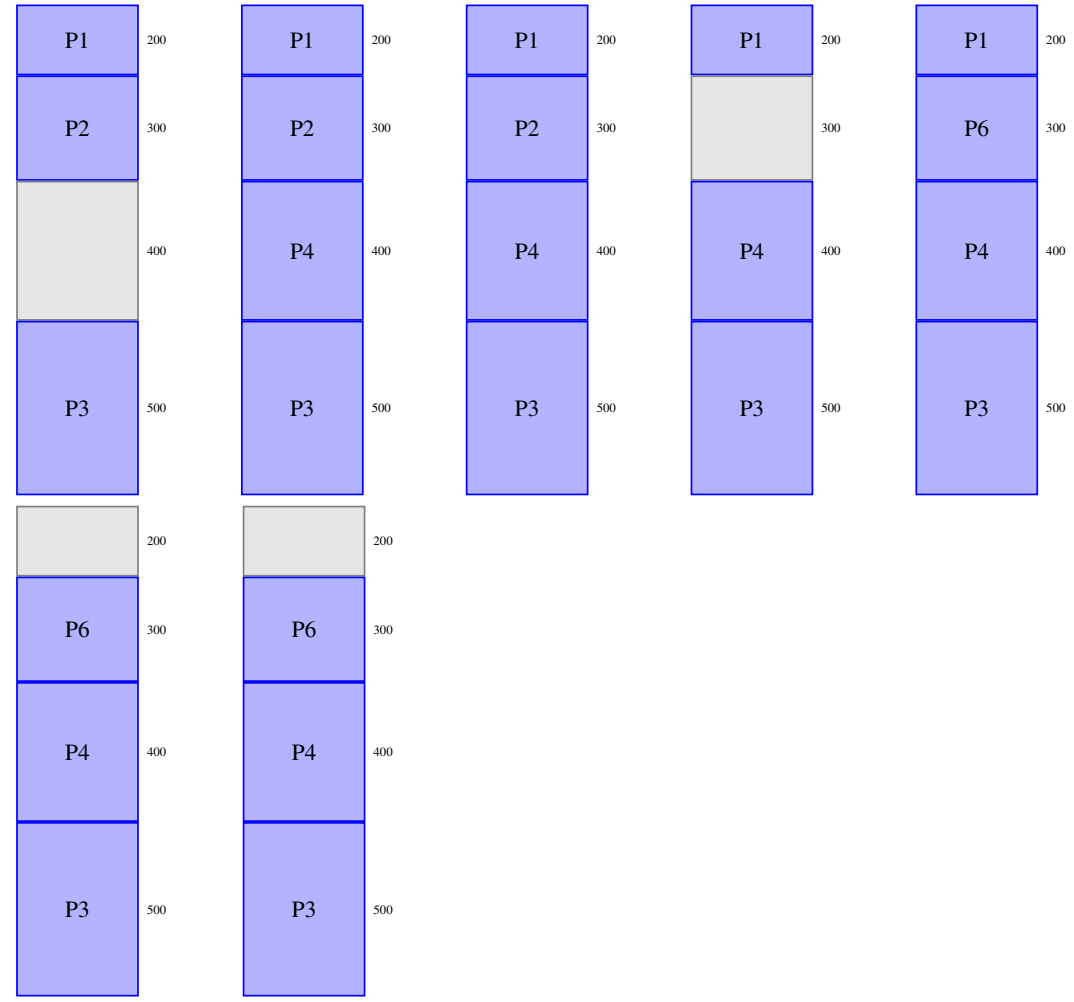



**SOLUZIONE**

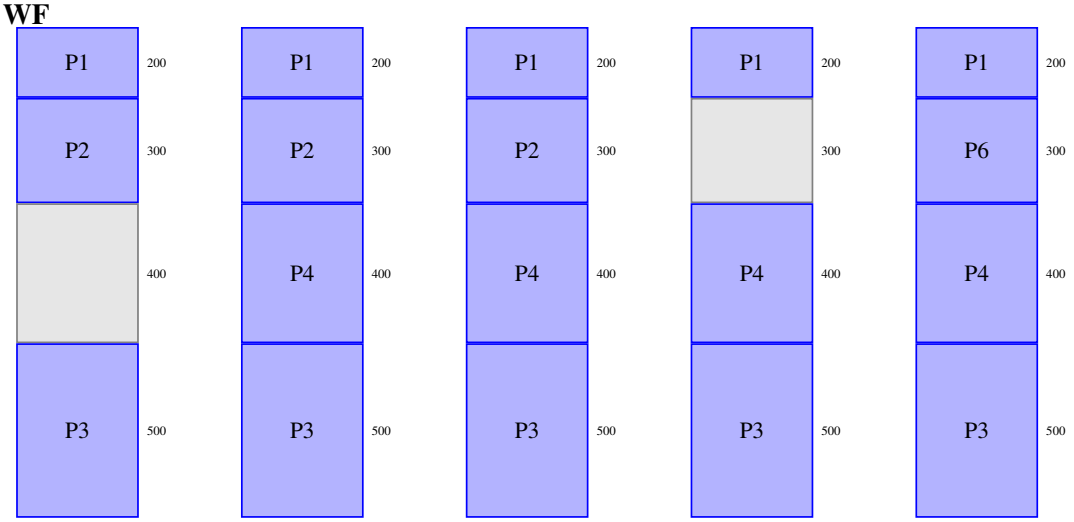
**FF**

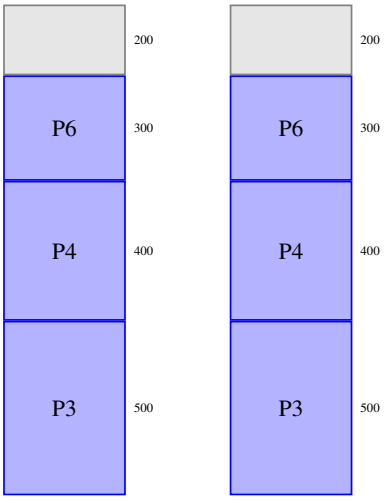



**BF**


P1
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P3
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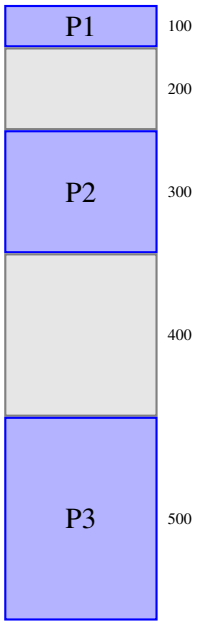
**WF**


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P3
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P6
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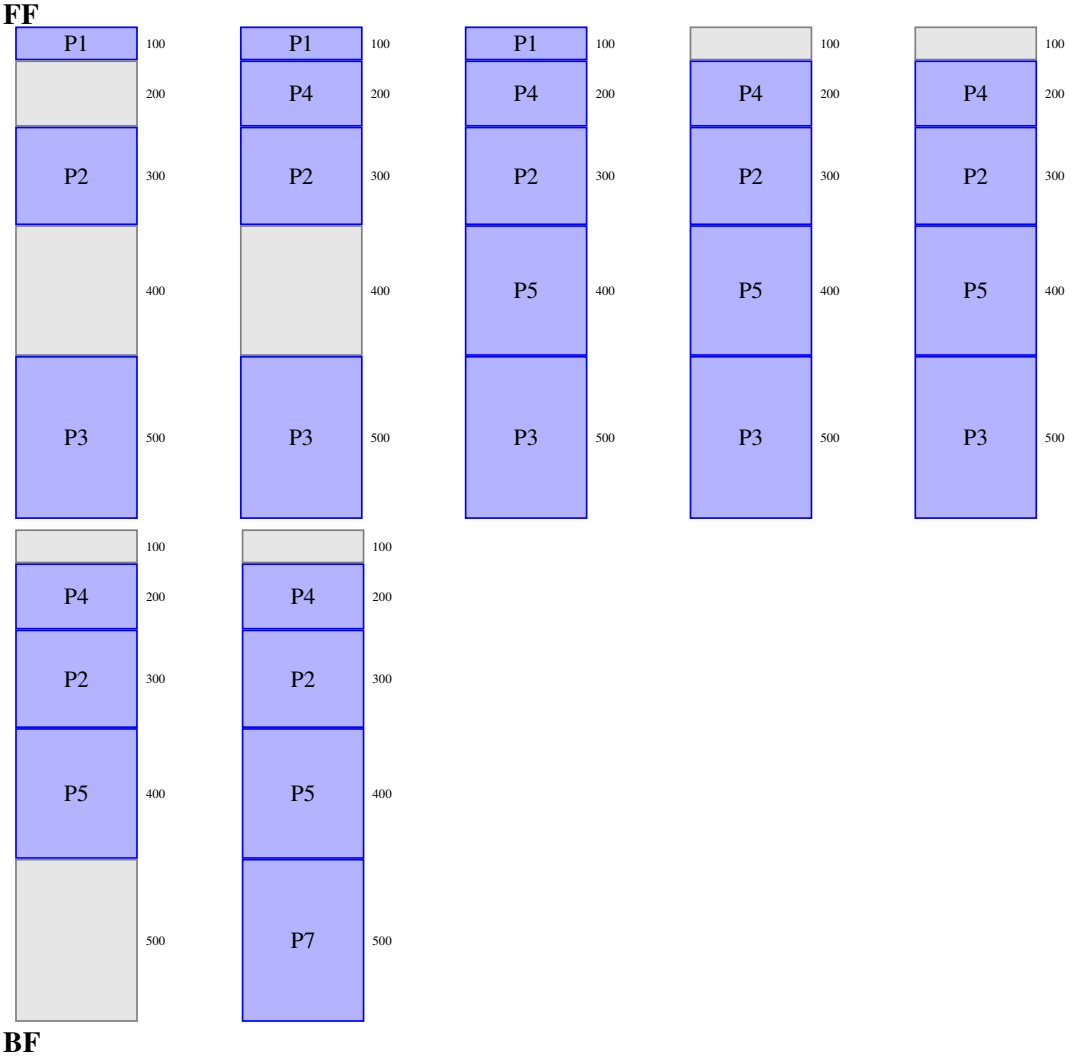
9

- •P4 di dimensione 150Kb
- •P5 di dimensione 250Kb
- •P1 esce
- •P6 di dimensione 180Kb
- •P3 esce
- •P7 di dimensione 350Kb

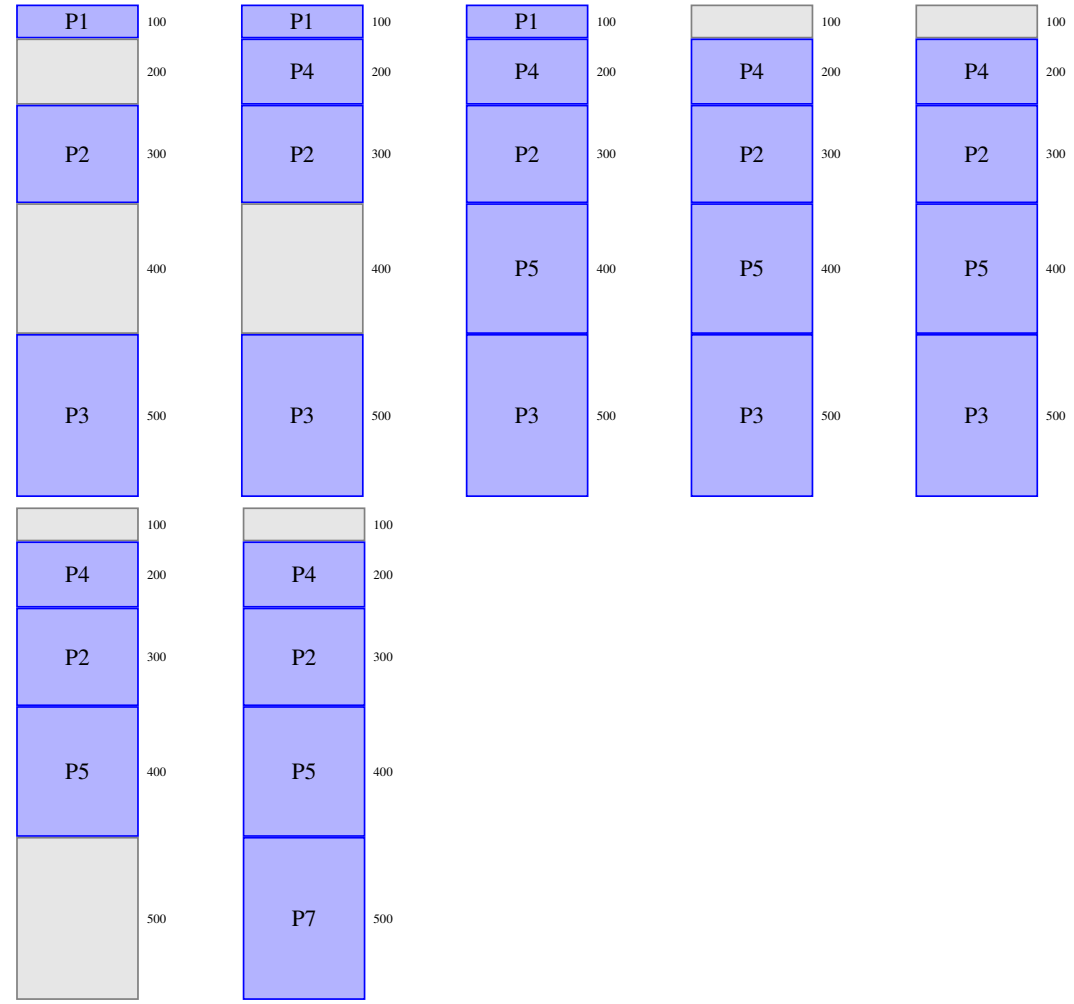



**SOLUZIONE**

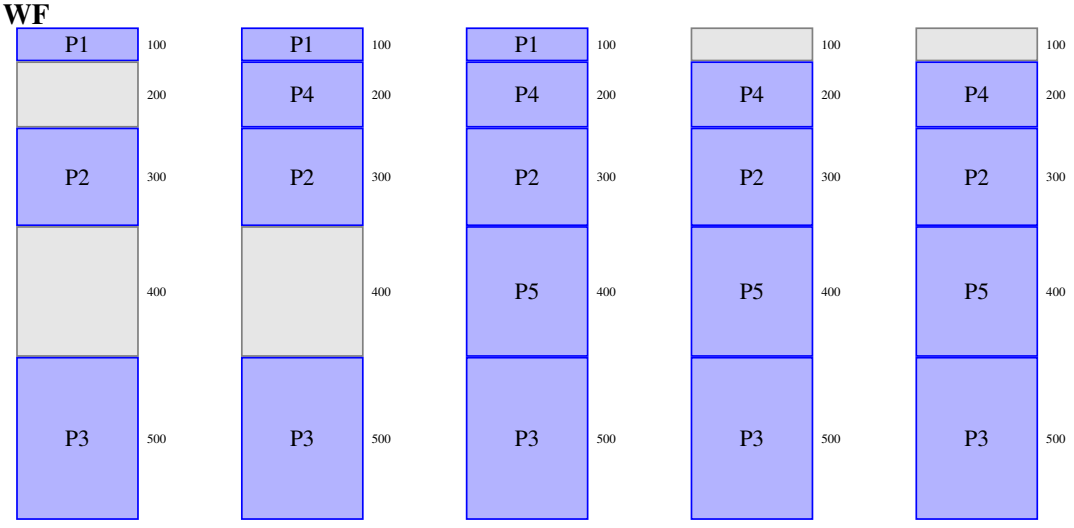
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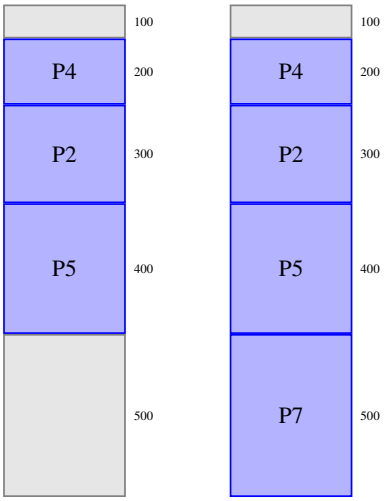



**BF**


P1
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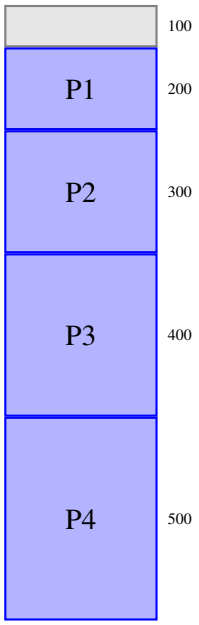
**WF**


P1
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P5
P3
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500


100
P4
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P2
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P5
400
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100
P4
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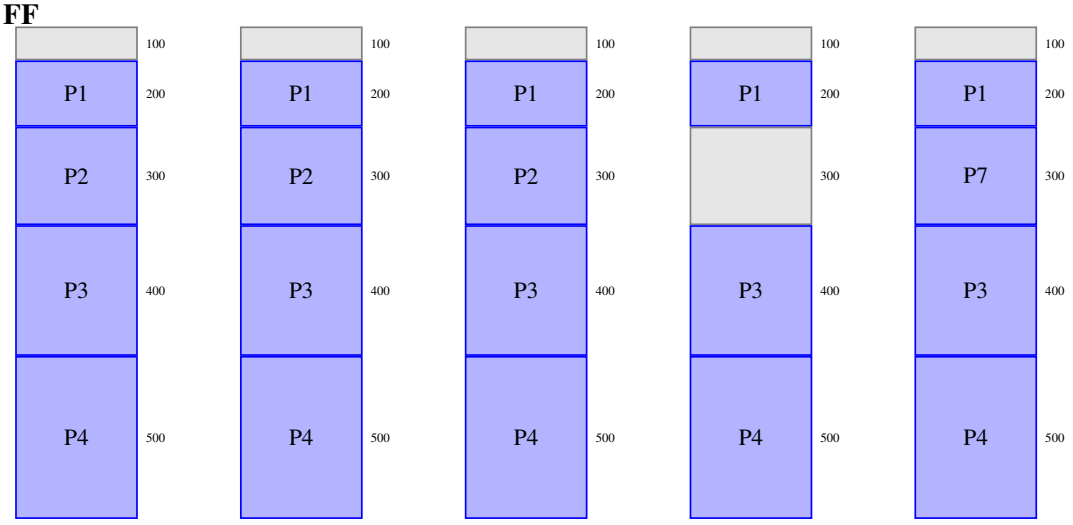
**10**

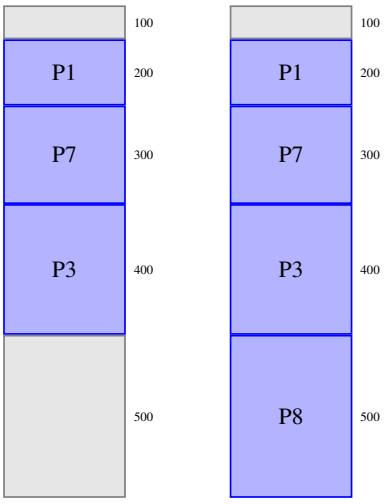
- P5 di dimensione 180Kb
- P6 di dimensione 250Kb
- P2 esce
- P7 di dimensione 190Kb
- P4 esce
- P8 di dimensione 380Kb



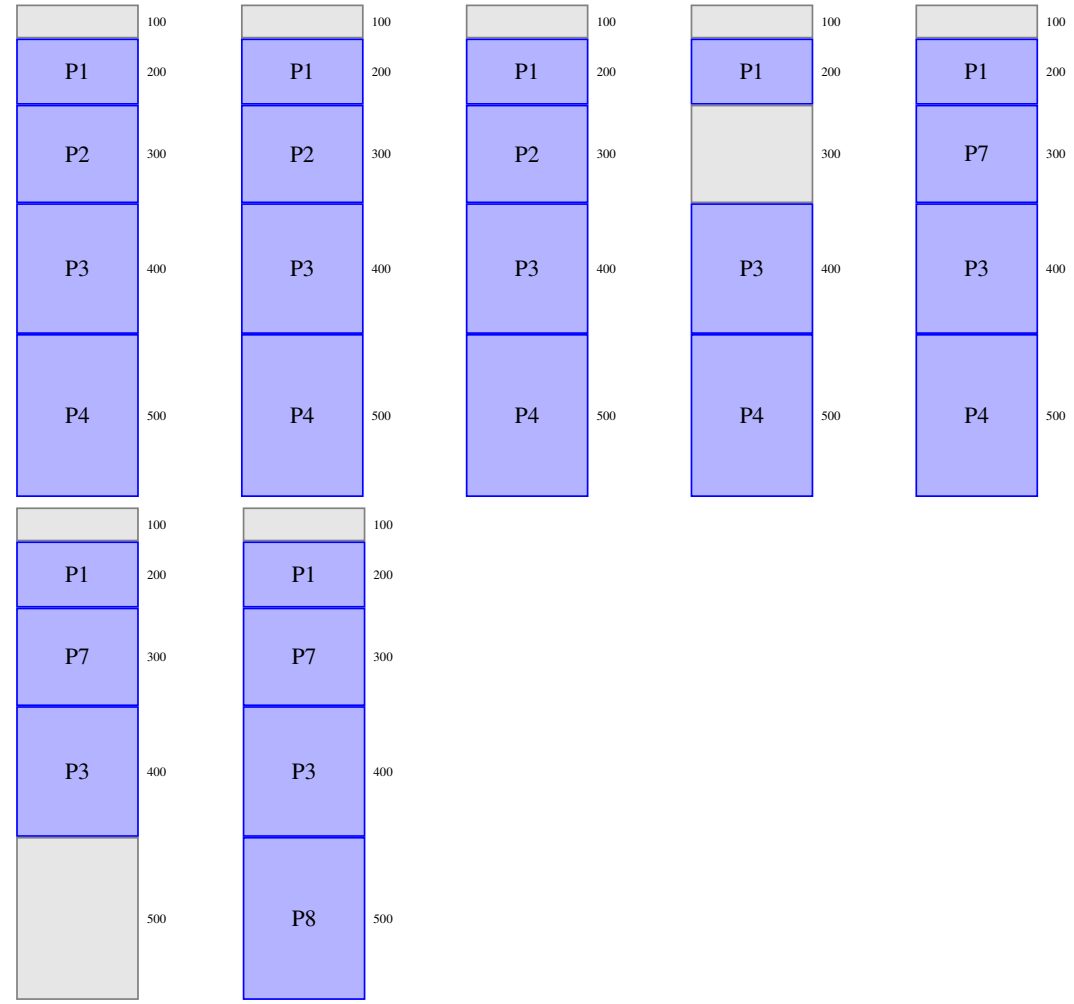

**SOLUZIONE**

**FF**

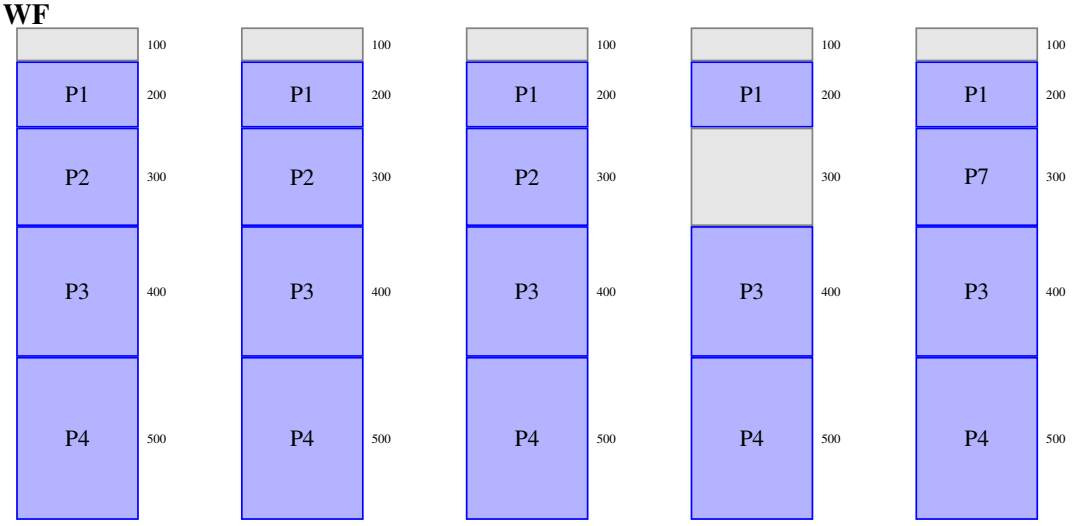


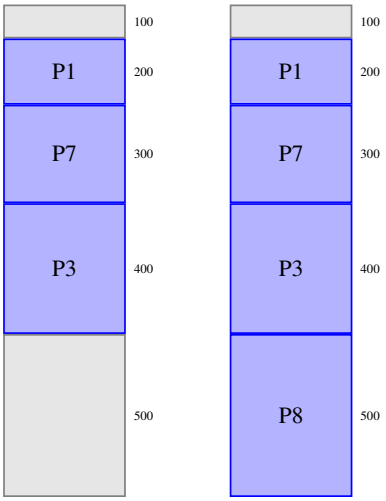




**BF**


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P3
P8

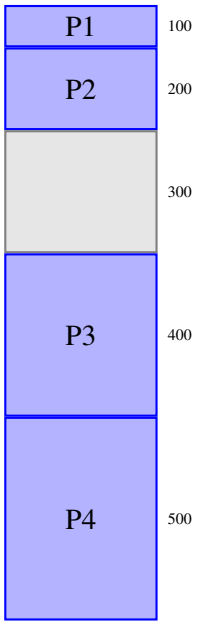
WF


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P4
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P7
P3
P4


100
P1
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P7
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P3
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100
P1
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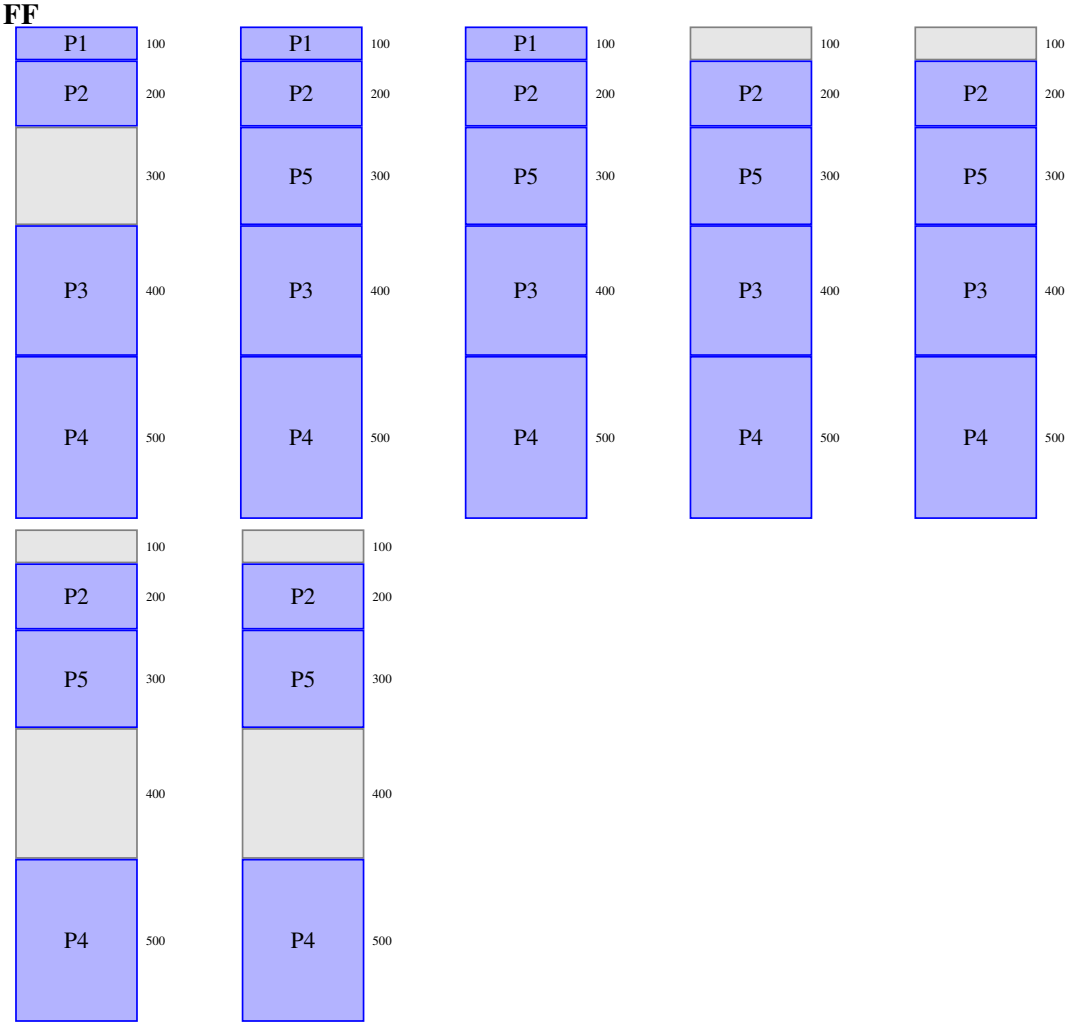
11

- P5 di dimensione 150Kb
- P6 di dimensione 250Kb
- P1 esce
- P7 di dimensione 280Kb
- P3 esce
- P8 di dimensione 450Kb

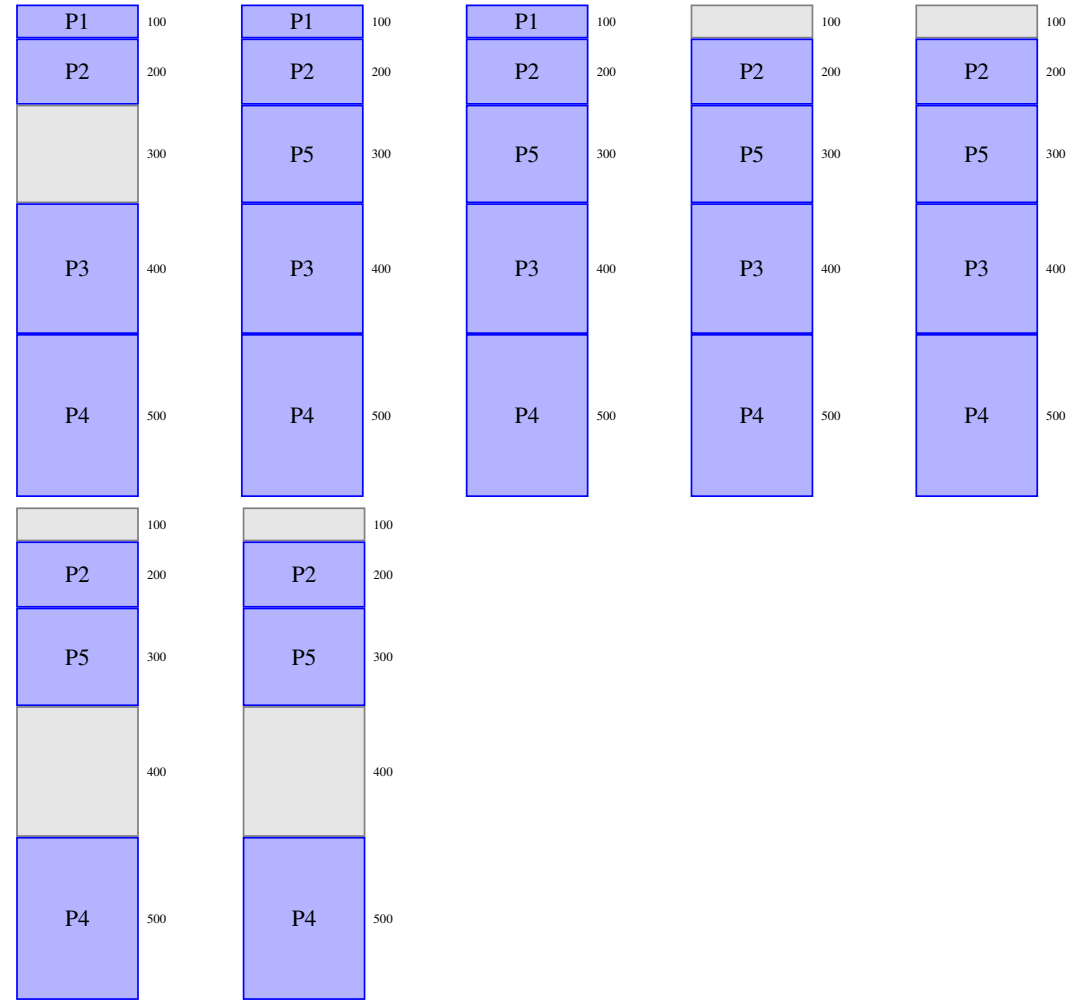



**SOLUZIONE**

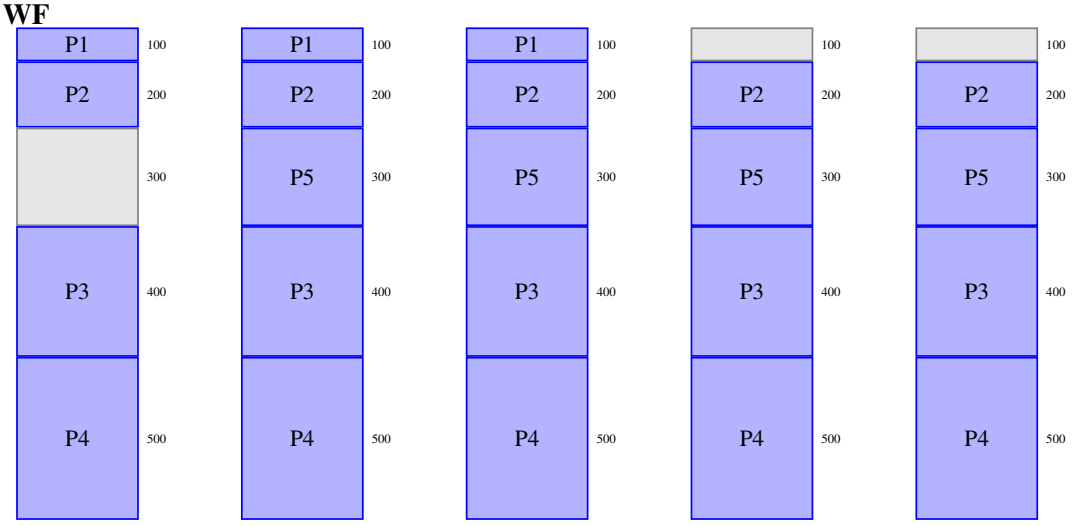
**FF**

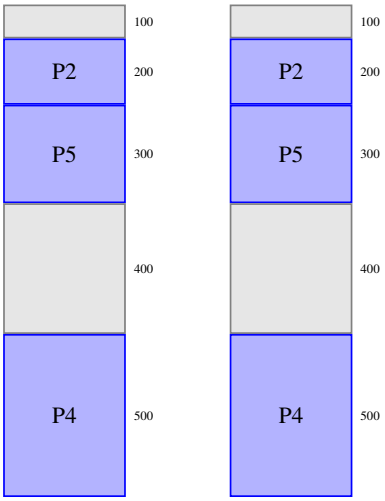



**BF**


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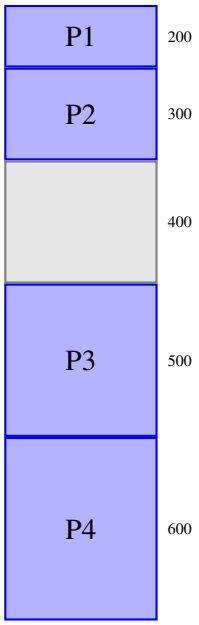
WF


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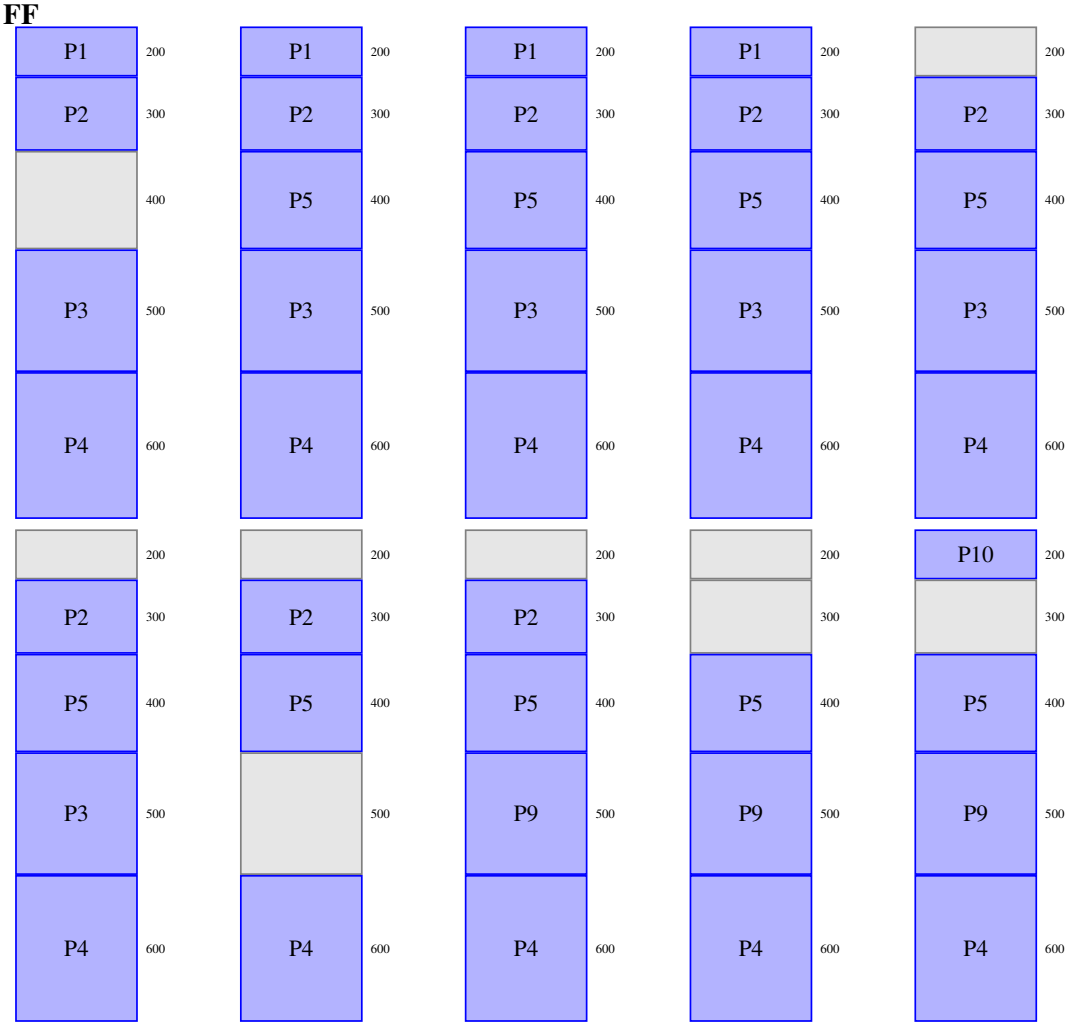
12

- •P5 di dimensione 180Kb
- •P6 di dimensione 250Kb
- •P7 di dimensione 350Kb
- •P1 esce
- •P8 di dimensione 280Kb
- •P3 esce
- •P9 di dimensione 480Kb
- •P2 esce
- •P10 di dimensione 190Kb

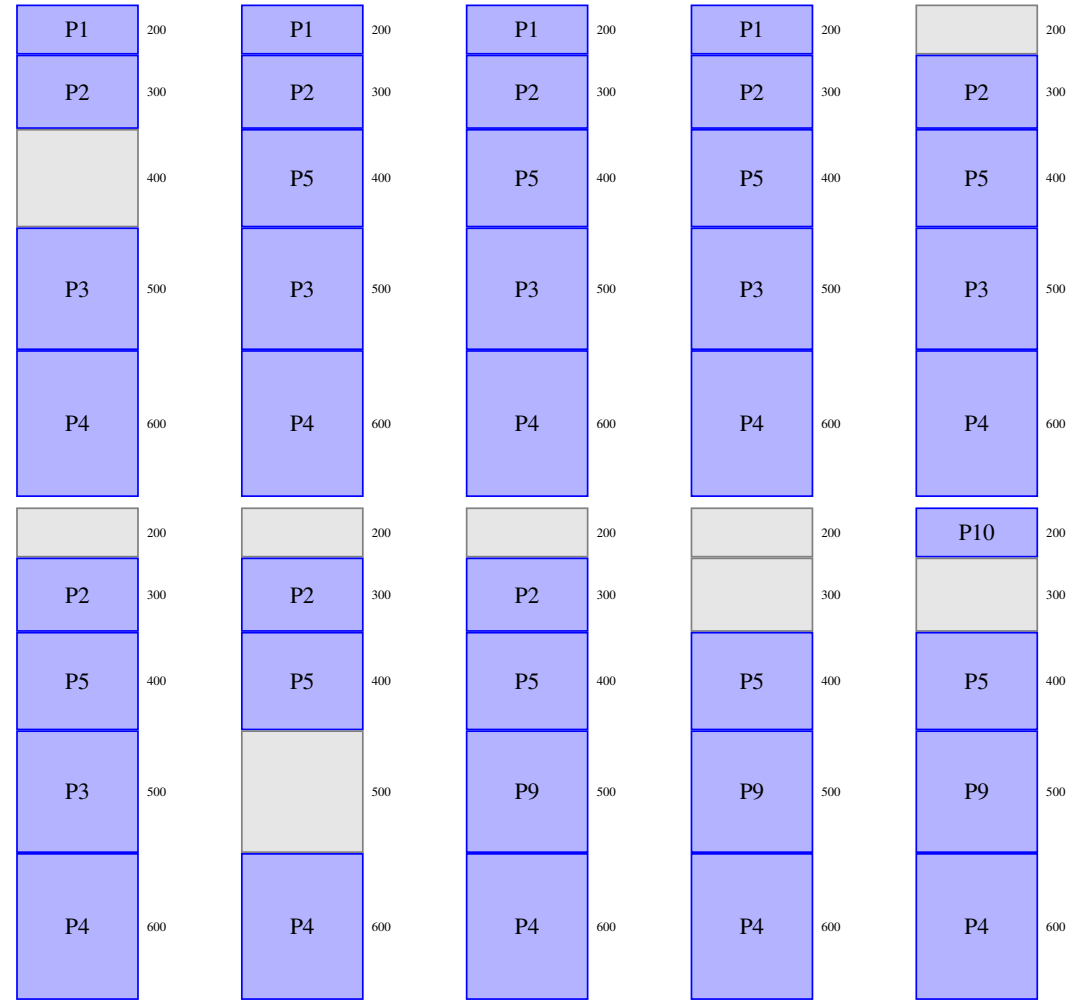



**SOLUZIONE**

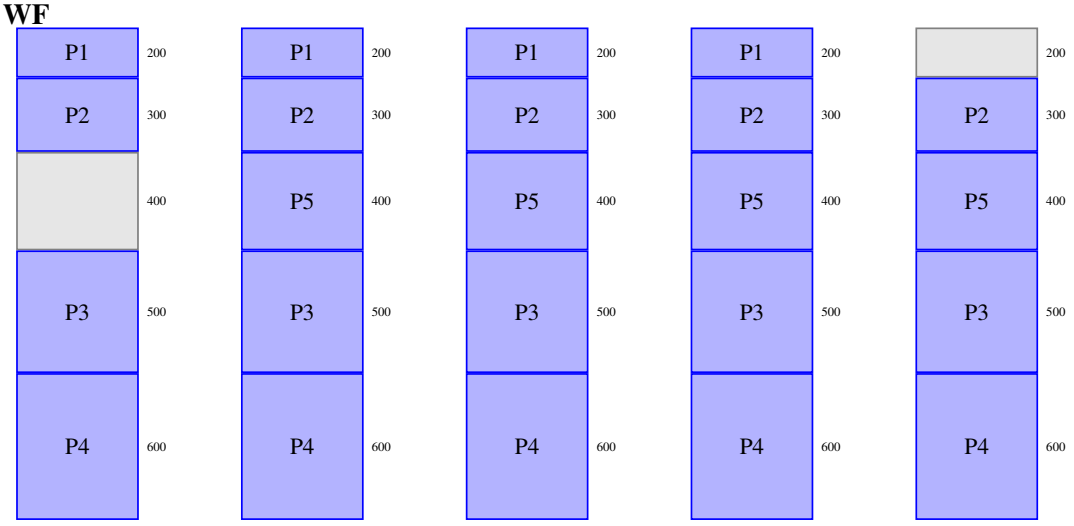
**FF**

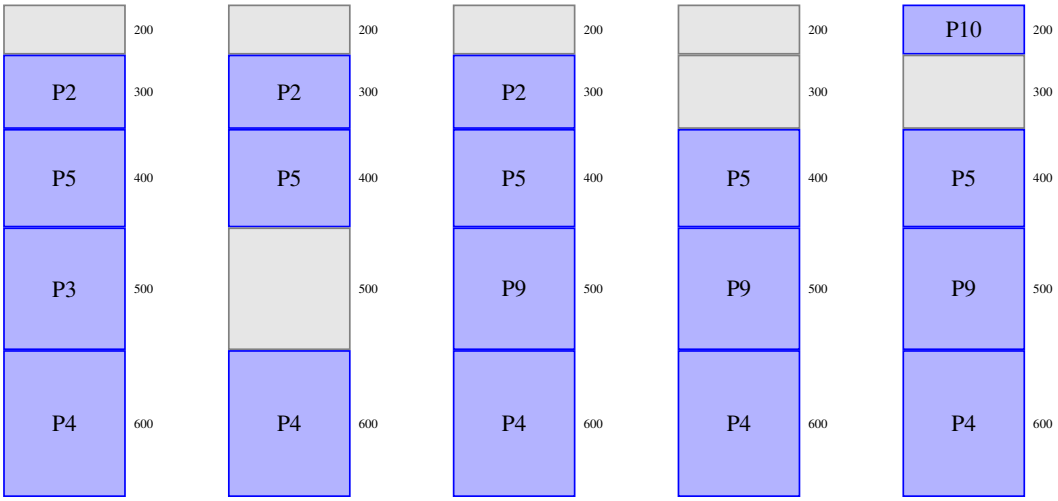



**BF**


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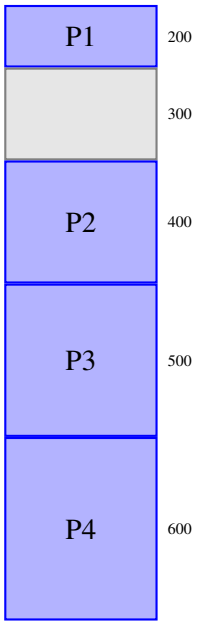
**WF**


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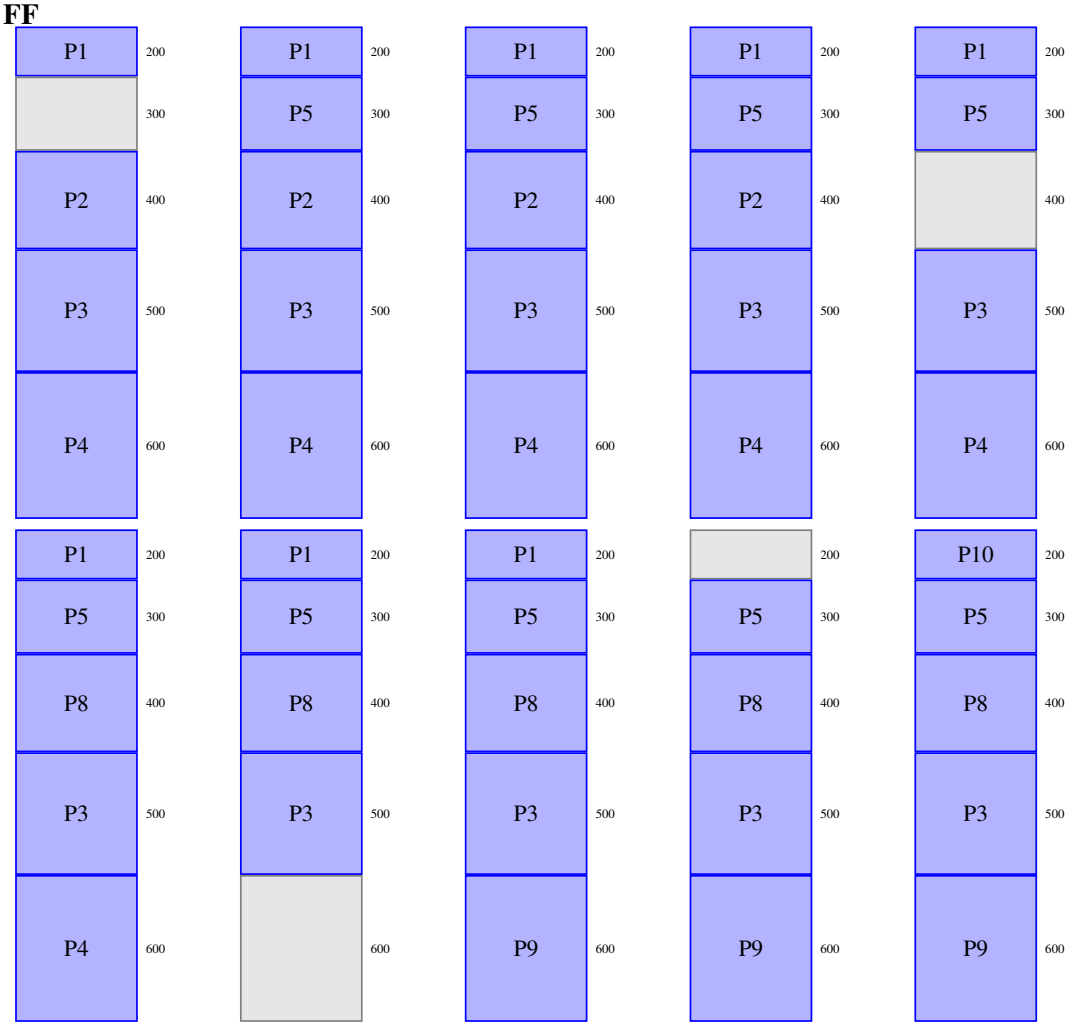
13

- •P5 di dimensione 150Kb
- •P6 di dimensione 250Kb
- •P7 di dimensione 350Kb
- •P2 esce
- •P8 di dimensione 280Kb
- •P4 esce
- •P9 di dimensione 450Kb
- •P1 esce
- •P10 di dimensione 190Kb

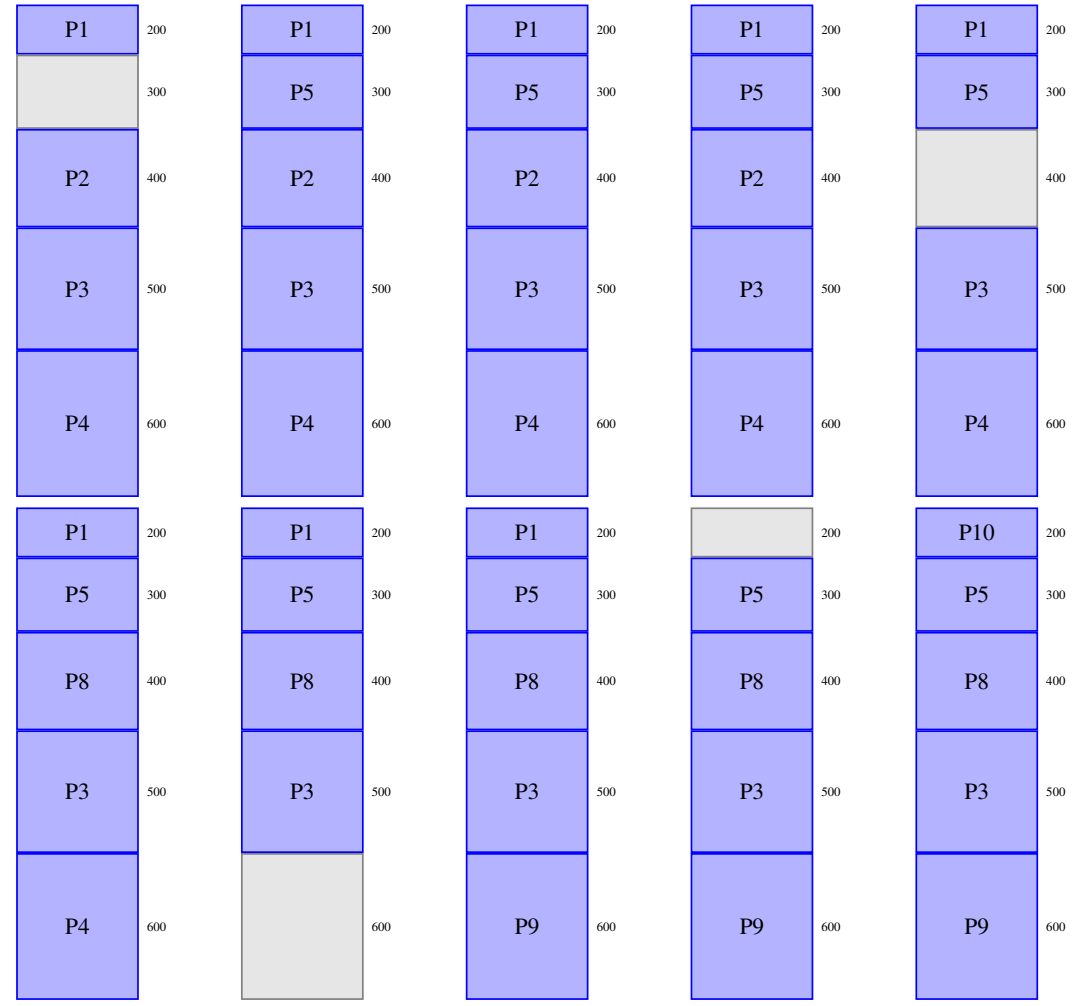



**SOLUZIONE**

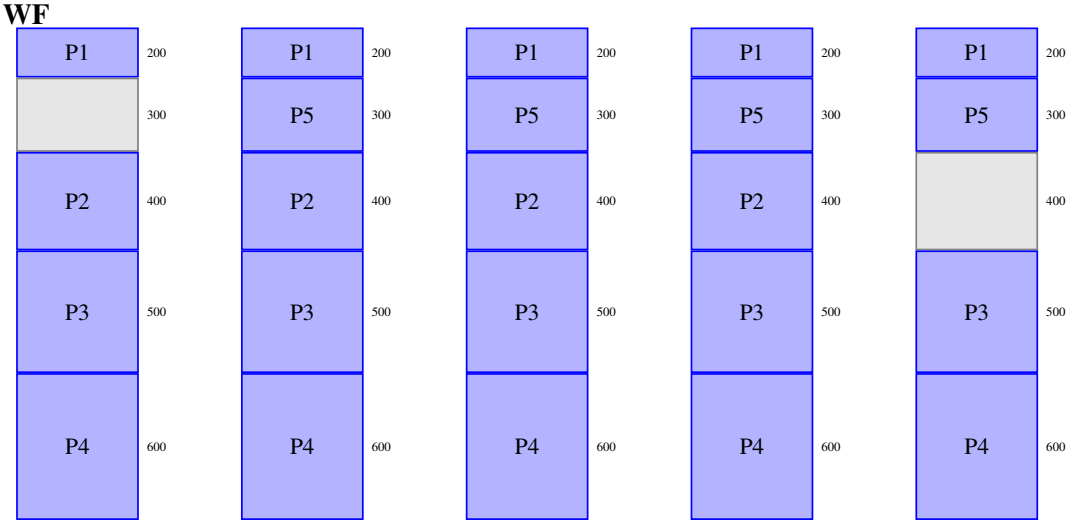
**FF**

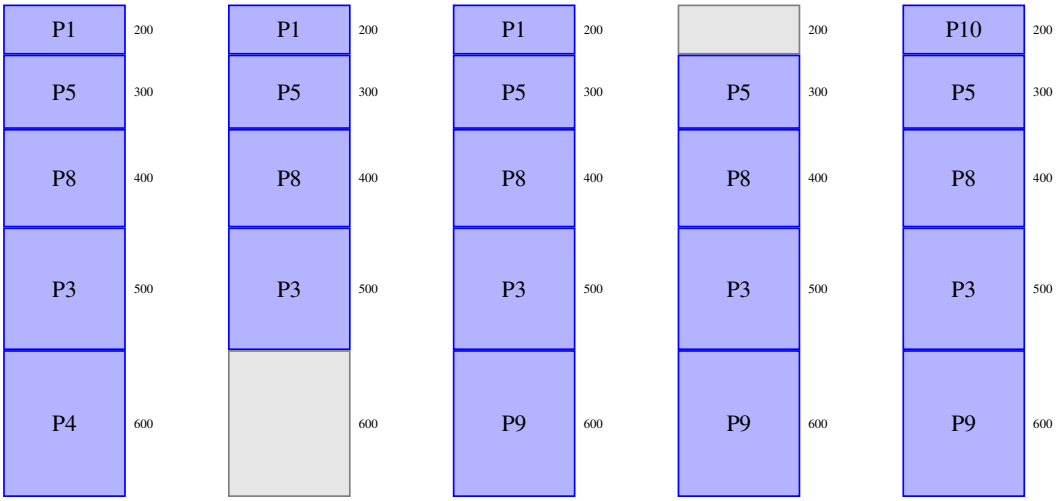



**BF**


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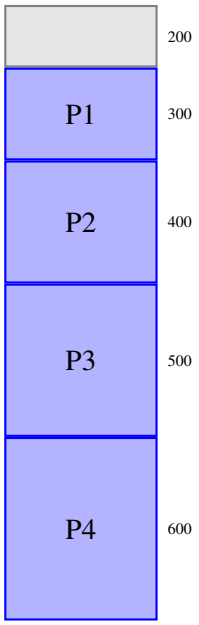
**WF**


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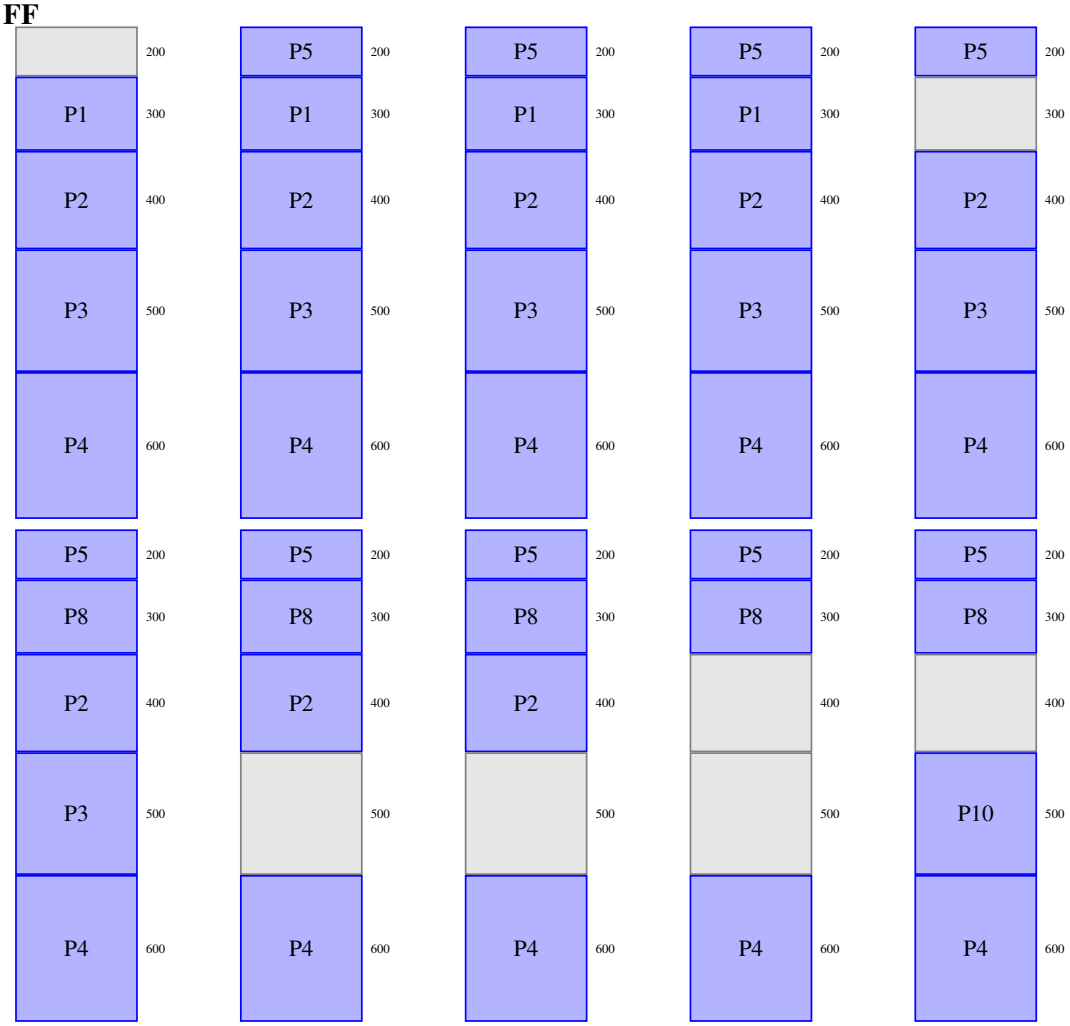
14

- •P5 di dimensione 200Kb
- •P6 di dimensione 300Kb
- •P7 di dimensione 400Kb
- •P1 esce
- •P8 di dimensione 250Kb
- •P3 esce
- •P9 di dimensione 550Kb
- •P2 esce
- •P10 di dimensione 350Kb

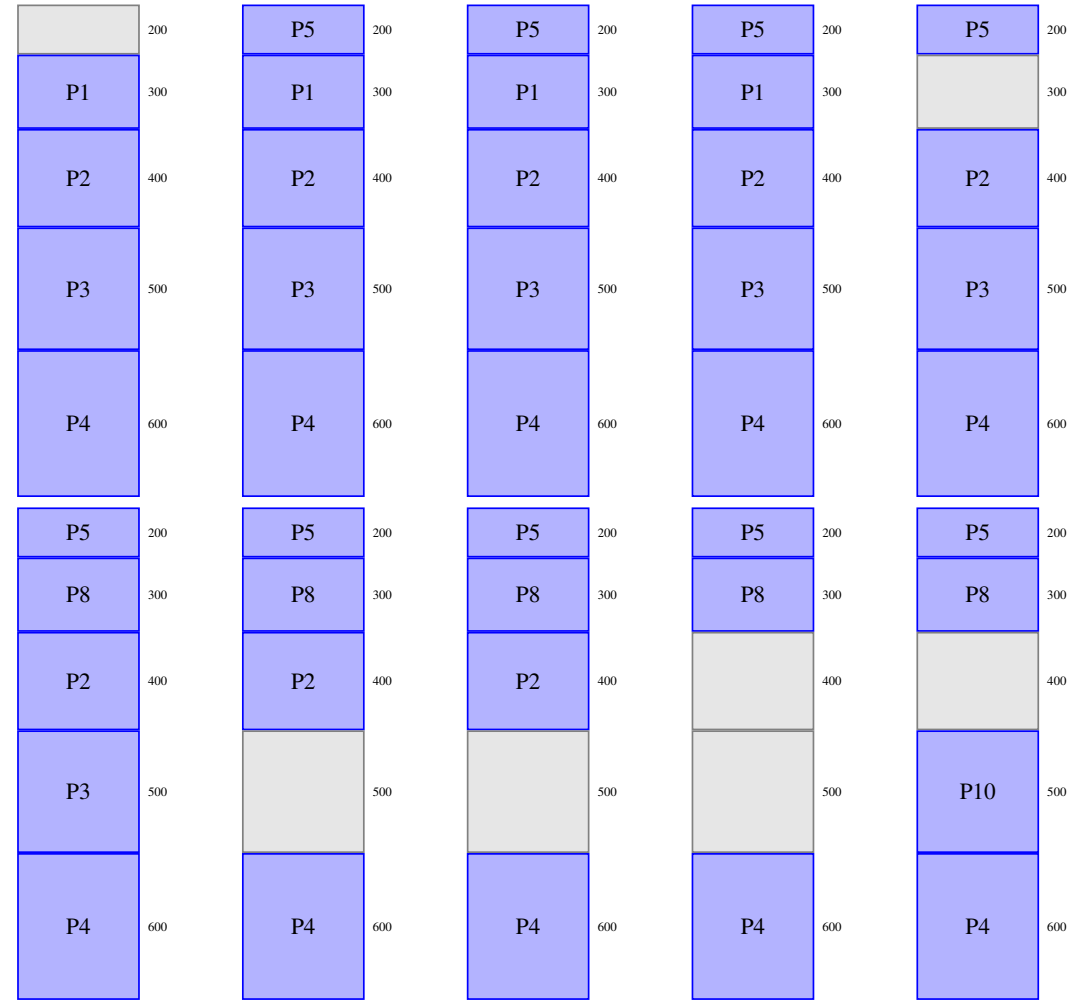



**SOLUZIONE**

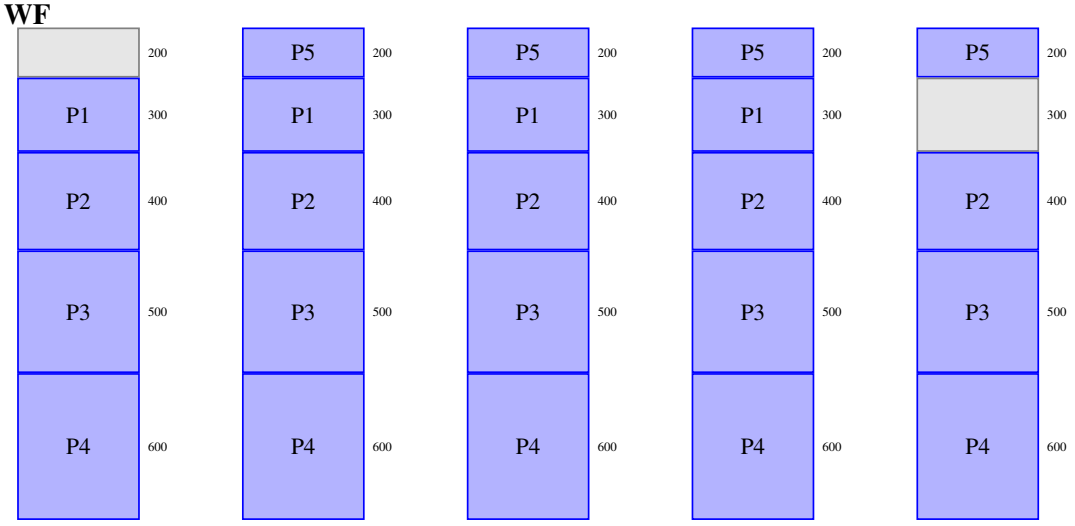
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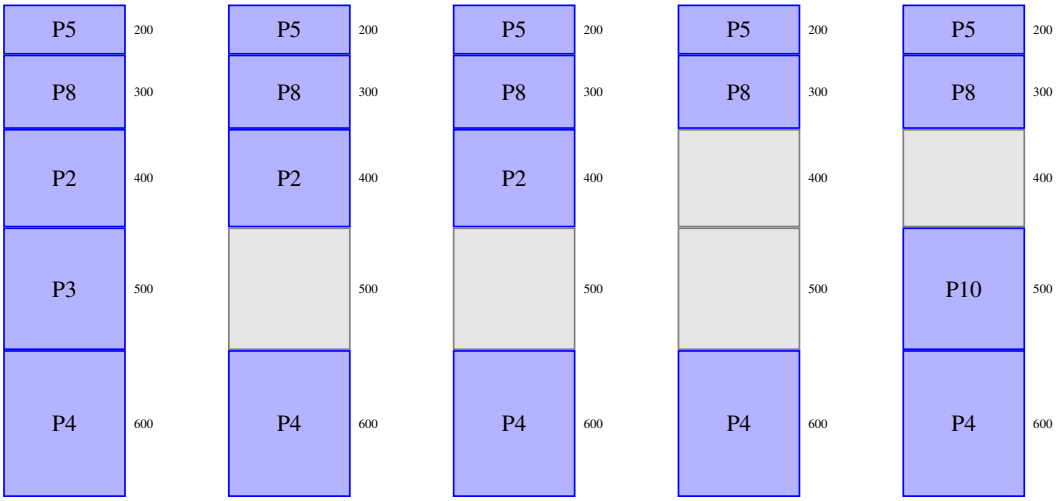



**BF**


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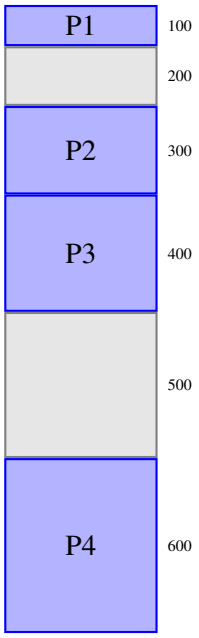
**WF**


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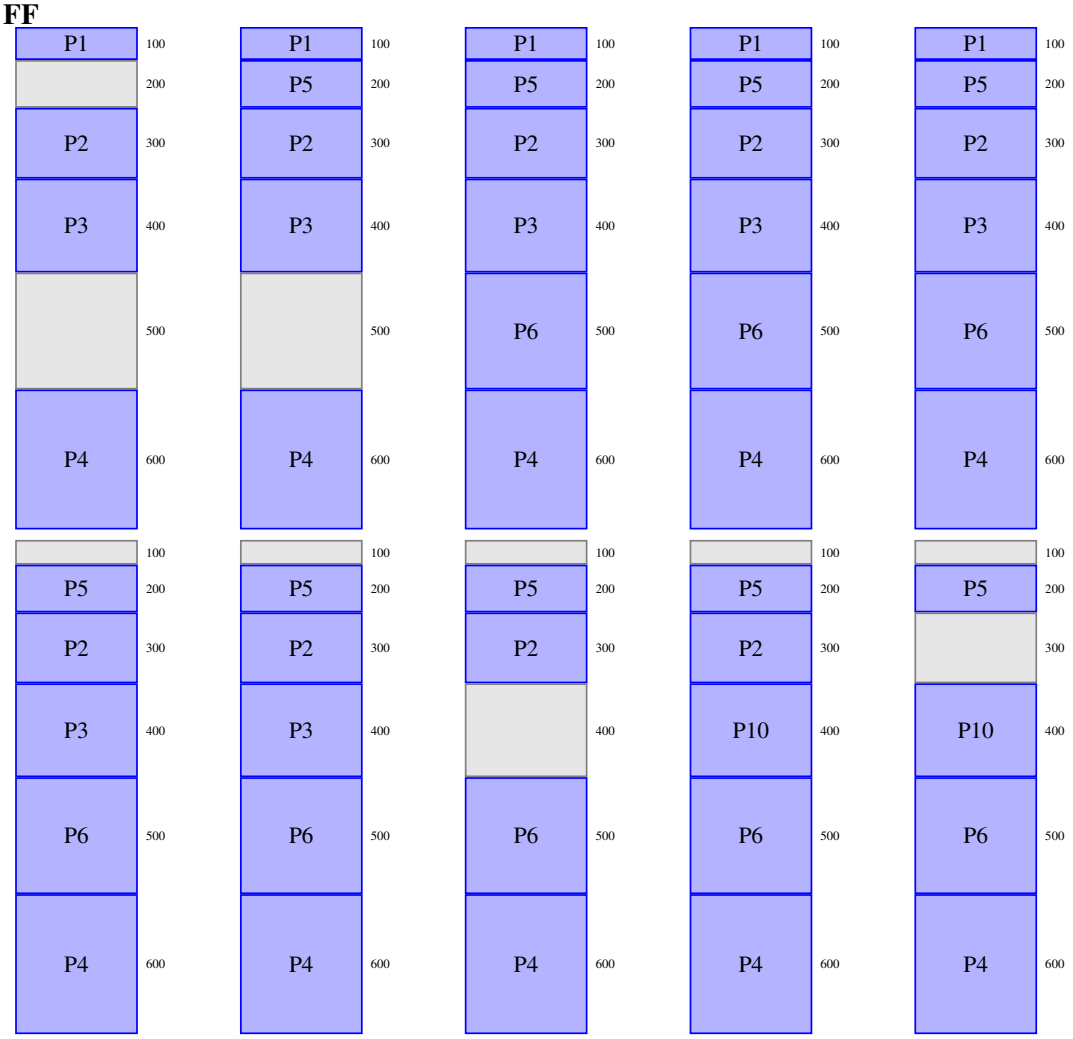
15

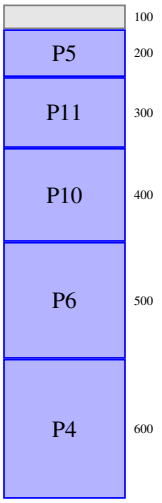
- •P5 di dimensione 150Kb
- •P6 di dimensione 250Kb
- •P7 di dimensione 350Kb
- •P8 di dimensione 450Kb
- •P1 esce
- •P9 di dimensione 180Kb
- •P3 esce
- •P10 di dimensione 380Kb
- •P2 esce
- •P11 di dimensione 280Kb



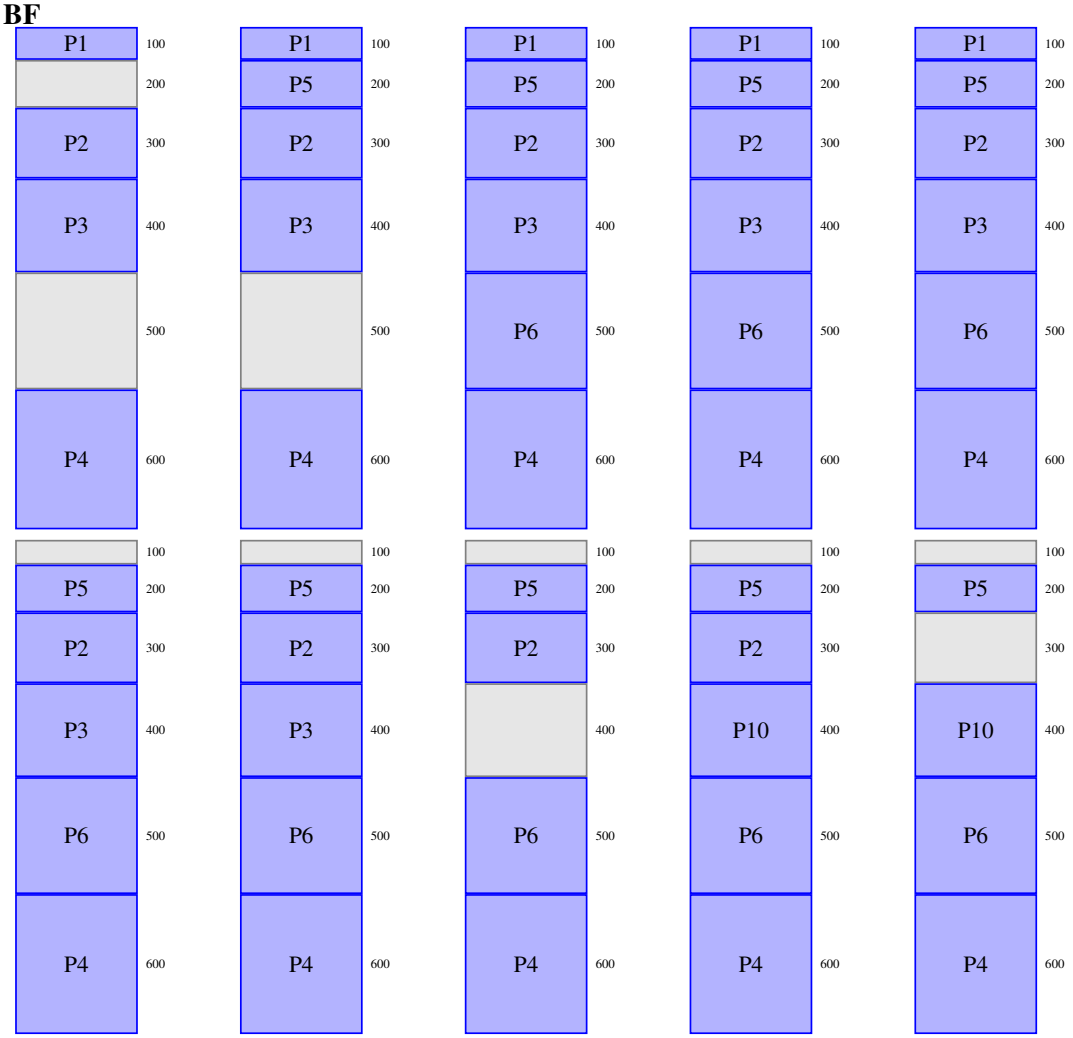

**SOLUZIONE**

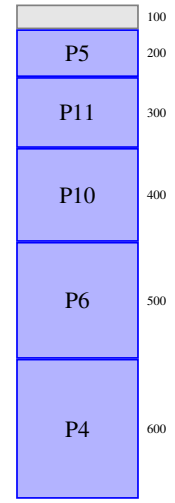
**FF**



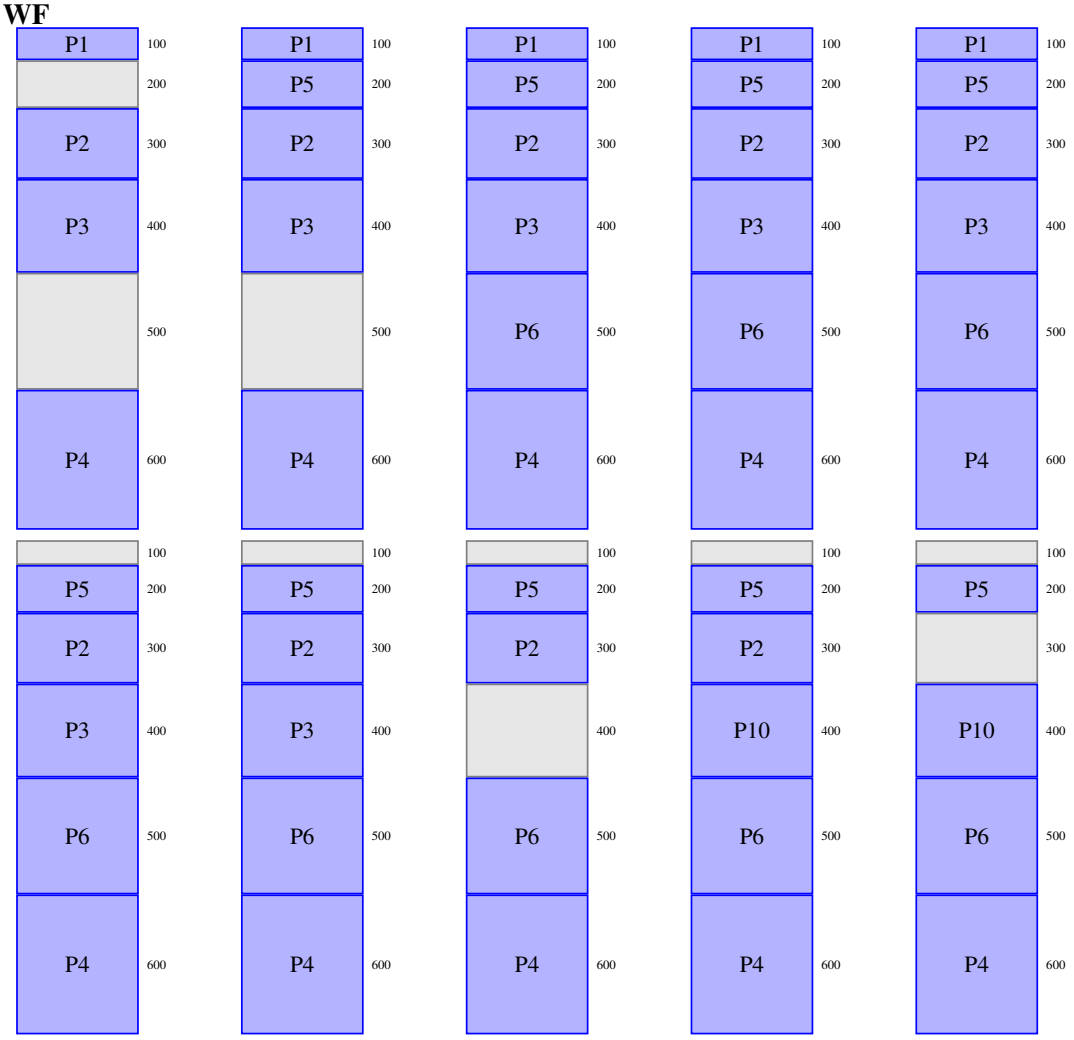

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P4
600

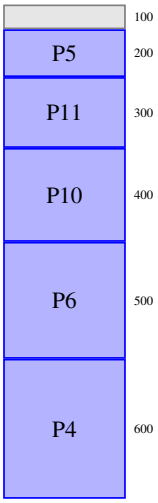
**BF**


P1
P5
P2
P3
P6
P4
P10
100
200
300
400
500
600


100
P5
200
P11
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P10
400
P6
500
P4
600

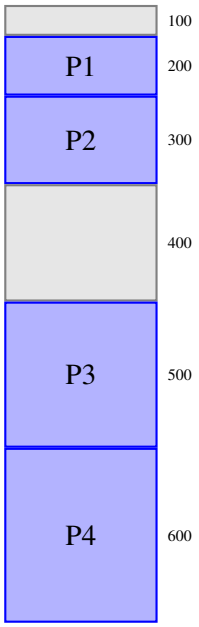
**WF**


P1
100
200
P2
300
P3
400
500
P4
600
P1
100
P5
200
P2
300
P3
400
500
P4
600
P1
100
P5
200
P2
300
P6
500
P4
600
P1
100
P5
200
P2
300
P3
400
P6
500
P4
600
P1
100
P5
200
P2
300
P3
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P6
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P4
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100
P5
200
P2
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P3
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P6
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P4
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P5
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P2
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P3
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P6
500
P4
600
100
P5
200
P2
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400
P6
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P4
600
100
P5
200
P2
300
P10
400
P6
500
P4
600
100
P5
200
300
P10
400
P6
500
P4
600


100
P5
200
P11
300
P10
400
P6
500
P4
600

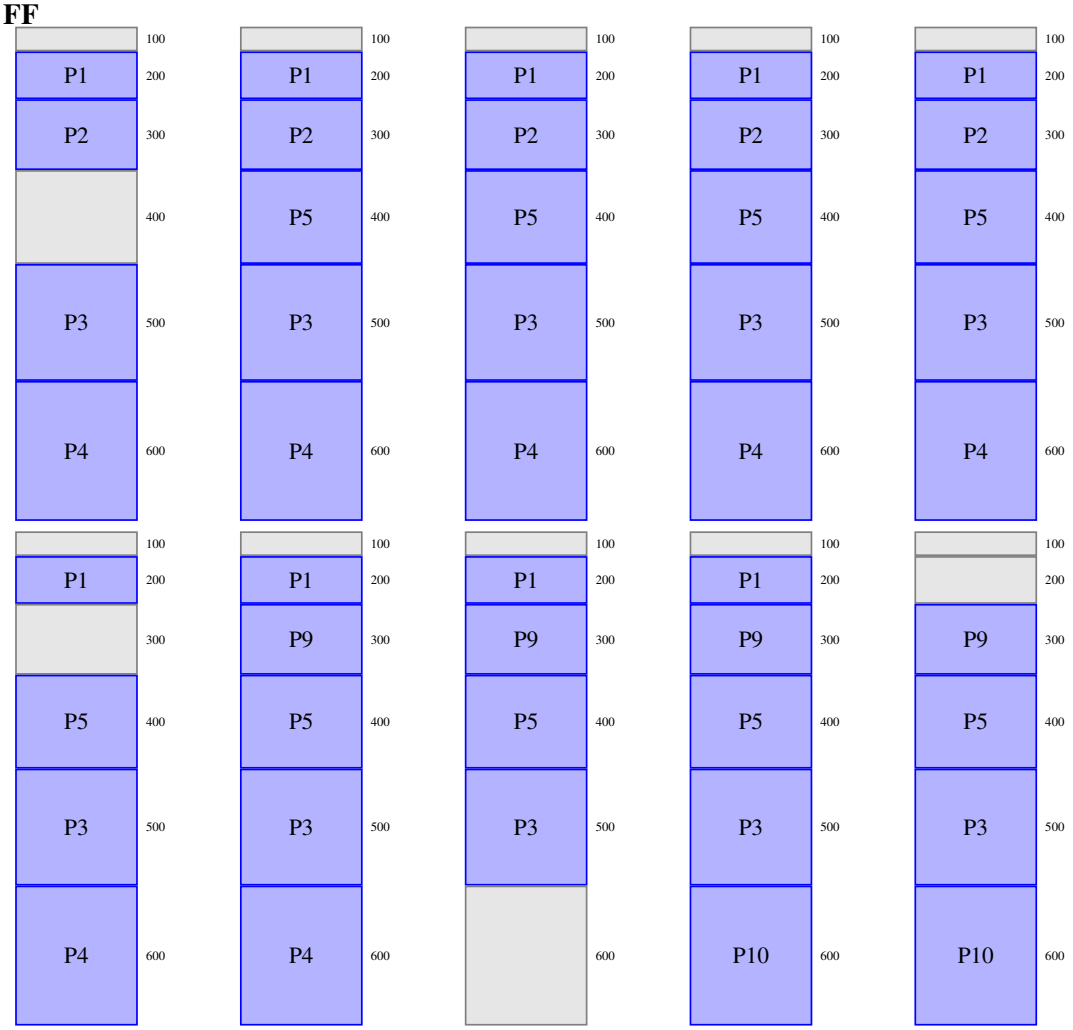
16

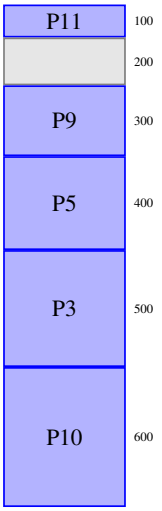
- •P5 di dimensione 180Kb
- •P6 di dimensione 280Kb
- •P7 di dimensione 380Kb
- •P8 di dimensione 480Kb
- •P2 esce
- •P9 di dimensione 190Kb
- •P4 esce
- •P10 di dimensione 450Kb
- •P1 esce
- •P11 di dimensione 90Kb



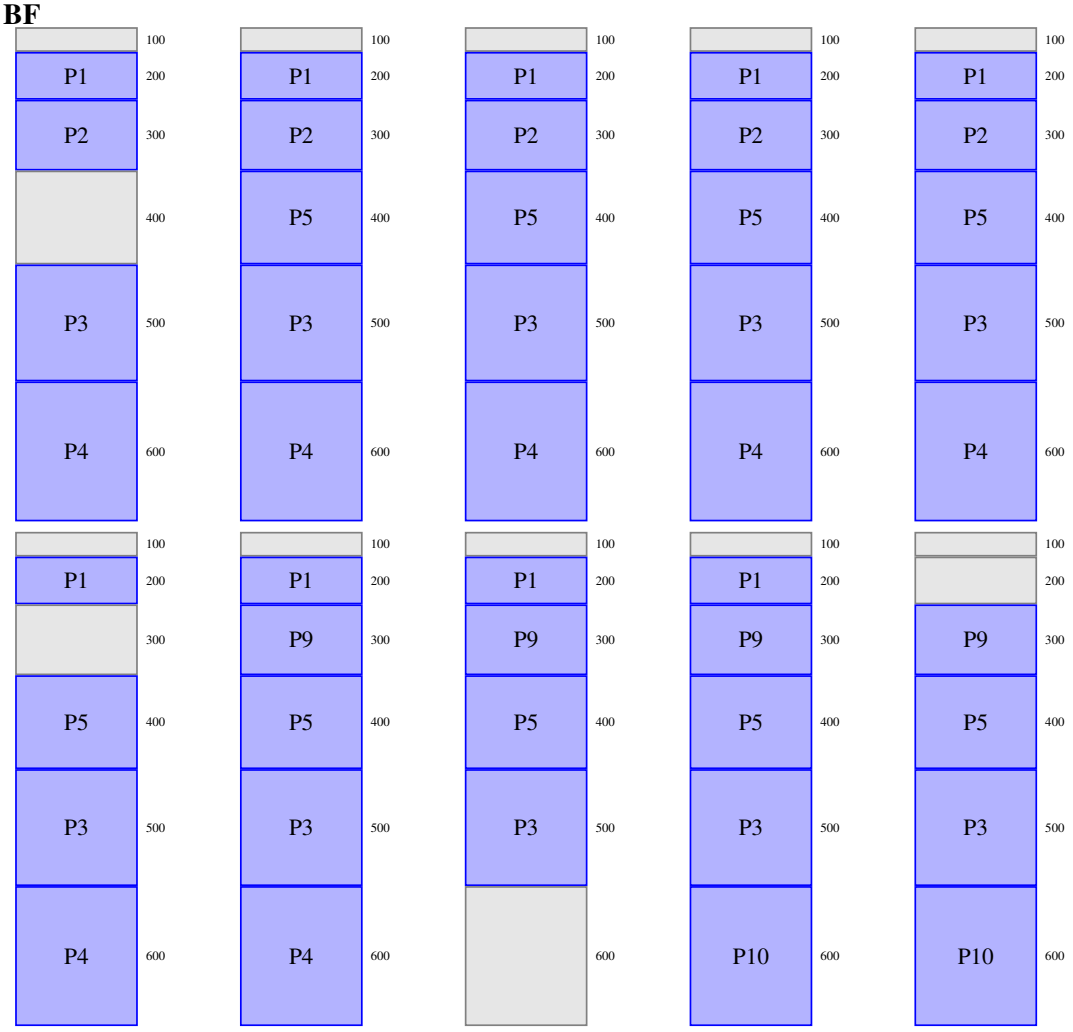

**SOLUZIONE**

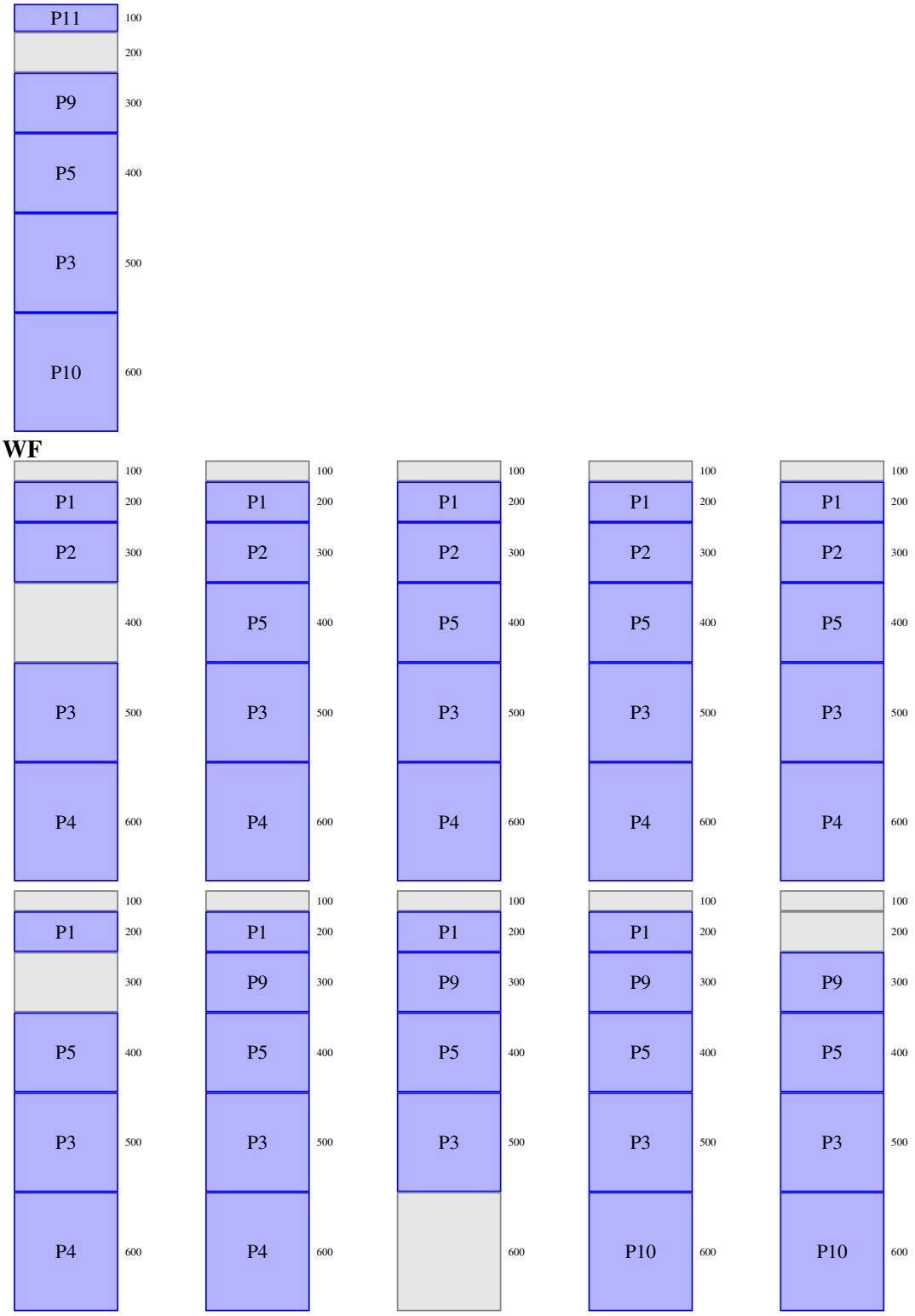
**FF**

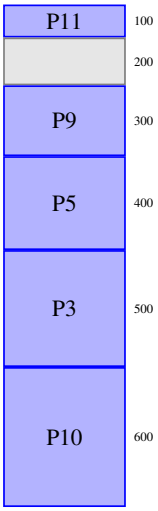



P11
100
200
P9
300
P5
400
P3
500
P10
600

**BF**

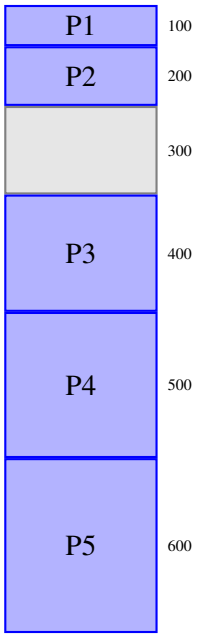

100
P1
200
P2
300
400
P3
500
P4
600
100
P1
200
P2
300
P5
400
P3
500
P4
600
100
P1
200
P2
300
P5
400
P3
500
600
100
P1
200
P2
300
P5
400
P3
500
P4
600
100
P1
200
P2
300
P5
400
P3
500
P4
600
100
P1
200
300
P5
400
P3
500
P4
600
100
P1
200
P9
300
P5
400
P3
500
P4
600
100
P1
200
P9
300
P5
400
P3
500
600
100
P1
200
P9
300
P5
400
P3
500
P10
600
100
200
P9
300
P5
400
P3
500
P10
600


P11
100
200
P9
300
P5
400
P3
500
P10
600
WF
P1
P2
P5
P3
P4
P9
P10


P11
100
200
P9
300
P5
400
P3
500
P10
600

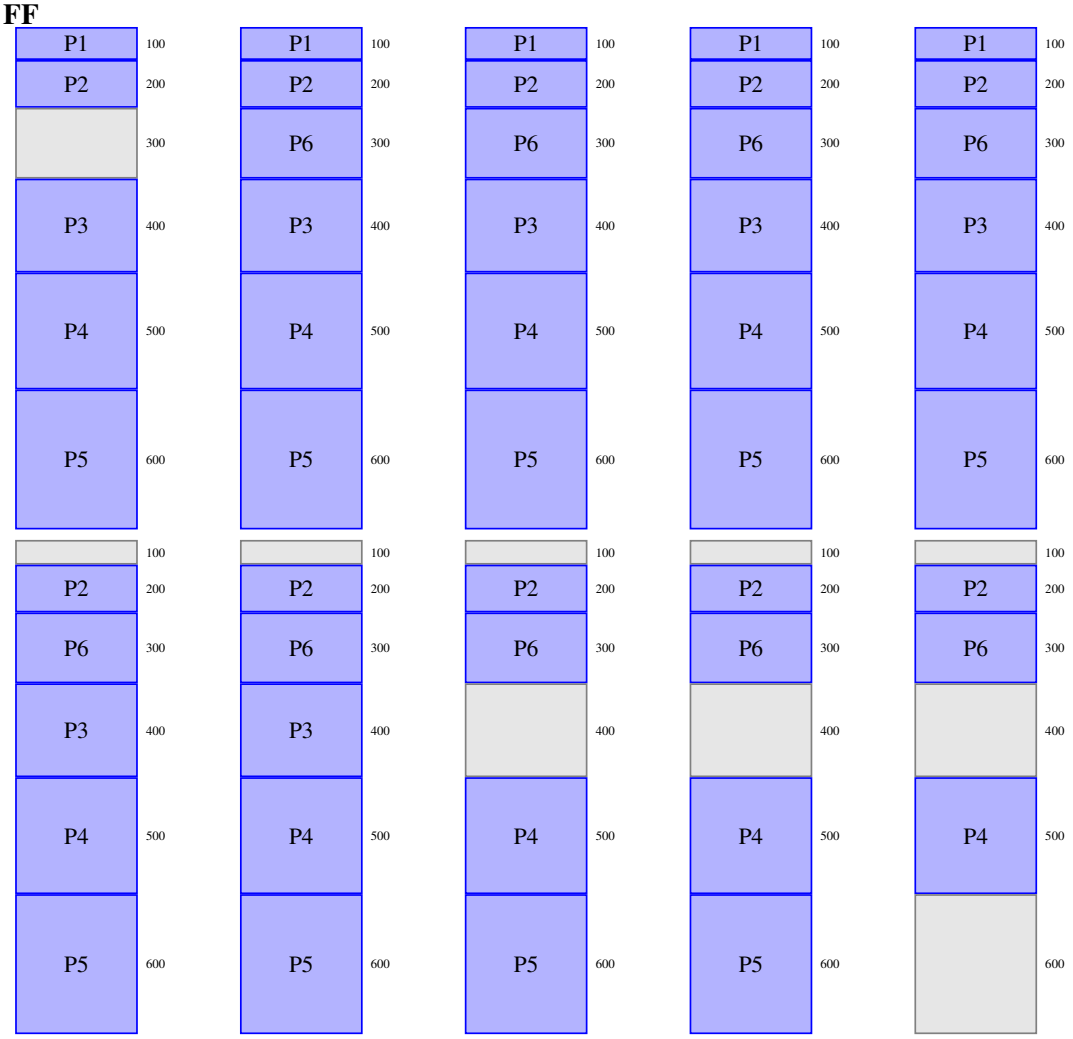
17

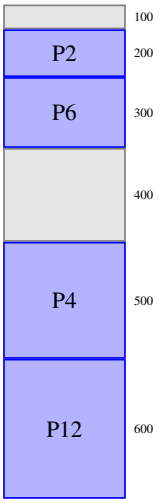
- P6 di dimensione 150Kb
- P7 di dimensione 250Kb
- P8 di dimensione 350Kb
- P9 di dimensione 450Kb
- P1 esce
- P10 di dimensione 280Kb
- P3 esce
- P11 di dimensione 550Kb
- P5 esce
- P12 di dimensione 580Kb



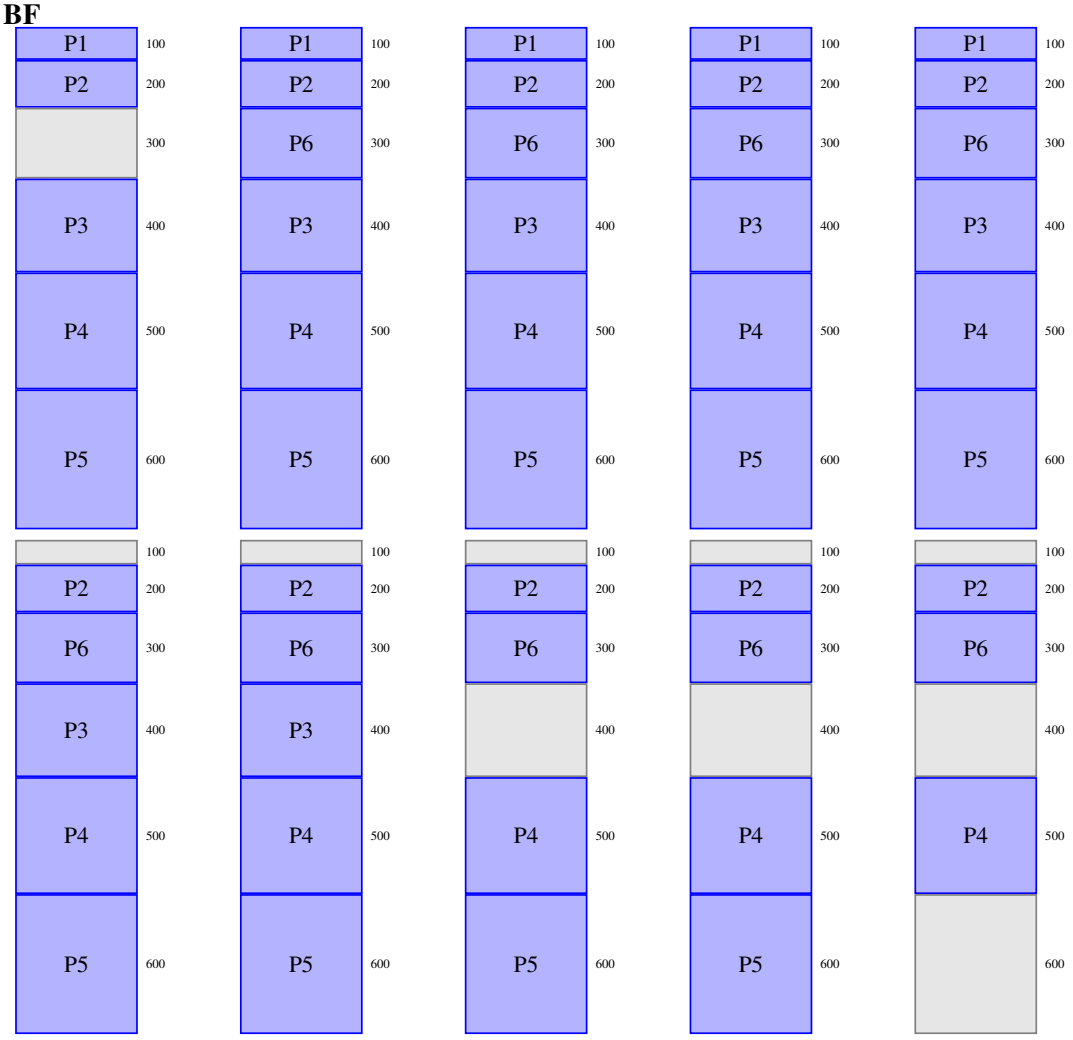

**SOLUZIONE**

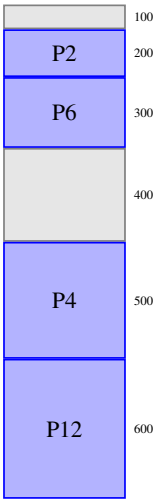
**FF**



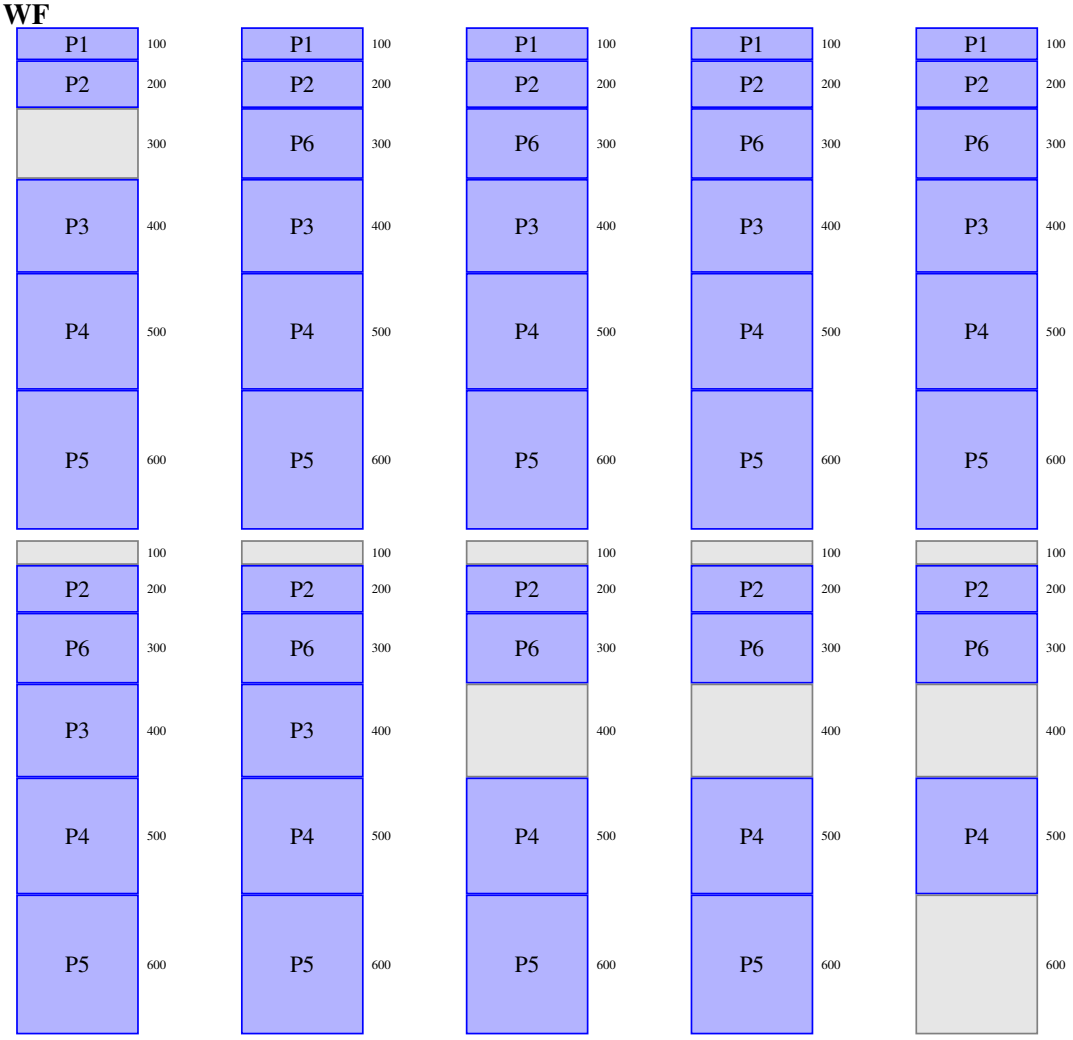

100
P2
200
P6
300
400
P4
500
P12
600

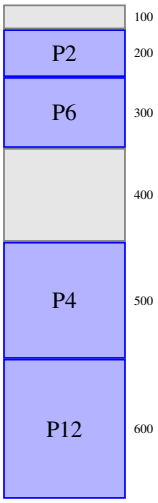
**BF**


P1
P2
P3
P4
P5
P6
100
200
300
400
500
600


100
P2
200
P6
300
400
P4
500
P12
600

**WF**


P1
P2
P3
P4
P5
P6
100
200
300
400
500
600

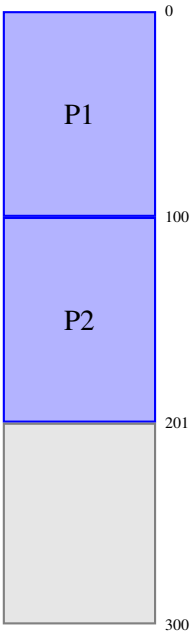

100
P2
200
P6
300
400
P4
500
P12
600

## 6.2 Memoria a partizione variabili

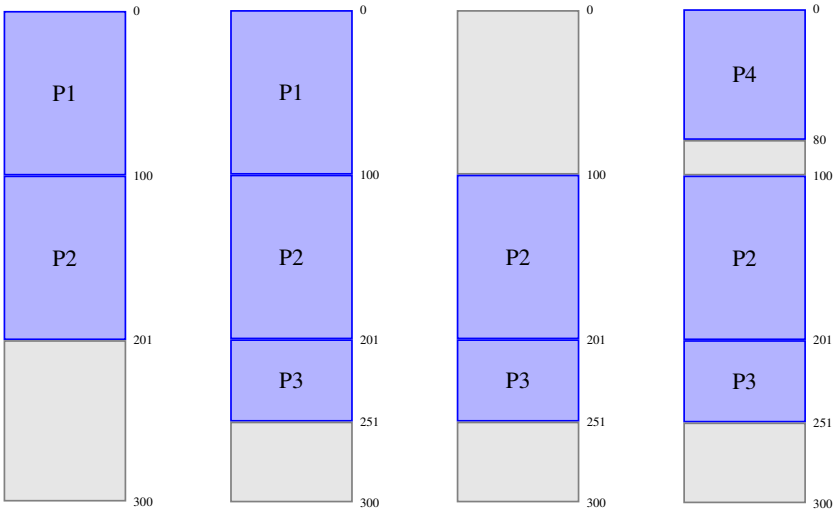
Mostra l'evoluzione della memoria dati i seguenti processi e la seguente memoria a partizioni variabili usando l'algoritmo FirstFit, BestFit e WorstFit

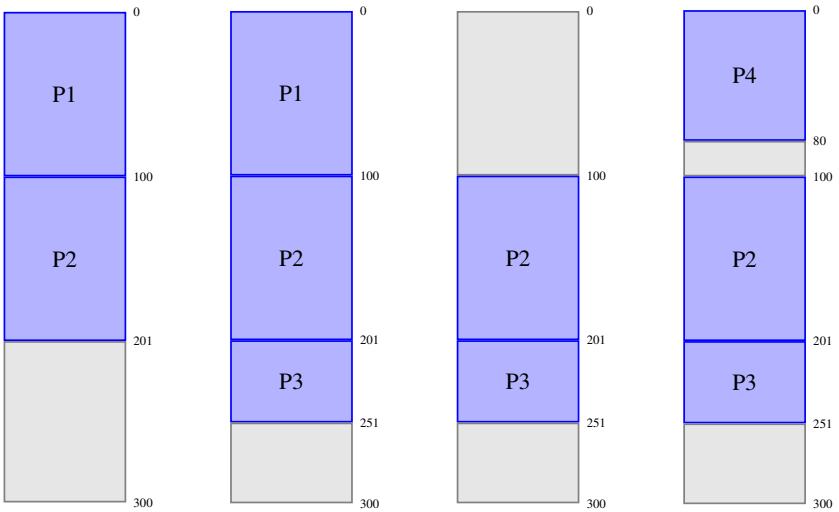
1

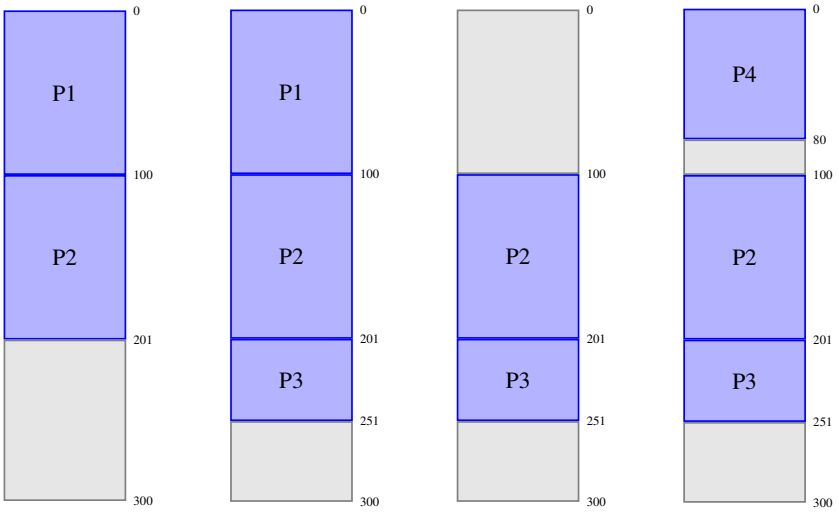
- P3 di dimensione 50Kb
- P1 esce
- P4 di dimensione 80Kb




**SOLUZIONE**

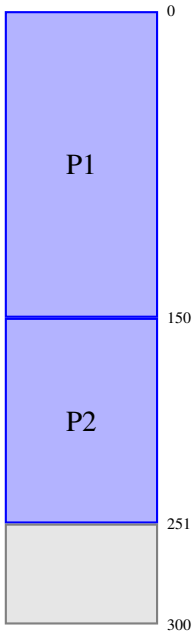


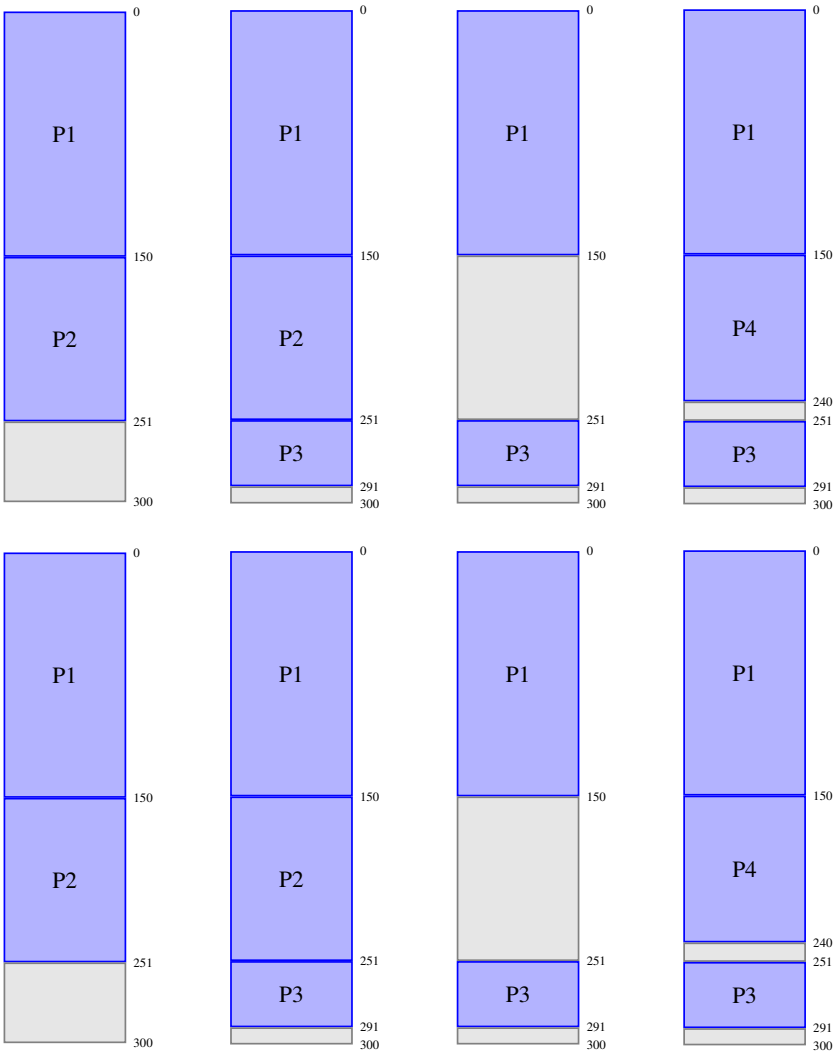

P1
P2
P1
P2
P3
P2
P3
P4
P2
P3
0
100
201
300
0
100
201
251
300
0
100
201
251
300
0
80
100
201
251
300

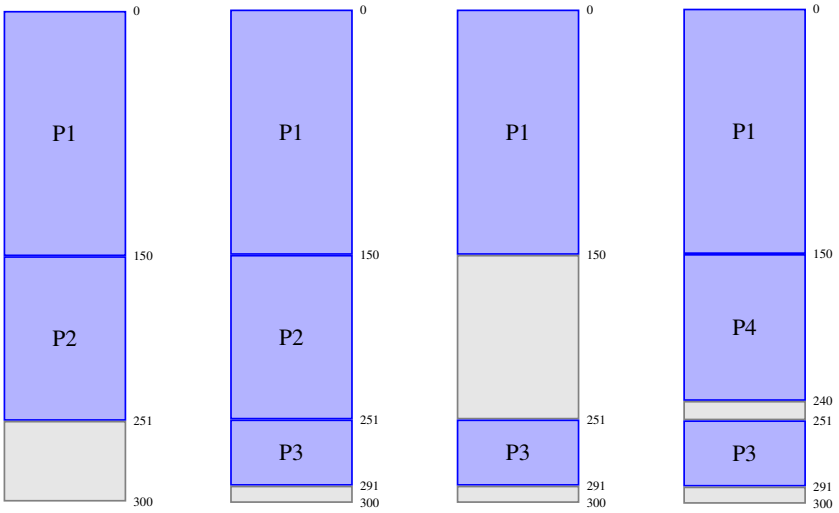
**2**

- P3 di dimensione 40Kb
- P2 esce
- P4 di dimensione 90Kb



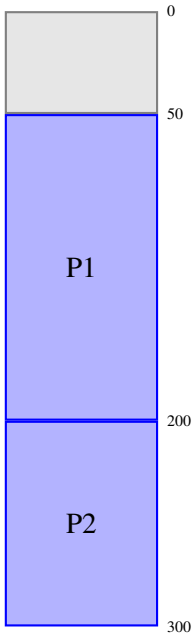

**SOLUZIONE**



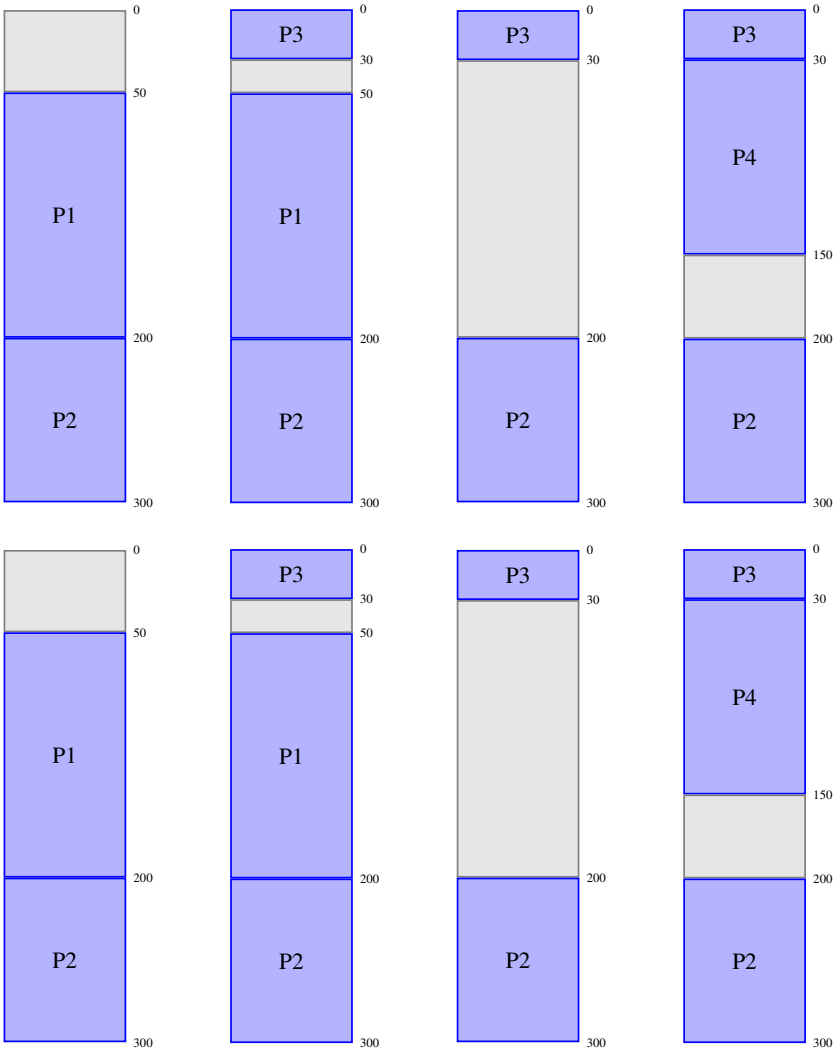

P1
P2
0
150
251
300
P1
P2
P3
0
150
251
291
300
P1
P3
0
150
251
291
300
P1
P4
P3
0
150
240
251
291
300

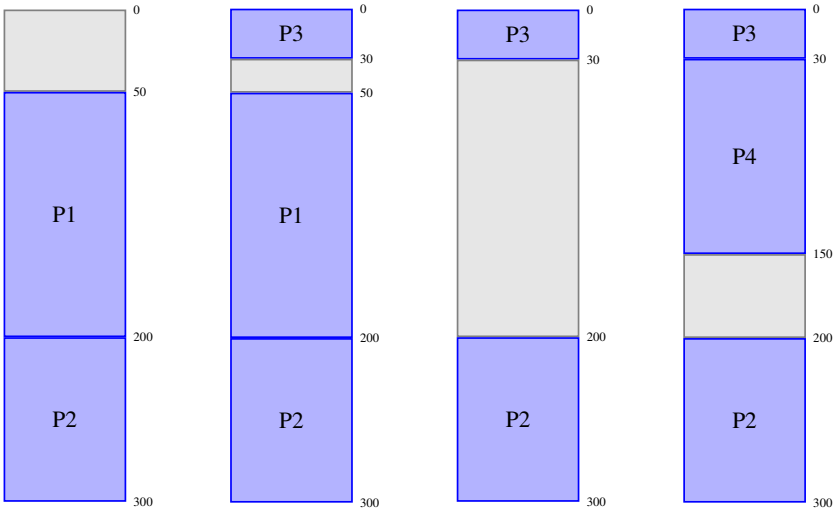
3

- P3 di dimensione 30Kb
- P1 esce
- P4 di dimensione 120Kb



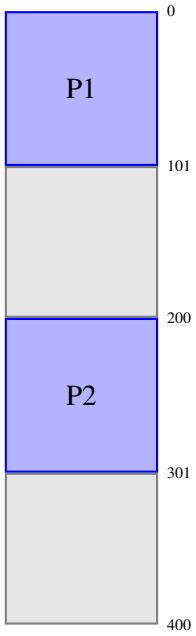

**SOLUZIONE**



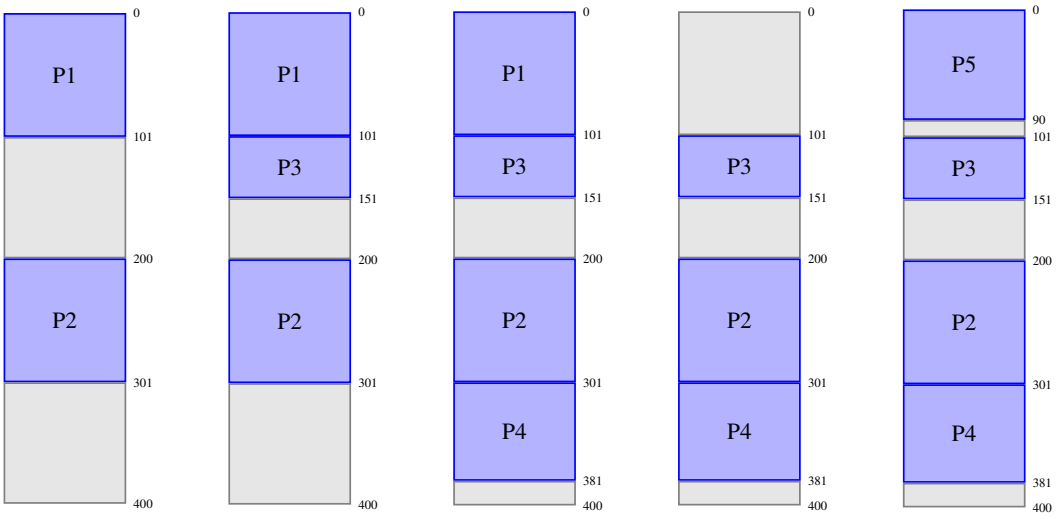

0
50
P1
200
P2
300
0
P3
30
50
P1
200
P2
300
0
P3
30
P2
200
300
0
P3
30
P4
150
200
P2
300

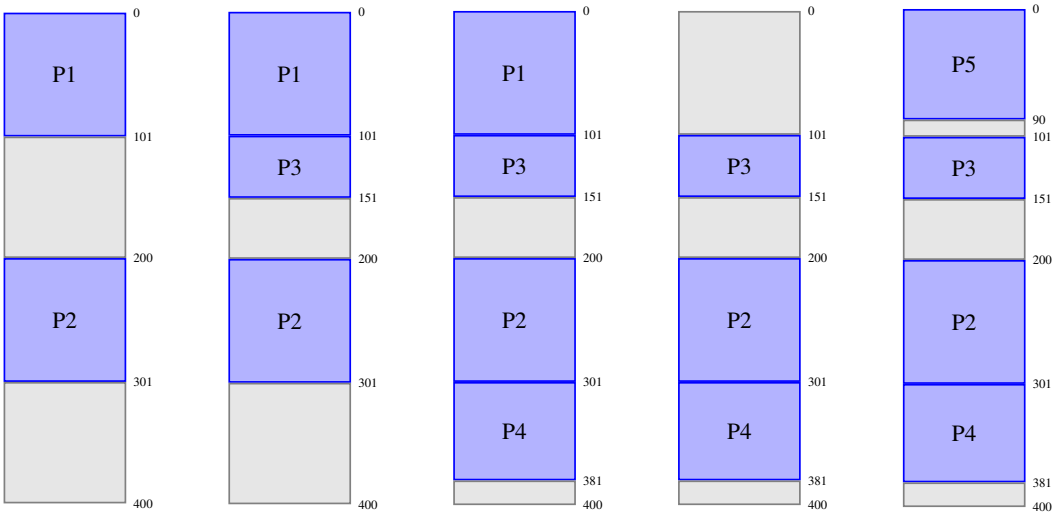
4

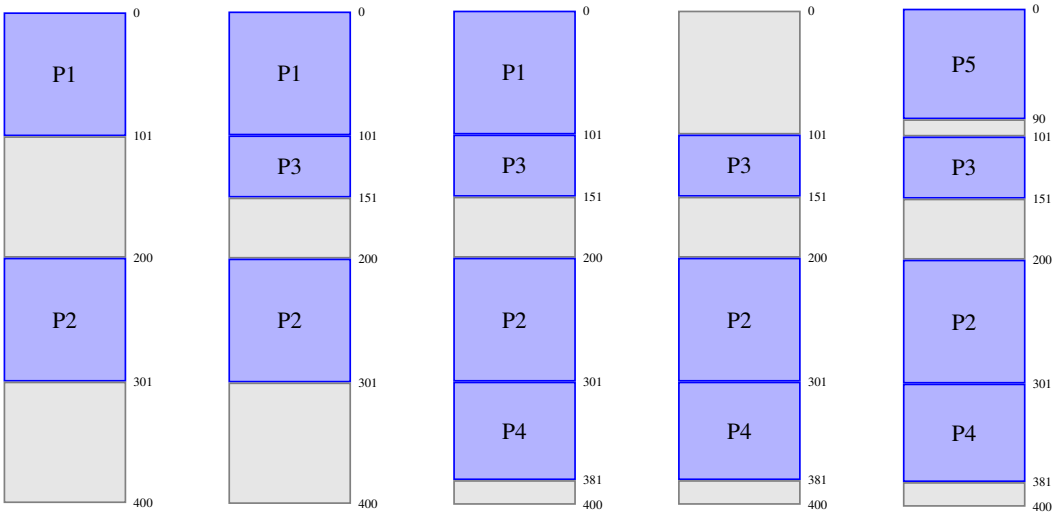
- P3 di dimensione 50Kb
- P4 di dimensione 80Kb
- P1 esce
- P5 di dimensione 90Kb




**SOLUZIONE**

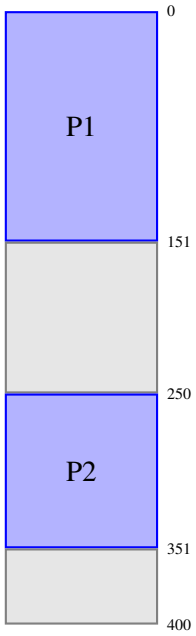


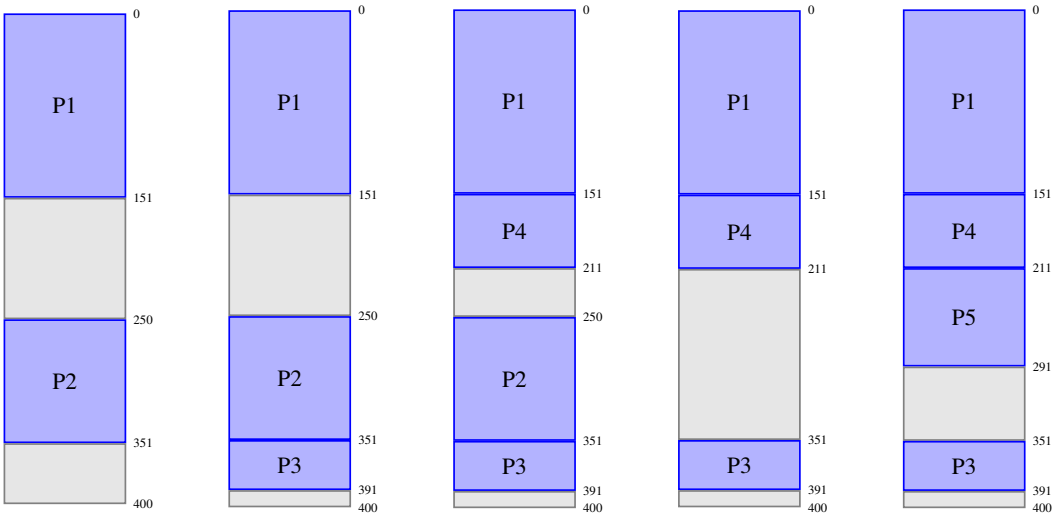

P1
P2
P3
P4
P5
0
101
200
301
400
151
381
90

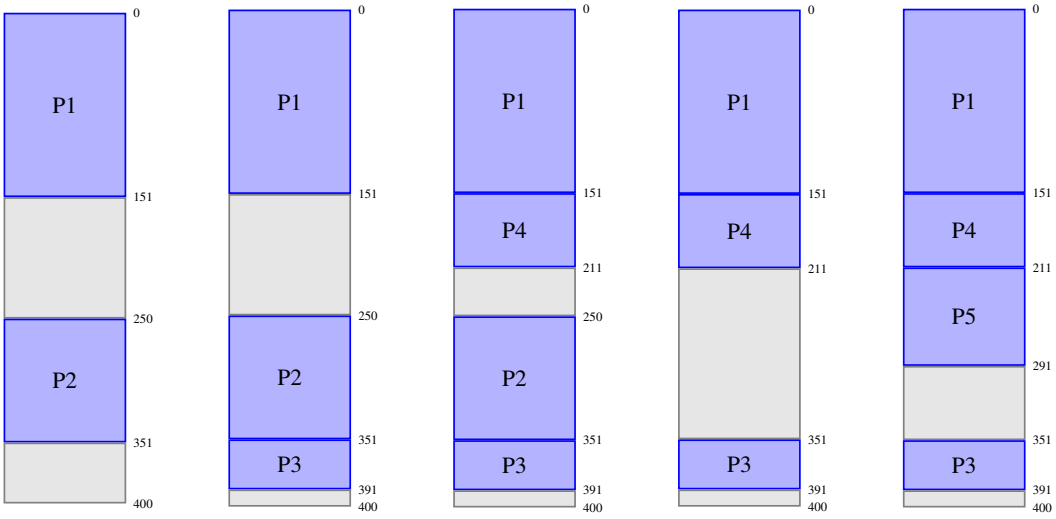
5

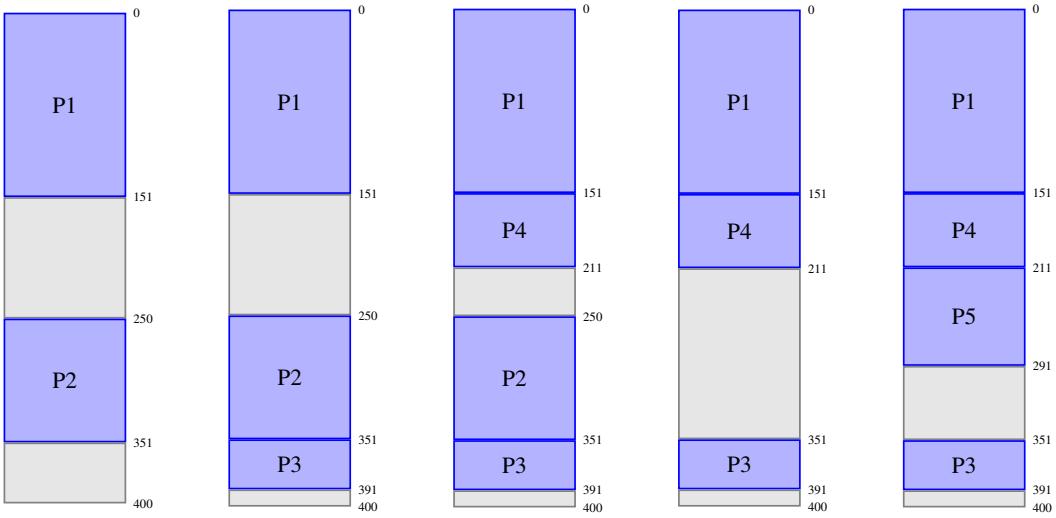
- P3 di dimensione 40Kb
- P4 di dimensione 60Kb
- P2 esce
- P5 di dimensione 80Kb




**SOLUZIONE**

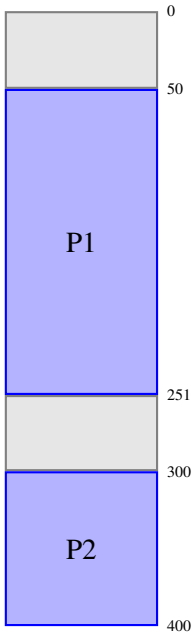


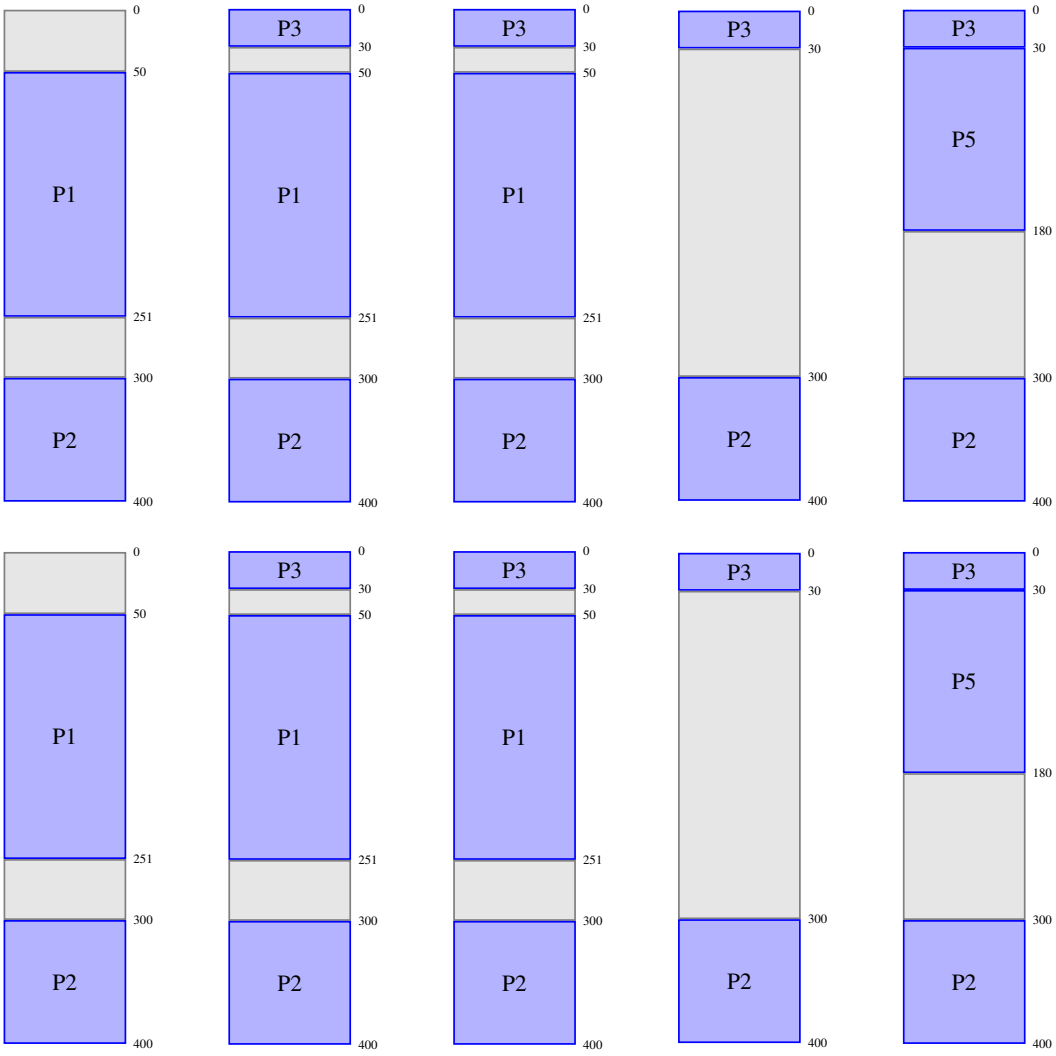

P1
P2
P1
P2
P3
P1
P4
P2
P3
P1
P4
P3
P1
P4
P5
P3
0
151
250
351
400
0
151
250
351
391
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0
151
211
291
351
391
400

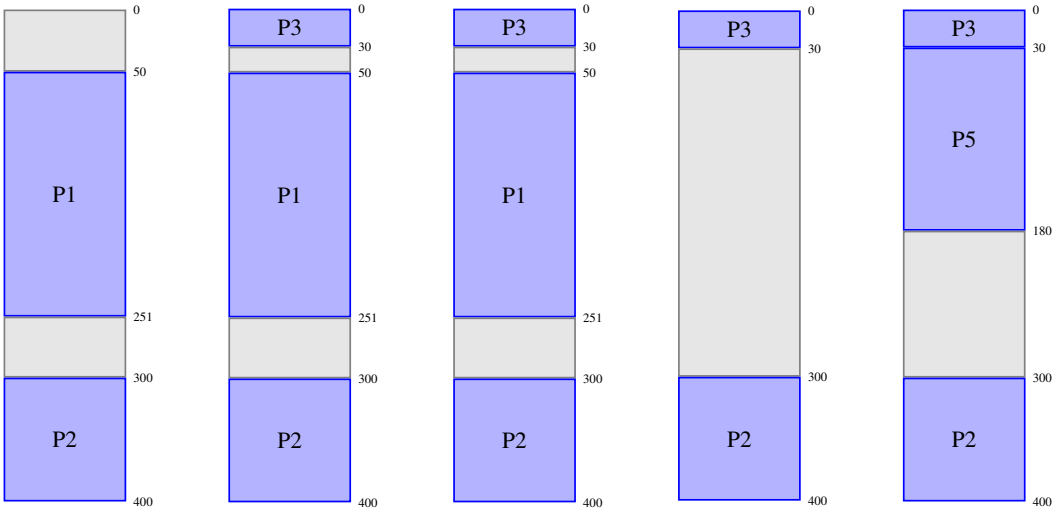
6

- P3 di dimensione 30Kb
- P4 di dimensione 50Kb
- P1 esce
- P5 di dimensione 150Kb



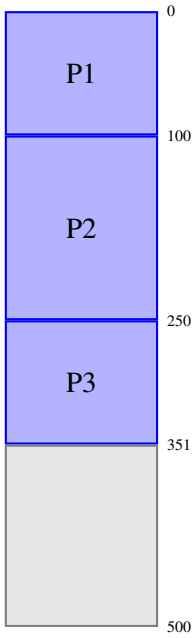

**SOLUZIONE**



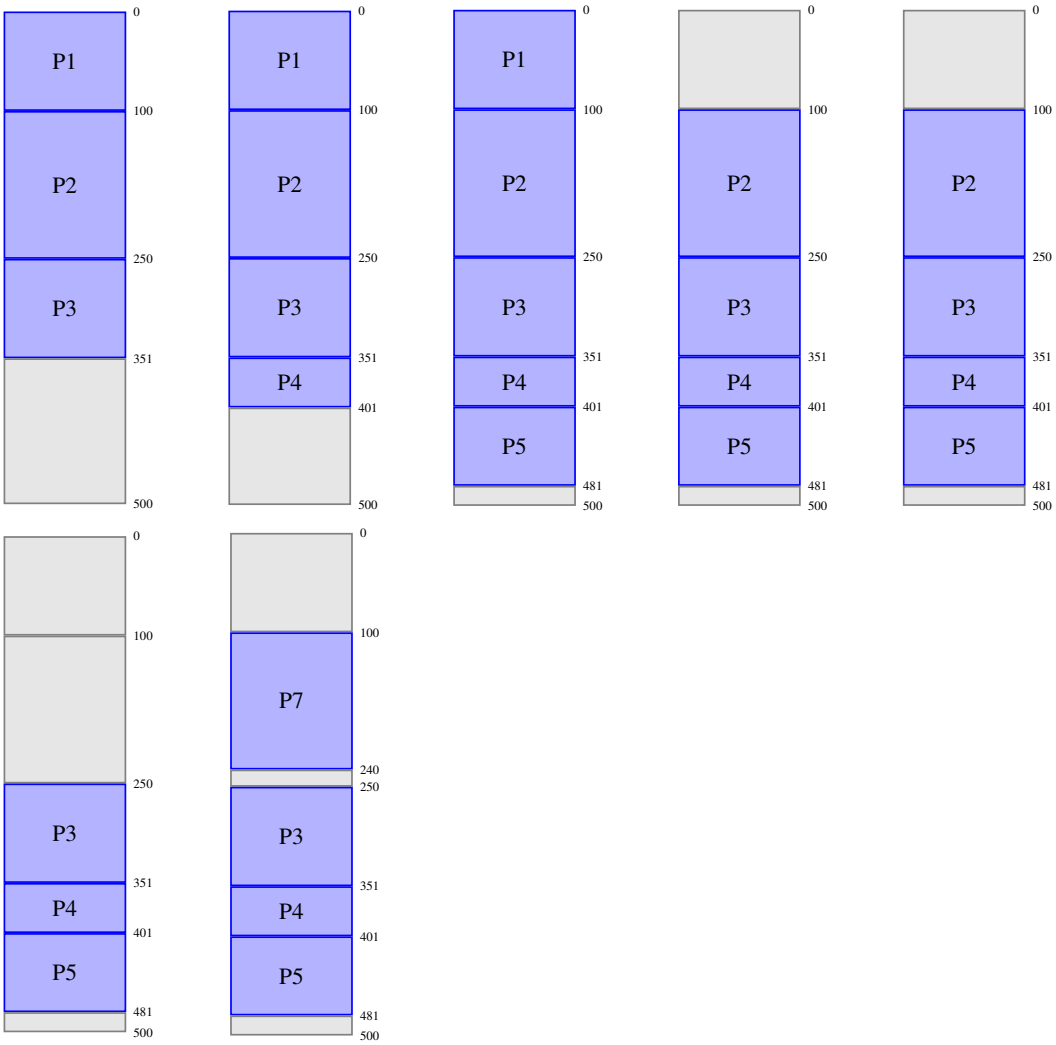

0
50
P1
251
300
P2
400
0
P3
30
50
P1
251
300
P2
400
0
P3
30
50
P1
251
300
P2
400
0
P3
30
300
P2
400
0
P3
30
P5
180
300
P2
400

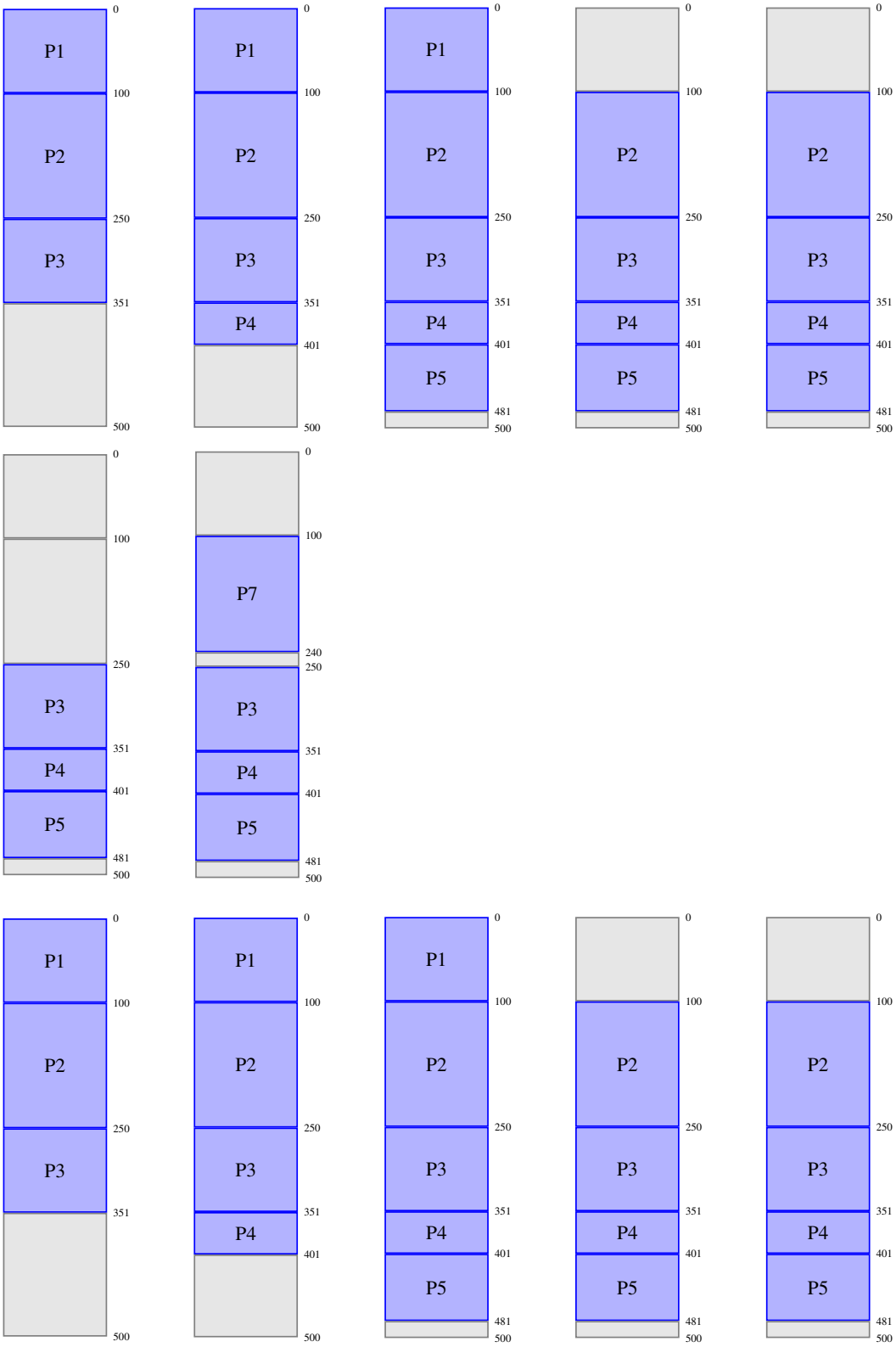
7

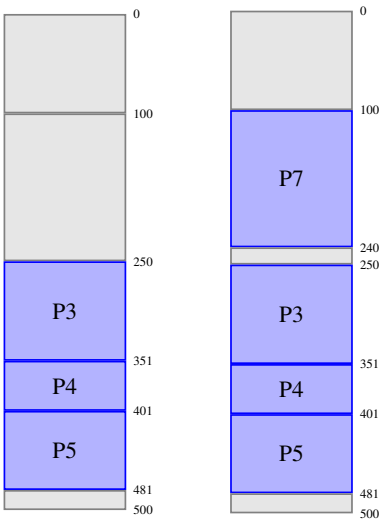
- P4 di dimensione 50Kb
- P5 di dimensione 80Kb
- P1 esce
- P6 di dimensione 120Kb
- P2 esce
- P7 di dimensione 140Kb




**SOLUZIONE**

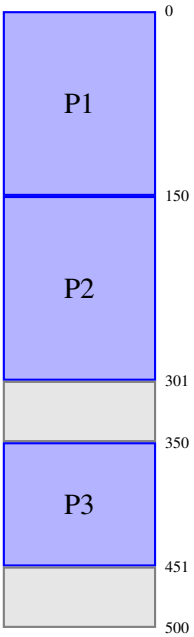



P1
P2
P3
P4
P5
P7
0
100
250
351
401
481
500
240

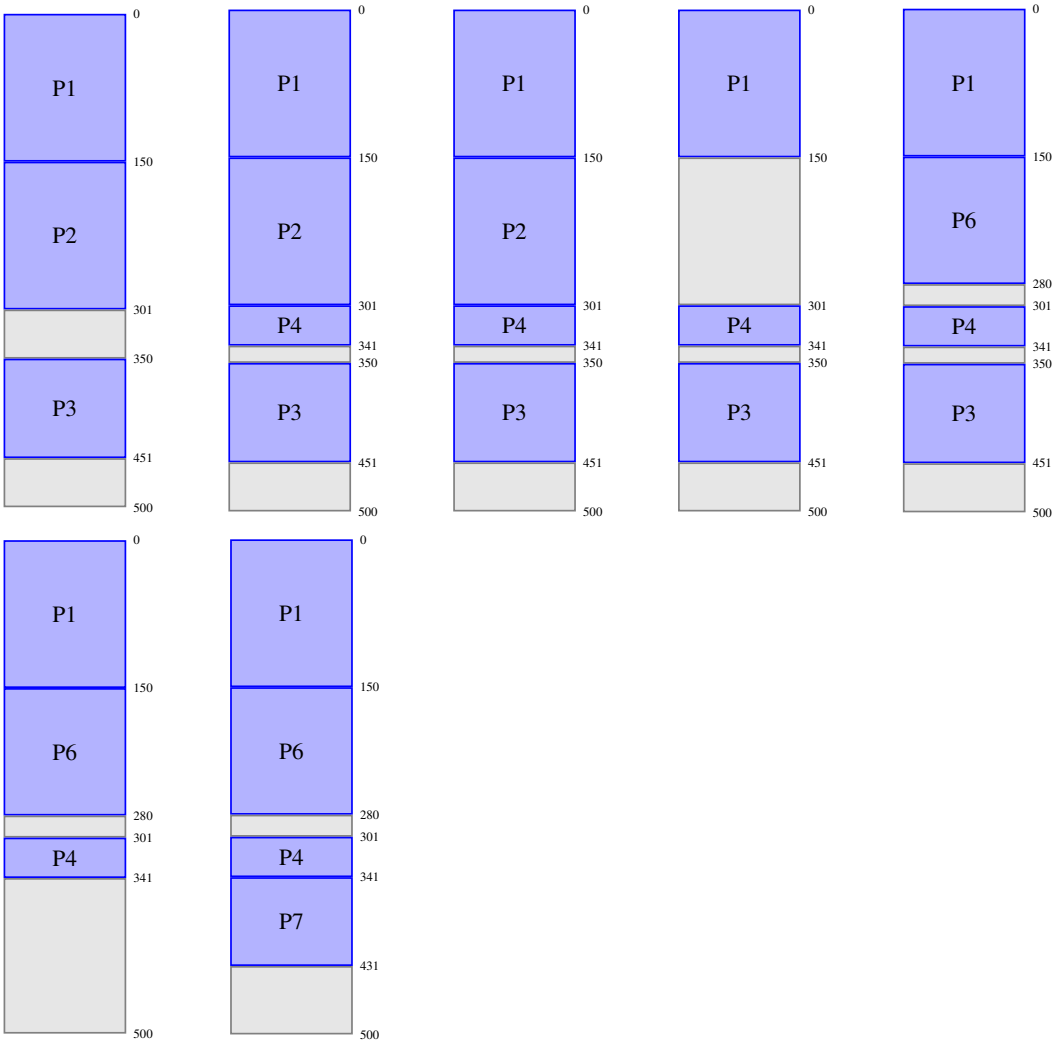

0
100
250
P3
351
P4
401
P5
481
500
0
100
P7
240
250
P3
351
P4
401
P5
481
500

8

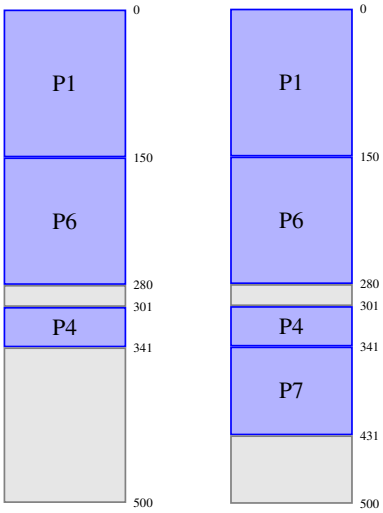
- P4 di dimensione 40Kb
- P5 di dimensione 60Kb
- P2 esce
- P6 di dimensione 130Kb
- P3 esce
- P7 di dimensione 90Kb




**SOLUZIONE**

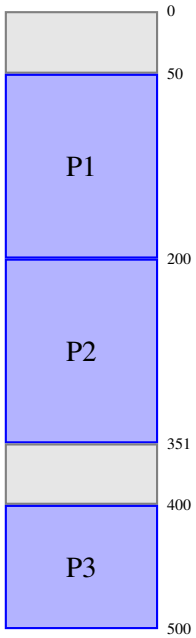



P1
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280
301
341
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431
451
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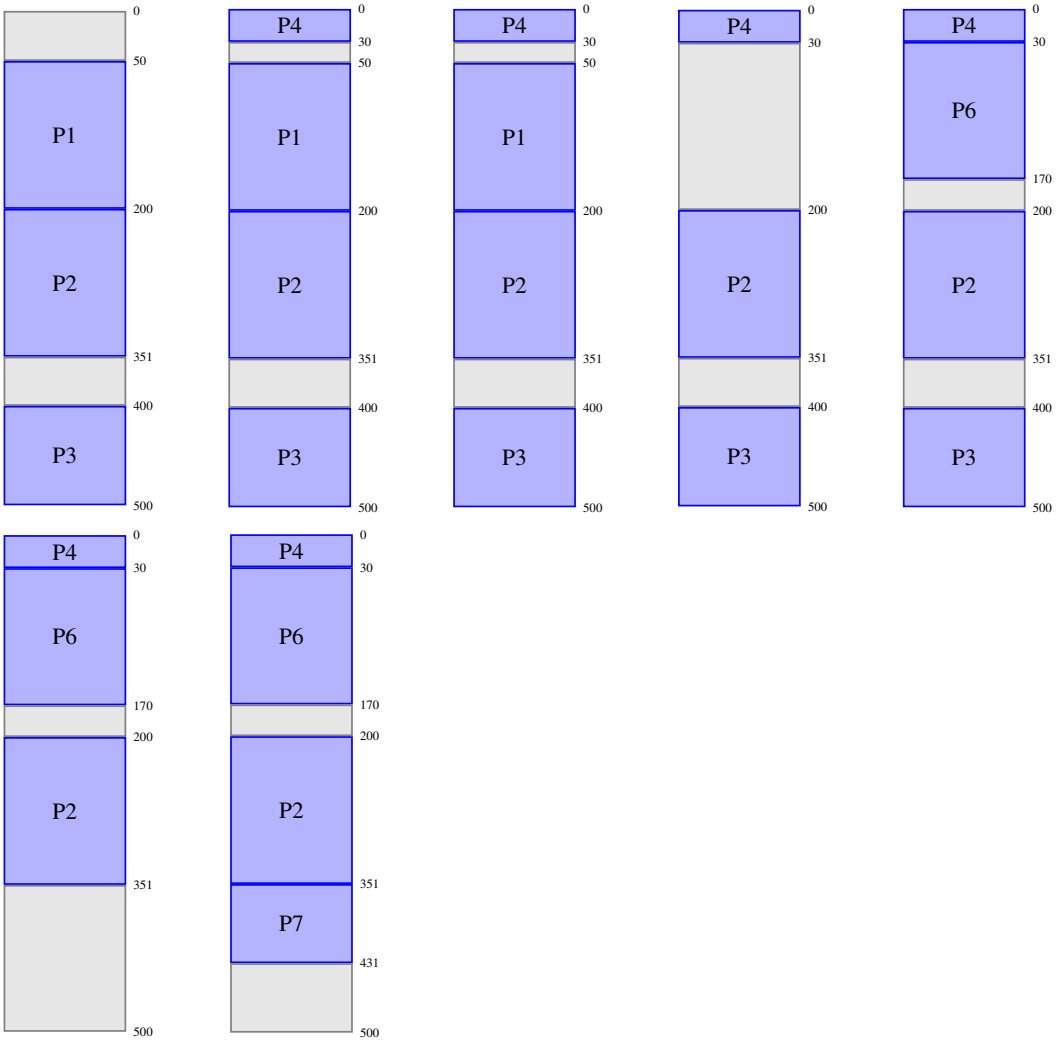

P1
P6
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P1
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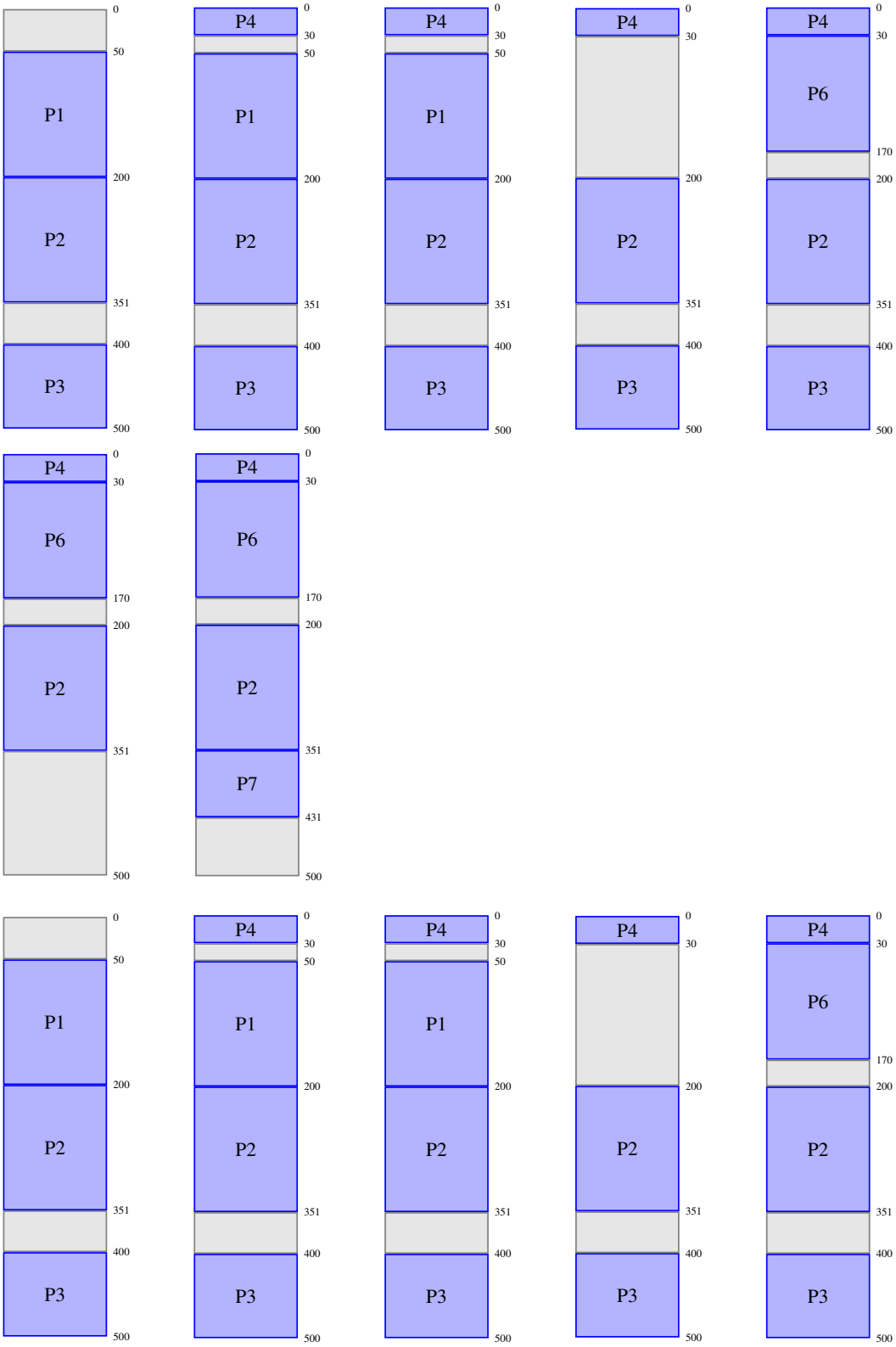
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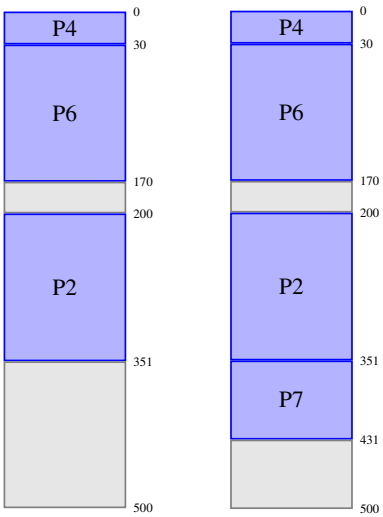
- P4 di dimensione 30Kb
- P5 di dimensione 50Kb
- P1 esce
- P6 di dimensione 140Kb
- P3 esce
- P7 di dimensione 80Kb




**SOLUZIONE**

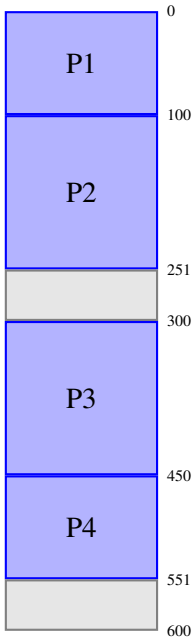



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351
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P3
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P4
30
P6
170
P7
431

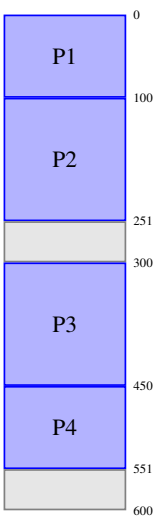

P4
P6
P2
P4
P6
P2
P7
0
30
170
200
351
500
0
30
170
200
351
431
500

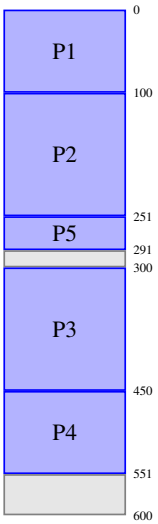
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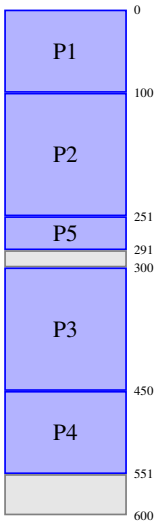
- P5 di dimensione 40Kb
- P6 di dimensione 60Kb
- P1 esce
- P7 di dimensione 120Kb
- P3 esce
- P8 di dimensione 140Kb

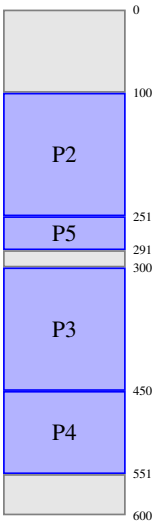


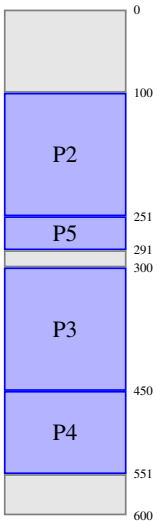

**SOLUZIONE**

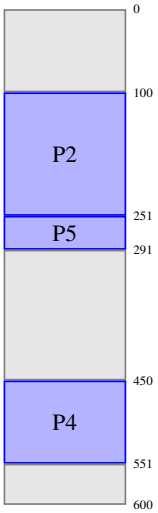


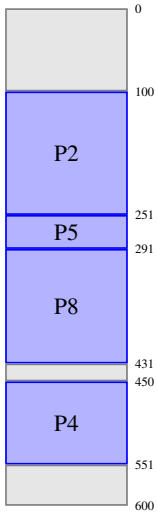



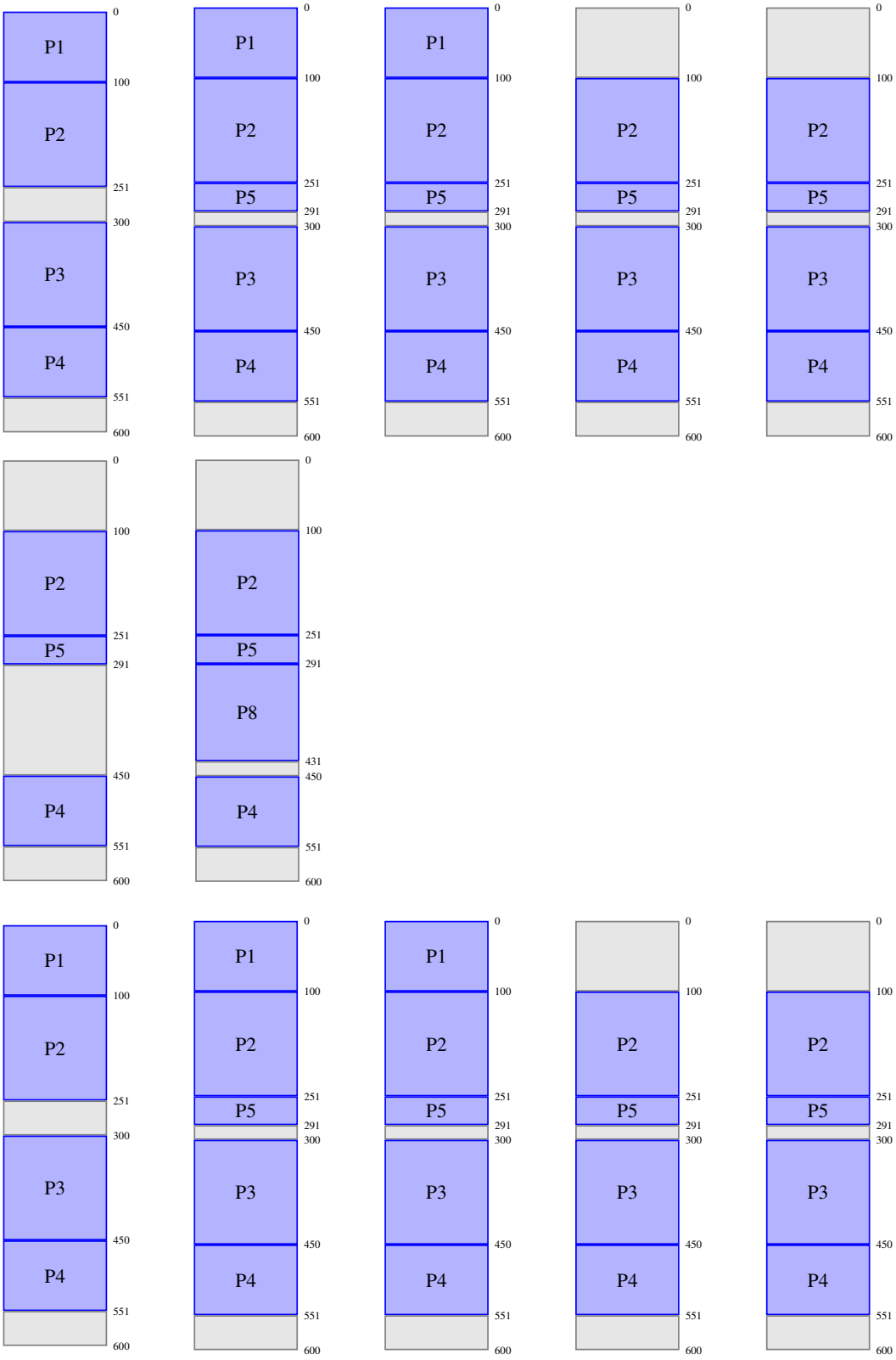



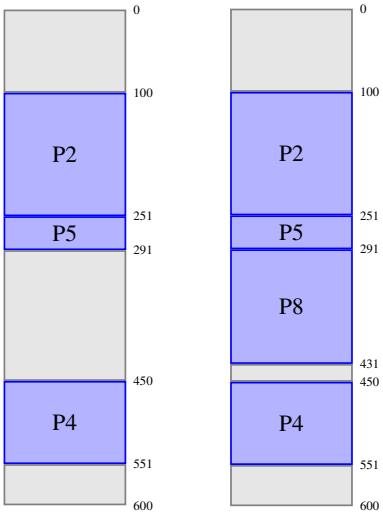



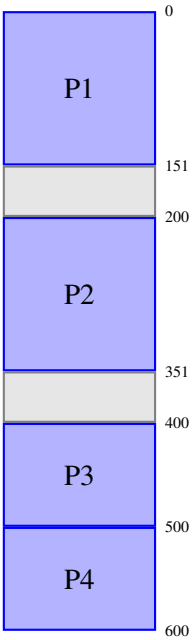




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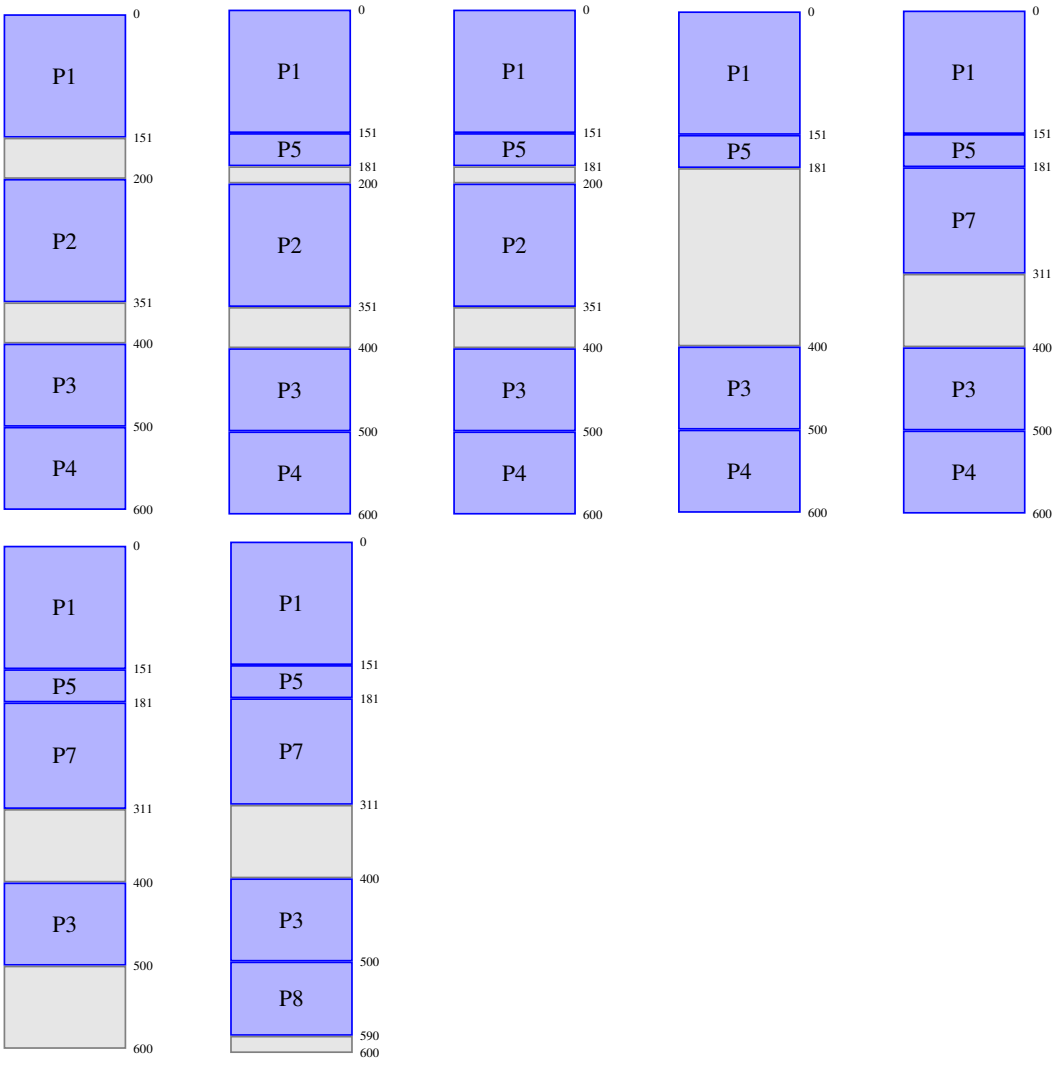

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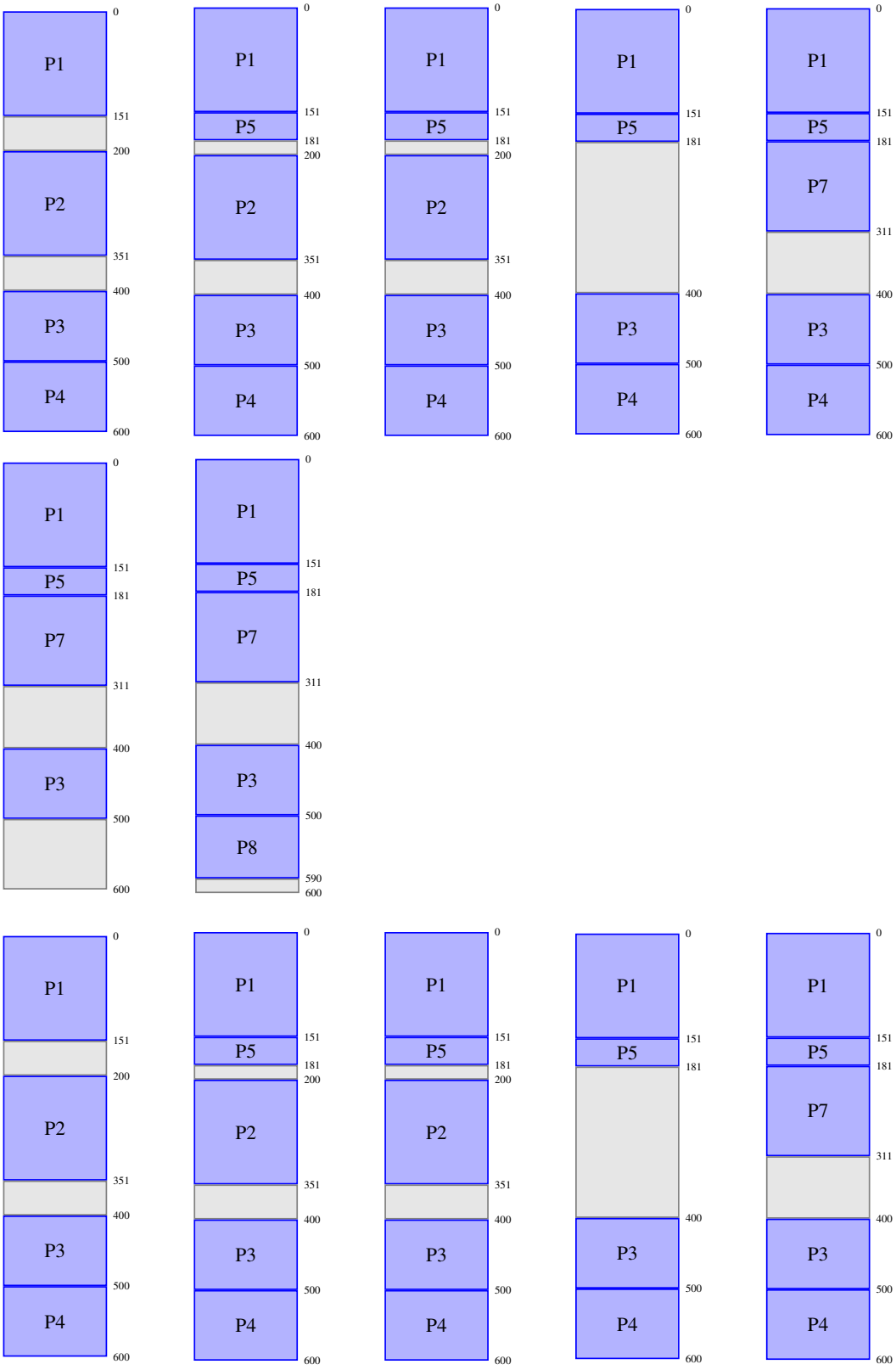
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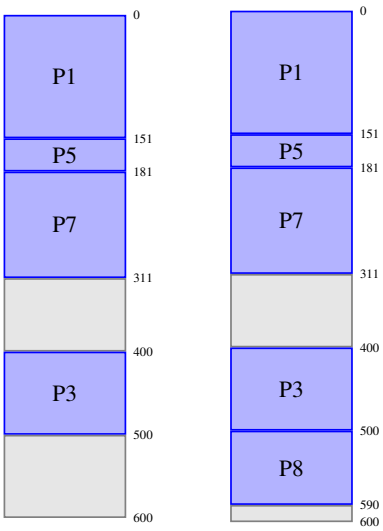
- P5 di dimensione 30Kb
- P6 di dimensione 50Kb
- P2 esce
- P7 di dimensione 130Kb
- P4 esce
- P8 di dimensione 90Kb




**SOLUZIONE**

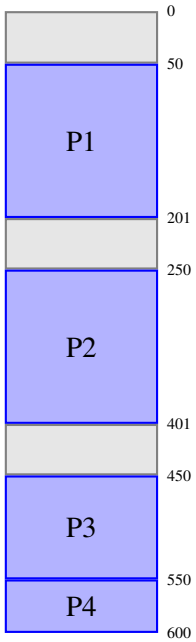



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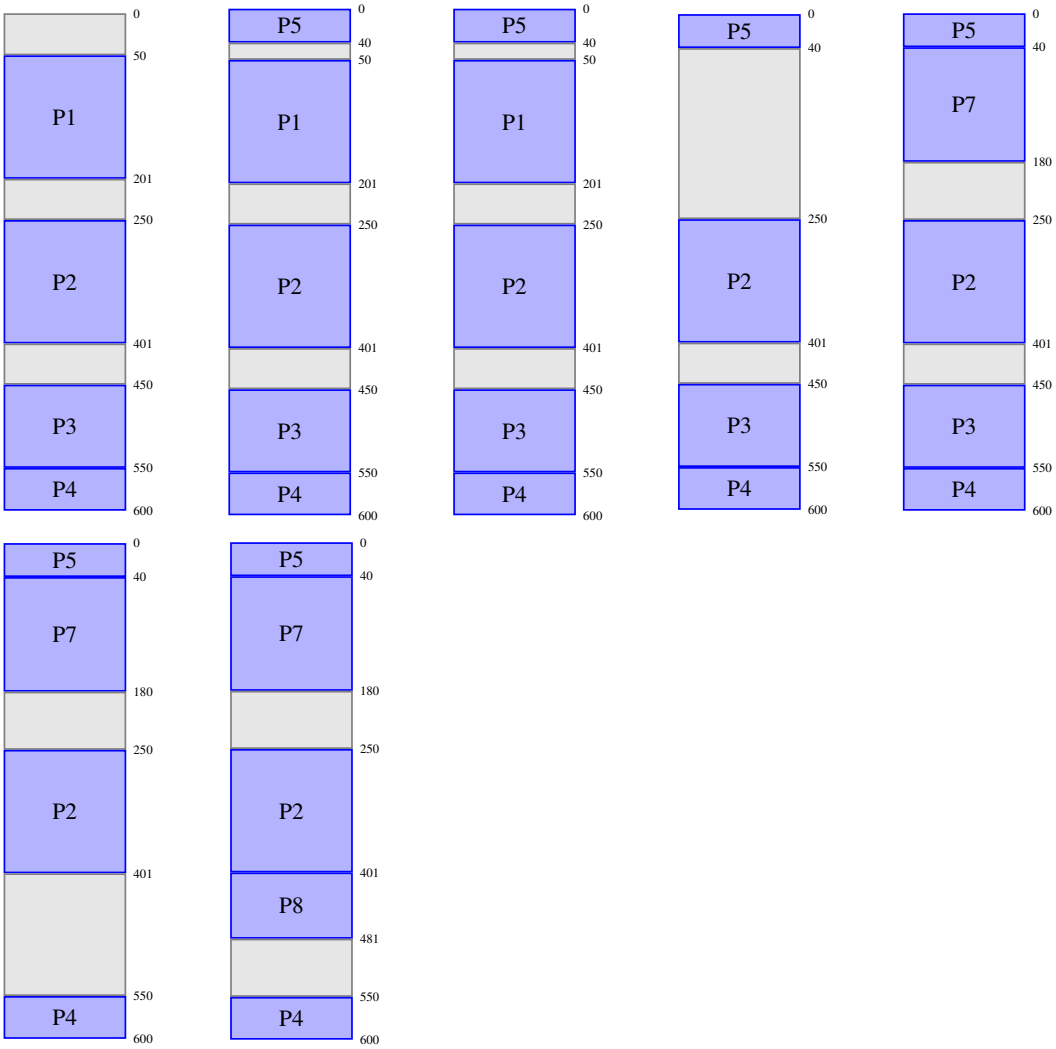

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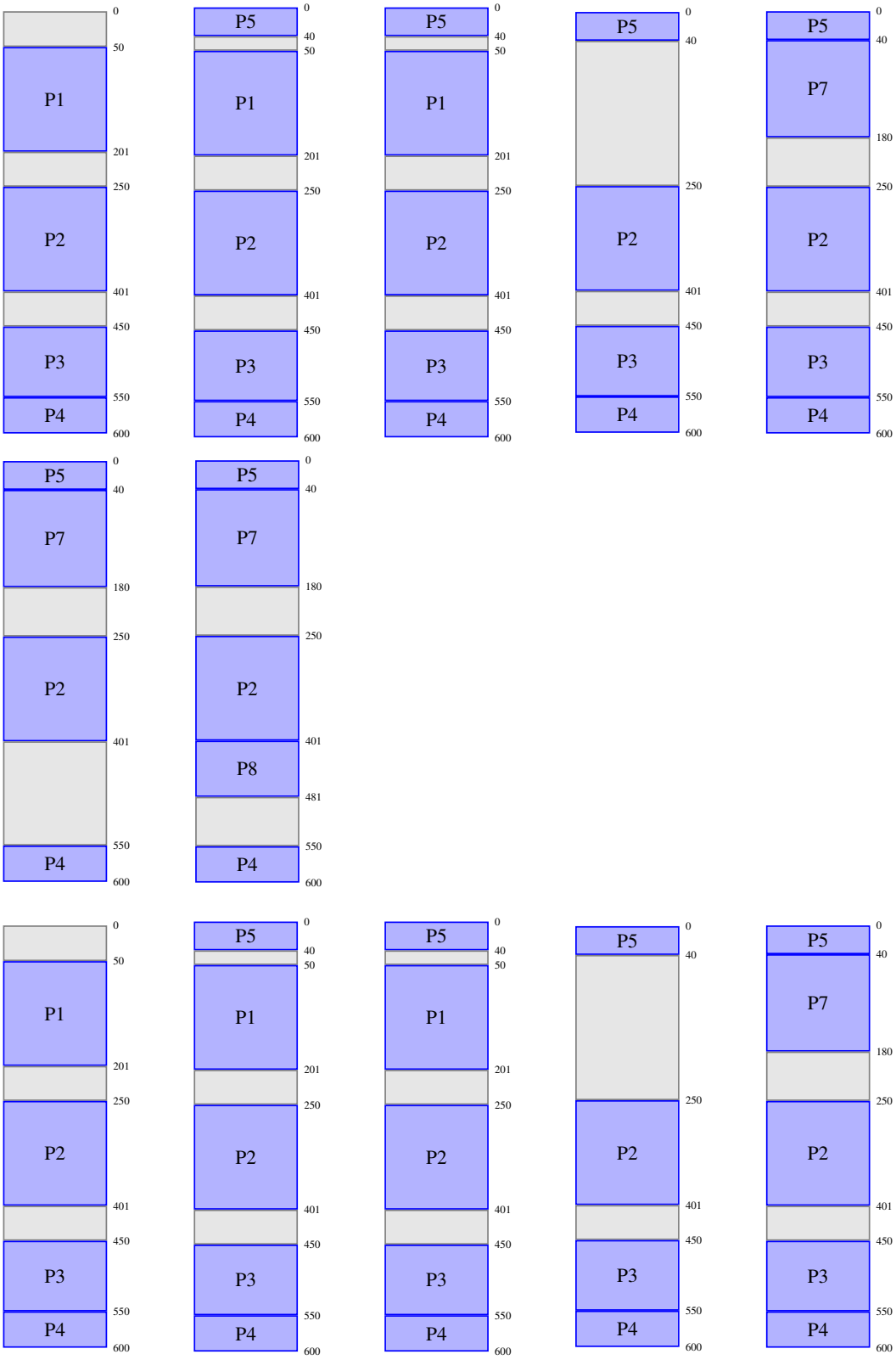
**12**

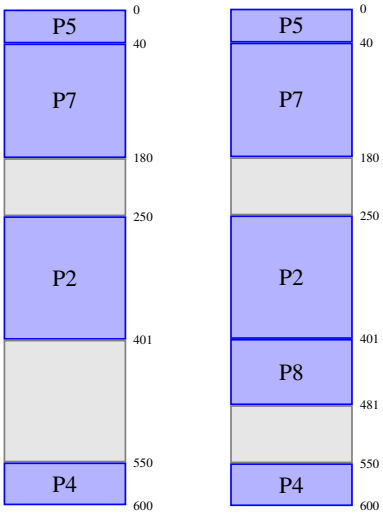
- P5 di dimensione 40Kb
- P6 di dimensione 60Kb
- P1 esce
- P7 di dimensione 140Kb
- P3 esce
- P8 di dimensione 80Kb




**SOLUZIONE**

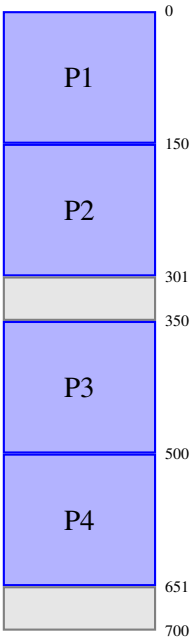



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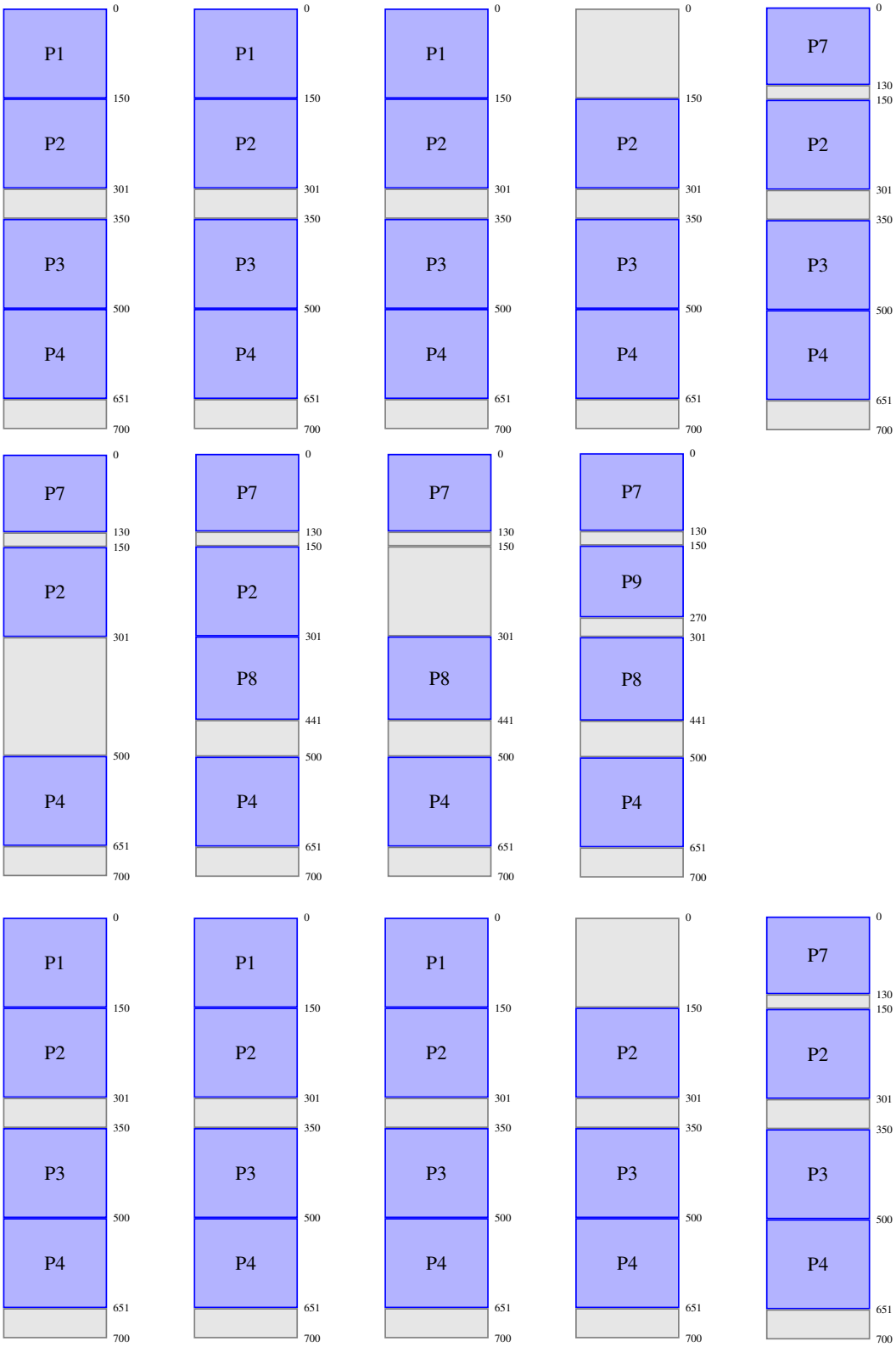
13

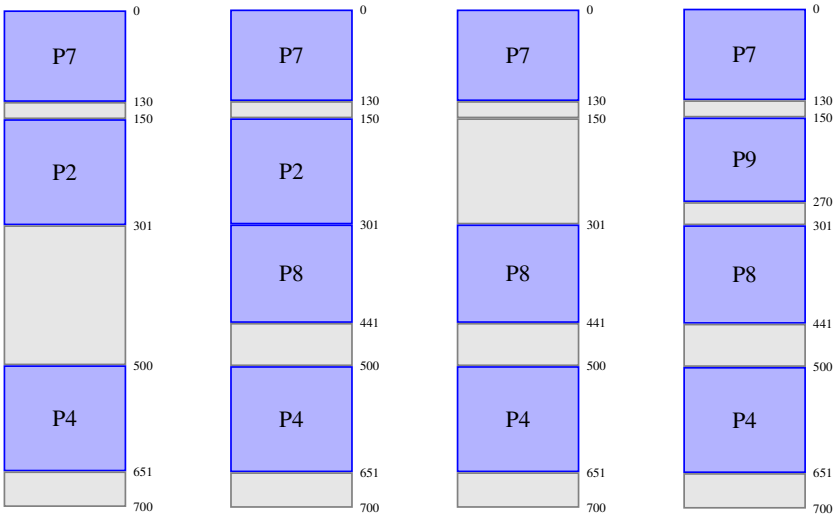
- P5 di dimensione 50Kb
- P6 di dimensione 70Kb
- P1 esce
- P7 di dimensione 130Kb
- P3 esce
- P8 di dimensione 140Kb
- P2 esce
- P9 di dimensione 120Kb




**SOLUZIONE**

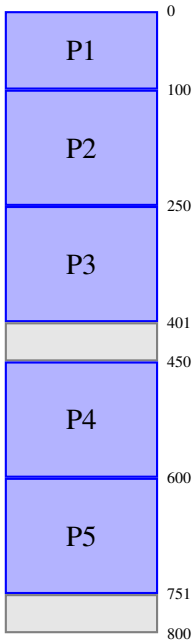



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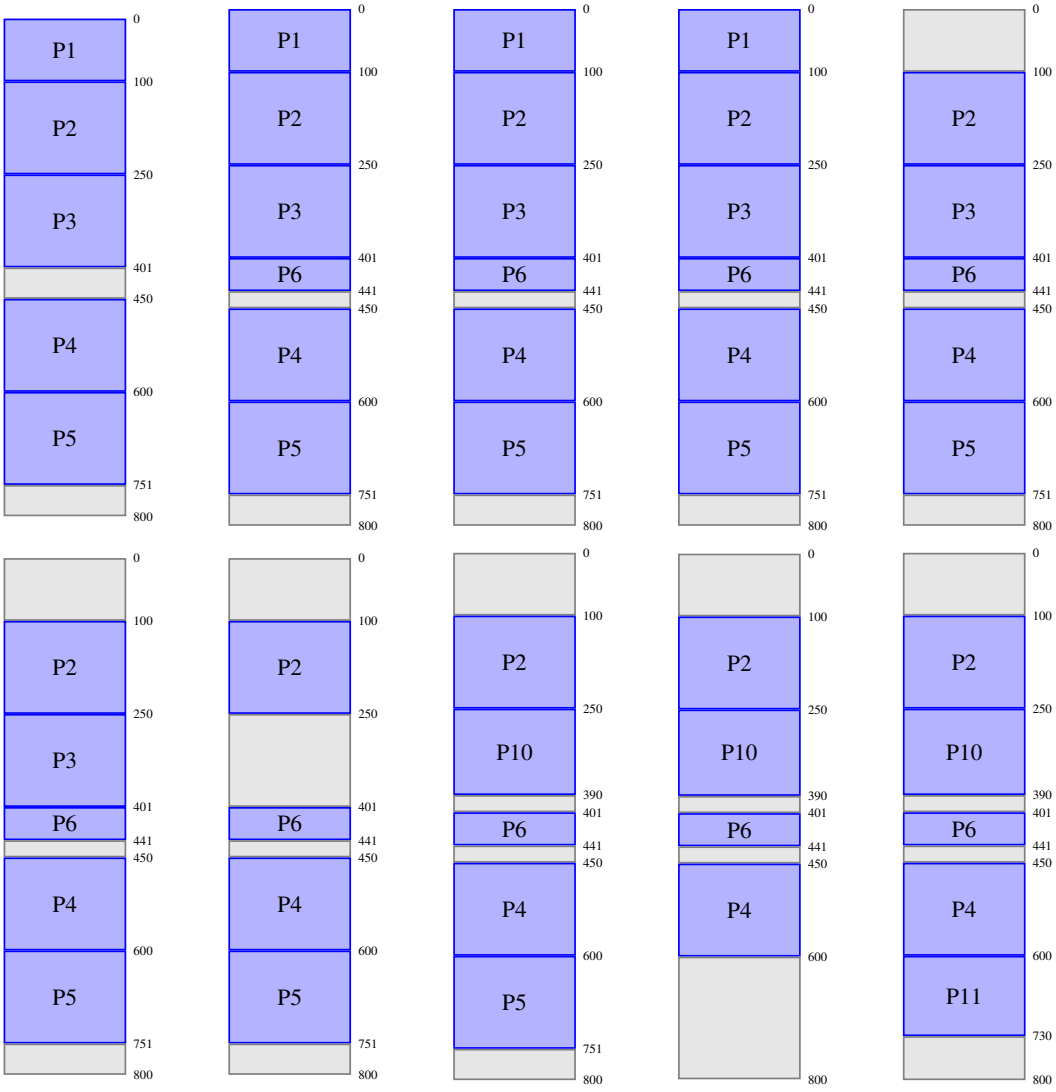

P7
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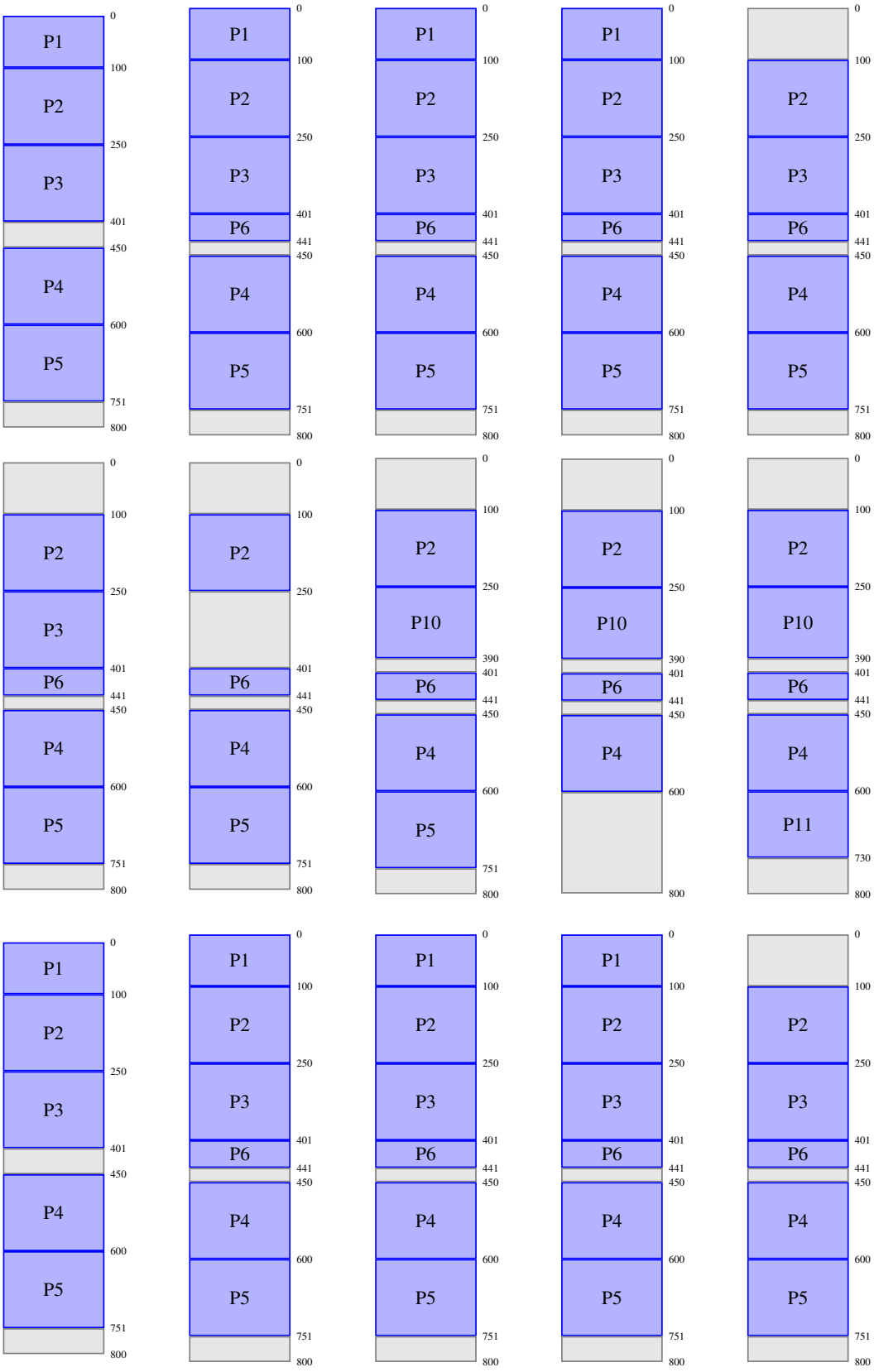
14

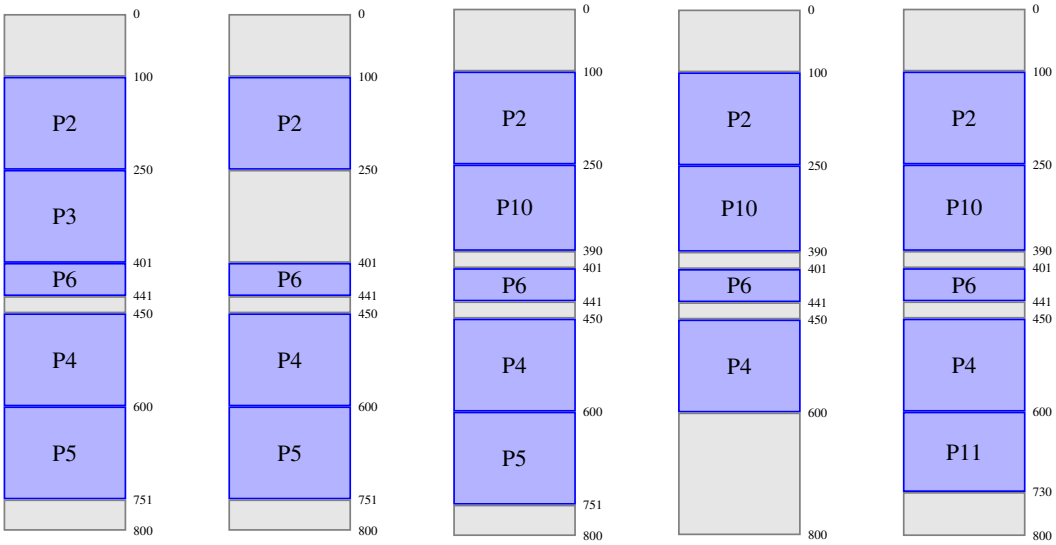
- P6 di dimensione 40Kb
- P7 di dimensione 60Kb
- P8 di dimensione 50Kb
- P1 esce
- P9 di dimensione 120Kb
- P3 esce
- P10 di dimensione 140Kb
- P5 esce
- P11 di dimensione 130Kb




**SOLUZIONE**

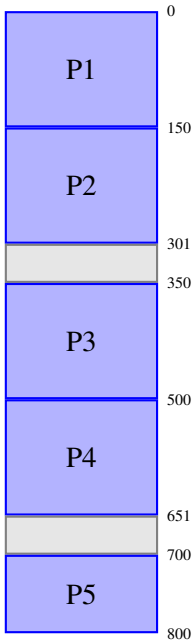



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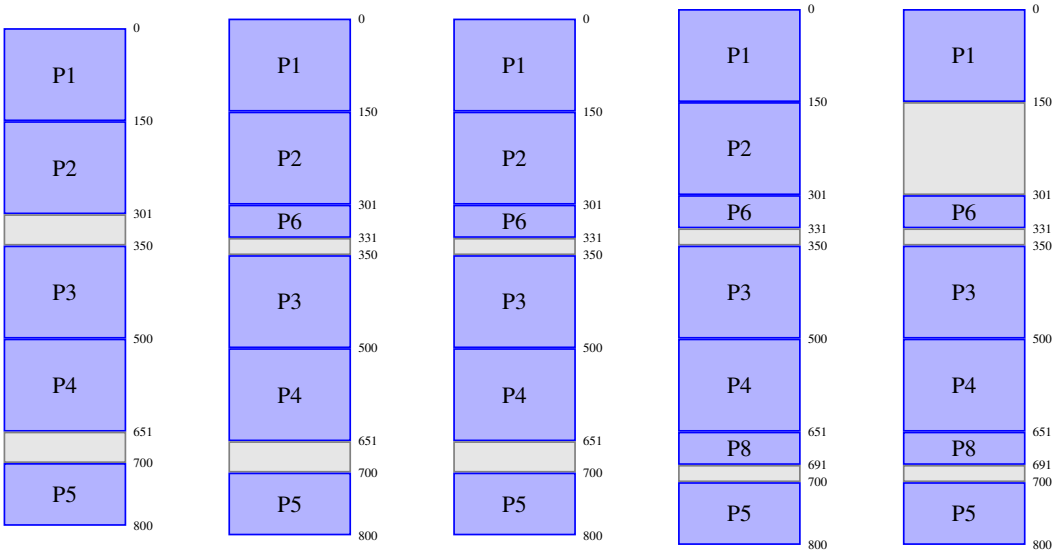

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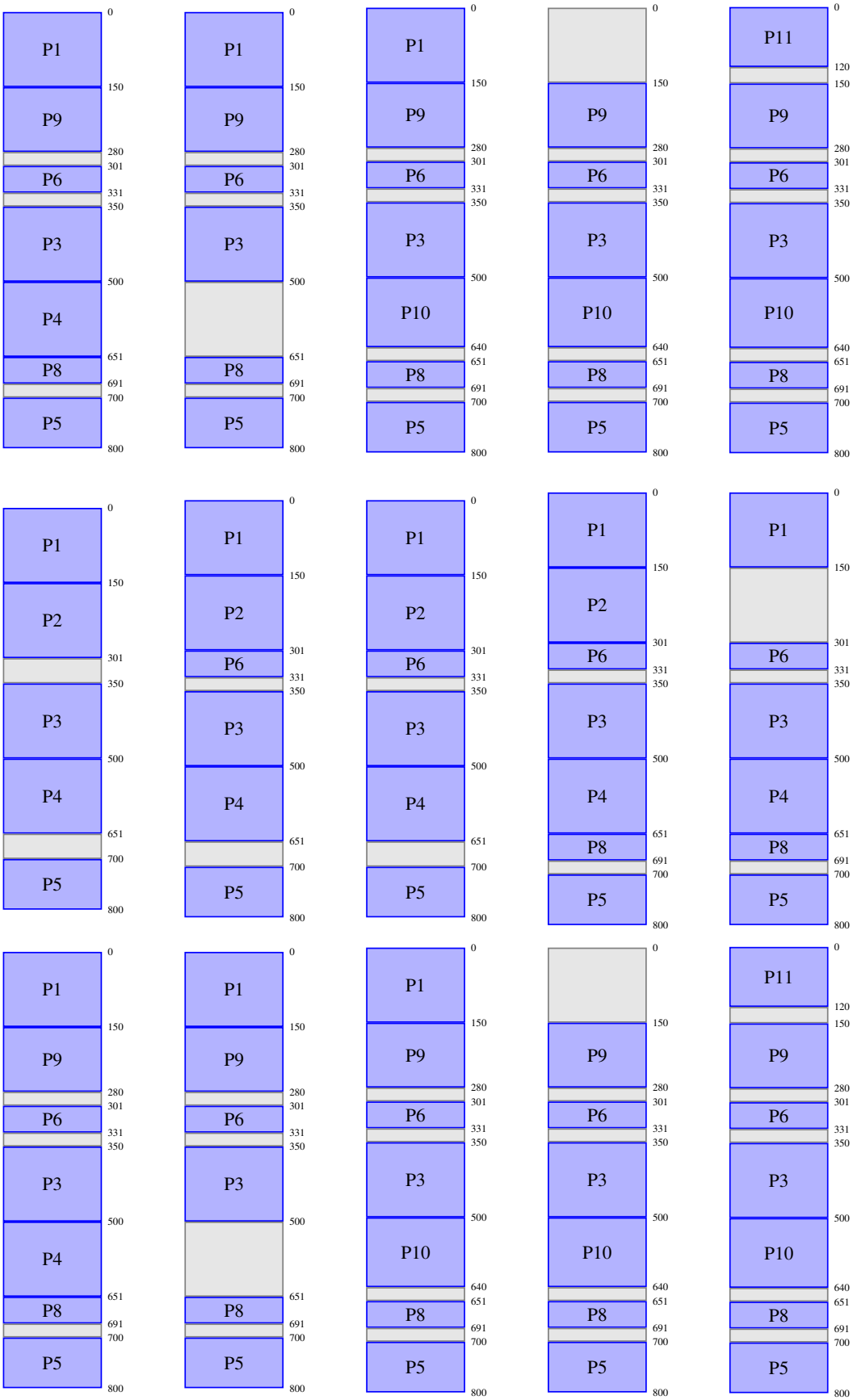
15

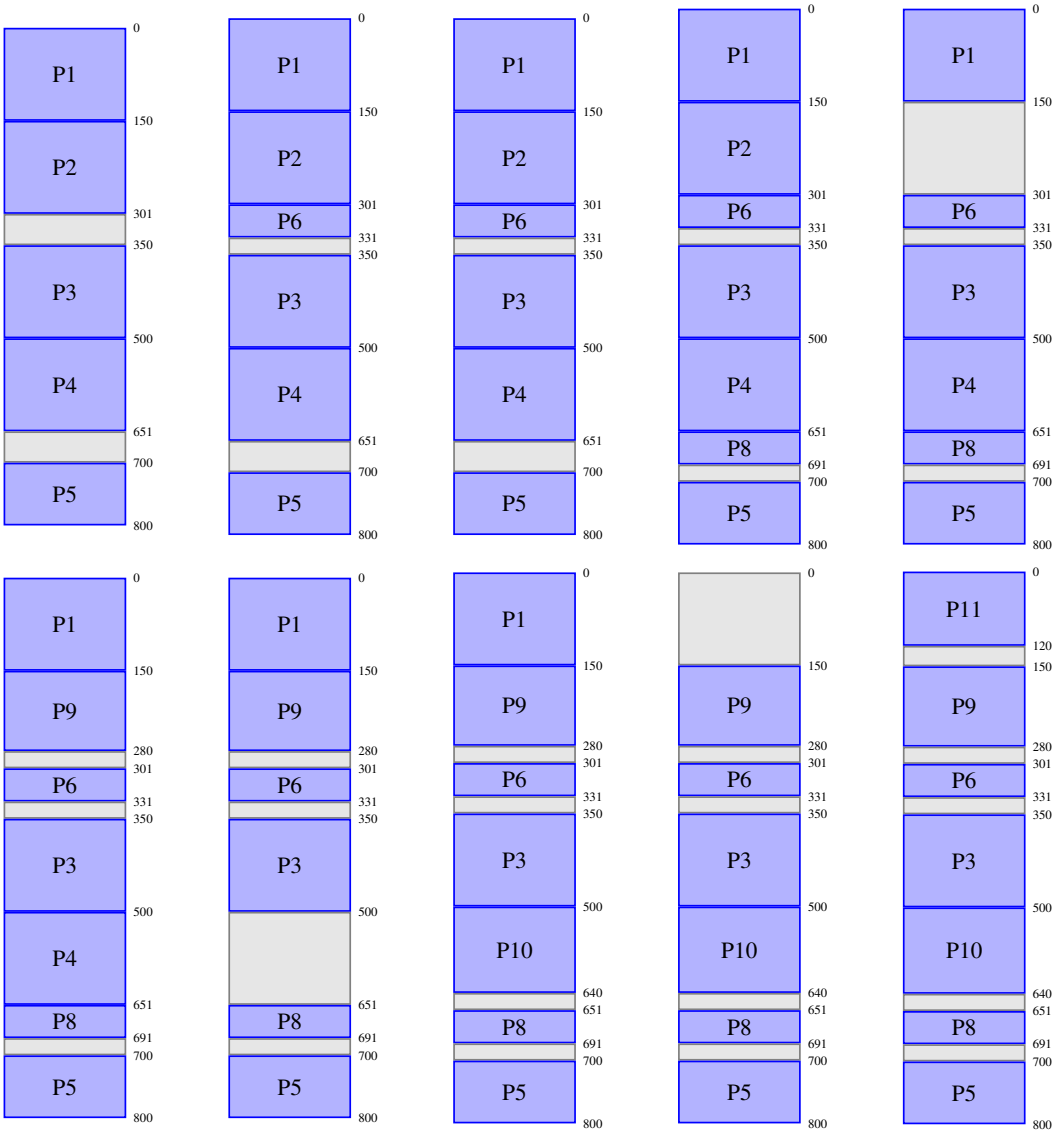
- •P6 di dimensione 30Kb
- •P7 di dimensione 50Kb
- •P8 di dimensione 40Kb
- •P2 esce
- •P9 di dimensione 130Kb
- •P4 esce
- •P10 di dimensione 140Kb
- •P1 esce
- •P11 di dimensione 120Kb




**SOLUZIONE**

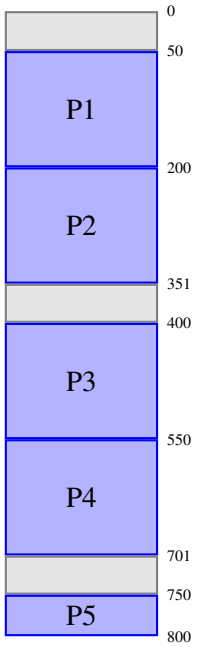



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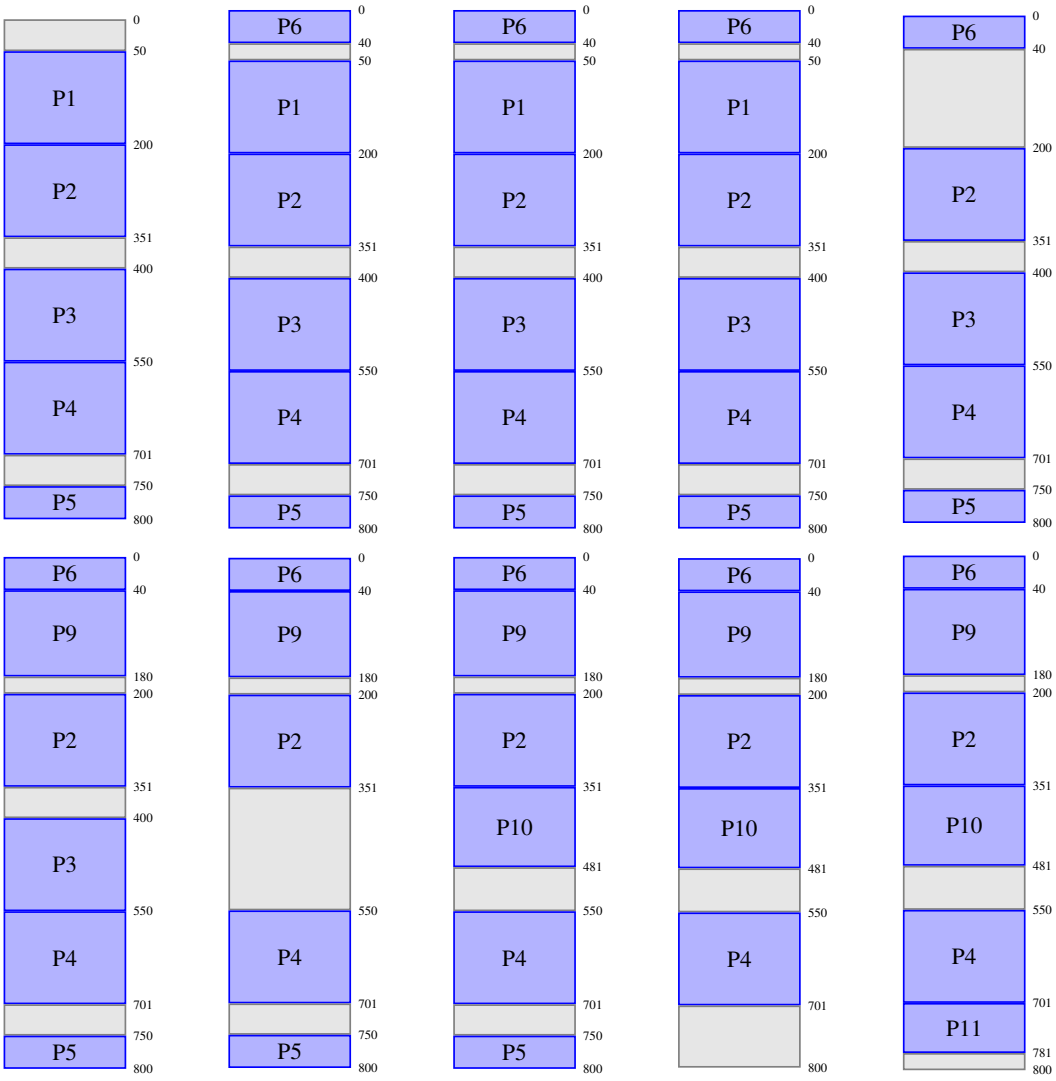

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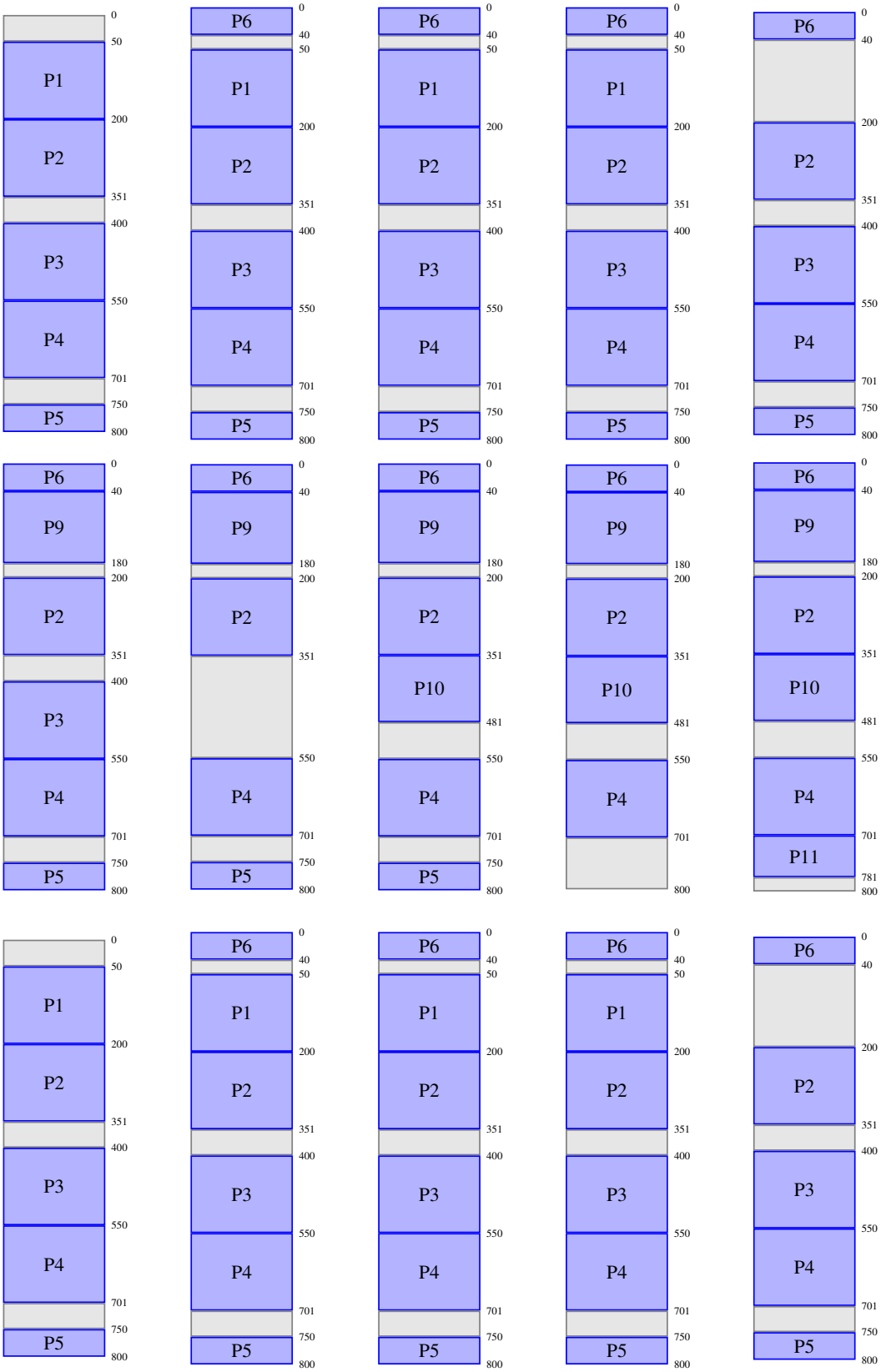
16

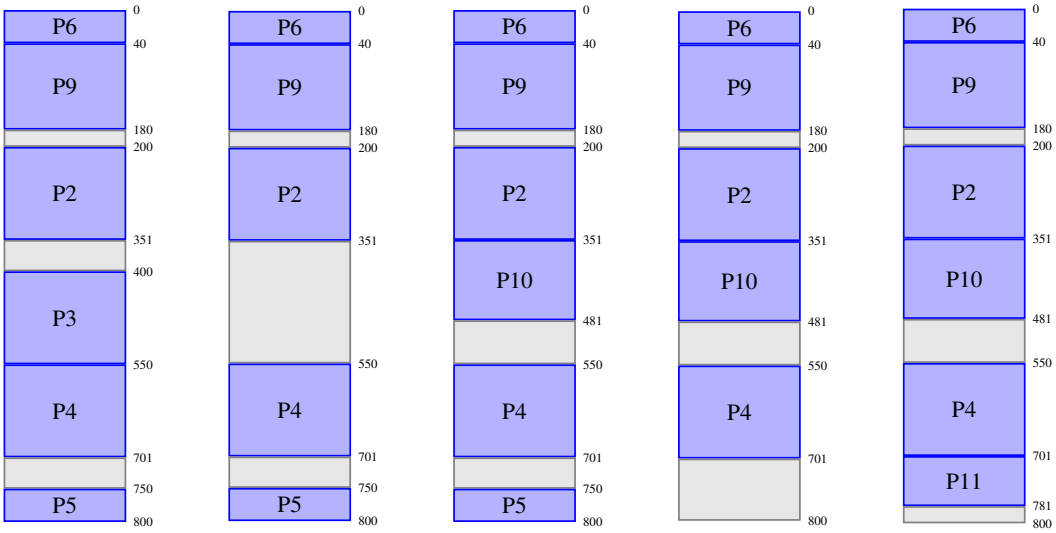
- P6 di dimensione 40Kb
- P7 di dimensione 60Kb
- P8 di dimensione 50Kb
- P1 esce
- P9 di dimensione 140Kb
- P3 esce
- P10 di dimensione 130Kb
- P5 esce
- P11 di dimensione 80Kb




**SOLUZIONE**

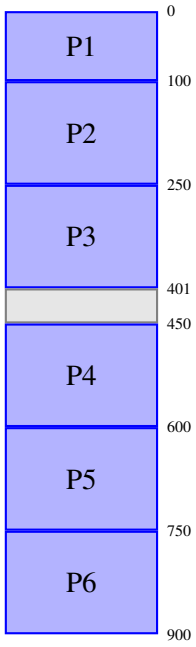



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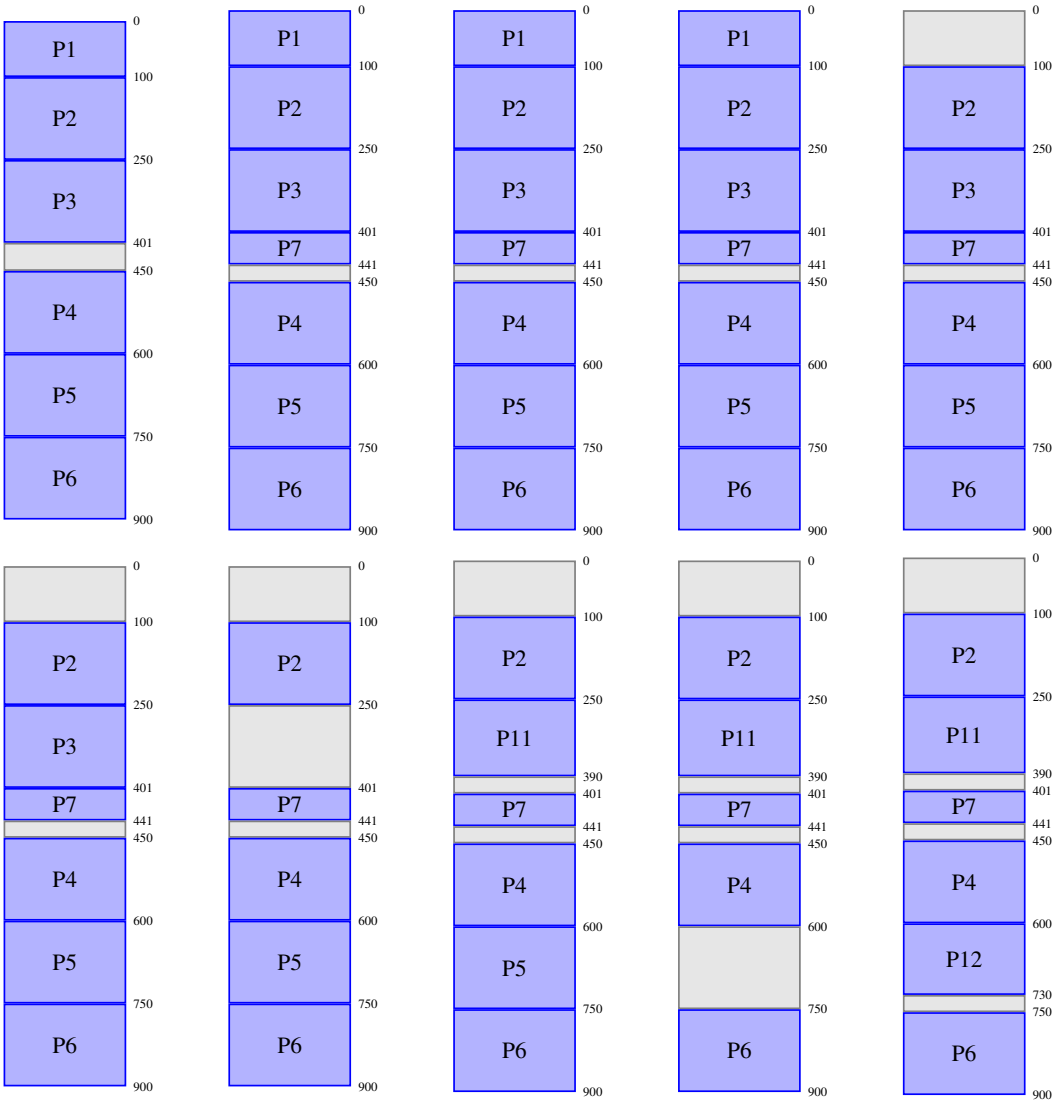

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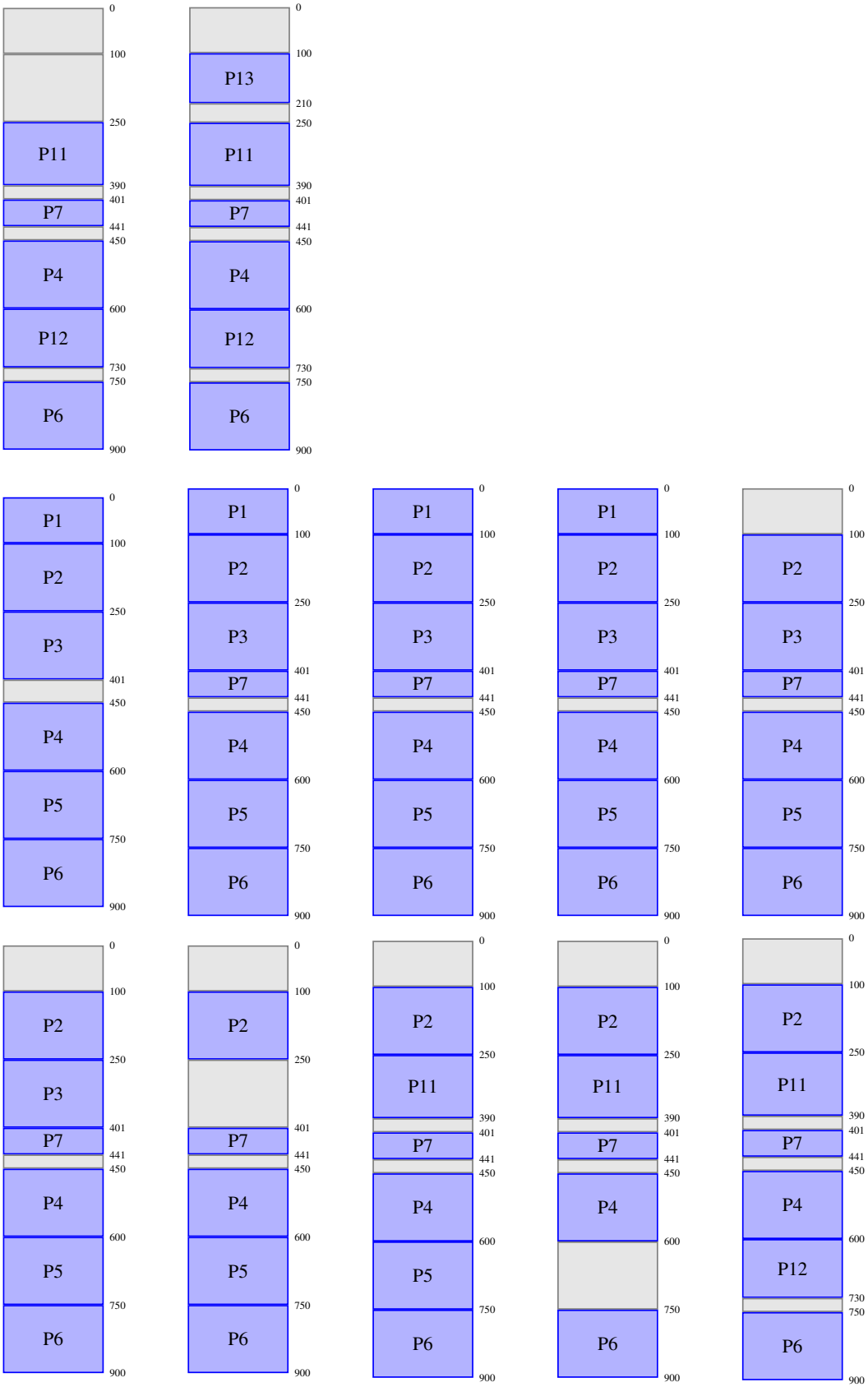
17

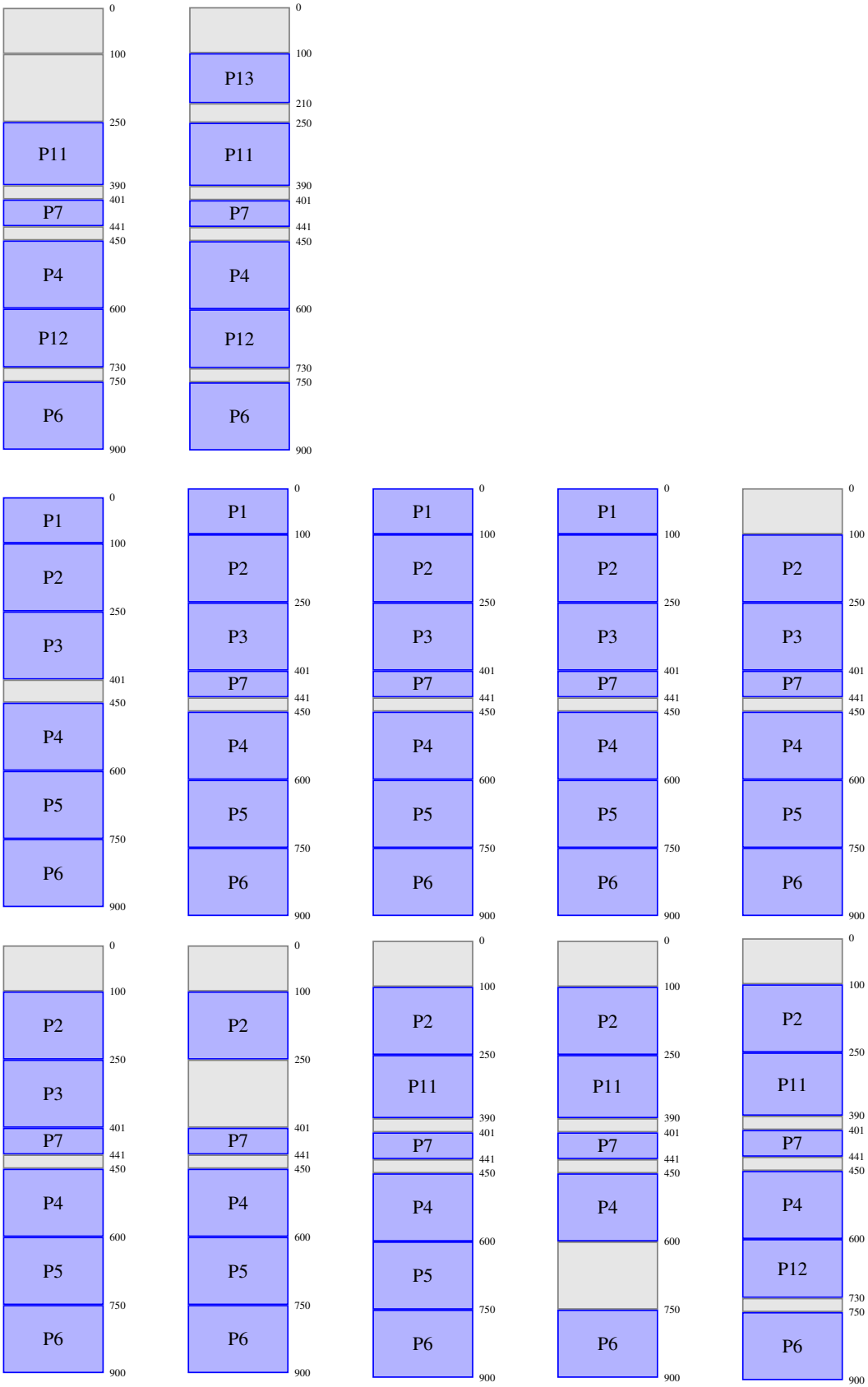
- •P7 di dimensione 40Kb
- •P8 di dimensione 60Kb
- •P9 di dimensione 50Kb
- •P1 esce
- •P10 di dimensione 120Kb
- •P3 esce
- •P11 di dimensione 140Kb
- •P5 esce
- •P12 di dimensione 130Kb
- •P2 esce
- •P13 di dimensione 110Kb

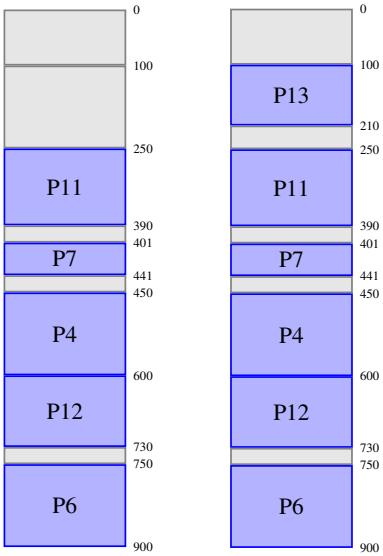



**SOLUZIONE**



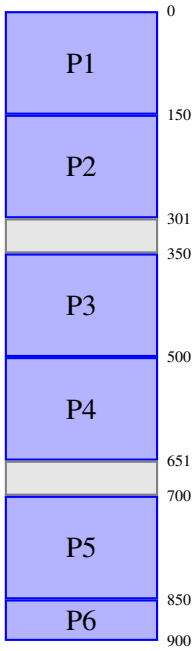

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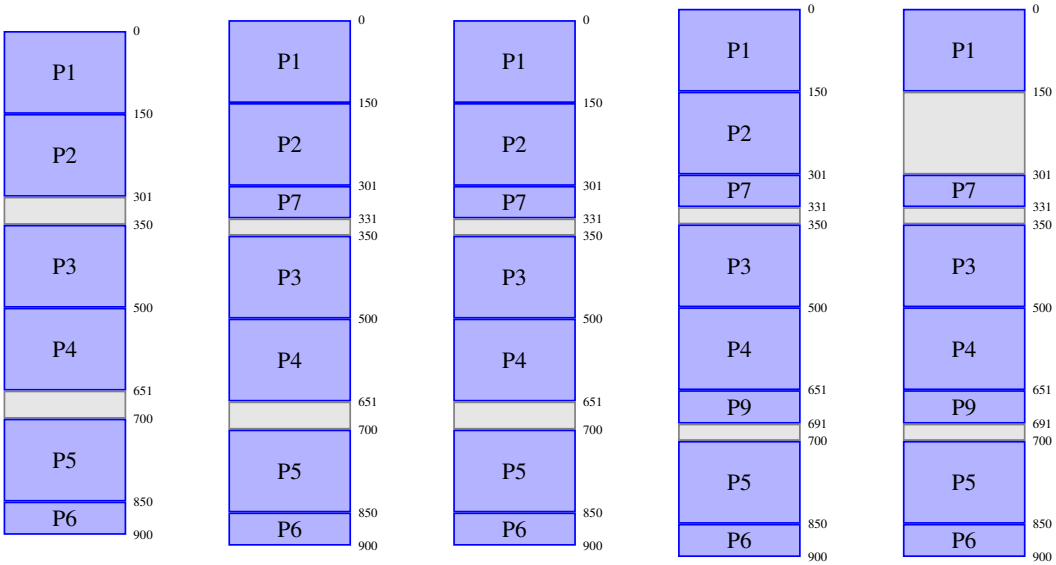

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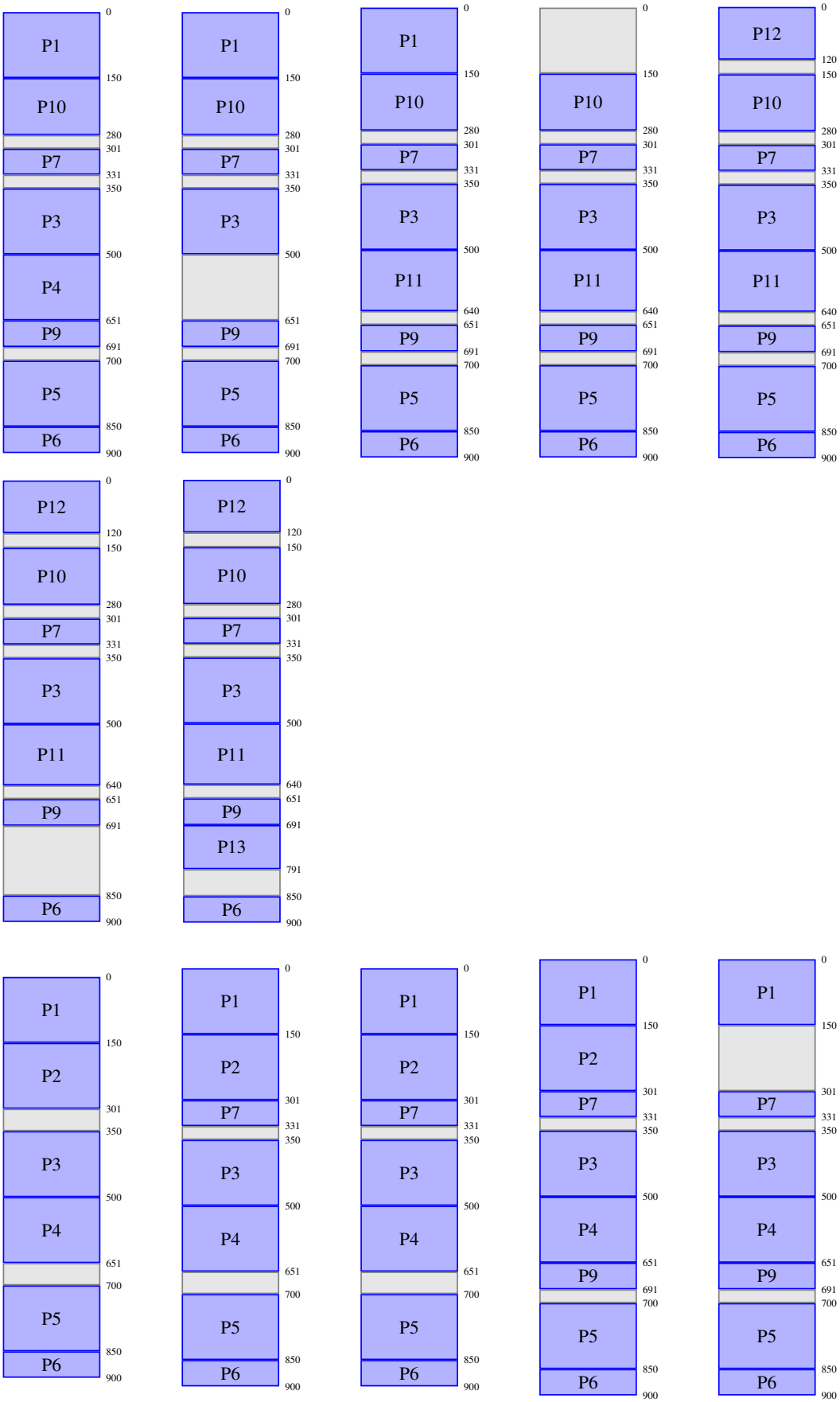
**18**

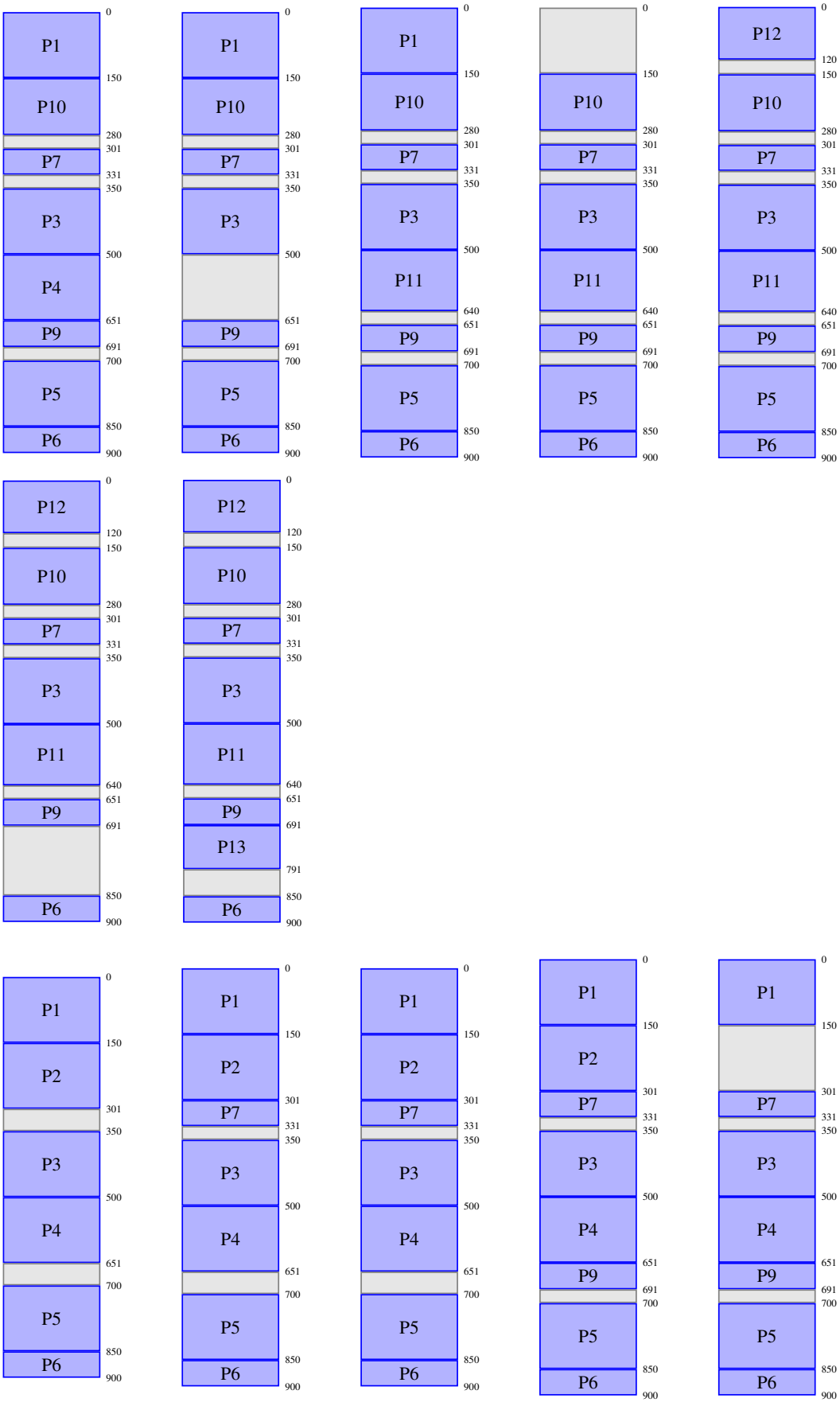
- P7 di dimensione 30Kb
- P8 di dimensione 50Kb
- P9 di dimensione 40Kb
- P2 esce
- P10 di dimensione 130Kb
- P4 esce
- P11 di dimensione 140Kb
- P1 esce
- P12 di dimensione 120Kb
- P5 esce
- P13 di dimensione 100Kb

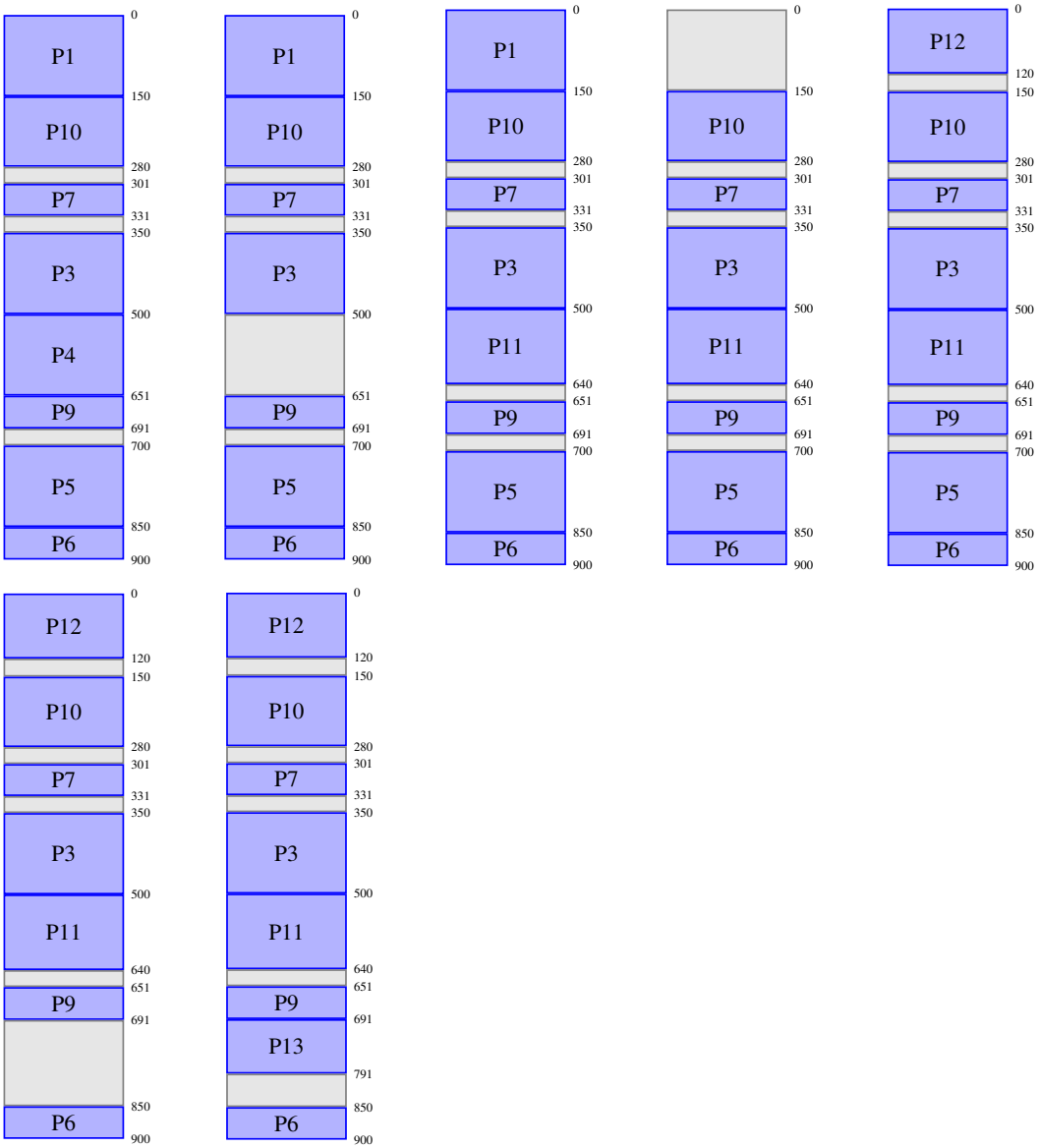



**SOLUZIONE**



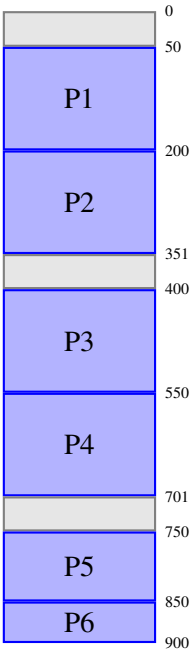

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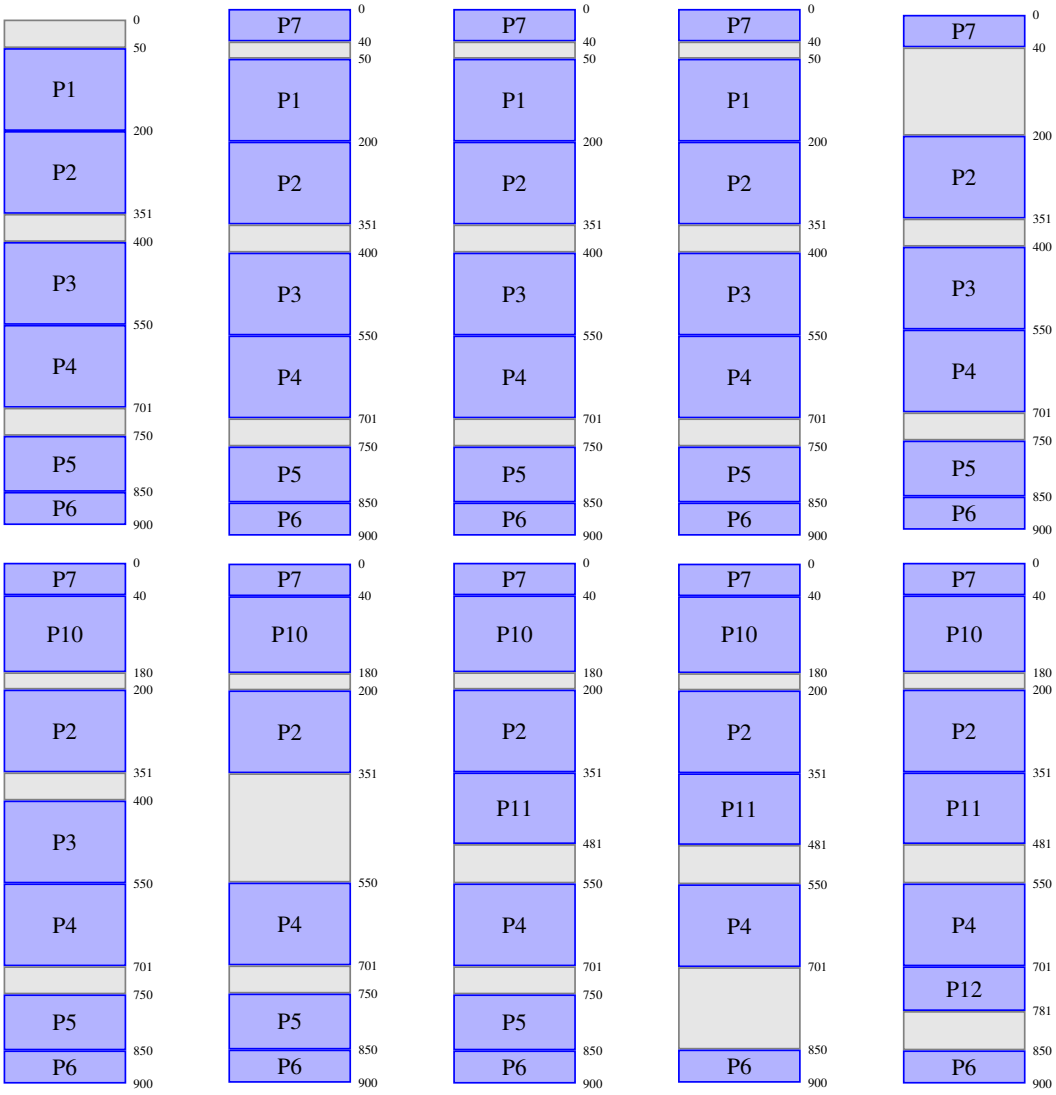

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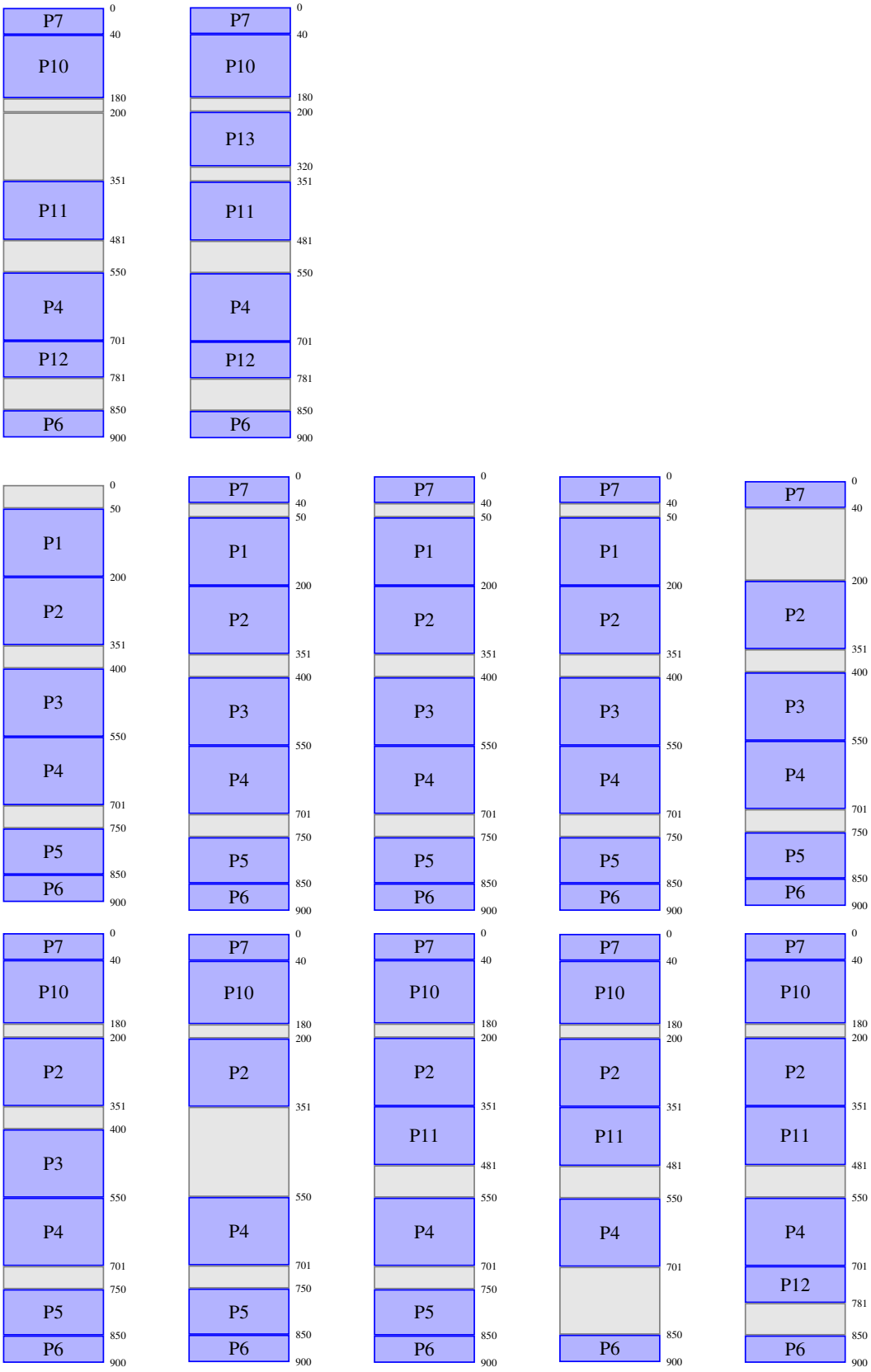
**19**

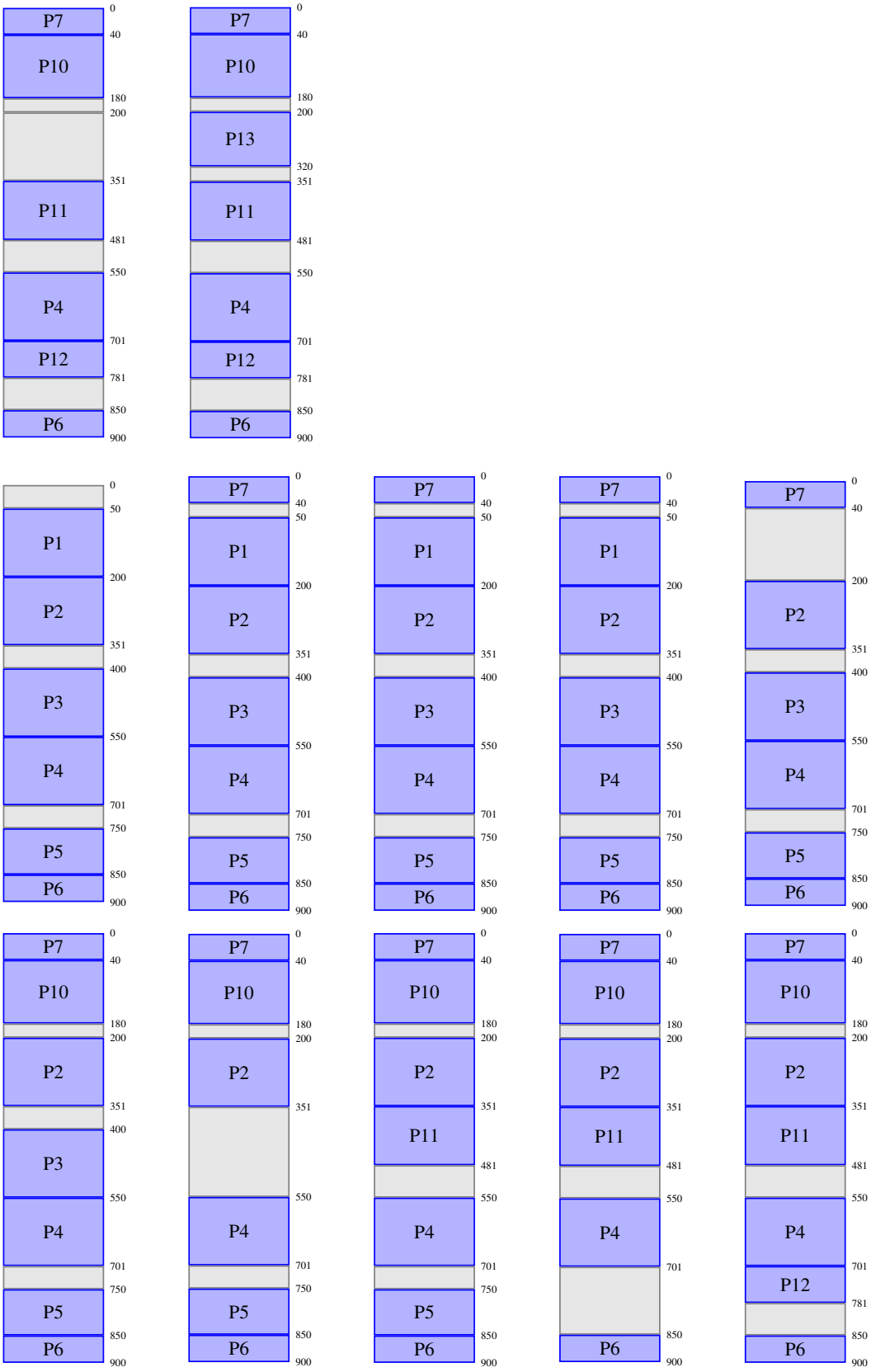
- P7 di dimensione 40Kb
- P8 di dimensione 60Kb
- P9 di dimensione 50Kb
- P1 esce
- P10 di dimensione 140Kb
- P3 esce
- P11 di dimensione 130Kb
- P5 esce
- P12 di dimensione 80Kb
- P2 esce
- P13 di dimensione 120Kb

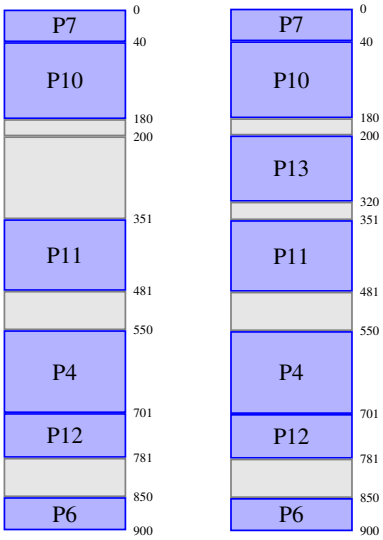



**SOLUZIONE**



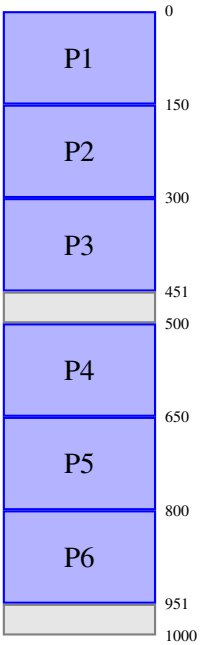

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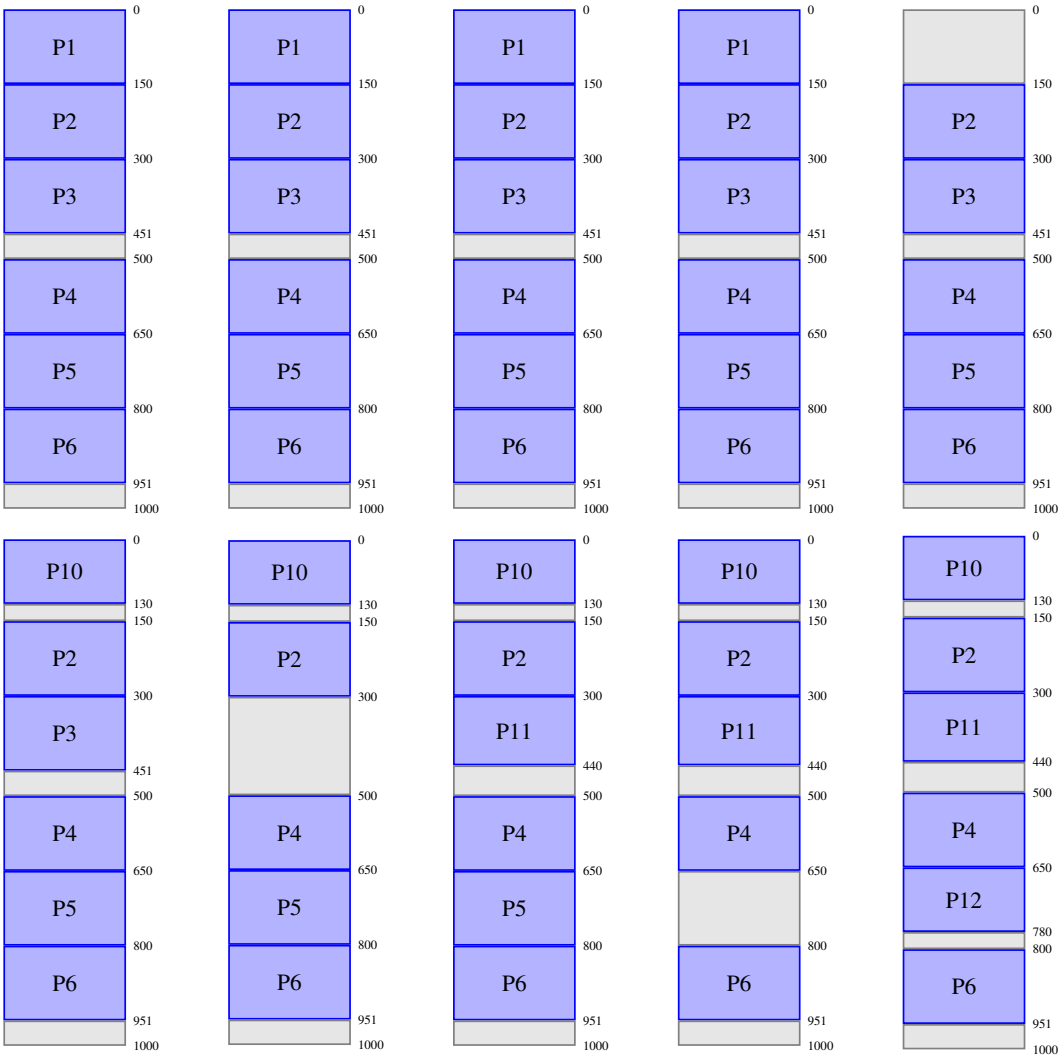

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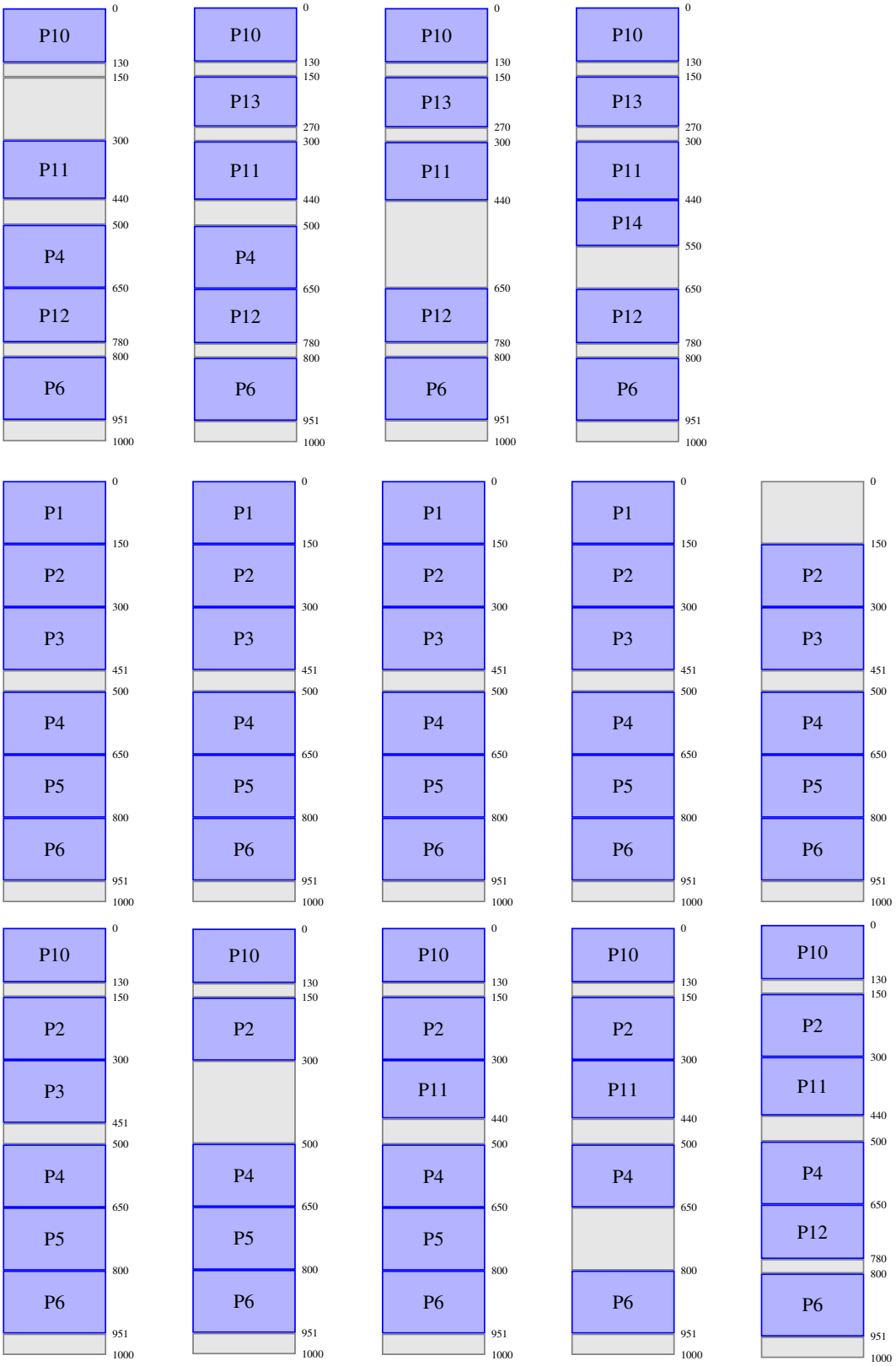
**20**

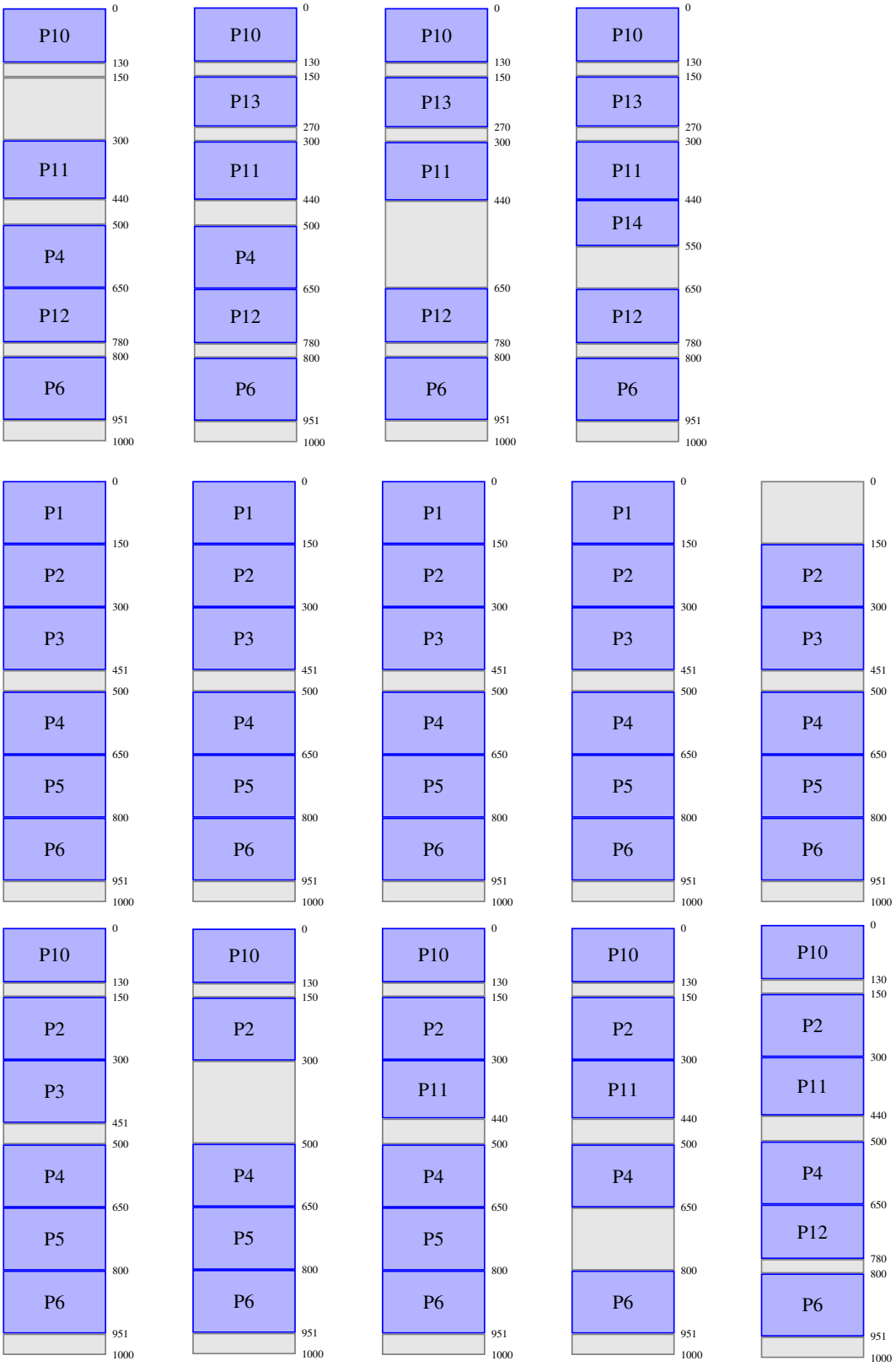
- P7 di dimensione 50Kb
- P8 di dimensione 70Kb
- P9 di dimensione 60Kb
- P1 esce
- P10 di dimensione 130Kb
- P3 esce
- P11 di dimensione 140Kb
- P5 esce
- P12 di dimensione 130Kb
- P2 esce
- P13 di dimensione 120Kb
- P4 esce
- P14 di dimensione 110Kb

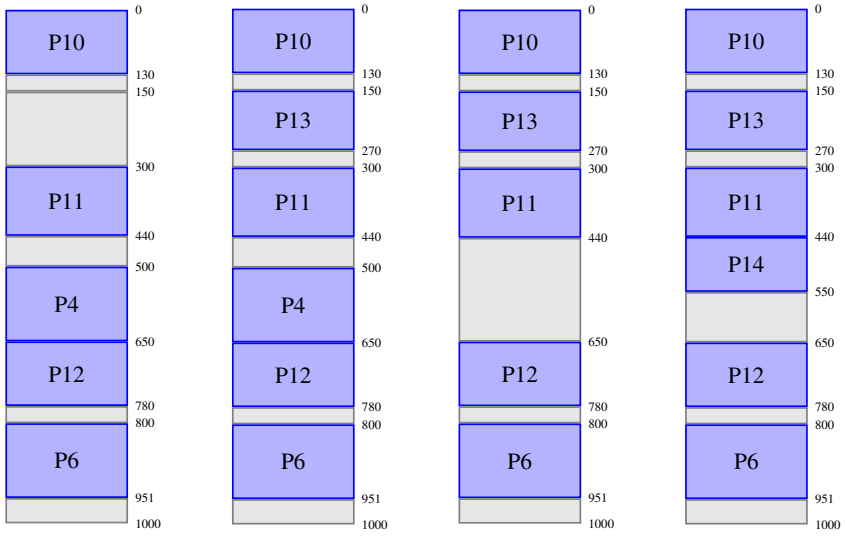



**SOLUZIONE**




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500
550
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780
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## 6.3 Allocazione delle pagine

Mostra l'evoluzione delle pagine allocate in memoria dati le seguenti pagine richieste usando gli algoritmi FIFO, LRU e OPT

**1**

Avendo a disposizione 3 frame

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 |
|---|---|---|---|---|---|---|---|---|

**SOLUZIONE**

**FIFO**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | |
|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 4<br>2<br>3 | 4<br>1<br>3 | 4<br>1<br>2 | 5<br>1<br>2 | 5<br>1<br>2 | 5<br>1<br>2 |
| F | F | F | F | F | F | F | | | |

**FIFO**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | |
|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 4<br>2<br>3 | 4<br>1<br>3 | 4<br>1<br>2 | 5<br>1<br>2 | 5<br>1<br>2 | 5<br>1<br>2 |
| F | F | F | F | F | F | F | | | |

**FIFO**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | |
|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 4<br>2<br>3 | 4<br>1<br>3 | 4<br>1<br>2 | 5<br>1<br>2 | 5<br>1<br>2 | 5<br>1<br>2 |
| F | F | F | F | F | F | F | | | |

**2**

Avendo a disposizione 3 frame

| 1 | 2 | 1 | 3 | 1 | 4 | 1 | 5 | 1 |
|---|---|---|---|---|---|---|---|---|

**SOLUZIONE**

**LRU**

| 1 | 2 | 1 | 3 | 1 | 4 | 1 | 5 | 1 | |
|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2 | 1<br>2<br>3 | 1<br>2<br>3 | 1<br>4<br>3 | 1<br>4<br>3 | 1<br>4<br>5 | 1<br>4<br>5 |
| F | F | | F | | F | | F | | |

**LRU**

| 1 | 2 | 1 | 3 | 1 | 4 | 1 | 5 | 1 | |
|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2 | 1<br>2<br>3 | 1<br>2<br>3 | 1<br>4<br>3 | 1<br>4<br>3 | 1<br>4<br>5 | 1<br>4<br>5 |
| F | F | | F | | F | | F | | |

**LRU**

| 1 | 2 | 1 | 3 | 1 | 4 | 1 | 5 | 1 | |
|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2 | 1<br>2<br>3 | 1<br>2<br>3 | 1<br>4<br>3 | 1<br>4<br>3 | 1<br>4<br>5 | 1<br>4<br>5 |
| F | F | | F | | F | | F | | |

3

Avendo a disposizione 3 frame

| 7 | 0 | 1 | 2 | 0 | 3 | 0 | 4 | 2 |
|---|---|---|---|---|---|---|---|---|

**SOLUZIONE**

**OPT**

| 7 | 0 | 1 | 2 | 0 | 3 | 0 | 4 | 2 | |
|---|---|---|---|---|---|---|---|---|---|
| | 7 | 7<br>0 | 7<br>0<br>1 | 2<br>0<br>1 | 2<br>0<br>1 | 2<br>0<br>3 | 2<br>0<br>3 | 2<br>4<br>3 | 2<br>4<br>3 |
| F | F | F | F | | F | | F | | |

**OPT**

| 7 | 0 | 1 | 2 | 0 | 3 | 0 | 4 | 2 | |
|---|---|---|---|---|---|---|---|---|---|
| | 7 | 7<br>0 | 7<br>0<br>1 | 2<br>0<br>1 | 2<br>0<br>1 | 2<br>0<br>3 | 2<br>0<br>3 | 2<br>4<br>3 | 2<br>4<br>3 |
| F | F | F | F | | F | | F | | |

**OPT**

| 7 | 0 | 1 | 2 | 0 | 3 | 0 | 4 | 2 | |
|---|---|---|---|---|---|---|---|---|---|
| | 7 | 7<br>0 | 7<br>0<br>1 | 2<br>0<br>1 | 2<br>0<br>1 | 2<br>0<br>3 | 2<br>0<br>3 | 2<br>4<br>3 | 2<br>4<br>3 |
| F | F | F | F | | F | | F | | |

**4**

Avendo a disposizione 4 frame

| 1 | 2 | 3 | 1 | 4 | 5 | 2 | 1 |
|---|---|---|---|---|---|---|---|

**SOLUZIONE**

**FIFO**

| 1 | 2 | 3 | 1 | 4 | 5 | 2 | 1 | |
|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 1<br>2<br>3 | 1<br>2<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>1<br>3<br>4 |
| F | F | F | | F | F | | F | |

**FIFO**

| 1 | 2 | 3 | 1 | 4 | 5 | 2 | 1 | |
|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 1<br>2<br>3 | 1<br>2<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>1<br>3<br>4 |
| F | F | F | | F | F | | F | |

**FIFO**

| 1 | 2 | 3 | 1 | 4 | 5 | 2 | 1 | |
|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 1<br>2<br>3 | 1<br>2<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>1<br>3<br>4 |
| F | F | F | | F | F | | F | |

**5**

Avendo a disposizione 4 frame

| 2 | 3 | 2 | 1 | 5 | 2 | 4 | 5 |
|---|---|---|---|---|---|---|---|

**SOLUZIONE**

**LRU**

| 2 | 3 | 2 | 1 | 5 | 2 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|
| | 2 | 23 | 23 | 231 | 2315 | 2315 | 2415 | 2415 |
| F | F | | F | F | | F | | |

**LRU**

| 2 | 3 | 2 | 1 | 5 | 2 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|
| | 2 | 23 | 23 | 231 | 2315 | 2315 | 2415 | 2415 |
| F | F | | F | F | | F | | |

**LRU**

| 2 | 3 | 2 | 1 | 5 | 2 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|
| | 2 | 23 | 23 | 231 | 2315 | 2315 | 2415 | 2415 |
| F | F | | F | F | | F | | |

6

Avendo a disposizione 3 frame

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|---|---|---|---|---|

**SOLUZIONE**

**LRU**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 4<br>2<br>3 | 4<br>1<br>3 | 4<br>1<br>2 | 5<br>1<br>2 | 5<br>1<br>2 | 5<br>1<br>2 | 3<br>1<br>2 | 3<br>4<br>2 | 3<br>4<br>5 |
| F | F | F | F | F | F | F | | | F | F | F | |

**LRU**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 4<br>2<br>3 | 4<br>1<br>3 | 4<br>1<br>2 | 5<br>1<br>2 | 5<br>1<br>2 | 5<br>1<br>2 | 3<br>1<br>2 | 3<br>4<br>2 | 3<br>4<br>5 |
| F | F | F | F | F | F | F | | | F | F | F | |

**LRU**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 4<br>2<br>3 | 4<br>1<br>3 | 4<br>1<br>2 | 5<br>1<br>2 | 5<br>1<br>2 | 5<br>1<br>2 | 3<br>1<br>2 | 3<br>4<br>2 | 3<br>4<br>5 |
| F | F | F | F | F | F | F | | | F | F | F | |

7

Avendo a disposizione 3 frame

| 7 | 0 | 1 | 2 | 0 | 3 | 0 | 4 | 2 | 3 | 0 | 3 |
|---|---|---|---|---|---|---|---|---|---|---|---|

**SOLUZIONE**

**FIFO**

| 7 | 0 | 1 | 2 | 0 | 3 | 0 | 4 | 2 | 3 | 0 | 3 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 7 | 7<br>0 | 7<br>0<br>1 | 2<br>0<br>1 | 2<br>0<br>1 | 2<br>3<br>1 | 2<br>3<br>0 | 4<br>3<br>0 | 4<br>2<br>0 | 4<br>2<br>3 | 0<br>2<br>3 | 0<br>2<br>3 |
| F | F | F | F | | F | F | F | F | F | F | | |

**FIFO**

| 7 | 0 | 1 | 2 | 0 | 3 | 0 | 4 | 2 | 3 | 0 | 3 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 7 | 7<br>0 | 7<br>0<br>1 | 2<br>0<br>1 | 2<br>0<br>1 | 2<br>3<br>1 | 2<br>3<br>0 | 4<br>3<br>0 | 4<br>2<br>0 | 4<br>2<br>3 | 0<br>2<br>3 | 0<br>2<br>3 |
| F | F | F | F | | F | F | F | F | F | F | | |

**FIFO**

| 7 | 0 | 1 | 2 | 0 | 3 | 0 | 4 | 2 | 3 | 0 | 3 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 7 | 7<br>0 | 7<br>0<br>1 | 2<br>0<br>1 | 2<br>0<br>1 | 2<br>3<br>1 | 2<br>3<br>0 | 4<br>3<br>0 | 4<br>2<br>0 | 4<br>2<br>3 | 0<br>2<br>3 | 0<br>2<br>3 |
| F | F | F | F | | F | F | F | F | F | F | | |

8

Avendo a disposizione 4 frame

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|---|---|---|---|---|

**SOLUZIONE**

**OPT**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>5 | 1<br>2<br>3<br>5 | 1<br>2<br>3<br>5 | 1<br>2<br>3<br>5 | 4<br>2<br>3<br>5 | 4<br>2<br>3<br>5 |
| F | F | F | F | | | F | | | | F | | | |

**OPT**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>5 | 1<br>2<br>3<br>5 | 1<br>2<br>3<br>5 | 1<br>2<br>3<br>5 | 4<br>2<br>3<br>5 | 4<br>2<br>3<br>5 |
| F | F | F | F | | | F | | | | F | | | |

**OPT**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>5 | 1<br>2<br>3<br>5 | 1<br>2<br>3<br>5 | 1<br>2<br>3<br>5 | 4<br>2<br>3<br>5 | 4<br>2<br>3<br>5 |
| F | F | F | F | | | F | | | | F | | | |

**9**

Avendo a disposizione 3 frame

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|---|---|---|---|---|

**SOLUZIONE**

**LFU**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 4<br>2<br>3 | 4<br>1<br>3 | 4<br>1<br>2 | 5<br>1<br>2 | 5<br>1<br>2 | 5<br>1<br>2 | 3<br>1<br>2 | 4<br>1<br>2 | 5<br>1<br>2 |
| F | F | F | F | F | F | F | | | F | F | F | |

**LFU**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 4<br>2<br>3 | 4<br>1<br>3 | 4<br>1<br>2 | 5<br>1<br>2 | 5<br>1<br>2 | 5<br>1<br>2 | 3<br>1<br>2 | 4<br>1<br>2 | 5<br>1<br>2 |
| F | F | F | F | F | F | F | | | F | F | F | |

**LFU**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 4<br>2<br>3 | 4<br>1<br>3 | 4<br>1<br>2 | 5<br>1<br>2 | 5<br>1<br>2 | 5<br>1<br>2 | 3<br>1<br>2 | 4<br>1<br>2 | 5<br>1<br>2 |
| F | F | F | F | F | F | F | | | F | F | F | |

**10** Avendo a disposizione 3 frame

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|---|---|---|---|---|

**SOLUZIONE**

**MFU**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 4<br>2<br>3 | 4<br>1<br>3 | 4<br>2<br>3 | 4<br>5<br>3 | 4<br>5<br>1 | 4<br>5<br>2 | 4<br>5<br>3 | 4<br>5<br>3 | 4<br>5<br>3 |
| F | F | F | F | F | F | F | F | F | F | | | |

**MFU**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 4<br>2<br>3 | 4<br>1<br>3 | 4<br>2<br>3 | 4<br>5<br>3 | 4<br>5<br>1 | 4<br>5<br>2 | 4<br>5<br>3 | 4<br>5<br>3 | 4<br>5<br>3 |
| F | F | F | F | F | F | F | F | F | F | | | |

**MFU**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 4<br>2<br>3 | 4<br>1<br>3 | 4<br>2<br>3 | 4<br>5<br>3 | 4<br>5<br>1 | 4<br>5<br>2 | 4<br>5<br>3 | 4<br>5<br>3 | 4<br>5<br>3 |
| F | F | F | F | F | F | F | F | F | F | | | |

**11**

Avendo a disposizione 4 frame

| 1 | 2 | 3 | 4 | 5 | 1 | 2 | 3 | 4 | 5 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

**SOLUZIONE**

**FIFO**

| 1 | 2 | 3 | 4 | 5 | 1 | 2 | 3 | 4 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 1<br>2<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>1<br>3<br>4 | 5<br>1<br>2<br>4 | 5<br>1<br>2<br>3 | 4<br>1<br>2<br>3 | 4<br>5<br>2<br>3 | 4<br>5<br>1<br>3 | 4<br>5<br>1<br>2 | 3<br>5<br>1<br>2 | 3<br>4<br>1<br>2 | 3<br>4<br>5<br>2 |
| F | F | F | F | F | F | F | F | F | F | F | F | F | F | F | |

**FIFO**

| 1 | 2 | 3 | 4 | 5 | 1 | 2 | 3 | 4 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 1<br>2<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>1<br>3<br>4 | 5<br>1<br>2<br>4 | 5<br>1<br>2<br>3 | 4<br>1<br>2<br>3 | 4<br>5<br>2<br>3 | 4<br>5<br>1<br>3 | 4<br>5<br>1<br>2 | 3<br>5<br>1<br>2 | 3<br>4<br>1<br>2 | 3<br>4<br>5<br>2 |
| F | F | F | F | F | F | F | F | F | F | F | F | F | F | F | |

**FIFO**

| 1 | 2 | 3 | 4 | 5 | 1 | 2 | 3 | 4 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 1<br>2<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>1<br>3<br>4 | 5<br>1<br>2<br>4 | 5<br>1<br>2<br>3 | 4<br>1<br>2<br>3 | 4<br>5<br>2<br>3 | 4<br>5<br>1<br>3 | 4<br>5<br>1<br>2 | 3<br>5<br>1<br>2 | 3<br>4<br>1<br>2 | 3<br>4<br>5<br>2 |
| F | F | F | F | F | F | F | F | F | F | F | F | F | F | F | |

**12**

Avendo a disposizione 4 frame

| 2 | 3 | 2 | 1 | 5 | 2 | 4 | 5 | 3 | 2 | 5 | 2 |
|---|---|---|---|---|---|---|---|---|---|---|---|

**SOLUZIONE**

**LRU**

| 2 | 3 | 2 | 1 | 5 | 2 | 4 | 5 | 3 | 2 | 5 | 2 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2 | 2<br>3 | 2<br>3 | 2<br>3<br>1 | 2<br>3<br>1<br>5 | 2<br>3<br>1<br>5 | 2<br>4<br>1<br>5 | 2<br>4<br>1<br>5 | 2<br>4<br>3<br>5 | 2<br>4<br>3<br>5 | 2<br>4<br>3<br>5 | 2<br>4<br>3<br>5 |
| F | F | | F | F | | F | | F | | | | |

**LRU**

| 2 | 3 | 2 | 1 | 5 | 2 | 4 | 5 | 3 | 2 | 5 | 2 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2 | 2<br>3 | 2<br>3 | 2<br>3<br>1 | 2<br>3<br>1<br>5 | 2<br>3<br>1<br>5 | 2<br>4<br>1<br>5 | 2<br>4<br>1<br>5 | 2<br>4<br>3<br>5 | 2<br>4<br>3<br>5 | 2<br>4<br>3<br>5 | 2<br>4<br>3<br>5 |
| F | F | | F | F | | F | | F | | | | |

**LRU**

| 2 | 3 | 2 | 1 | 5 | 2 | 4 | 5 | 3 | 2 | 5 | 2 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2 | 2<br>3 | 2<br>3 | 2<br>3<br>1 | 2<br>3<br>1<br>5 | 2<br>3<br>1<br>5 | 2<br>4<br>1<br>5 | 2<br>4<br>1<br>5 | 2<br>4<br>3<br>5 | 2<br>4<br>3<br>5 | 2<br>4<br>3<br>5 | 2<br>4<br>3<br>5 |
| F | F | | F | F | | F | | F | | | | |

**13**

Avendo a disposizione 3 frame

| 2 | 3 | 2 | 1 | 5 | 2 | 4 | 5 | 3 | 2 | 5 | 2 |
|---|---|---|---|---|---|---|---|---|---|---|---|

**SOLUZIONE**

**OPT**

| 2 | 3 | 2 | 1 | 5 | 2 | 4 | 5 | 3 | 2 | 5 | 2 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2 | 2<br>3 | 2<br>3 | 2<br>3<br>1 | 2<br>3<br>5 | 2<br>3<br>5 | 4<br>3<br>5 | 4<br>3<br>5 | 4<br>3<br>5 | 2<br>3<br>5 | 2<br>3<br>5 | 2<br>3<br>5 |
| F | F | | F | F | | F | | | F | | | |

**OPT**

| 2 | 3 | 2 | 1 | 5 | 2 | 4 | 5 | 3 | 2 | 5 | 2 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2 | 2<br>3 | 2<br>3 | 2<br>3<br>1 | 2<br>3<br>5 | 2<br>3<br>5 | 4<br>3<br>5 | 4<br>3<br>5 | 4<br>3<br>5 | 2<br>3<br>5 | 2<br>3<br>5 | 2<br>3<br>5 |
| F | F | | F | F | | F | | | F | | | |

**OPT**

| 2 | 3 | 2 | 1 | 5 | 2 | 4 | 5 | 3 | 2 | 5 | 2 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2 | 2<br>3 | 2<br>3 | 2<br>3<br>1 | 2<br>3<br>5 | 2<br>3<br>5 | 4<br>3<br>5 | 4<br>3<br>5 | 4<br>3<br>5 | 2<br>3<br>5 | 2<br>3<br>5 | 2<br>3<br>5 |
| F | F | | F | F | | F | | | F | | | |

**14**

Avendo a disposizione 4 frame

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|---|---|---|---|---|

**SOLUZIONE**

**LFU**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 1<br>2<br>5<br>4 | 1<br>2<br>5<br>4 | 1<br>2<br>5<br>4 | 1<br>2<br>5<br>3 | 1<br>2<br>4<br>3 | 1<br>2<br>4<br>5 |
| F | F | F | F | | | F | | | F | F | F | |

**LFU**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 1<br>2<br>5<br>4 | 1<br>2<br>5<br>4 | 1<br>2<br>5<br>4 | 1<br>2<br>5<br>3 | 1<br>2<br>4<br>3 | 1<br>2<br>4<br>5 |
| F | F | F | F | | | F | | | F | F | F | |

**LFU**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 1<br>2<br>5<br>4 | 1<br>2<br>5<br>4 | 1<br>2<br>5<br>4 | 1<br>2<br>5<br>3 | 1<br>2<br>4<br>3 | 1<br>2<br>4<br>5 |
| F | F | F | F | | | F | | | F | F | F | |

**15**

Avendo a disposizione 4 frame

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|---|---|---|---|---|

**SOLUZIONE**

**MFU**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>1<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>2<br>3<br>4 |
| F | F | F | F | | | F | F | F | | | | |

**MFU**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>1<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>2<br>3<br>4 |
| F | F | F | F | | | F | F | F | | | | |

**MFU**

| 1 | 2 | 3 | 4 | 1 | 2 | 5 | 1 | 2 | 3 | 4 | 5 | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 1<br>2 | 1<br>2<br>3 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 1<br>2<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>1<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>2<br>3<br>4 | 5<br>2<br>3<br>4 |
| F | F | F | F | | | F | F | F | | | | |